

P-type LST bases

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Contents

1	Description	4
1.1	Summary	4
2	List of P-type LSTs	6
2.1	0 node(506)	6
2.1.1	non-instanton Side Links + Noble Molecules with extra curve(123)	6
2.1.2	A,D series like LSTs(383)	7
2.2	1 node(947)	7
2.2.1	Sg(43)	7
2.2.2	SgS(388)	8
2.2.3	SSgS(333)	13
2.2.4	SSgIS(183)	17
2.3	2 node(1504)	20
2.3.1	gLg(4)	20
2.3.2	111gLg(3)	21
2.3.3	SgLg(115)	21
2.3.4	SgLgS(552)	23
2.3.5	SSgLg(73)	29
2.3.6	SSgLgS(576)	31
2.3.7	SSgLgIS(181)	38
2.4	3 node(1907)	41
2.4.1	gLgLg(11)	41
2.4.2	gLIgLg(8)	42
2.4.3	SgLgLg(170)	42
2.4.4	SgLgLgS(589)	46
2.4.5	SgLIgLg(65)	54
2.4.6	SgLIgLgS(128)	56
2.4.7	SSgLgLg(80)	58
2.4.8	SSgLgLgS(548)	61
2.4.9	SSgLgLIS(165)	69
2.4.10	SSgLIgLg(25)	72
2.4.11	SSgLIgLgS(96)	73
2.4.12	SSgLIgLIS(22)	75
2.5	4 node(1730)	75
2.5.1	gLgLgLg(16)	75
2.5.2	gLIgLgLg(11)	76
2.5.3	gLIgLIS(3)	77
2.5.4	SgLgLgLg(175)	77
2.5.5	SgLgLgLgS(500)	82
2.5.6	SgLgLIS(57)	90

2.5.7	SgLIgLg(28)	92
2.5.8	SgLIgLgS(93)	93
2.5.9	SgLIgLIg(3)	95
2.5.10	SgLIgLIgS(1)	95
2.5.11	SSgLgLg(76)	95
2.5.12	SSgLgLgS(471)	98
2.5.13	SSgLgLIgS(149)	105
2.5.14	SSgLgLIg(21)	108
2.5.15	SSgLgLIgS(43)	109
2.5.16	SSgLgLIgLIgS(15)	110
2.5.17	SSgLIgLg(13)	111
2.5.18	SSgLIgLgS(38)	111
2.5.19	SSgLIgLIgS(15)	112
2.5.20	SSgLIgLIg(2)	112
2.6	5 node(1332)	113
2.6.1	gLgLgLg(17)	113
2.6.2	gLgLIgLg(4)	114
2.6.3	gLIgLgLg(10)	114
2.6.4	gLIgLIg(1)	115
2.6.5	SgLgLgLg(133)	115
2.6.6	SgLgLgLgS(407)	119
2.6.7	SgLgLIg(27)	126
2.6.8	SgLgLIg(4)	127
2.6.9	SgLgLIgS(1)	128
2.6.10	SgLIgLg(12)	128
2.6.11	SgLIgLgS(43)	128
2.6.12	SgLIgLIg(1)	129
2.6.13	SgLIgLIgS(1)	129
2.6.14	SSgLgLg(63)	130
2.6.15	SSgLgLgS(402)	132
2.6.16	SSgLgLgLIgS(137)	139
2.6.17	SSgLgLIg(9)	141
2.6.18	SSgLgLIgS(25)	142
2.6.19	SSgLgLIgLIgS(6)	143
2.6.20	SSgLgLIg(2)	143
2.6.21	SSgLIgLg(4)	143
2.6.22	SSgLIgLgS(16)	143
2.6.23	SSgLIgLgLIgS(6)	144
2.6.24	SSgLIgLIg(1)	144
2.7	6 node(1127)	144
2.7.1	g6(12)	144
2.7.2	Sg6(81)	145
2.7.3	Sg6S(376)	147
2.7.4	SSg6(44)	153
2.7.5	SSg6S(383)	155
2.7.6	SSg6IS(135)	161
2.7.7	gLIg5(3)	164
2.7.8	SgLIg5(5)	164
2.7.9	SgLIg3LIg(1)	164
2.7.10	SgLIg3LIgS(1)	165
2.7.11	SSgLIg5(3)	165
2.7.12	SSgLIg3LIg(1)	165
2.7.13	SSgLIg5S(12)	165

2.7.14	SSgLIg5IS(4)	165
2.7.15	SSg4LIgLg(3)	166
2.7.16	SSg4LIgLgS(20)	166
2.7.17	SSg4LIgLIgS(4)	166
2.7.18	Sg4LIgLg(7)	167
2.7.19	SgLIg5S(32)	167
2.8	$n \geq 7$ node(10,217)	168
2.8.1	g(n)(47)	168
2.8.2	Sg(n)(549)	171
2.8.3	Sg(n)S(3795)	187
2.8.4	SSg(n)(356)	246
2.8.5	SSg(n)S(3769)	257
2.8.6	SSg(n)IS(1294)	315
2.8.7	gLIg(n-1)(3)	340
2.8.8	gLIgLg(n-4)LIgLg(1)	340
2.8.9	SgLIg(n-1)(26)	340
2.8.10	SgLIg(n-3)LIgLg(1)	342
2.8.11	SgLIg(n-3)LIgLgS(11)	342
2.8.12	SSgLIg(n-1)(1)	342
2.8.13	SSgLIg(n-1)S(1)	343
2.8.14	SSg(n-2)LIgLg(6)	343
2.8.15	SSg(n-2)LIgLgS(148)	343
2.8.16	Sg(n-2)LIgLg(15)	347
2.8.17	SgLIg(n-1)S(194)	348

1 Description

LSTs with P-type endpoint bases with $T \leq 192$ are presented.

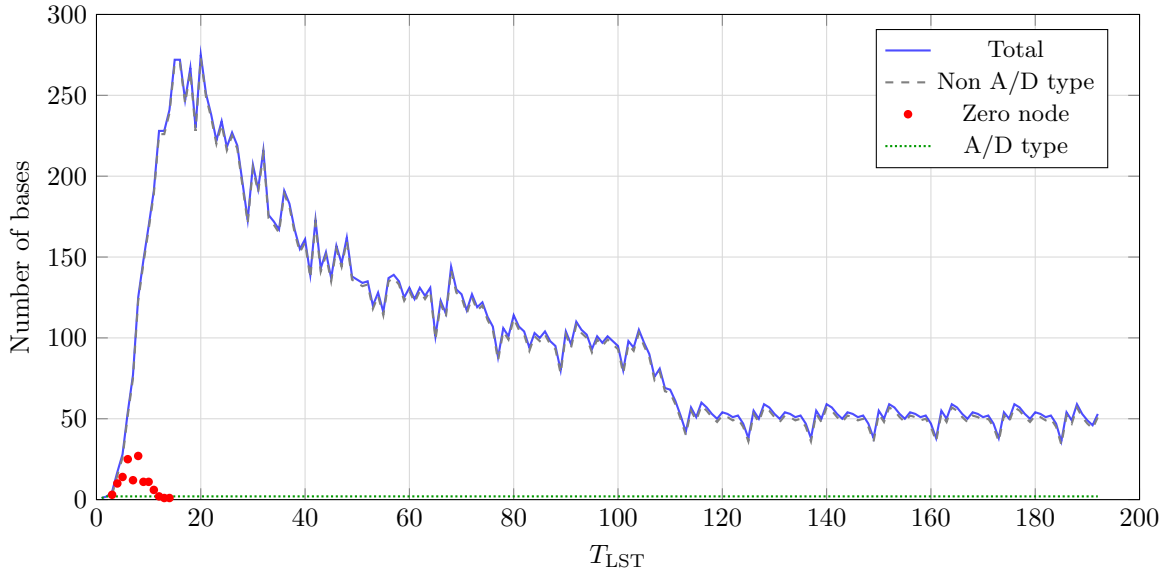
1.1 Summary

Summary

Category	Count
0 node	123
non-zero node	18,764
A/D type	383
Total	19,270

$T_{\text{LST}} \in [1, 192]$. A-type: $A(n) = \overbrace{12 \cdots 21}^n$ ($T = n - 1$, $T \geq 1$, 192 total). D-type: $D(n) = \overbrace{12^2 \cdots 21}^n$ ($T = n - 1$, $T \geq 2$, 191 total).

Distribution



Full Table (Catalog + A/D)

T	Count	T	Count	T	Count	T	Count
1	1	49	138	97	97	145	53
2	2	50	136	98	101	146	51
3	5	51	134	99	98	147	52
4	17	52	135	100	95	148	47
5	28	53	120	101	80	149	38
6	54	54	128	102	98	150	55
7	77	55	116	103	94	151	50
8	125	56	137	104	105	152	59
9	148	57	139	105	97	153	57
10	169	58	135	106	90	154	53
11	191	59	125	107	76	155	50
12	228	60	131	108	81	156	54
13	228	61	124	109	69	157	53
14	241	62	131	110	68	158	51
15	272	63	126	111	61	159	52
16	272	64	131	112	52	160	47
17	247	65	101	113	42	161	38
18	267	66	123	114	57	162	55
19	230	67	115	115	51	163	50
20	275	68	144	116	60	164	59
21	251	69	130	117	57	165	57
22	238	70	127	118	53	166	53
23	222	71	117	119	50	167	50
24	234	72	127	120	54	168	54
25	218	73	119	121	53	169	53
26	227	74	122	122	51	170	51
27	219	75	113	123	52	171	52
28	196	76	107	124	47	172	47
29	173	77	88	125	38	173	38
30	207	78	106	126	55	174	54
31	192	79	101	127	50	175	50
32	216	80	114	128	59	176	59
33	176	81	107	129	57	177	57
34	172	82	104	130	53	178	53
35	167	83	94	131	50	179	50
36	191	84	103	132	54	180	54
37	183	85	100	133	53	181	53
38	167	86	104	134	51	182	51
39	155	87	98	135	52	183	52
40	161	88	95	136	47	184	47
41	139	89	80	137	38	185	36
42	173	90	104	138	55	186	54
43	143	91	96	139	50	187	49
44	153	92	110	140	59	188	59
45	137	93	105	141	57	189	53
46	157	94	102	142	53	190	49
47	146	95	93	143	50	191	46
48	162	96	101	144	54	192	53

2 List of P-type LSTs

2.1 0 node(506)

2.1.1 non-instanton Side Links + Noble Molecules with extra curve(123)

221, ³	22221, ⁵	131 ²
131, ¹	12321, ³	12321 ¹
123151321, ¹	12231513221, ⁵	12231513221 ¹
1315131, ³	1315131, ²	1315131 ¹
1321, ²	1321, ¹	1321 ²
1321, ³	13221, ²	13221 ¹
13151321, ³	13151321, ²	13151321 ¹
131513221, ³	131513221, ²	131513221 ¹
131513221, ⁵	1231513221, ¹	1231513221 ⁵
231, ³	231, ¹	1231 ²³
231, ²²	2313221, ¹	2313221 ²
22313221, ¹	313221, ¹	313221 ⁵
231321, ¹	231321, ²	231321 ⁵
31321, ¹	23131, ²	223131 ⁵
3131, ¹	3131, ³	15131 ²¹
15131, ¹	15131, ¹²	15131 ¹³
151321, ²¹	151321, ¹	1513221 ²¹
1513221, ¹	1513221, ⁵	315131 ³
3151321, ³	3151321, ¹⁵	31513221 ³
2315131, ¹⁵	1512321, ²	1512321 ¹
321513221, ¹	2321513221, ¹	32151321 ¹
32151321, ⁵	232151321, ¹	31512321 ¹
31512321, ⁵	3215131, ¹	23215131 ⁵
151231, ²	151231, ¹	151231 ²
151231, ³	512231, ¹	215131 ²

$215131,$	$215131,$	2151321
$21513221,$	$21513221,$	15131513221
$15131513221,$	$1513151321,$	1513151321
$1513151321,$	$151315131,$	151315131
$51231513221,$	$2315131513221,$	22315131513221
$315131513221,$	$5123151321,$	5123151321
$231513151321,$	$31513151321,$	512315131
$3151315131,$	$3151315131,$	2132
$1511,$	$1511,$	15132
$1512,$	$1512,$	31512
$31512,$	$31513,$	13215132
$22215,$	$231315,$	215123
$221513,$	$21512,$	21513215
$31512315,$	$32151315,$	1513151
$1513151,$	$1513151,$	15132151
$21513151,$	$1513151315,$	51232151

2.1.2 A,D series like LSTs(383)

$$\overbrace{1222 \cdots 2}^n 1 \quad 1 \leq n \leq 192$$

$$\overbrace{222 \cdots 21}^n \quad 2 \leq n \leq 192$$

2.2 1 node(947)

2.2.1 Sg(43)

$132150,$	$132150,$	1231513150
$322150,$	$22150,$	313150
$223150,$	$51232150,$	2315132150
$22315132150,$	$315132150,$	31513150
$15123150,$	$21513150,$	5131513150
$15132150,$	15132150	

$\overset{5}{1}231513221\mathfrak{e}_6,$
 $\overset{2}{2}2221\mathfrak{e}_6,$
 $\overset{1}{2}3151321\mathfrak{e}_6,$
 $\overset{3}{2}1513221\mathfrak{e}_6,$
 $\overset{3}{1}513151321\mathfrak{e}_6$

$\overset{2}{3}13221\mathfrak{e}_6,$
 $\overset{1}{3}1513221\mathfrak{e}_6,$
 $\overset{2}{3}1512321\mathfrak{e}_6,$
 $\overset{1}{5}131513221\mathfrak{e}_6,$

$\overset{2}{2}31321\mathfrak{e}_6$
 $\overset{2}{3}2151321\mathfrak{e}_6$
 $\overset{1}{2}2315131\mathfrak{e}_6$
 $\overset{2}{3}151315131\mathfrak{e}_6$

$\overset{2}{2}313221\mathfrak{e}'_7,$
 $\overset{1}{2}31513221\mathfrak{e}'_7,$
 $\overset{2}{3}1513151321\mathfrak{e}'_7,$

$\overset{2}{2}2221\mathfrak{e}'_7,$
 $\overset{1}{2}23151321\mathfrak{e}'_7,$
 $\overset{3}{1}5131513221\mathfrak{e}'_7$

$\overset{2}{3}21513221\mathfrak{e}'_7$
 $\overset{2}{2}31512321\mathfrak{e}'_7$

$\overset{2}{2}22221\mathfrak{e}_7,$
 $\overset{2}{2}31513151321\mathfrak{e}_7$

$\overset{1}{2}231513221\mathfrak{e}_7,$

$\overset{2}{3}15131513221\mathfrak{e}_7$

$\overset{2}{2}222222221\mathfrak{e}_8$

2.2.2 SgS(388)

$\overset{1}{2}2313221\mathfrak{e}_7,$
 $\overset{1}{2}231512321\mathfrak{e}_7,$
 $\overset{1}{2}31513221\mathfrak{e}_7,$
 $\overset{2,3}{4,5} \otimes \mathfrak{e}_7,$
 $\overset{2,3}{2}31321\mathfrak{e}_7,$
 $\overset{2,3}{3}1512321\mathfrak{e}_7,$
 $\overset{2,3}{3}151321\mathfrak{e}_7,$
 $\overset{2,4}{2,4} \otimes \mathfrak{e}_7,$
 $\overset{2,4}{5,4} \otimes \mathfrak{e}_7,$
 $\overset{2,4}{2}151321\mathfrak{e}_7,$

$I^{\oplus 7} \mathfrak{e}_7,$
 $\overset{1}{3}15131513221\mathfrak{e}_7,$
 $\overset{1}{2}23151321\mathfrak{e}_7,$
 $\overset{2,3}{5,5} \otimes \mathfrak{e}_7,$
 $I^{\oplus 5} \mathfrak{e}_7,$
 $\overset{2,3}{2}1513221\mathfrak{e}_7,$
 $\overset{2,3}{1}513151321\mathfrak{e}_7,$
 $\overset{2,4}{3,4} \otimes \mathfrak{e}_7,$
 $I^{\oplus 4} \mathfrak{e}_7,$
 $\overset{2,4}{1}2321\mathfrak{e}_7,$

$\overset{1}{2}2321513221\mathfrak{e}_7$
 $\overset{1}{2}31513151321\mathfrak{e}_7$
 $\overset{2,3}{3,5} \otimes \mathfrak{e}_7$
 $\overset{2,3}{3}13221\mathfrak{e}_7$
 $\overset{2,3}{3}2151321\mathfrak{e}_7$
 $\overset{2,3}{5}123151321\mathfrak{e}_7$
 $\overset{2,3}{1}513221\mathfrak{e}_7$
 $\overset{2,4}{4,4} \otimes \mathfrak{e}_7$
 $\overset{2,4}{3}1321\mathfrak{e}_7$
 $\overset{2,4}{1}512321\mathfrak{e}_7$

$$\begin{array}{lll}
1 & 2,4 & \\
151321 \otimes \mathfrak{e}_7, & 3,5 \otimes \mathfrak{e}_7, & 4,5 \otimes \mathfrak{e}_7 \\
5,5 \otimes \mathfrak{e}_7, & 3,3 & 3,3 \\
I^{\oplus 5} \otimes \mathfrak{e}_7, & 32151321 \otimes \mathfrak{e}_7, & 231321 \otimes \mathfrak{e}_7 \\
21513221 \otimes \mathfrak{e}_7, & 5123151321 \otimes \mathfrak{e}_7, & 31512321 \otimes \mathfrak{e}_7 \\
1513151321 \otimes \mathfrak{e}_7, & 1 & 1 \\
31321 \otimes \mathfrak{e}_7, & 1513221 \otimes \mathfrak{e}_7, & 3151321 \otimes \mathfrak{e}_7 \\
1512321 \otimes \mathfrak{e}_7, & 5,4 \otimes \mathfrak{e}_7, & 3,4 \otimes \mathfrak{e}_7 \\
4,4 \otimes \mathfrak{e}_7, & 2151321 \otimes \mathfrak{e}_7, & I^{\oplus 4} \otimes \mathfrak{e}_7 \\
31321 \otimes \mathfrak{e}_7, & 151321 \otimes \mathfrak{e}_7, & 2321 \otimes \mathfrak{e}_7 \\
1512321 \otimes \mathfrak{e}_7, & 5,4 \otimes \mathfrak{e}_7, & I^{\oplus 3} \otimes \mathfrak{e}_7 \\
4,4 \otimes \mathfrak{e}_7, & 2151321 \otimes \mathfrak{e}_7, & I^{\oplus 4} \otimes \mathfrak{e}_7 \\
31321 \otimes \mathfrak{e}_7, & 151321 \otimes \mathfrak{e}_7, & 2321 \otimes \mathfrak{e}_7 \\
1512321 \otimes \mathfrak{e}_7, & I^{\oplus 4} \otimes \mathfrak{e}_7, & I^{\oplus 3} \otimes \mathfrak{e}_7 \\
5,4 \otimes \mathfrak{e}_7, & 2321 \otimes \mathfrak{e}_7, & 31321 \otimes \mathfrak{e}_7 \\
2151321 \otimes \mathfrak{e}_7, & I^{\oplus 3} \otimes \mathfrak{e}_7, & 1512321 \otimes \mathfrak{e}_7 \\
151321 \otimes \mathfrak{e}_7, & I^{\oplus 2} \otimes \mathfrak{e}_7, & 513151321 \otimes \mathfrak{e}_7 \\
1513221 \otimes \mathfrak{e}_7, & I^{\oplus 3} \otimes \mathfrak{e}_7, & I^{\oplus 3} \otimes \mathfrak{e}_7 \\
5131513221 \otimes \mathfrak{e}_7, & I^{\oplus 3} \otimes \mathfrak{e}_7, & I^{\oplus 6} \otimes \mathfrak{e}_7 \\
2231321 \otimes \mathfrak{e}_7, & 321513221 \otimes \mathfrak{e}_7, & 232151321 \otimes \mathfrak{e}_7 \\
231512321 \otimes \mathfrak{e}_7, & 51231513221 \otimes \mathfrak{e}_7, & 31513151321 \otimes \mathfrak{e}_7 \\
31513221 \otimes \mathfrak{e}_7, & 23151321 \otimes \mathfrak{e}_7, & 15131513221 \otimes \mathfrak{e}_7 \\
I^{\oplus 5} \otimes \mathfrak{e}_7, & 32151321 \otimes \mathfrak{e}_7, & 31512321 \otimes \mathfrak{e}_7 \\
21513221 \otimes \mathfrak{e}_7, & 5123151321 \otimes \mathfrak{e}_7, & 3151321 \otimes \mathfrak{e}_7 \\
1513151321 \otimes \mathfrak{e}_7, & 1513221 \otimes \mathfrak{e}_7, & I^{\oplus 4} \otimes \mathfrak{e}_7 \\
31321 \otimes \mathfrak{e}_7, & 2151321 \otimes \mathfrak{e}_7, & 2321 \otimes \mathfrak{e}_7 \\
1512321 \otimes \mathfrak{e}_7, & 151321 \otimes \mathfrak{e}_7, & 31321 \otimes \mathfrak{e}_7 \\
2151321 \otimes \mathfrak{e}_7, & 2321 \otimes \mathfrak{e}_7, & 1512321 \otimes \mathfrak{e}_7 \\
151321 \otimes \mathfrak{e}_7, & 3151321 \otimes \mathfrak{e}_7, & 51321 \otimes \mathfrak{e}_7 \\
513151321 \otimes \mathfrak{e}_7, & 1513221 \otimes \mathfrak{e}_7, & 2151321 \otimes \mathfrak{e}_7
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
\overset{1}{2321} \overset{2151321}{\mathfrak{e}_7}, \\
151321 \overset{513151321}{\mathfrak{e}_7}, \\
\overset{1}{151321} \overset{2321}{\mathfrak{e}_7}, \\
\overset{1}{151321} \overset{1512321}{\mathfrak{e}_7},
\end{array}
&
\begin{array}{l}
1512321 \overset{2151321}{\mathfrak{e}_7}, \\
\overset{1}{2321} \overset{2321}{\mathfrak{e}_7}, \\
151321 \overset{1513221}{\mathfrak{e}_7}, \\
\overset{1}{151321} \overset{151321}{\mathfrak{e}_7}
\end{array}
&
\begin{array}{l}
\overset{1}{151321} \overset{2151321}{\mathfrak{e}_7} \\
1512321 \overset{2321}{\mathfrak{e}_7} \\
1512321 \overset{1512321}{\mathfrak{e}_7}
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
\overset{3,5}{\otimes} \overset{1}{\mathfrak{e}_6}, \\
313221 \overset{1}{\mathfrak{e}_6}, \\
223131 \overset{1}{\mathfrak{e}_6}, \\
23215131 \overset{1}{\mathfrak{e}_6}, \\
3151315131 \overset{1}{\mathfrak{e}_6}, \\
1513151321 \overset{1}{\mathfrak{e}_6}, \\
\overset{3,4}{\otimes} \overset{2,2}{\mathfrak{e}_6}, \\
I^{\oplus 4} \overset{2,2}{\otimes} \mathfrak{e}_6, \\
3215131 \overset{2,2}{\otimes} \mathfrak{e}_6, \\
\overset{1}{2321} \overset{2,2}{\otimes} \mathfrak{e}_6, \\
151315131 \overset{2,2}{\otimes} \mathfrak{e}_6, \\
\overset{3,3}{\otimes} \overset{2,3}{\mathfrak{e}_6}, \\
\overset{3,3}{\otimes} \overset{2,3}{\mathfrak{e}_6}, \\
\bigcirc \overset{2,3}{\mathfrak{e}_6}, \\
215131 \overset{2,3}{\otimes} \mathfrak{e}_6, \\
\overset{3,3}{\otimes} \overset{3,3}{\mathfrak{e}_6}, \\
\overset{3,3}{\otimes} \overset{3,3}{\mathfrak{e}_6}, \\
\bigcirc \overset{3,3}{\mathfrak{e}_6}, \\
215131 \overset{3,3}{\otimes} \mathfrak{e}_6, \\
\overset{4,3}{\otimes} \overset{4,3}{\mathfrak{e}_6}, \\
I^{\oplus 3} \overset{4,3}{\otimes} \mathfrak{e}_6, \\
\overset{1}{15131} \overset{4,3}{\otimes} \mathfrak{e}_6, \\
\overset{3,3}{\otimes} \overset{5,3}{\mathfrak{e}_6}, \\
\bigcirc \mathfrak{e}_6,
\end{array}
&
\begin{array}{l}
\overset{4,5}{\otimes} \overset{1}{\mathfrak{e}_6}, \\
231321 \overset{1}{\mathfrak{e}_6}, \\
32151321 \overset{1}{\mathfrak{e}_6}, \\
21513221 \overset{1}{\mathfrak{e}_6}, \\
\overset{1}{3151321} \overset{1}{\mathfrak{e}_6}, \\
\overset{1}{1513221} \overset{1}{\mathfrak{e}_6}, \\
\overset{4,4}{\otimes} \overset{2,2}{\mathfrak{e}_6}, \\
31321 \overset{2,2}{\otimes} \mathfrak{e}_6, \\
2151321 \overset{2,2}{\otimes} \mathfrak{e}_6, \\
\overset{1}{315131} \overset{2,2}{\otimes} \mathfrak{e}_6, \\
\overset{1}{151321} \overset{2,2}{\otimes} \mathfrak{e}_6, \\
\overset{4,3}{\otimes} \overset{2,3}{\mathfrak{e}_6}, \\
I^{\oplus 3} \overset{2,3}{\otimes} \mathfrak{e}_6, \\
\overset{1}{15131} \overset{2,3}{\otimes} \mathfrak{e}_6, \\
\overset{4,3}{\otimes} \overset{3,3}{\mathfrak{e}_6}, \\
I^{\oplus 3} \overset{3,3}{\otimes} \mathfrak{e}_6, \\
\overset{1}{15131} \overset{3,3}{\otimes} \mathfrak{e}_6, \\
\overset{5,3}{\otimes} \overset{4,3}{\mathfrak{e}_6}, \\
\overset{4,3}{\otimes} \mathfrak{e}_6, \\
3131 \overset{4,3}{\otimes} \mathfrak{e}_6, \\
I^{\oplus 2} \overset{4,4}{\otimes} \mathfrak{e}_6, \\
I^{\oplus 3} \overset{5,3}{\otimes} \mathfrak{e}_6,
\end{array}
&
\begin{array}{l}
\overset{5,5}{\otimes} \overset{1}{\mathfrak{e}_6} \\
I^{\oplus 5} \overset{1}{\mathfrak{e}_6} \\
31512321 \overset{1}{\mathfrak{e}_6} \\
5123151321 \overset{1}{\mathfrak{e}_6} \\
2315131 \overset{1}{\mathfrak{e}_6} \\
\overset{2,4}{\otimes} \overset{2,2}{\mathfrak{e}_6} \\
\overset{5,4}{\otimes} \overset{2,2}{\mathfrak{e}_6} \\
23131 \overset{2,2}{\otimes} \mathfrak{e}_6 \\
512315131 \overset{2,2}{\otimes} \mathfrak{e}_6 \\
1512321 \overset{2,2}{\otimes} \mathfrak{e}_6 \\
\overset{2,3}{\otimes} \overset{2,3}{\mathfrak{e}_6} \\
\overset{5,3}{\otimes} \overset{2,3}{\mathfrak{e}_6} \\
3131 \overset{2,3}{\otimes} \mathfrak{e}_6 \\
I^{\oplus 2} \overset{2,4}{\otimes} \mathfrak{e}_6 \\
\overset{5,3}{\otimes} \overset{3,3}{\mathfrak{e}_6} \\
\overset{3,3}{\otimes} \mathfrak{e}_6 \\
3131 \overset{3,3}{\otimes} \mathfrak{e}_6 \\
I^{\oplus 2} \overset{3,4}{\otimes} \mathfrak{e}_6 \\
\overset{3,3}{\otimes} \overset{4,3}{\mathfrak{e}_6} \\
\bigcirc \mathfrak{e}_6 \\
215131 \overset{4,3}{\otimes} \mathfrak{e}_6 \\
\overset{5,3}{\otimes} \overset{5,3}{\mathfrak{e}_6} \\
\overset{5,3}{\otimes} \mathfrak{e}_6 \\
3131 \overset{5,3}{\otimes} \mathfrak{e}_6
\end{array}
\end{array}$$

$$\begin{aligned}
& 215131 \overset{5,3}{\otimes} \mathfrak{e}_6, \\
& 321 \overset{3221}{\mathfrak{e}_6}, \\
& 51315131 \overset{321}{\mathfrak{e}_6}, \\
& \overset{3,3}{\bigcirc} \mathfrak{e}_6, \\
& 215131 \overset{3,3}{\bigcirc} \mathfrak{e}_6, \\
& 31321 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 2151321 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& \overset{1}{315131} \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& \overset{1}{151321} \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 215131 \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
& 1513221 \overset{31}{\mathfrak{e}_6}, \\
& \overset{1}{15131} \overset{3131}{\mathfrak{e}_6}, \\
& 5131 \overset{3151321}{\mathfrak{e}_6}, \\
& 15131 \overset{51315131}{\mathfrak{e}_6},
\end{aligned}$$

$$\begin{aligned}
& \overset{1}{15131} \overset{5,3}{\otimes} \mathfrak{e}_6, \\
& 15131 \overset{3221}{\mathfrak{e}_6}, \\
& 151321 \overset{321}{\mathfrak{e}_6}, \\
& \overset{3,3}{\bigcirc} \mathfrak{e}_6, \\
& \overset{1}{15131} \overset{3,3}{\bigcirc} \mathfrak{e}_6, \\
& 23131 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 512315131 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 1512321 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
& \overset{1}{15131} \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
& 3131 \overset{3131}{\mathfrak{e}_6}, \\
& 315131 \overset{51321}{\mathfrak{e}_6}, \\
& 215131 \overset{215131}{\mathfrak{e}_6}, \\
& 151321 \overset{15131}{\mathfrak{e}_6},
\end{aligned}$$

$$\begin{aligned}
& \overset{5,4}{\otimes} \mathfrak{e}_6, \\
& 512321 \overset{321}{\mathfrak{e}_6}, \\
& 51321 \overset{2321}{\mathfrak{e}_6}, \\
& 3131 \overset{3,3}{\bigcirc} \mathfrak{e}_6, \\
& \overset{I^{\oplus 4}}{\mathfrak{e}_6}, \\
& 3215131 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& \overset{1}{2321} \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 151315131 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 3131 \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
& 513151321 \overset{31}{\mathfrak{e}_6}, \\
& 215131 \overset{3131}{\mathfrak{e}_6}, \\
& 15131 \overset{512321}{\mathfrak{e}_6}, \\
& 15131 \overset{1}{\mathfrak{e}_6} \overset{215131}{\mathfrak{e}_6}, \\
& 15131 \overset{1}{\mathfrak{e}_6} \overset{15131}{\mathfrak{e}_6},
\end{aligned}$$

$$\begin{aligned}
& \overset{I^{\oplus 11}}{\mathfrak{e}_8}, \\
& \overset{2,4}{\otimes} \mathfrak{e}_8, \\
& 2321513221 \overset{3,5}{\otimes} \mathfrak{e}_8, \\
& 22313221 \overset{4,5}{\otimes} \mathfrak{e}_8, \\
& 315131513221 \overset{4,5}{\otimes} \mathfrak{e}_8, \\
& \overset{5,5}{\otimes} \mathfrak{e}_8, \\
& \overset{1}{231513221} \overset{5,5}{\otimes} \mathfrak{e}_8, \\
& 321513221 \overset{2313221}{\mathfrak{e}_8}, \\
& 15131513221 \overset{2313221}{\mathfrak{e}_8}, \\
& 21513221 \overset{22313221}{\mathfrak{e}_8}, \\
& 2321513221 \overset{313221}{\mathfrak{e}_8}, \\
& \overset{I^{\oplus 10}}{\mathfrak{e}_8}, \\
& 2315131513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8},
\end{aligned}$$

$$\begin{aligned}
& 2315131513221 \overset{2,4}{\otimes} \mathfrak{e}_8, \\
& 22313221 \overset{3,5}{\otimes} \mathfrak{e}_8, \\
& 315131513221 \overset{3,5}{\otimes} \mathfrak{e}_8, \\
& \overset{4,5}{\otimes} \mathfrak{e}_8, \\
& \overset{1}{231513221} \overset{4,5}{\otimes} \mathfrak{e}_8, \\
& 2321513221 \overset{5,5}{\otimes} \mathfrak{e}_8, \\
& 2313221 \overset{2313221}{\mathfrak{e}_8}, \\
& 51231513221 \overset{2313221}{\mathfrak{e}_8}, \\
& 313221 \overset{22313221}{\mathfrak{e}_8}, \\
& \overset{1}{1513221} \overset{22313221}{\mathfrak{e}_8}, \\
& 315131513221 \overset{313221}{\mathfrak{e}_8}, \\
& \overset{I^{\oplus 9}}{\mathfrak{e}_8}, \\
& 2231513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8},
\end{aligned}$$

$$\begin{aligned}
& 2231513221 \overset{2,4}{\otimes} \mathfrak{e}_8, \\
& \overset{3,5}{\otimes} \mathfrak{e}_8, \\
& \overset{1}{231513221} \overset{3,5}{\otimes} \mathfrak{e}_8, \\
& 2321513221 \overset{4,5}{\otimes} \mathfrak{e}_8, \\
& 22313221 \overset{5,5}{\otimes} \mathfrak{e}_8, \\
& 315131513221 \overset{5,5}{\otimes} \mathfrak{e}_8, \\
& \overset{I^{\oplus 6}}{\mathfrak{e}_8}, \\
& \overset{1}{31513221} \overset{2313221}{\mathfrak{e}_8}, \\
& \overset{I^{\oplus 5}}{\mathfrak{e}_8}, \\
& \overset{I^{\oplus 7}}{\mathfrak{e}_8}, \\
& \overset{1}{231513221} \overset{313221}{\mathfrak{e}_8}, \\
& 22315131513221 \overset{I^{\oplus 3}}{\mathfrak{e}_8}, \\
& \overset{I^{\oplus 8}}{\mathfrak{e}_8},
\end{aligned}$$

$$\begin{array}{lll}
I^{\oplus 7} \mathfrak{e}_8, & 2321513221 \mathfrak{e}_8^{I^{\oplus 5}}, & 315131513221 \mathfrak{e}_8^{I^{\oplus 5}} \\
231513221 \mathfrak{e}_8^{I^{\oplus 5}}, & I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 6}}, & 321513221 \mathfrak{e}_8^{I^{\oplus 6}} \\
51231513221 \mathfrak{e}_8^{I^{\oplus 6}}, & 31513221 \mathfrak{e}_8^{I^{\oplus 6}}, & 15131513221 \mathfrak{e}_8^{I^{\oplus 6}} \\
21513221 \mathfrak{e}_8^{I^{\oplus 7}}, & 1513221 \mathfrak{e}_8^{I^{\oplus 7}}, & 321513221 \mathfrak{e}_8^{321513221} \\
51231513221 \mathfrak{e}_8^{321513221}, & 31513221 \mathfrak{e}_8^{321513221}, & 15131513221 \mathfrak{e}_8^{321513221} \\
21513221 \mathfrak{e}_8^{2321513221}, & 1513221 \mathfrak{e}_8^{2321513221}, & 315131513221 \mathfrak{e}_8^{21513221} \\
231513221 \mathfrak{e}_8^{21513221}, & 51231513221 \mathfrak{e}_8^{51231513221}, & 31513221 \mathfrak{e}_8^{51231513221} \\
15131513221 \mathfrak{e}_8^{51231513221}, & 1513221 \mathfrak{e}_8^{315131513221}, & 31513221 \mathfrak{e}_8^{31513221} \\
15131513221 \mathfrak{e}_8^{31513221}, & 1513221 \mathfrak{e}_8^{231513221}, & 15131513221 \mathfrak{e}_8^{15131513221}
\end{array}$$

$$\begin{array}{lll}
2313221 \mathfrak{e}'_7, & I^{\oplus 6} \mathfrak{e}'_7, & 2231321 \mathfrak{e}'_7 \\
321513221 \mathfrak{e}'_7, & 232151321 \mathfrak{e}'_7, & 231512321 \mathfrak{e}'_7 \\
51231513221 \mathfrak{e}'_7, & 31513151321 \mathfrak{e}'_7, & 31513221 \mathfrak{e}'_7 \\
23151321 \mathfrak{e}'_7, & 15131513221 \mathfrak{e}'_7, & \begin{array}{c} 2,3 \\ 2,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array} \\
\begin{array}{c} 2,3 \\ 3,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 2,3 \\ 4,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 2,3 \\ 5,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array} \\
I^{\oplus 4} \mathfrak{e}'_7, & \begin{array}{c} 2,3 \\ \otimes \end{array}, & \begin{array}{c} 2,3 \\ \otimes \end{array} \\
\begin{array}{c} 2,3 \\ \otimes \end{array}, & \begin{array}{c} 2,3 \\ \otimes \end{array}, & \begin{array}{c} 2,3 \\ \otimes \end{array} \\
2321 \mathfrak{e}'_7, & 1512321 \mathfrak{e}'_7, & 151321 \mathfrak{e}'_7 \\
\begin{array}{c} 2,4 \\ 3,3 \otimes \\ \otimes \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 2,4 \\ I^{\oplus 3} \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 3,3 \\ 3,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array} \\
\begin{array}{c} 3,3 \\ 4,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 3,3 \\ 5,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 3,3 \\ I^{\oplus 4} \mathfrak{e}'_7 \end{array} \\
\begin{array}{c} 3,3 \\ \otimes \end{array}, & \begin{array}{c} 3,3 \\ \otimes \end{array}, & \begin{array}{c} 3,3 \\ \otimes \end{array} \\
31321 \mathfrak{e}'_7, & 2151321 \mathfrak{e}'_7, & 2321 \mathfrak{e}'_7 \\
\begin{array}{c} 3,3 \\ \otimes \end{array}, & \begin{array}{c} 3,3 \\ \otimes \end{array}, & \begin{array}{c} 3,4 \\ \otimes \end{array} \\
1512321 \mathfrak{e}'_7, & 151321 \mathfrak{e}'_7, & I^{\oplus 3} \mathfrak{e}'_7 \\
\begin{array}{c} 3,5 \\ \otimes \end{array}, & \begin{array}{c} 4,4 \\ \otimes \end{array}, & \begin{array}{c} 4,5 \\ \otimes \end{array} \\
I^{\oplus 2} \mathfrak{e}'_7, & I^{\oplus 3} \mathfrak{e}'_7, & I^{\oplus 2} \mathfrak{e}'_7 \\
\begin{array}{c} 5,4 \\ \otimes \end{array}, & \begin{array}{c} 5,5 \\ \otimes \end{array}, & \\
I^{\oplus 3} \mathfrak{e}'_7, & I^{\oplus 2} \mathfrak{e}'_7, & \begin{array}{c} 3221 \\ 3221 \mathfrak{e}'_7 \end{array} \\
512321 \mathfrak{e}'_7, & 151321 \mathfrak{e}'_7, & I^{\oplus 2} \mathfrak{e}'_7 \\
513151321 \mathfrak{e}'_7, & 1513221 \mathfrak{e}'_7, & 513221 \mathfrak{e}'_7 \\
I^{\oplus 2} \mathfrak{e}'_7, & I^{\oplus 5} \mathfrak{e}'_7, & 32151321 \mathfrak{e}'_7
\end{array}$$

$$\begin{aligned}
& 31512321 \overset{I^{\oplus 2}}{\mathfrak{e}'_7}, \\
& 3151321 \overset{1}{\overset{I^{\oplus 2}}{\mathfrak{e}'_7}}, \\
& I^{\oplus 4} \overset{I^{\oplus 3}}{\mathfrak{e}'_7}, \\
& 2321 \overset{1}{\overset{I^{\oplus 3}}{\mathfrak{e}'_7}}, \\
& 3151321 \overset{51321}{\mathfrak{e}'_7}, \\
& 151321 \overset{151321}{\mathfrak{e}'_7}
\end{aligned}$$

$$\begin{aligned}
& 21513221 \overset{I^{\oplus 2}}{\mathfrak{e}'_7}, \\
& 1513151321 \overset{I^{\oplus 2}}{\mathfrak{e}'_7}, \\
& 31321 \overset{I^{\oplus 3}}{\mathfrak{e}'_7}, \\
& 1512321 \overset{I^{\oplus 3}}{\mathfrak{e}'_7}, \\
& 512321 \overset{512321}{\mathfrak{e}'_7},
\end{aligned}$$

$$\begin{aligned}
& 5123151321 \overset{I^{\oplus 2}}{\mathfrak{e}'_7} \\
& 1513221 \overset{1}{\overset{I^{\oplus 2}}{\mathfrak{e}'_7}} \\
& 2151321 \overset{I^{\oplus 3}}{\mathfrak{e}'_7} \\
& 151321 \overset{1}{\overset{I^{\oplus 3}}{\mathfrak{e}'_7}} \\
& 151321 \overset{512321}{\mathfrak{e}'_7}
\end{aligned}$$

$$\begin{aligned}
& \overset{2,3}{\otimes} \overset{1}{\mathfrak{so}}, \\
& \overset{5,3}{\otimes} \overset{1}{\mathfrak{so}}, \\
& 3131 \overset{1}{\mathfrak{so}}, \\
& 215131 \overset{1}{\mathfrak{so}}, \\
& 15131 \overset{1}{\mathfrak{so}}, \\
& \overset{4,2}{\otimes} \overset{2,2}{\mathfrak{so}}, \\
& \overset{3,2}{\otimes} \overset{3,2}{\mathfrak{so}}, \\
& I^{\oplus 2} \overset{3,2}{\otimes} \mathfrak{so}, \\
& I^{\oplus 2} \overset{4,2}{\otimes} \mathfrak{so}, \\
& I^{\oplus 2} \overset{I^{\oplus 2}}{\mathfrak{so}}, \\
& 5131 \overset{231}{\mathfrak{so}}
\end{aligned}$$

$$\begin{aligned}
& \overset{3,3}{\otimes} \overset{1}{\mathfrak{so}}, \\
& \overset{3,3}{\bigcirc} \overset{1}{\mathfrak{so}}, \\
& 151231 \overset{1}{\mathfrak{so}}, \\
& 231 \overset{1}{\overset{1}{\mathfrak{so}}}, \\
& \overset{2,2}{\otimes} \overset{2,2}{\mathfrak{so}}, \\
& 231 \overset{2}{\overset{2,2}{\otimes} \mathfrak{so}}, \\
& \overset{4,2}{\otimes} \overset{3,2}{\mathfrak{so}}, \\
& \overset{4,2}{\otimes} \overset{4,2}{\mathfrak{so}}, \\
& I^{\oplus 2} \overset{2}{\overset{231}{\mathfrak{so}}}, \\
& 51231 \overset{31}{\mathfrak{so}},
\end{aligned}$$

$$\begin{aligned}
& \overset{4,3}{\otimes} \overset{1}{\mathfrak{so}} \\
& I^{\oplus 3} \overset{1}{\mathfrak{so}} \\
& 512231 \overset{1}{\mathfrak{so}} \\
& 1231 \overset{2}{\overset{1}{\mathfrak{so}}} \\
& \overset{3,2}{\otimes} \overset{2,2}{\mathfrak{so}} \\
& I^{\oplus 2} \overset{2}{\overset{2,2}{\otimes} \mathfrak{so}} \\
& 231 \overset{2}{\overset{3,2}{\otimes} \mathfrak{so}} \\
& 231 \overset{2}{\overset{4,2}{\otimes} \mathfrak{so}} \\
& 31 \overset{321}{\mathfrak{so}} \\
& 15131 \overset{31}{\mathfrak{so}}
\end{aligned}$$

2.2.3 SSgS(333)

$$I^{\oplus 10} \overset{1}{\mathfrak{e}_8} 1,$$

$$2321513221 \overset{2,4}{\otimes} \mathfrak{e}_8 1,$$

$$2313221 \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$51231513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$2313221 \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$51231513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$2313221 \overset{5,5}{\otimes} \mathfrak{e}_8 1,$$

$$22313221 \overset{2,4}{\otimes} \mathfrak{e}_8 1,$$

$$315131513221 \overset{2,4}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 6} \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$31513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 6} \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$31513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 6} \overset{5,5}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 7} \overset{2,4}{\otimes} \mathfrak{e}_8 1$$

$$231513221 \overset{1}{\overset{2,4}{\otimes} \mathfrak{e}_8} 1$$

$$321513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1$$

$$15131513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1$$

$$321513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1$$

$$15131513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1$$

$$321513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1$$

$$\begin{array}{lll}
51231513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1, & 31513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1, & 15131513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1 \\
313221 \overset{2313221}{\mathfrak{e}_8} 1, & I^{\oplus 5} \overset{2313221}{\mathfrak{e}_8} 1, & 21513221 \overset{2313221}{\mathfrak{e}_8} 1 \\
1513221 \overset{1}{\mathfrak{e}_8} \overset{2313221}{1}, & I^{\oplus 4} \overset{22313221}{\mathfrak{e}_8} 1, & I^{\oplus 6} \overset{313221}{\mathfrak{e}_8} 1 \\
321513221 \overset{313221}{\mathfrak{e}_8} 1, & 51231513221 \overset{313221}{\mathfrak{e}_8} 1, & 31513221 \overset{1}{\mathfrak{e}_8} \overset{313221}{1} \\
15131513221 \overset{313221}{\mathfrak{e}_8} 1, & I^{\oplus 9} \overset{I^{\oplus 2}}{\mathfrak{e}_8} 1, & 22315131513221 \overset{I^{\oplus 2}}{\mathfrak{e}_8} 1 \\
2315131513221 \overset{I^{\oplus 3}}{\mathfrak{e}_8} 1, & 2231513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 3}}{1}, & I^{\oplus 8} \overset{I^{\oplus 3}}{\mathfrak{e}_8} 1 \\
I^{\oplus 7} \overset{I^{\oplus 4}}{\mathfrak{e}_8} 1, & 2321513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8} 1, & 315131513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8} 1 \\
231513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 4}}{1}, & I^{\oplus 6} \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1, & 321513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1 \\
51231513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1, & 31513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 5}}{1}, & 15131513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1 \\
21513221 \overset{I^{\oplus 6}}{\mathfrak{e}_8} 1, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 6}}{1}, & 21513221 \overset{321513221}{\mathfrak{e}_8} 1 \\
1513221 \overset{1}{\mathfrak{e}_8} \overset{321513221}{1}, & 51231513221 \overset{21513221}{\mathfrak{e}_8} 1, & 31513221 \overset{1}{\mathfrak{e}_8} \overset{21513221}{1} \\
15131513221 \overset{21513221}{\mathfrak{e}_8} 1, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{51231513221}{1}, & 5131513221 \overset{5131513221}{\mathfrak{e}_8} 1 \\
1513221 \overset{1}{\mathfrak{e}_8} \overset{31513221}{1}, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{15131513221}{1}, & \overset{2,4}{\otimes} \overset{2,4}{\mathfrak{e}_8} \overset{4,2}{\otimes} \\
I^{\oplus 4} \overset{2,4}{\mathfrak{e}_8} \overset{4,2}{\otimes}, & I^{\oplus 3} \overset{3,5}{\mathfrak{e}_8} \overset{4,2}{\otimes}, & I^{\oplus 3} \overset{4,5}{\mathfrak{e}_8} \overset{4,2}{\otimes} \\
I^{\oplus 3} \overset{5,5}{\mathfrak{e}_8} \overset{4,2}{\otimes}, & 1513221 \overset{32214,2}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{23132214,2}{\mathfrak{e}_8} \otimes \\
I^{\oplus 3} \overset{3132214,2}{\mathfrak{e}_8} \otimes, & I^{\oplus 6} \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes, & 321513221 \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes \\
51231513221 \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes, & 31513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 24,2}}{\otimes}, & 15131513221 \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes \\
I^{\oplus 5} \overset{I^{\oplus 34,2}}{\mathfrak{e}_8} \otimes, & 21513221 \overset{I^{\oplus 34,2}}{\mathfrak{e}_8} \otimes, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 34,2}}{\otimes} \\
I^{\oplus 4} \overset{I^{\oplus 44,2}}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{3,5}{\mathfrak{e}_8} \overset{5,3}{\otimes}, & I^{\oplus 2} \overset{4,5}{\mathfrak{e}_8} \overset{5,3}{\otimes} \\
I^{\oplus 2} \overset{5,5}{\mathfrak{e}_8} \overset{5,3}{\otimes}, & 3221 \overset{32215,3}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{3132215,3}{\mathfrak{e}_8} \otimes \\
I^{\oplus 5} \overset{I^{\oplus 25,3}}{\mathfrak{e}_8} \otimes, & 21513221 \overset{I^{\oplus 25,3}}{\mathfrak{e}_8} \otimes, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 25,3}}{\otimes} \\
I^{\oplus 4} \overset{I^{\oplus 35,3}}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{4,5}{\mathfrak{e}_8} \overset{5,4}{\otimes}, & I^{\oplus 2} \overset{5,5}{\mathfrak{e}_8} \overset{5,4}{\otimes} \\
3221 \overset{32215,4}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{3132215,4}{\mathfrak{e}_8} \otimes, & I^{\oplus 5} \overset{I^{\oplus 25,4}}{\mathfrak{e}_8} \otimes \\
21513221 \overset{I^{\oplus 25,4}}{\mathfrak{e}_8} \otimes, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 25,4}}{\otimes}, & I^{\oplus 4} \overset{I^{\oplus 35,4}}{\mathfrak{e}_8} \otimes \\
I^{\oplus 2} \overset{5,5}{\mathfrak{e}_8} \overset{5,5}{\otimes}, & 3221 \overset{32215,5}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{3132215,5}{\mathfrak{e}_8} \otimes \\
I^{\oplus 5} \overset{I^{\oplus 25,5}}{\mathfrak{e}_8} \otimes, & 21513221 \overset{I^{\oplus 25,5}}{\mathfrak{e}_8} \otimes, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 25,5}}{\otimes} \\
I^{\oplus 4} \overset{I^{\oplus 35,5}}{\mathfrak{e}_8} \otimes, & 313221 \overset{3221}{\mathfrak{e}_8} 1223, & I^{\oplus 5} \overset{3221}{\mathfrak{e}_8} 1223 \\
21513221 \overset{3221}{\mathfrak{e}_8} 1223, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{3221}{1223}, & 5131513221 \overset{I^{\oplus 3}}{\mathfrak{e}_8} 1223
\end{array}$$

$$\begin{array}{lll}
1513221 \mathfrak{e}_8^{I^{\oplus 4}} 1223, & I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} 1223132, & I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}} 1223132 \\
I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2}} 12231322, & I^{\oplus 2} \mathfrak{e}_8^{313221} 122313, & I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 2}} 122313 \\
21513221 \mathfrak{e}_8^{I^{\oplus 2}} 122313, & \mathfrak{e}_8^{11513221} I^{\oplus 2} 122313, & I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3}} 122313 \\
2315131513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2}, & 2231513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2}, & I^{\oplus 8} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2} \\
I^{\oplus 7} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2}, & 2321513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2}, & 315131513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2} \\
231513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2}, & I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2}, & 321513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2} \\
51231513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2}, & \mathfrak{e}_8^{131513221} I^{\oplus 4} I^{\oplus 2}, & 15131513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2} \\
I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 5}} I^{\oplus 2}, & 21513221 \mathfrak{e}_8^{I^{\oplus 5}} I^{\oplus 2}, & \mathfrak{e}_8^{11513221} I^{\oplus 5} I^{\oplus 2} \\
21513221 \mathfrak{e}_8^{21513221} I^{\oplus 2}, & \mathfrak{e}_8^{11513221} I^{\oplus 2} I^{\oplus 2}, & 1513221 \mathfrak{e}_8^{5131513221} I^{\oplus 2} \\
\mathfrak{e}_8^{11513221} I^{\oplus 2} I^{\oplus 2}, & I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3}, & 321513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3} \\
51231513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3}, & \mathfrak{e}_8^{131513221} I^{\oplus 3} I^{\oplus 3}, & 15131513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3} \\
I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 3}, & 21513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 3}, & \mathfrak{e}_8^{11513221} I^{\oplus 4} I^{\oplus 3} \\
31513221 \mathfrak{e}_8^{513221} I^{\oplus 3}, & 1513221 \mathfrak{e}_8^{1513221} I^{\oplus 3}, & I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 4}
\end{array}$$

$$\begin{array}{lll}
2313221 \mathfrak{e}_7^1 1, & I^{\oplus 6} \mathfrak{e}_7^1 1, & 2231321 \mathfrak{e}_7^1 1 \\
321513221 \mathfrak{e}_7^1 1, & 232151321 \mathfrak{e}_7^1 1, & 231512321 \mathfrak{e}_7^1 1 \\
51231513221 \mathfrak{e}_7^1 1, & 31513151321 \mathfrak{e}_7^1 1, & \mathfrak{e}_7^{131513221} 1 \\
\mathfrak{e}_7^{123151321} 1, & 15131513221 \mathfrak{e}_7^1 1, & \mathfrak{e}_7^{2,4} \otimes \mathfrak{e}_7^{2,3} 1 \\
\mathfrak{e}_7^{3,4} \otimes \mathfrak{e}_7^{2,3} 1, & \mathfrak{e}_7^{4,4} \otimes \mathfrak{e}_7^{2,3} 1, & \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{2,3} 1 \\
I^{\oplus 4} \mathfrak{e}_7^{2,3} 1, & 31321 \mathfrak{e}_7^{2,3} 1, & 2151321 \mathfrak{e}_7^{2,3} 1 \\
\mathfrak{e}_7^{12321} 1, & 1512321 \mathfrak{e}_7^{2,3} 1, & \mathfrak{e}_7^{1151321} 1 \\
\mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{2,4} 1, & I^{\oplus 3} \mathfrak{e}_7^{2,4} 1, & \mathfrak{e}_7^{3,4} \otimes \mathfrak{e}_7^{3,3} 1 \\
\mathfrak{e}_7^{4,4} \otimes \mathfrak{e}_7^{3,3} 1, & \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} 1, & I^{\oplus 4} \mathfrak{e}_7^{3,3} 1 \\
31321 \mathfrak{e}_7^{3,3} 1, & 2151321 \mathfrak{e}_7^{3,3} 1, & \mathfrak{e}_7^{12321} 1 \\
1512321 \mathfrak{e}_7^{3,3} 1, & \mathfrak{e}_7^{1151321} 1, & I^{\oplus 3} \mathfrak{e}_7^{3,4} 1 \\
I^{\oplus 2} \mathfrak{e}_7^{3,5} 1, & I^{\oplus 3} \mathfrak{e}_7^{4,4} 1, & I^{\oplus 2} \mathfrak{e}_7^{4,5} 1 \\
I^{\oplus 3} \mathfrak{e}_7^{5,4} 1, & I^{\oplus 2} \mathfrak{e}_7^{5,5} 1, & 3221 \mathfrak{e}_7^{3221} 1 \\
512321 \mathfrak{e}_7^{3221} 1, & 151321 \mathfrak{e}_7^{3221} 1, & I^{\oplus 2} \mathfrak{e}_7^{313221} 1
\end{array}$$

$513151321 \mathfrak{e}_7^{321} 1,$	$1513221 \mathfrak{e}_7^{321} 1,$	$513221 \mathfrak{e}_7^{2321} 1$
$I^{\oplus 2} \mathfrak{e}_7^{231321} 1,$	$I^{\oplus 5} \mathfrak{e}_7^{I^{\oplus 2}},$	$32151321 \mathfrak{e}_7^{I^{\oplus 2}} 1$
$31512321 \mathfrak{e}_7^{I^{\oplus 2}} 1,$	$21513221 \mathfrak{e}_7^{I^{\oplus 2}} 1,$	$5123151321 \mathfrak{e}_7^{I^{\oplus 2}} 1$
$\overset{1}{3151321} \mathfrak{e}_7^{I^{\oplus 2}} 1,$	$1513151321 \mathfrak{e}_7^{I^{\oplus 2}} 1,$	$\overset{1}{1513221} \mathfrak{e}_7^{I^{\oplus 2}} 1$
$I^{\oplus 4} \mathfrak{e}_7^{I^{\oplus 3}} 1,$	$31321 \mathfrak{e}_7^{I^{\oplus 3}} 1,$	$2151321 \mathfrak{e}_7^{I^{\oplus 3}} 1$
$\overset{1}{2321} \mathfrak{e}_7^{I^{\oplus 3}} 1,$	$1512321 \mathfrak{e}_7^{I^{\oplus 3}} 1,$	$\overset{1}{151321} \mathfrak{e}_7^{I^{\oplus 3}} 1$
$3151321 \mathfrak{e}_7^{51321} 1,$	$512321 \mathfrak{e}_7^{512321} 1,$	$151321 \mathfrak{e}_7^{512321} 1$
$151321 \mathfrak{e}_7^{151321} 1,$	$I^{\oplus 2} \mathfrak{e}_7^{\otimes 2,3,2},$	$I^{\oplus 2} \mathfrak{e}_7^{\otimes 3,3,2}$
$321 \mathfrak{e}_7^{\otimes 321,2},$	$I^{\oplus 3} \mathfrak{e}_7^{\otimes 23,2},$	$I^{\oplus 2} \mathfrak{e}_7^{\otimes 24,2}$
$I^{\oplus 2} \mathfrak{e}_7^{\otimes 3,3,3},$	$321 \mathfrak{e}_7^{\otimes 321,3,3},$	$I^{\oplus 3} \mathfrak{e}_7^{\otimes 23,3}$
$I^{\oplus 2} \mathfrak{e}_7^{\otimes 24,3},$	$I^{\oplus 2} \mathfrak{e}_7^{\otimes 24,4},$	$I^{\oplus 2} \mathfrak{e}_7^{\otimes 24,5}$
$I^{\oplus 2} \mathfrak{e}_7^{321} 1223,$	$I^{\oplus 3} \mathfrak{e}_7^{321} 123,$	$512321 \mathfrak{e}_7^{I^{\oplus 2}} 123$
$151321 \mathfrak{e}_7^{I^{\oplus 2}} 123,$	$51321 \mathfrak{e}_7^{I^{\oplus 2}} 1232,$	$I^{\oplus 4} \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2}$
$31321 \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2},$	$2151321 \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2},$	$\overset{1}{2321} \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2}$
$1512321 \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2},$	$\overset{1}{151321} \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2},$	$I^{\oplus 3} \mathfrak{e}_7^{I^{\oplus 3}} I^{\oplus 2}$
$\overset{2,4}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{3,4}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{4,4}{\otimes} \mathfrak{e}_6^1 1$
$\overset{5,4}{\otimes} \mathfrak{e}_6^1 1,$	$I^{\oplus 4} \mathfrak{e}_6^1 1,$	$31321 \mathfrak{e}_6^1 1$
$23131 \mathfrak{e}_6^1 1,$	$3215131 \mathfrak{e}_6^1 1,$	$2151321 \mathfrak{e}_6^1 1$
$512315131 \mathfrak{e}_6^1 1,$	$\overset{1}{2321} \mathfrak{e}_6^1 1,$	$\overset{1}{315131} \mathfrak{e}_6^1 1$
$1512321 \mathfrak{e}_6^1 1,$	$151315131 \mathfrak{e}_6^1 1,$	$\overset{1}{151321} \mathfrak{e}_6^1 1$
$\overset{2,3}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{3,3}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{4,3}{\otimes} \mathfrak{e}_6^1 1$
$\overset{5,3}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{3,3}{\odot} \mathfrak{e}_6^1 1,$	$I^{\oplus 3} \mathfrak{e}_6^1 1$
$\overset{2,2}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{2,2}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{1}{15131} \mathfrak{e}_6^1 1$
$I^{\oplus 2} \mathfrak{e}_6^1 1,$	$I^{\oplus 2} \mathfrak{e}_6^1 1,$	$I^{\oplus 2} \mathfrak{e}_6^1 1$
$I^{\oplus 2} \mathfrak{e}_6^1 1,$	$31 \mathfrak{e}_6^1 1,$	$321 \mathfrak{e}_6^1 1$
$15131 \mathfrak{e}_6^1 1,$	$5131 \mathfrak{e}_6^1 1,$	$I^{\oplus 2} \mathfrak{e}_6^1 1$
$I^{\oplus 3} \mathfrak{e}_6^1 1,$	$3131 \mathfrak{e}_6^1 1,$	$215131 \mathfrak{e}_6^1 1$

$15131 \overset{1}{\mathfrak{e}}_6 I^{\oplus 2},$	$512321 \overset{31}{\mathfrak{e}}_6 1,$	$51315131 \overset{31}{\mathfrak{e}}_6 1$
$151321 \overset{31}{\mathfrak{e}}_6 1,$	$315131 \overset{5131}{\mathfrak{e}}_6 1,$	$15131 \overset{15131}{\mathfrak{e}}_6 1$
$2,2 \overset{2,2}{\otimes} 2,2$ $\otimes \mathfrak{e}_6 \otimes,$	$I^{\oplus 2} \overset{2,2}{\otimes} 2,2$ $\mathfrak{e}_6 \otimes,$	$31 \overset{3212,2}{\mathfrak{e}}_6 \otimes$
$I^{\oplus 2} \overset{I^{\oplus 2} 2,2}{\mathfrak{e}}_6 \otimes,$	$15131 \overset{312,2}{\mathfrak{e}}_6 \otimes,$	$31 \overset{313,2}{\mathfrak{e}}_6 \otimes$
$31 \overset{313,3}{\mathfrak{e}}_6 \otimes,$	$31 \overset{313,4}{\mathfrak{e}}_6 \otimes,$	$31 \overset{313,5}{\mathfrak{e}}_6 \otimes$
$31 \overset{I^{\oplus 2}}{\mathfrak{e}}_6 123,$	$31 \overset{313,3}{\mathfrak{e}}_6 \bigcirc,$	$I^{\oplus 2} \overset{I^{\oplus 2}}{\mathfrak{e}}_6 I^{\oplus 2}$
$15131 \overset{31}{\mathfrak{e}}_6 I^{\oplus 2},$	$31 \overset{31}{\mathfrak{e}}_6 I^{\oplus 3},$	$3131 \overset{31}{\mathfrak{e}}_6 13$
$215131 \overset{31}{\mathfrak{e}}_6 13,$	$15131 \overset{31}{\mathfrak{e}}_6 13$	
$3,5 \overset{1}{\otimes} \mathfrak{e}'_7 1,$	$4,5 \overset{1}{\otimes} \mathfrak{e}'_7 1,$	$5,5 \overset{1}{\otimes} \mathfrak{e}'_7 1$
$313221 \overset{1}{\mathfrak{e}}'_7 1,$	$231321 \overset{1}{\mathfrak{e}}'_7 1,$	$I^{\oplus 5} \overset{1}{\mathfrak{e}}'_7 1$
$32151321 \overset{1}{\mathfrak{e}}'_7 1,$	$31512321 \overset{1}{\mathfrak{e}}'_7 1,$	$21513221 \overset{1}{\mathfrak{e}}'_7 1$
$5123151321 \overset{1}{\mathfrak{e}}'_7 1,$	$3151321 \overset{1}{\mathfrak{e}}'_7 1,$	$1513151321 \overset{1}{\mathfrak{e}}'_7 1$
$1513221 \overset{1}{\mathfrak{e}}'_7 1,$	$2,3 \overset{2,3}{\otimes} \mathfrak{e}'_7 1,$	$3,3 \overset{2,3}{\otimes} \mathfrak{e}'_7 1$
$I^{\oplus 3} \overset{2,3}{\otimes} \mathfrak{e}'_7 1,$	$I^{\oplus 2} \overset{2,4}{\otimes} \mathfrak{e}'_7 1,$	$3,3 \overset{3,3}{\otimes} \mathfrak{e}'_7 1$
$I^{\oplus 3} \overset{3,3}{\otimes} \mathfrak{e}'_7 1,$	$I^{\oplus 2} \overset{3,4}{\otimes} \mathfrak{e}'_7 1,$	$I^{\oplus 2} \overset{4,4}{\otimes} \mathfrak{e}'_7 1$
$I^{\oplus 2} \overset{5,4}{\otimes} \mathfrak{e}'_7 1,$	$321 \overset{3221}{\mathfrak{e}}'_7 1,$	$512321 \overset{321}{\mathfrak{e}}'_7 1$
$151321 \overset{321}{\mathfrak{e}}'_7 1,$	$51321 \overset{2321}{\mathfrak{e}}'_7 1,$	$I^{\oplus 4} \overset{I^{\oplus 2}}{\mathfrak{e}}'_7 1$
$31321 \overset{I^{\oplus 2}}{\mathfrak{e}}'_7 1,$	$2151321 \overset{I^{\oplus 2}}{\mathfrak{e}}'_7 1,$	$2321 \overset{1}{\mathfrak{e}}'_7 I^{\oplus 2}$
$1512321 \overset{I^{\oplus 2}}{\mathfrak{e}}'_7 1,$	$151321 \overset{1}{\mathfrak{e}}'_7 I^{\oplus 2},$	$I^{\oplus 3} \overset{I^{\oplus 3}}{\mathfrak{e}}'_7 1$
$I^{\oplus 2} \overset{I^{\oplus 2} 3,2}{\mathfrak{e}}'_7 \otimes,$	$I^{\oplus 2} \overset{I^{\oplus 2} 3,3}{\mathfrak{e}}'_7 \otimes,$	$I^{\oplus 2} \overset{321}{\mathfrak{e}}'_7 123$
$I^{\oplus 3} \overset{I^{\oplus 2}}{\mathfrak{e}}'_7 I^{\oplus 2}$		
$2,2 \overset{1}{\otimes} \mathfrak{so} 1,$	$3,2 \overset{1}{\otimes} \mathfrak{so} 1,$	$4,2 \overset{1}{\otimes} \mathfrak{so} 1$
$231 \overset{2}{\mathfrak{so}} 1,$	$I^{\oplus 2} \overset{1}{\mathfrak{so}} 1,$	$31 \overset{31}{\mathfrak{so}} 1$

2.2.4 SSgIS(183)

$I^{\oplus 9} \overset{1}{\mathfrak{e}}_8 1,$	$22315131513221 \overset{1}{\mathfrak{e}}_8 1,$	$2313221 \overset{1}{\mathfrak{e}}_8 \overset{4,2}{\otimes} 1$
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$$\begin{array}{l}
I^{\oplus 6} \mathfrak{e}_8^{1\ 4,2} \otimes, \\
31513221 \mathfrak{e}_8^{1\ 4,2} \otimes, \\
\otimes \mathfrak{e}_8^{4,5\ 1\ 5,3} \otimes, \\
I^{\oplus 5} \mathfrak{e}_8^{1\ 5,3} \otimes, \\
\otimes \mathfrak{e}_8^{4,5\ 1\ 5,4} \otimes, \\
I^{\oplus 5} \mathfrak{e}_8^{1\ 5,4} \otimes, \\
\otimes \mathfrak{e}_8^{5,5\ 1\ 5,5} \otimes, \\
21513221 \mathfrak{e}_8^{1\ 5,5} \otimes, \\
22313221 \mathfrak{e}_8^{I^{\oplus 3}} 1, \\
21513221 \mathfrak{e}_8^{1\ 122313}, \\
2231513221 \mathfrak{e}_8^{I^{\oplus 2}} 1, \\
2321513221 \mathfrak{e}_8^{I^{\oplus 3}} 1, \\
I^{\oplus 6} \mathfrak{e}_8^{1\ 1} I^{\oplus 4}, \\
31513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 4}, \\
21513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 5}, \\
1513221 \mathfrak{e}_8^{1\ 1} 12231512, \\
\otimes \mathfrak{e}_8^{2,4\ I^{\oplus 3} 4,2} \otimes, \\
\otimes \mathfrak{e}_8^{5,5\ I^{\oplus 2} 4,2} \otimes, \\
I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3} 4,2} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes, \\
I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,5} \otimes, \\
1513221 \mathfrak{e}_8^{I^{\oplus 3}} 1223, \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} 122313, \\
321513221 \mathfrak{e}_8^{1\ 4,2} \otimes, \\
15131513221 \mathfrak{e}_8^{1\ 4,2} \otimes, \\
\otimes \mathfrak{e}_8^{5,5\ 1\ 5,3} \otimes, \\
21513221 \mathfrak{e}_8^{1\ 5,3} \otimes, \\
\otimes \mathfrak{e}_8^{5,5\ 1\ 5,4} \otimes, \\
21513221 \mathfrak{e}_8^{1\ 5,4} \otimes, \\
313221 \mathfrak{e}_8^{1\ 5,5} \otimes, \\
1513221 \mathfrak{e}_8^{1\ 5,5} \otimes, \\
313221 \mathfrak{e}_8^{1\ 122313}, \\
1513221 \mathfrak{e}_8^{1\ 1} 122313, \\
I^{\oplus 8} \mathfrak{e}_8^{I^{\oplus 2}} 1, \\
315131513221 \mathfrak{e}_8^{I^{\oplus 3}} 1, \\
321513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 4}, \\
15131513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 4}, \\
1513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 5}, \\
1513221 \mathfrak{e}_8^{1\ 1} 1223151315, \\
\otimes \mathfrak{e}_8^{3,5\ I^{\oplus 2} 4,2} \otimes, \\
3221 \mathfrak{e}_8^{1\ 1} 1223, \\
21513221 \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,3} \otimes, \\
I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,4} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{1\ 1} 1223, \\
I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} 1223132, \\
I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}} 122313, \\
51231513221 \mathfrak{e}_8^{1\ 4,2} \otimes, \\
\otimes \mathfrak{e}_8^{3,5\ 1\ 5,3} \otimes, \\
313221 \mathfrak{e}_8^{1\ 5,3} \otimes, \\
1513221 \mathfrak{e}_8^{1\ 5,3} \otimes, \\
313221 \mathfrak{e}_8^{1\ 5,4} \otimes, \\
1513221 \mathfrak{e}_8^{1\ 5,4} \otimes, \\
I^{\oplus 5} \mathfrak{e}_8^{1\ 5,5} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{1\ 1} 1223132, \\
I^{\oplus 5} \mathfrak{e}_8^{1\ 1} 122313, \\
2315131513221 \mathfrak{e}_8^{I^{\oplus 2}} 1, \\
I^{\oplus 7} \mathfrak{e}_8^{I^{\oplus 3}} 1, \\
231513221 \mathfrak{e}_8^{I^{\oplus 3}} 1, \\
51231513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 4}, \\
I^{\oplus 5} \mathfrak{e}_8^{1\ 1} I^{\oplus 5}, \\
21513221 \mathfrak{e}_8^{1\ 1} 12231512, \\
1513221 \mathfrak{e}_8^{1\ 1} 1223151, \\
\otimes \mathfrak{e}_8^{4,5\ I^{\oplus 2} 4,2} \otimes, \\
313221 \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes, \\
1513221 \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes, \\
I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,3} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,5} \otimes, \\
5131513221 \mathfrak{e}_8^{I^{\oplus 2}} 1223, \\
I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2}} 12231322, \\
I^{\oplus 7} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2}
\end{array}$$

$$2321513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$31513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$21513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 4},$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 2}} 1223151,$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3},$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$31513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$21513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3},$$

$$315131513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$321513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$15131513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 4},$$

$$I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3},$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 4},$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$321513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$15131513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3},$$

$$231513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$51231513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 4},$$

$$31513221 \mathfrak{e}_8^{I^{\oplus 2}} 122315,$$

$$21513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 24,2}} \otimes$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} 1223$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$51231513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 4},$$

$$\otimes \mathfrak{e}_7^{3,5} 1,$$

$$313221 \mathfrak{e}_7^{1,1},$$

$$32151321 \mathfrak{e}_7^{1,1},$$

$$5123151321 \mathfrak{e}_7^{1,1},$$

$$1513221 \mathfrak{e}_7^{1,1},$$

$$\otimes \mathfrak{e}_7^{2,3} I^{\oplus 3},$$

$$\otimes \mathfrak{e}_7^{3,3} I^{\oplus 3},$$

$$\otimes \mathfrak{e}_7^{5,4} I^{\oplus 2},$$

$$151321 \mathfrak{e}_7^{1,1} 123,$$

$$31321 \mathfrak{e}_7^{I^{\oplus 2}},$$

$$\otimes \mathfrak{e}_7^{4,5} 1,$$

$$231321 \mathfrak{e}_7^{1,1},$$

$$31512321 \mathfrak{e}_7^{1,1},$$

$$3151321 \mathfrak{e}_7^{1,1},$$

$$\otimes \mathfrak{e}_7^{2,3} 1 \otimes \mathfrak{e}_7^{3,2},$$

$$\otimes \mathfrak{e}_7^{2,4} I^{\oplus 2},$$

$$\otimes \mathfrak{e}_7^{3,4} I^{\oplus 2},$$

$$321 \mathfrak{e}_7^{1,1} 1223,$$

$$51321 \mathfrak{e}_7^{1,1} 1232,$$

$$2151321 \mathfrak{e}_7^{I^{\oplus 2}},$$

$$\otimes \mathfrak{e}_7^{5,5} 1,$$

$$I^{\oplus 5} \mathfrak{e}_7^{1,1},$$

$$21513221 \mathfrak{e}_7^{1,1},$$

$$1513151321 \mathfrak{e}_7^{1,1},$$

$$\otimes \mathfrak{e}_7^{3,3} 1 \otimes \mathfrak{e}_7^{3,2},$$

$$\otimes \mathfrak{e}_7^{3,3} 1 \otimes \mathfrak{e}_7^{3,3},$$

$$\otimes \mathfrak{e}_7^{4,4} I^{\oplus 2},$$

$$512321 \mathfrak{e}_7^{1,1} 123,$$

$$I^{\oplus 4} \mathfrak{e}_7^{I^{\oplus 2}},$$

$$2321 \mathfrak{e}_7^{I^{\oplus 2}},$$

$$\begin{aligned}
&1512321 \overset{I^{\oplus 2}}{\underset{1}{\mathfrak{e}_7}} 1, \\
&I^{\oplus 2} \overset{I^{\oplus 23,2}}{\underset{1}{\mathfrak{e}_7}} \otimes, \\
&I^{\oplus 2} \overset{I^{\oplus 3}}{\underset{1}{\mathfrak{e}_7}} I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
&151321 \overset{I^{\oplus 2}}{\underset{1}{\mathfrak{e}_7}} 1, \\
&I^{\oplus 2} \overset{I^{\oplus 23,3}}{\underset{1}{\mathfrak{e}_7}} \otimes, \\
&I^{\oplus 2} \overset{I^{\oplus 2}}{\underset{I^{\oplus 2}}{\mathfrak{e}_7}} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
&I^{\oplus 3} \overset{I^{\oplus 3}}{\underset{1}{\mathfrak{e}_7}} 1 \\
&321 \overset{I^{\oplus 2}}{\underset{1}{\mathfrak{e}_7}} 123
\end{aligned}$$

$$\overset{2,3}{\underset{1}{\otimes}} \overset{1}{\mathfrak{e}_6} 1,$$

$$\overset{3,3}{\underset{1}{\otimes}} \overset{1}{\mathfrak{e}_6} 1,$$

$$\overset{4,3}{\underset{1}{\otimes}} \overset{1}{\mathfrak{e}_6} 1$$

$$\overset{5,3}{\underset{1}{\otimes}} \overset{1}{\mathfrak{e}_6} 1,$$

$$\overset{3,3}{\underset{1}{\bigcirc}} \overset{1}{\mathfrak{e}_6} 1,$$

$$1 \overset{I^{\oplus 3}}{\underset{1}{\mathfrak{e}_6}} 1$$

$$3131 \overset{1}{\underset{1}{\mathfrak{e}_6}} 1,$$

$$215131 \overset{1}{\underset{1}{\mathfrak{e}_6}} 1,$$

$$15131 \overset{1}{\underset{1}{\mathfrak{e}_6}} 1$$

$$\overset{2,2}{\underset{1}{\otimes}} \overset{1}{\mathfrak{e}_6} \overset{2,2}{\underset{1}{\otimes}},$$

$$\overset{2,2}{\underset{1}{\otimes}} \overset{I^{\oplus 2}}{\mathfrak{e}_6} 1,$$

$$31 \overset{1}{\underset{1}{\mathfrak{e}_6}} 123$$

$$I^{\oplus 2} \overset{I^{\oplus 2}}{\underset{1}{\mathfrak{e}_6}} 1,$$

$$15131 \overset{1}{\underset{1}{\mathfrak{e}_6}} 13,$$

$$31 \overset{1}{\underset{\overset{2,2}{\otimes}}{\mathfrak{e}_6}} 13$$

$$31 \overset{I^{\oplus 2}}{\underset{1}{\mathfrak{e}_6}} 13$$

$$\overset{2,4}{\underset{1}{\otimes}} \overset{1}{\mathfrak{e}_7'} 1,$$

$$\overset{3,4}{\underset{1}{\otimes}} \overset{1}{\mathfrak{e}_7'} 1,$$

$$\overset{4,4}{\underset{1}{\otimes}} \overset{1}{\mathfrak{e}_7'} 1$$

$$\overset{5,4}{\underset{1}{\otimes}} \overset{1}{\mathfrak{e}_7'} 1,$$

$$I^{\oplus 4} \overset{1}{\underset{1}{\mathfrak{e}_7'}} 1,$$

$$31321 \overset{1}{\underset{1}{\mathfrak{e}_7'}} 1$$

$$2151321 \overset{1}{\underset{1}{\mathfrak{e}_7'}} 1,$$

$$2321 \overset{1}{\underset{1}{\mathfrak{e}_7'}} 1,$$

$$1512321 \overset{1}{\underset{1}{\mathfrak{e}_7'}} 1$$

$$151321 \overset{1}{\underset{1}{\mathfrak{e}_7'}} 1,$$

$$\overset{2,3}{\underset{1}{\otimes}} \overset{I^{\oplus 2}}{\mathfrak{e}_7'} 1,$$

$$\overset{3,3}{\underset{1}{\otimes}} \overset{I^{\oplus 2}}{\mathfrak{e}_7'} 1$$

$$321 \overset{1}{\underset{1}{\mathfrak{e}_7'}} 123,$$

$$I^{\oplus 2} \overset{I^{\oplus 3}}{\underset{1}{\mathfrak{e}_7'}} 1,$$

$$I^{\oplus 2} \overset{I^{\oplus 2}}{\underset{1}{\mathfrak{e}_7'}} I^{\oplus 2}$$

$$1 \overset{1}{\underset{1}{\mathfrak{so}}} 1$$

2.3 2 node(1504)

2.3.1 gLg(4)

$$\overset{3,3}{\mathfrak{so}} \otimes \mathfrak{so},$$

$$\overset{3,3}{\mathfrak{so}} \bigcirc \mathfrak{so}$$

$$\overset{3,5}{\mathfrak{so}} \otimes \mathfrak{e}_6$$

$$\overset{5,5}{\mathfrak{e}_6} \otimes \mathfrak{e}_6$$

2.3.2 111gLg(3)

$$\begin{smallmatrix} 1 & 4,3 \\ 1\mathfrak{e}_7 & \otimes \mathfrak{so} \\ 1 \end{smallmatrix}$$

$$\begin{smallmatrix} 1 & 4,5 \\ 1\mathfrak{e}_7 & \otimes \mathfrak{e}_6 \\ 1 \end{smallmatrix}$$

$$\begin{smallmatrix} 1 & 3,3 \\ 1\mathfrak{e}'_7 & \otimes \mathfrak{so} \\ 1 \end{smallmatrix}$$

2.3.3 SgLg(115)

$$\begin{aligned} & 513151321\mathfrak{e}_7 \otimes \mathfrak{so}, \\ & 2231\mathfrak{so} \otimes \mathfrak{e}_7, \\ & \begin{smallmatrix} 2,4 & 3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix} \\ & \begin{smallmatrix} 5,4 & 3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix} \\ & 2151321\mathfrak{e}_7 \otimes \mathfrak{so}, \\ & \begin{smallmatrix} 1 & 3,3 \\ 151321\mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix} \\ & I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{so}, \end{aligned}$$

$$\begin{aligned} & 1513221\mathfrak{e}_7 \otimes \mathfrak{so}, \\ & 512321\mathfrak{e}_7 \otimes \mathfrak{so}, \\ & \begin{smallmatrix} 3,4 & 3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix} \\ & I^{\oplus 4}\mathfrak{e}_7 \otimes \mathfrak{so}, \\ & \begin{smallmatrix} 1 & 3,3 \\ 2321\mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix} \\ & \begin{smallmatrix} 2,3 & 4,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix} \\ & I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{so} \end{aligned}$$

$$\begin{aligned} & 3221\mathfrak{e}_7 \otimes \mathfrak{so} \\ & 151321\mathfrak{e}_7 \otimes \mathfrak{so} \\ & \begin{smallmatrix} 4,4 & 3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so} \end{smallmatrix} \\ & 31321\mathfrak{e}_7 \otimes \mathfrak{so} \\ & 1512321\mathfrak{e}_7 \otimes \mathfrak{so} \\ & \begin{smallmatrix} 3,3 & 4,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so} \end{smallmatrix} \end{aligned}$$

$$\begin{aligned} & 3221\mathfrak{e}_6 \otimes \mathfrak{so}, \\ & 51315131\mathfrak{e}_6 \otimes \mathfrak{so}, \\ & 231\mathfrak{so} \otimes \mathfrak{e}_6, \\ & 31\mathfrak{e}_6 \otimes \mathfrak{so}, \\ & 1\mathfrak{e}_6 \otimes \mathfrak{so}, \end{aligned}$$

$$\begin{aligned} & 2231\mathfrak{so} \otimes \mathfrak{e}_6, \\ & 151321\mathfrak{e}_6 \otimes \mathfrak{so}, \\ & 15131\mathfrak{e}_6 \otimes \mathfrak{so}, \\ & \begin{smallmatrix} 2,2 & 3,3 \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{so}, \end{smallmatrix} \\ & \begin{smallmatrix} 2,2 & 3,3 \\ \otimes \mathfrak{e}_6 & \bigcirc \mathfrak{so}, \end{smallmatrix} \end{aligned}$$

$$\begin{aligned} & 512321\mathfrak{e}_6 \otimes \mathfrak{so} \\ & 321\mathfrak{e}_6 \otimes \mathfrak{so} \\ & 31\mathfrak{so} \otimes \mathfrak{e}_6 \\ & I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{so} \\ & I^{\oplus 2}\mathfrak{e}_6 \bigcirc \mathfrak{so} \end{aligned}$$

$$\begin{aligned} & 31513221\mathfrak{e}_7 \otimes \mathfrak{e}_6, \\ & 2315131\mathfrak{e}_6 \otimes \mathfrak{e}_7, \\ & 315131\mathfrak{e}_6 \otimes \mathfrak{e}_7, \\ & 151321\mathfrak{e}_7 \otimes \mathfrak{e}_6, \\ & 321\mathfrak{e}_7 \otimes \mathfrak{e}_6, \end{aligned}$$

$$\begin{aligned} & 22315131\mathfrak{e}_6 \otimes \mathfrak{e}_7, \\ & 2321\mathfrak{e}_6 \otimes \mathfrak{e}_7, \\ & 3221\mathfrak{e}_7 \otimes \mathfrak{e}_6, \\ & \begin{smallmatrix} 2,3 & 4,5 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6, \end{smallmatrix} \\ & I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{e}_6, \end{aligned}$$

$$\begin{aligned} & 3151321\mathfrak{e}_7 \otimes \mathfrak{e}_6 \\ & 2321\mathfrak{e}_7 \otimes \mathfrak{e}_6 \\ & 512321\mathfrak{e}_7 \otimes \mathfrak{e}_6 \\ & \begin{smallmatrix} 3,3 & 4,5 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6 \end{smallmatrix} \\ & I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_6 \end{aligned}$$

$$3221\mathfrak{e}'_7 \otimes \mathfrak{so},$$

$$2231\mathfrak{so} \otimes \mathfrak{e}'_7,$$

$$512321\mathfrak{e}'_7 \otimes \mathfrak{so}$$

$$\begin{aligned}
&151321\mathfrak{e}'_7 \overset{3,2}{\otimes} \mathfrak{so}, \\
&\overset{2,3}{\otimes} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{so}, \\
&I^{\oplus 2}\mathfrak{e}'_7 \overset{4,3}{\otimes} \mathfrak{so},
\end{aligned}$$

$$\begin{aligned}
&321\mathfrak{e}'_7 \overset{4,2}{\otimes} \mathfrak{so}, \\
&\overset{3,3}{\otimes} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{so}, \\
&1\mathfrak{e}'_7 \overset{5,3}{\otimes} \mathfrak{so}
\end{aligned}$$

$$\begin{aligned}
&231\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}'_7 \\
&I^{\oplus 3}\mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{so}
\end{aligned}$$

$$\begin{aligned}
&3151321\mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}_6, \\
&2321\mathfrak{e}'_7 \overset{4,3}{\otimes} \mathfrak{e}_6, \\
&15131\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}'_7, \\
&31\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}'_7,
\end{aligned}$$

$$\begin{aligned}
&2315131\mathfrak{e}_6 \overset{3,3}{\otimes} \mathfrak{e}'_7, \\
&315131\mathfrak{e}_6 \overset{3,4}{\otimes} \mathfrak{e}'_7, \\
&321\mathfrak{e}'_7 \overset{4,4}{\otimes} \mathfrak{e}_6, \\
&1\mathfrak{e}'_7 \overset{5,5}{\otimes} \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
&2321\mathfrak{e}_6 \overset{3,4}{\otimes} \mathfrak{e}'_7 \\
&321\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}'_7 \\
&I^{\oplus 2}\mathfrak{e}'_7 \overset{4,5}{\otimes} \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
&23151321\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6, \\
&321\mathfrak{e}_6 \overset{3,4}{\otimes} \mathfrak{e}_6, \\
&I^{\oplus 2}\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_6, \\
&2321\mathfrak{e}_6 \overset{3,3}{\bigcirc} \mathfrak{e}_6,
\end{aligned}$$

$$\begin{aligned}
&2321\mathfrak{e}_6 \overset{3,3}{\otimes} \mathfrak{e}_6, \\
&15131\mathfrak{e}_6 \overset{3,4}{\otimes} \mathfrak{e}_6, \\
&31\mathfrak{e}_6 \overset{4,4}{\otimes} \mathfrak{e}_6, \\
&315131\mathfrak{e}_6 \overset{3,3}{\bigcirc} \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
&315131\mathfrak{e}_6 \overset{3,3}{\otimes} \mathfrak{e}_6 \\
&\overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_6 \\
&1\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
&231513221\mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}'_7, \\
&3221\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}'_7, \\
&512321\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}'_7,
\end{aligned}$$

$$\begin{aligned}
&223151321\mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
&2321\mathfrak{e}'_7 \overset{4,5}{\otimes} \mathfrak{e}_7, \\
&151321\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}'_7,
\end{aligned}$$

$$\begin{aligned}
&3151321\mathfrak{e}_7 \overset{4,4}{\otimes} \mathfrak{e}'_7 \\
&2321\mathfrak{e}_7 \overset{5,4}{\otimes} \mathfrak{e}'_7 \\
&321\mathfrak{e}_7 \overset{5,5}{\otimes} \mathfrak{e}'_7
\end{aligned}$$

$$\begin{aligned}
&2313221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6, \\
&51231513221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6,
\end{aligned}$$

$$\begin{aligned}
&I^{\oplus 6}\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6, \\
&\overset{1}{31513221}\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6,
\end{aligned}$$

$$\begin{aligned}
&321513221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6 \\
&15131513221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
&2313221\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so}, \\
&51231513221\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so},
\end{aligned}$$

$$\begin{aligned}
&I^{\oplus 6}\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so}, \\
&\overset{1}{31513221}\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so},
\end{aligned}$$

$$\begin{aligned}
&321513221\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so} \\
&15131513221\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so}
\end{aligned}$$

$$\begin{aligned}
&2321\mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so}, \\
&1\mathfrak{so} \overset{2,3}{\otimes} \mathfrak{so}
\end{aligned}$$

$$315131\mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so},$$

$$31\mathfrak{so} \overset{2,2}{\otimes} \mathfrak{so}$$

$$\begin{aligned}
&2231513221\mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
&2321\mathfrak{e}_7 \overset{5,5}{\otimes} \mathfrak{e}_7
\end{aligned}$$

$$23151321\mathfrak{e}_7 \overset{4,4}{\otimes} \mathfrak{e}_7,$$

$$3151321\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_7$$

$$23151321\mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7,$$

$$2321\mathfrak{e}'_7 \overset{4,4}{\otimes} \mathfrak{e}'_7,$$

$$321\mathfrak{e}'_7 \overset{4,5}{\otimes} \mathfrak{e}'_7$$

$$\begin{aligned}
& 1\epsilon_6^{2,2} \otimes \epsilon_6^{2,2} 1231513, \\
& \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{2,2} 13151315, \\
& 2,3 \quad 2,2 \quad 3,3 \\
& \otimes \epsilon_6 \otimes \epsilon_6 \otimes, \\
& 2,3 \quad 2,2 \quad 3,3 \\
& \otimes \epsilon_6 \otimes \epsilon_6 \bigcirc, \\
& 2,3 \quad 2,2 \\
& \otimes \epsilon_6 \otimes \epsilon_6 131512, \\
& 3,3 \quad 2,2 \quad 3,4 \\
& \otimes \epsilon_6 \otimes \epsilon_6 \otimes, \\
& I^{\oplus 3} \epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \otimes, \\
& 3,3 \quad 2,2 \quad 1 \\
& \otimes \epsilon_6 \otimes \epsilon_6 13151, \\
& 3,3 \quad 2,2 \quad 3,4 \\
& \bigcirc \epsilon_6 \otimes \epsilon_6 \otimes, \\
& 4,3 \quad 2,2 \\
& \otimes \epsilon_6 \otimes \epsilon_6 131512, \\
& 3,3 \quad 2,2 \quad 3,5 \\
& \bigcirc \epsilon_6 \otimes \epsilon_6 \otimes, \\
& 5,3 \quad 2,2 \\
& \otimes \epsilon_6 \otimes \epsilon_6 131512, \\
& 321\epsilon_6^{2,2} \otimes \epsilon_6^{2,2} 12315, \\
& 3131\epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \bigcirc, \\
& I^{\oplus 2} \epsilon_6^{2,2} \otimes \epsilon_6^{2,2} 123215, \\
& I^{\oplus 3} \epsilon_6^{2,2} \otimes \epsilon_6^{2,2} I^{\oplus 3}, \\
& I^{\oplus 3} \epsilon_6^{2,2} \otimes \epsilon_6^{1,1} 13151, \\
& 3131\epsilon_6^{2,2} \otimes \epsilon_6^{1,1} 13151, \\
& 15131\epsilon_6^{1,1} \otimes \epsilon_6^{2,2} 131512 \\
& 1\epsilon_6^{3,3} \otimes \epsilon_6^{3,3} 13151, \\
& I^{\oplus 2} \epsilon_6^{3,3} \otimes \epsilon_6^{3,3} I^{\oplus 2}, \\
& 1\epsilon_6^{4,3} \otimes \epsilon_6^{4,3} I^{\oplus 2}, \\
& 1\epsilon_6^{3,3} \bigcirc \epsilon_6^{3,3} 123, \\
& I^{\oplus 2} \epsilon_6^{3,3} \bigcirc \epsilon_6^{2,2} \otimes,
\end{aligned}$$

$$\begin{aligned}
& \overset{2,2}{\otimes} \epsilon_6 \otimes \overset{2,2}{\otimes} \epsilon_6 1223, \\
& \overset{2,2}{\otimes} \epsilon_6 \otimes \overset{2,2}{\otimes} \epsilon_6 123151, \\
& \overset{2,3}{\otimes} \epsilon_6 \otimes \overset{2,2}{\otimes} \epsilon_6 \overset{3,4}{\otimes}, \\
& I^{\oplus 3} \epsilon_6 \overset{2,2}{\otimes} \epsilon_6 \overset{3,2}{\otimes}, \\
& \overset{2,3}{\otimes} \epsilon_6 \otimes \overset{2,2}{\otimes} \epsilon_6 \overset{1}{13151}, \\
& \overset{3,3}{\otimes} \epsilon_6 \otimes \overset{2,2}{\otimes} \epsilon_6 \overset{3,5}{\otimes}, \\
& \overset{3,3}{\otimes} \epsilon_6 \otimes \overset{2,2}{\otimes} \epsilon_6 1313, \\
& \overset{4,3}{\otimes} \epsilon_6 \otimes \overset{2,2}{\otimes} \epsilon_6 \overset{3,4}{\otimes}, \\
& I^{\oplus 3} \epsilon_6 \overset{2,2}{\otimes} \epsilon_6 \overset{3,4}{\otimes}, \\
& \overset{4,3}{\otimes} \epsilon_6 \otimes \overset{2,2}{\otimes} \epsilon_6 \overset{1}{13151}, \\
& I^{\oplus 3} \epsilon_6 \overset{2,2}{\otimes} \epsilon_6 \overset{3,5}{\otimes}, \\
& \overset{5,3}{\otimes} \epsilon_6 \otimes \overset{2,2}{\otimes} \epsilon_6 \overset{1}{13151}, \\
& \overset{3,3}{\circlearrowleft} \epsilon_6 \otimes \overset{2,2}{\otimes} \epsilon_6 \overset{3,3}{\circlearrowleft}, \\
& \overset{3,3}{\circlearrowleft} \epsilon_6 \otimes \overset{2,2}{\otimes} \epsilon_6 131512, \\
& I^{\oplus 2} \epsilon_6 \overset{2,2}{\otimes} \epsilon_6 13151315, \\
& I^{\oplus 3} \epsilon_6 \overset{2,2}{\otimes} \epsilon_6 1313, \\
& 3131 \epsilon_6 \overset{2,2}{\otimes} \epsilon_6 1313, \\
& 51321 \epsilon_6 \overset{2,2}{\otimes} \epsilon_6 13151, \\
& \overset{1}{15131} \epsilon_6 \overset{2,2}{\otimes} \epsilon_6 \overset{1}{13151}, \\
& \overset{2,2}{\otimes} \epsilon_6 \otimes \overset{3,3}{\otimes} \epsilon_6 \overset{2,2}{\otimes}, \\
& 31 \epsilon_6 \overset{3,3}{\otimes} \epsilon_6 1315, \\
& 31 \epsilon_6 \overset{4,3}{\otimes} \epsilon_6 1, \\
& 1 \epsilon_6 \overset{3,3}{\circlearrowleft} \epsilon_6 13151, \\
& I^{\oplus 2} \epsilon_6 \overset{3,3}{\circlearrowleft} \epsilon_6 I^{\oplus 2},
\end{aligned}$$

$$\begin{array}{l}
{}^{2,2}_3 \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} 123215 \\
{}^{2,3}_3 \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{3,2}_{\mathbf{e}_6} \otimes \\
{}^{2,3}_3 \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{3,5}_{\mathbf{e}_6} \otimes \\
{}^{2,3}_3 \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} 1313 \\
{}^{3,3}_3 \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{3,3}_{\mathbf{e}_6} \otimes \\
{}^{3,3}_3 \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{3,3}_{\mathbf{e}_6} \bigcirc \\
{}^{3,3}_3 \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} 131512 \\
{}^{4,3}_3 \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{3,5}_{\mathbf{e}_6} \otimes \\
3131 {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes {}^{3,4}_{\mathbf{e}_6} \otimes \\
{}^{5,3}_3 \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{3,5}_{\mathbf{e}_6} \otimes \\
3131 {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes {}^{3,5}_{\mathbf{e}_6} \otimes \\
I^{\oplus 2} {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} 1223 \\
I^{\oplus 3} {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes {}^{3,3}_{\mathbf{e}_6} \bigcirc \\
{}^{3,3}_3 \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes {}^1_{\mathbf{e}_6} 13151 \\
I^{\oplus 2} {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} 123151 \\
I^{\oplus 3} {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} 131512 \\
3131 {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} 131512 \\
215131 {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} 131512 \\
1 {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{3,3}_{\mathbf{e}_6} \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} 123 \\
I^{\oplus 2} {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{3,3}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes \\
1 {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{4,3}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes \\
1 {}^{\mathbf{e}_6}_{\mathbf{e}_6} \otimes {}^{4,4}_{\mathbf{e}_6} \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} 1 \\
{}^{2,2}_3 \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} \bigcirc {}^{3,3}_{\mathbf{e}_6} \otimes {}^{2,2}_{\mathbf{e}_6} \otimes \\
31 {}^{\mathbf{e}_6}_{\mathbf{e}_6} \bigcirc {}^{3,3}_{\mathbf{e}_6} \otimes {}^{\mathbf{e}_6}_{\mathbf{e}_6} 1315
\end{array}$$

$$\begin{aligned}
& 2315131\epsilon_6 \otimes \epsilon_7^{\overline{3,3}}, \\
& \otimes \epsilon_6^{\overline{2,2}} \otimes \epsilon_7^{\overline{3,3}} \otimes, \\
& \otimes \epsilon_6^{\overline{2,2}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{4,5}}, \\
& \otimes \epsilon_6^{\overline{2,2}} \otimes \epsilon_7^{\overline{3,3}} 1231512, \\
& \otimes \epsilon_6^{\overline{2,2}} \otimes \epsilon_7^{\overline{3,3}} 123151^1, \\
& I^{\oplus 2} \epsilon_6^{\overline{3,3}} \otimes \epsilon_7^{\overline{4,2}}, \\
& I^{\oplus 2} \epsilon_6^{\overline{3,3}} \otimes \epsilon_7^{\overline{4,3}}, \\
& 5131\epsilon_6^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} 1223, \\
& I^{\oplus 2} \epsilon_6^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} I^{\oplus 4}, \\
& I^{\oplus 2} \epsilon_6^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} 1231512, \\
& I^{\oplus 2} \epsilon_6^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} 123151^1, \\
& 5131\epsilon_6^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} 123215,
\end{aligned}$$

$$\begin{aligned}
& 1\mathbf{e}_6^{3,3} \otimes \mathbf{e}_7 123151315, \\
& \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_7^{3,3} \otimes, \\
& \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_7^{3,3} I^{\oplus 4}, \\
& \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_7^{3,3} 1232, \\
& 321\mathbf{e}_6^{3,3} \otimes \mathbf{e}_7^{3,2} \otimes, \\
& 321\mathbf{e}_6^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes, \\
& I^{\oplus 2} \mathbf{e}_6^{3,3} \otimes \mathbf{e}_7^{4,4} \otimes, \\
& 321\mathbf{e}_6^{3,3} \otimes \mathbf{e}_7 I^{\oplus 3}, \\
& I^{\oplus 2} \mathbf{e}_6^{3,3} \otimes \mathbf{e}_7 12313, \\
& I^{\oplus 2} \mathbf{e}_6^{3,3} \otimes \mathbf{e}_7 1232, \\
& 15131\mathbf{e}_6^{3,3} \otimes \mathbf{e}_7 I^{\oplus 3}, \\
& 5131\mathbf{e}_6^{3,3} \otimes \mathbf{e}_7 123151,
\end{aligned}$$

$$\begin{array}{l}
1\epsilon_6^{3,3} \otimes \epsilon_7 1223151 \\
2,2 \otimes \epsilon_6^{3,3} \otimes \epsilon_7^{4,4} \\
2,2 \otimes \epsilon_6^{3,3} \otimes \epsilon_7 12313 \\
2,2 \otimes \epsilon_6^{3,3} \otimes \epsilon_7 1232151 \\
15131\epsilon_6^{3,3} \otimes \epsilon_7^{3,2} \\
15131\epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \\
I^{\oplus 2} \epsilon_6^{3,3} \otimes \epsilon_7^{4,5} \\
2321\epsilon_6^{3,3} \otimes \epsilon_7 I^{\oplus 2} \\
315131\epsilon_6^{3,3} \otimes \epsilon_7 I^{\oplus 2} \\
I^{\oplus 2} \epsilon_6^{3,3} \otimes \epsilon_7 1232151 \\
31\epsilon_6^{3,3} \otimes \epsilon_7 122315 \\
1\epsilon_6^{3,4} \otimes \epsilon_7 1223
\end{array}$$

$$\begin{array}{lll}
321\mathfrak{e}'_7 \otimes \mathfrak{e}_8 I^{\oplus 5}, & 321\mathfrak{e}'_7 \otimes \mathfrak{e}_8 12231512, & 321\mathfrak{e}'_7 \otimes \mathfrak{e}_8 1223151^1 \\
2321\mathfrak{e}'_7 \otimes \mathfrak{e}_8 I^{\oplus 4}, & I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}_8 I^{\oplus 6}, & I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}_8 122315123 \\
I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}_8 12231513215, & I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}_8 12231513^1, & I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}_8 12231513151 \\
1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 1223132, & 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 I^{\oplus 6}, & 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 122315123 \\
1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 12231513215, & 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 12231513^1, & 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 12231513151
\end{array}$$

$$\begin{array}{lll}
2313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 1223132, & I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 1223132, & 2313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 122315123 \\
2313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513215, & 2313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513^1, & 2313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513151 \\
I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 I^{\oplus 6}, & I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 122315123, & I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513215 \\
I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513^1, & I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513151, & 321513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 122315123 \\
321513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513215, & 31513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 122315123, & 321513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513151 \\
51231513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513215, & 31513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513215, & 15131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513215 \\
31513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513^1, & 31513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513151, & 15131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 12231513151
\end{array}$$

$$\begin{array}{lll}
1\mathfrak{so} \otimes \mathfrak{e}_8 12231322, & 1\mathfrak{so} \otimes \mathfrak{e}_8 I^{\oplus 7}, & 1\mathfrak{so} \otimes \mathfrak{e}_8 1223151232 \\
1\mathfrak{so} \otimes \mathfrak{e}_8 122315131513, & 1\mathfrak{so} \otimes \mathfrak{e}_8 122315132^1, & 2231\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \\
231\mathfrak{so} \otimes \mathfrak{e}_8 \otimes, & 231\mathfrak{so} \otimes \mathfrak{e}_8 \otimes, & 231\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \\
31\mathfrak{so} \otimes \mathfrak{e}_8 1223132, & 231\mathfrak{so} \otimes \mathfrak{e}_8 122313, & 2231\mathfrak{so} \otimes \mathfrak{e}_8 I^{\oplus 4} \\
231\mathfrak{so} \otimes \mathfrak{e}_8 I^{\oplus 5}, & 31\mathfrak{so} \otimes \mathfrak{e}_8 I^{\oplus 6}, & 31\mathfrak{so} \otimes \mathfrak{e}_8 122315123 \\
31\mathfrak{so} \otimes \mathfrak{e}_8 12231513215, & 31\mathfrak{so} \otimes \mathfrak{e}_8 12231513^1, & 31\mathfrak{so} \otimes \mathfrak{e}_8 12231513151 \\
231\mathfrak{so} \otimes \mathfrak{e}_8 12231512, & 231\mathfrak{so} \otimes \mathfrak{e}_8 1223151^1, &
\end{array}$$

$$\begin{array}{lll}
1\mathfrak{so} \otimes \mathfrak{e}_7 \otimes, & 1\mathfrak{so} \otimes \mathfrak{e}_7 \otimes, & 1\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \\
1\mathfrak{so} \otimes \mathfrak{e}_7 \otimes, & 1\mathfrak{so} \otimes \mathfrak{e}_7 I^{\oplus 4}, & 1\mathfrak{so} \otimes \mathfrak{e}_7 12313 \\
2231\mathfrak{so} \otimes \mathfrak{e}_7 1, & 1\mathfrak{so} \otimes \mathfrak{e}_7 1231512, & 1\mathfrak{so} \otimes \mathfrak{e}_7 1232^1 \\
1\mathfrak{so} \otimes \mathfrak{e}_7 1232151, & 1\mathfrak{so} \otimes \mathfrak{e}_7 123151^1, & 31\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \\
31\mathfrak{so} \otimes \mathfrak{e}_7 \otimes, & 231\mathfrak{so} \otimes \mathfrak{e}_7 I^{\oplus 2}, & 31\mathfrak{so} \otimes \mathfrak{e}_7 I^{\oplus 3} \\
1\mathfrak{so} \otimes \mathfrak{e}_7 \otimes, & 1\mathfrak{so} \otimes \mathfrak{e}_7 \otimes, & 1\mathfrak{so} \otimes \mathfrak{e}_7 I^{\oplus 3} \\
231\mathfrak{so} \otimes \mathfrak{e}_7 1, & 31\mathfrak{so} \otimes \mathfrak{e}_7 I^{\oplus 2}
\end{array}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{so}123, \\
& \begin{smallmatrix} 2,2 & 1,1 & 2,2 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix}, \\
& \begin{smallmatrix} 2,2 & 1,1 & 2 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \mathbf{132}, \end{smallmatrix} \\
& \begin{smallmatrix} 3,2 & 1,1 & 2,4 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix}, \\
& \begin{smallmatrix} 4,2 & 1,1 & 2,4 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix}, \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so}132, \\
& \mathbf{1so} \otimes \mathbf{so}1
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{so}13215, \\
& \begin{smallmatrix} 2,2 & 1,1 & 2,3 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix}, \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so} \otimes \mathbf{so}, \\
& \begin{smallmatrix} 3,2 & 1,1 & 2 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \mathbf{132}, \end{smallmatrix} \\
& \begin{smallmatrix} 4,2 & 1,1 & 2 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \mathbf{132}, \end{smallmatrix} \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so} I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{so}13151 \\
& \begin{smallmatrix} 2,2 & 1,1 & 2,4 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix} \\
& \begin{smallmatrix} 3,2 & 1,1 & 2,3 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix} \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so} \otimes \mathbf{so} \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so} \otimes \mathbf{so} \\
& \mathbf{31so} \otimes \mathbf{so}1315
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_6 \otimes, \\
& \mathbf{1so} \otimes \mathbf{e}_6 \otimes, \\
& \mathbf{1so} \otimes \mathbf{e}_61313, \\
& \mathbf{1so} \otimes \mathbf{e}_613151, \\
& \mathbf{1so} \otimes \mathbf{e}_6 \otimes, \\
& \mathbf{1so} \otimes \mathbf{e}_61
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_6 \otimes, \\
& \mathbf{1so} \otimes \mathbf{e}_6 \bigcirc, \\
& 231\mathbf{so} \otimes \mathbf{e}_61, \\
& 31\mathbf{so} \otimes \mathbf{e}_6 \otimes, \\
& \mathbf{1so} \otimes \mathbf{e}_6 I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_6 \otimes \\
& \mathbf{1so} \otimes \mathbf{e}_6 I^{\oplus 3} \\
& \mathbf{1so} \otimes \mathbf{e}_6131512 \\
& 31\mathbf{so} \otimes \mathbf{e}_6 I^{\oplus 2} \\
& 31\mathbf{so} \otimes \mathbf{e}_61
\end{aligned}$$

$$\begin{aligned}
& 1\mathbf{e}'_7 \otimes \mathbf{e}'_71231513, \\
& I^{\oplus 3} \mathbf{e}'_7 \otimes \mathbf{e}'_7 \otimes, \\
& I^{\oplus 2} \mathbf{e}'_7 \otimes \mathbf{e}'_71223, \\
& I^{\oplus 2} \mathbf{e}'_7 \otimes \mathbf{e}'_7123151, \\
& I^{\oplus 2} \mathbf{e}'_7 \otimes \mathbf{e}'_7 I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \begin{smallmatrix} 2,3 & 3,3 & 3,2 \\ \otimes \mathbf{e}'_7 & \otimes \mathbf{e}'_7 & \otimes \end{smallmatrix}, \\
& \begin{smallmatrix} 3,3 & 3,3 & 3,3 \\ \otimes \mathbf{e}'_7 & \otimes \mathbf{e}'_7 & \otimes \end{smallmatrix}, \\
& 321\mathbf{e}'_7 \otimes \mathbf{e}'_712315, \\
& I^{\oplus 3} \mathbf{e}'_7 \otimes \mathbf{e}'_7 I^{\oplus 3}, \\
& 1\mathbf{e}'_7 \otimes \mathbf{e}'_7 I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \begin{smallmatrix} 2,3 & 3,3 & 3,3 \\ \otimes \mathbf{e}'_7 & \otimes \mathbf{e}'_7 & \otimes \end{smallmatrix} \\
& I^{\oplus 3} \mathbf{e}'_7 \otimes \mathbf{e}'_7 \otimes \mathbf{e}'_7 \\
& I^{\oplus 2} \mathbf{e}'_7 \otimes \mathbf{e}'_7123215 \\
& 1\mathbf{e}'_7 \otimes \mathbf{e}'_7123 \\
& 1\mathbf{e}'_7 \otimes \mathbf{e}'_71
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}'_7 \otimes, \\
& 231\mathbf{so} \otimes \mathbf{e}'_71, \\
& 31\mathbf{so} \otimes \mathbf{e}'_71
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}'_7 \otimes, \\
& 31\mathbf{so} \otimes \mathbf{e}'_7 I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}'_7 I^{\oplus 3} \\
& \mathbf{1so} \otimes \mathbf{e}'_7 I^{\oplus 2}
\end{aligned}$$

2.3.5 SSgLg(73)

$$\begin{aligned}
& 3221\mathbf{e}_8 \otimes \mathbf{so}, \\
& 1513221\mathbf{e}_8 \otimes \mathbf{so}, \\
& \mathbf{e}_8 \otimes \mathbf{so},
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 4} \mathbf{e}_8 \otimes \mathbf{so}, \\
& \mathbf{e}_8 \otimes \mathbf{so}, \\
& 313221\mathbf{e}_8 \otimes \mathbf{so},
\end{aligned}$$

$$\begin{aligned}
& 5131513221\mathbf{e}_8 \otimes \mathbf{so} \\
& \mathbf{e}_8 \otimes \mathbf{so} \\
& I^{\oplus 5} \mathbf{e}_8 \otimes \mathbf{so}
\end{aligned}$$

$$21513221 \mathfrak{e}_8^{1 \ 5,3} \otimes \mathfrak{so},$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} \ 5,3} \otimes \mathfrak{so},$$

$$1513221 \mathfrak{e}_8^{1 \ 5,3} \otimes \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} \ 5,3} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_8^{2,4 \ 5,3} \otimes \mathfrak{so}$$

$$31513221 \mathfrak{e}_8^{I^{\oplus 2} \ 5,3} \otimes \mathfrak{e}_6,$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 2} \ 5,4} \otimes \mathfrak{e}_6,$$

$$\mathfrak{e}_8^{5,5 \ 1 \ 5,5} \otimes \mathfrak{e}_6,$$

$$5131513221 \mathfrak{e}_8^{1 \ 5,4} \otimes \mathfrak{e}_6,$$

$$\mathfrak{e}_8^{3,5 \ 1 \ 5,5} \otimes \mathfrak{e}_6,$$

$$313221 \mathfrak{e}_8^{1 \ 5,5} \otimes \mathfrak{e}_6,$$

$$I^{\oplus 3} \mathfrak{e}_8^{3221 \ 5,4} \otimes \mathfrak{e}_6$$

$$\mathfrak{e}_8^{4,5 \ 1 \ 5,5} \otimes \mathfrak{e}_6$$

$$I^{\oplus 5} \mathfrak{e}_8^{1 \ 5,5} \otimes \mathfrak{e}_6$$

$$21513221 \mathfrak{e}_8^{1 \ 5,5} \otimes \mathfrak{e}_6,$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} \ 5,5} \otimes \mathfrak{e}_6,$$

$$1513221 \mathfrak{e}_8^{1 \ 5,5} \otimes \mathfrak{e}_6,$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} \ 5,5} \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_8^{2,4 \ 5,5} \otimes \mathfrak{e}_6$$

$$3221 \mathfrak{e}_7^{1 \ 3,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7^{321 \ 3,2} \otimes \mathfrak{so},$$

$$\mathfrak{e}_7^{3,3 \ 1 \ 3,3} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7^{1 \ 4,3} \otimes \mathfrak{so},$$

$$512321 \mathfrak{e}_7^{1 \ 3,2} \otimes \mathfrak{so},$$

$$321 \mathfrak{e}_7^{1 \ 4,2} \otimes \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_7^{1 \ 3,3} \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7^{1 \ 5,3} \otimes \mathfrak{so}$$

$$151321 \mathfrak{e}_7^{1 \ 3,2} \otimes \mathfrak{so}$$

$$\mathfrak{e}_7^{2,3 \ 1 \ 3,3} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7^{I^{\oplus 2} \ 3,3} \otimes \mathfrak{so}$$

$$321 \mathfrak{e}_6^{1 \ 2,2} \otimes \mathfrak{so},$$

$$31 \mathfrak{e}_6^{I^{\oplus 2} \ 2,2} \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_6^{1 \ 3,3} \bigcirc \mathfrak{so}$$

$$15131 \mathfrak{e}_6^{1 \ 2,2} \otimes \mathfrak{so},$$

$$31 \mathfrak{e}_6^{1 \ 3,2} \otimes \mathfrak{so},$$

$$31 \mathfrak{e}_6^{2,2 \ 2,2} \otimes \mathfrak{so}$$

$$1 \mathfrak{e}_6^{1 \ 3,3} \otimes \mathfrak{so}$$

$$3151321 \mathfrak{e}_7^{1 \ 3,3} \otimes \mathfrak{e}_6,$$

$$321 \mathfrak{e}_7^{1 \ 4,4} \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2} \mathfrak{e}_7^{2321 \ 3,3} \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2} \mathfrak{e}_7^{1 \ 4,5} \otimes \mathfrak{e}_6,$$

$$2321 \mathfrak{e}_7^{1 \ 4,3} \otimes \mathfrak{e}_6$$

$$1 \mathfrak{e}_7^{1 \ 5,5} \otimes \mathfrak{e}_6$$

$$31513221 \mathfrak{e}_8^{I^{\oplus 2} \ 5,4} \otimes \mathfrak{e}_7',$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 2} \ 5,5} \otimes \mathfrak{e}_7'$$

$$5131513221 \mathfrak{e}_8^{1 \ 5,5} \otimes \mathfrak{e}_7',$$

$$I^{\oplus 3} \mathfrak{e}_8^{3221 \ 5,5} \otimes \mathfrak{e}_7'$$

$$321 \mathfrak{e}_7'^{1 \ 3,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 \ 3,3} \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7'^{1 \ 4,3} \otimes \mathfrak{so}$$

$$2321 \mathfrak{e}_7'^{1 \ 3,3} \otimes \mathfrak{e}_6,$$

$$31 \mathfrak{e}_6^{1 \ 3,5} \otimes \mathfrak{e}_7',$$

$$1 \mathfrak{e}_7'^{1 \ 4,5} \otimes \mathfrak{e}_6$$

$$23151321 \mathfrak{e}_7^{1 \ 3,3} \otimes \mathfrak{e}_7',$$

$$2321 \mathfrak{e}_7^{1 \ 4,4} \otimes \mathfrak{e}_7',$$

$$321 \mathfrak{e}_7^{1 \ 4,5} \otimes \mathfrak{e}_7'$$

$$2315131\mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2},$$

$$31\mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{3,4},$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{3,5}$$

$$231513221\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7,$$

$$31513221\mathfrak{e}_8^{I^{\oplus 2} 5,5} \otimes \mathfrak{e}_7$$

$$223151321\mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3},$$

$$2321\mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{4,5}$$

$$231\mathfrak{so}^1 \otimes \mathfrak{so}^{1,1}$$

2.3.6 SSgLgS(576)

$$1\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} 123151322,$$

$$\otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$\otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2}$$

$$\otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$313221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$I^{\oplus 5}\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2}$$

$$21513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$1513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$\otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{1223}$$

$$2321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_7^{4,2},$$

$$\otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{123215},$$

$$\otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{123151}$$

$$\otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{3,3},$$

$$\otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{3,3},$$

$$\otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$313221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3},$$

$$I^{\oplus 5}\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3},$$

$$21513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$1513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3},$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,3},$$

$$\otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{I^{\oplus 3}}$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,4},$$

$$\otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{I^{\oplus 3}},$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,5}$$

$$\otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{I^{\oplus 3}},$$

$$I^{\oplus 4}\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{1223},$$

$$I^{\oplus 2}\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1223132}$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{122313},$$

$$313221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{I^{\oplus 3}},$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{I^{\oplus 5}}$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{12231512},$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1223151},$$

$$2321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{I^{\oplus 4}}$$

$$I^{\oplus 2}\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{I^{\oplus 6}},$$

$$I^{\oplus 2}\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{122315123},$$

$$31513221\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^1$$

$$I^{\oplus 2}\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{12315132},$$

$$I^{\oplus 2}\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{12231513215},$$

$$5131513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{1,5,4} \otimes \mathfrak{e}_7^{I^{\oplus 2}}$$

$$I^{\oplus 2}\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{12231513},$$

$$I^{\oplus 2}\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{12231513151},$$

$$I^{\oplus 5}\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{I^{\oplus 3}}$$

$$I^{\oplus 3}\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{1231513},$$

$$21513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{I^{\oplus 3}},$$

$$1513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{I^{\oplus 3}}$$

$$I^{\oplus 4}\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{123215},$$

$$I^{\oplus 4}\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{123151},$$

$$1513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{12315}$$

$$I^{\oplus 2}\mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$I^{\oplus 4}\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$I^{\oplus 3}\mathfrak{e}_8^{I^{\oplus 3} 5,4} \otimes \mathfrak{e}_7^{3,2}$$

$$I^{\oplus 2}\mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3},$$

$$I^{\oplus 2}\mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{I^{\oplus 3}},$$

$$I^{\oplus 4}\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$I^{\oplus 3}\mathfrak{e}_8^{I^{\oplus 3} 5,4} \otimes \mathfrak{e}_7^{3,3},$$

$$I^{\oplus 3}\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{1223},$$

$$I^{\oplus 3}\mathfrak{e}_8^{3221 5,4} \otimes \mathfrak{e}_7^{I^{\oplus 2}}$$

$$I^{\oplus 2}\mathfrak{e}_8^{3221 5,4} \otimes \mathfrak{e}_7^{12315},$$

$$513221\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{123},$$

$$I^{\oplus 2}\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{1231513}$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_6 1,$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_6 13$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,4} \otimes \mathfrak{e}_6 1,$$

$$I^{\oplus 4} \mathfrak{e}_8^{1 5,4} \otimes \mathfrak{e}_6 13$$

$$1 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 1223,$$

$$315131 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 1,$$

$$\mathfrak{e}_6^{2,2 1 2,2 3,2} \otimes \mathfrak{e}_6 \otimes,$$

$$\mathfrak{e}_6^{2,2 1 2,2 3,5} \otimes \mathfrak{e}_6 \otimes,$$

$$\mathfrak{e}_6^{2,2 1 2,2} \otimes \mathfrak{e}_6 I^{\oplus 3},$$

$$15131 \mathfrak{e}_6^{1 2,2 2,2} \otimes \mathfrak{e}_6 \otimes,$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2 3,3} \otimes \mathfrak{e}_6 \otimes,$$

$$321 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 I^{\oplus 3},$$

$$15131 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$5131 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 13151,$$

$$31 \mathfrak{e}_6^{2,2 2,2} \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$31 \mathfrak{e}_6^{31 2,2 3,4} \otimes \mathfrak{e}_6 \otimes,$$

$$31 \mathfrak{e}_6^{I^{\oplus 2} 2,2} \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$31 \mathfrak{e}_6^{31 2,2} \otimes \mathfrak{e}_6 131512,$$

$$1 \mathfrak{e}_6^{1 3,3} \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$1 \mathfrak{e}_6^{1 3,3 2,2} \bigcirc \mathfrak{e}_6 \otimes,$$

$$1 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' 1231513,$$

$$1 \mathfrak{e}_7'^{1 3,3} \otimes \mathfrak{e}_7 1223151,$$

$$\mathfrak{e}_7^{3,3 1 3,3 3,2} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' \otimes,$$

$$\mathfrak{e}_7^{2,3 1 3,3} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 3},$$

$$321 \mathfrak{e}_7'^{1 3,3 3,3} \otimes \mathfrak{e}_7 \otimes,$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 3,3 4,3} \otimes \mathfrak{e}_7 \otimes,$$

$$I^{\oplus 2} \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' 1223,$$

$$321 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' 12315,$$

$$2321 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 1,$$

$$1 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 13151315,$$

$$\mathfrak{e}_6^{2,2 1 2,2 3,3} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes,$$

$$321 \mathfrak{e}_6^{1 2,2 2,2} \otimes \mathfrak{e}_6 \otimes,$$

$$\mathfrak{e}_6^{2,2 1 2,2} \otimes \mathfrak{e}_6 \mathfrak{e}_6 1313,$$

$$\mathfrak{e}_6^{2,2 1 2,2} \otimes \mathfrak{e}_6 \mathfrak{e}_6 13151,$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2 3,4} \otimes \mathfrak{e}_6 \otimes,$$

$$5131 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 123,$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 1313,$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 13151,$$

$$31 \mathfrak{e}_6^{2,2 2,2 2,2} \otimes \mathfrak{e}_6 \otimes,$$

$$31 \mathfrak{e}_6^{31 2,2 3,2} \otimes \mathfrak{e}_6 \otimes,$$

$$31 \mathfrak{e}_6^{31 2,2 3,5} \otimes \mathfrak{e}_6 \otimes,$$

$$31 \mathfrak{e}_6^{31 2,2} \otimes \mathfrak{e}_6 I^{\oplus 3},$$

$$31 \mathfrak{e}_6^{31 2,2} \otimes \mathfrak{e}_6 13151,$$

$$31 \mathfrak{e}_6^{1 3,3} \otimes \mathfrak{e}_6 1,$$

$$1 \mathfrak{e}_6^{1 3,3} \bigcirc \mathfrak{e}_6 I^{\oplus 2},$$

$$3151321 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' 1,$$

$$\mathfrak{e}_7^{2,3 1 3,3 3,2} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' \otimes,$$

$$321 \mathfrak{e}_7'^{1 3,3 3,2} \otimes \mathfrak{e}_7 \otimes,$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 3,3 4,2} \otimes \mathfrak{e}_7 \otimes,$$

$$I^{\oplus 3} \mathfrak{e}_7^{1 3,3 3,3} \otimes \mathfrak{e}_7' \otimes,$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 3,3 4,4} \otimes \mathfrak{e}_7 \otimes,$$

$$3221 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' I^{\oplus 2},$$

$$51321 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' 123,$$

$$1 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 123215$$

$$1 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 123151$$

$$\mathfrak{e}_6^{2,2 1 2,2 3,4} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes$$

$$\mathfrak{e}_6^{2,2 1 2,2 3,3} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \bigcirc$$

$$\mathfrak{e}_6^{2,2 1 2,2} \otimes \mathfrak{e}_6 \mathfrak{e}_6 131512$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2 3,2} \otimes \mathfrak{e}_6 \otimes$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2 3,5} \otimes \mathfrak{e}_6 \otimes$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2 3,3} \otimes \mathfrak{e}_6 \bigcirc$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 131512$$

$$31 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 12315$$

$$31 \mathfrak{e}_6^{I^{\oplus 2} 2,2 2,2} \otimes \mathfrak{e}_6 \otimes$$

$$31 \mathfrak{e}_6^{31 2,2 3,3} \otimes \mathfrak{e}_6 \otimes$$

$$31 \mathfrak{e}_6^{31 2,2 3,3} \otimes \mathfrak{e}_6 \bigcirc$$

$$31 \mathfrak{e}_6^{31 2,2} \otimes \mathfrak{e}_6 1313$$

$$1 \mathfrak{e}_6^{1 3,3 2,2} \otimes \mathfrak{e}_6 \otimes$$

$$1 \mathfrak{e}_6^{1 3,4} \otimes \mathfrak{e}_6 1$$

$$31 \mathfrak{e}_6^{1 3,3} \bigcirc \mathfrak{e}_6 1$$

$$1 \mathfrak{e}_7'^{1 3,3} \otimes \mathfrak{e}_7 123151315$$

$$\mathfrak{e}_7^{2,3 1 3,3 3,3} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' \otimes$$

$$I^{\oplus 3} \mathfrak{e}_7^{1 3,3 3,2} \otimes \mathfrak{e}_7' \otimes$$

$$\mathfrak{e}_7^{3,3 1 3,3 3,3} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' \otimes$$

$$\mathfrak{e}_7^{3,3 1 3,3} \otimes \mathfrak{e}_7' I^{\oplus 3}$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 3,3 4,5} \otimes \mathfrak{e}_7 \otimes$$

$$321 \mathfrak{e}_7'^{1 3,3} \otimes \mathfrak{e}_7 I^{\oplus 3}$$

$$2321 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$\begin{aligned}
& I^{\oplus 2} \mathbf{e}_7^{2321 \ 3,3} \otimes \mathbf{e}_7' 1, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 123215, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7^{1232}, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7 1232151, \\
& I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2} \ 3,3} \otimes \mathbf{e}_7^{3,2}, \\
& I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2} \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 3}, \\
& 1 \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_7' 123, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_7' I^{\oplus 2}, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 4,5} \otimes \mathbf{e}_7' 1,
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7 I^{\oplus 4}, \\
& 512321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 2}, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 123151, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7 123151, \\
& I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2} \ 3,3} \otimes \mathbf{e}_7^{3,3}, \\
& 1 \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_7^{3,2}, \\
& 321 \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_7' 1, \\
& 1 \mathbf{e}_7^{1 \ 4,5} \otimes \mathbf{e}_7 I^{\oplus 2}, \\
& 1 \mathbf{e}_7^{1 \ 5,5} \otimes \mathbf{e}_7' 1
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7 12313 \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7 1231512 \\
& 151321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 2} \\
& I^{\oplus 3} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 3} \\
& I^{\oplus 2} \mathbf{e}_7^{321 \ 3,3} \otimes \mathbf{e}_7' I^{\oplus 2} \\
& 1 \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_7^{3,3} \\
& 1 \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_7 I^{\oplus 3} \\
& 1 \mathbf{e}_7^{1 \ 5,4} \otimes \mathbf{e}_7' I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7^{4,2}, \\
& 1 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7^{4,5}, \\
& 1 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7 I^{\oplus 4}, \\
& 1 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_6 131513, \\
& 151321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_6 1, \\
& \mathbf{e}_7^{2,3 \ 1 \ 3,3} \otimes \mathbf{e}_6^{2,2}, \\
& \mathbf{e}_7^{2,3 \ 1 \ 3,3} \otimes \mathbf{e}_6 I^{\oplus 2}, \\
& 31 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7^{3,3}, \\
& 321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_6 1315, \\
& 31 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7 I^{\oplus 3}, \\
& I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2} \ 3,3} \otimes \mathbf{e}_6 I^{\oplus 2}, \\
& 1 \mathbf{e}_7^{1 \ 4,3} \otimes \mathbf{e}_6 123, \\
& 1 \mathbf{e}_7^{1 \ 4,3} \otimes \mathbf{e}_6 13151, \\
& 31 \mathbf{e}_6^{1 \ 3,4} \otimes \mathbf{e}_7 I^{\oplus 2}, \\
& 1 \mathbf{e}_7^{1 \ 5,3} \otimes \mathbf{e}_6 I^{\oplus 2}, \\
& 1 \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_6 13,
\end{aligned}$$

$$\begin{aligned}
& 1 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7^{4,3}, \\
& 3221 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_6 1, \\
& 1 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7 12313, \\
& 1 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7 1231512, \\
& 1 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7 1232151, \\
& \mathbf{e}_7^{3,3 \ 1 \ 3,3} \otimes \mathbf{e}_6^{2,2}, \\
& 31 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7^{3,2}, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_6 123, \\
& I^{\oplus 3} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_6 I^{\oplus 2}, \\
& 51321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_6 13, \\
& 1 \mathbf{e}_6^{1 \ 3,4} \otimes \mathbf{e}_7^{3,2}, \\
& 321 \mathbf{e}_7^{1 \ 4,3} \otimes \mathbf{e}_6 1, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 4,3} \otimes \mathbf{e}_6^{2,2}, \\
& 1 \mathbf{e}_7^{1 \ 5,3} \otimes \mathbf{e}_6^{2,2}, \\
& 31 \mathbf{e}_6^{1 \ 3,5} \otimes \mathbf{e}_7 1, \\
& 1 \mathbf{e}_7^{1 \ 5,4} \otimes \mathbf{e}_6 1
\end{aligned}$$

$$\begin{aligned}
& 1 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7^{4,4} \\
& 1 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_6 1232 \\
& 512321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_6 1 \\
& 1 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7 1232 \\
& 1 \mathbf{e}_6^{1 \ 3,3} \otimes \mathbf{e}_7 123151 \\
& I^{\oplus 3} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_6^{2,2} \\
& \mathbf{e}_7^{3,3 \ 1 \ 3,3} \otimes \mathbf{e}_6 I^{\oplus 2} \\
& I^{\oplus 2} \mathbf{e}_7^{321 \ 3,3} \otimes \mathbf{e}_6 1 \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_6 13151 \\
& I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2} \ 3,3} \otimes \mathbf{e}_6^{2,2} \\
& 1 \mathbf{e}_6^{1 \ 3,4} \otimes \mathbf{e}_7^{3,3} \\
& 1 \mathbf{e}_6^{1 \ 3,4} \otimes \mathbf{e}_7 I^{\oplus 3} \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 4,3} \otimes \mathbf{e}_6 I^{\oplus 2} \\
& 1 \mathbf{e}_6^{1 \ 3,5} \otimes \mathbf{e}_7 I^{\oplus 2} \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_6 1
\end{aligned}$$

$$\begin{aligned}
& 1 \mathbf{e}_7^{1 \ 4,5} \otimes \mathbf{e}_8 1223132, \\
& 1 \mathbf{e}_7^{1 \ 4,5} \otimes \mathbf{e}_8 12231513215,
\end{aligned}$$

$$\begin{aligned}
& 1 \mathbf{e}_7^{1 \ 4,5} \otimes \mathbf{e}_8 I^{\oplus 6}, \\
& 5131513221 \mathbf{e}_8^{1 \ 5,4} \otimes \mathbf{e}_7' 1,
\end{aligned}$$

$$\begin{aligned}
& 1 \mathbf{e}_7^{1 \ 4,5} \otimes \mathbf{e}_8 122315123 \\
& 1 \mathbf{e}_7^{1 \ 4,5} \otimes \mathbf{e}_8 12231513
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8^{1,4,5} 12231513151, \\
& \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& 313221\mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& I^{\oplus 5} \mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& 1513221\mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}'_7 1232, \\
& \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}'_7 1, \\
& 313221\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}'_7 1, \\
& 1513221\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}'_7 1, \\
& I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,5} \otimes \mathfrak{e}'_7 1
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}'_7 123, \\
& \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& I^{\oplus 4} \mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}'_7 123, \\
& 21513221\mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& I^{\oplus 2} \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}'_7 1, \\
& I^{\oplus 5} \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}'_7 1, \\
& I^{\oplus 2} \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}'_7 1,
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& I^{\oplus 3} \mathfrak{e}_8^{3221 5,4} \otimes \mathfrak{e}'_7 1, \\
& I^{\oplus 3} \mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}'_7 1232, \\
& 1513221\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}'_7 1, \\
& I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}'_7 123, \\
& I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}'_7 1, \\
& 21513221\mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}'_7 1, \\
& I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,5} \otimes \mathfrak{e}'_7 1
\end{aligned}$$

$$\begin{aligned}
& 2313221\mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}1, \\
& 51231513221\mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}1, \\
& I^{\oplus 3} \mathfrak{e}_8^{2,4} \otimes \mathfrak{so}1, \\
& \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{so}13, \\
& I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}1, \\
& I^{\oplus 2} \mathfrak{e}_8^{313221 4,2} \otimes \mathfrak{so}1, \\
& 21513221\mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes \mathfrak{so}1, \\
& I^{\oplus 3} \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}1322, \\
& 21513221\mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}13, \\
& I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes \mathfrak{so}1322, \\
& I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 4,2} \otimes \mathfrak{so}13
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 6} \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}1, \\
& 31513221\mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}1, \\
& \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{so}132, \\
& I^{\oplus 2} \mathfrak{e}_8^{4,5} \otimes \mathfrak{so}1, \\
& \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}13, \\
& 313221\mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}13, \\
& 1513221\mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes \mathfrak{so}1, \\
& I^{\oplus 4} \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}132, \\
& 1513221\mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}13, \\
& I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes \mathfrak{so}132,
\end{aligned}$$

$$\begin{aligned}
& 321513221\mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}1, \\
& 15131513221\mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}1, \\
& I^{\oplus 2} \mathfrak{e}_8^{3,5} \otimes \mathfrak{so}1, \\
& \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{so}13, \\
& 3221\mathfrak{e}_8^{3221 4,2} \otimes \mathfrak{so}1, \\
& I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes \mathfrak{so}1, \\
& I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3} 4,2} \otimes \mathfrak{so}1, \\
& I^{\oplus 5} \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}13, \\
& I^{\oplus 2} \mathfrak{e}_8^{2,4} \otimes \mathfrak{so}13, \\
& I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes \mathfrak{so}13
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 \otimes, \\
& 321\mathfrak{e}'_7 \otimes \mathfrak{e}_6 1, \\
& I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}_6^{2,2}, \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_6^{4,3}, \\
& 31\mathfrak{e}_6 \otimes \mathfrak{e}'_7 1,
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 \otimes, \\
& 1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 I^{\oplus 3}, \\
& I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}_6 I^{\oplus 2}, \\
& 1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& 1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 1,
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_6 123, \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_6 13151, \\
& 31\mathfrak{e}_6 \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_6 I^{\oplus 2}, \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_6 1
\end{aligned}$$

$$\begin{aligned}
& 1\epsilon'_7 \otimes \epsilon'_7{}^{1,3,3}1223, \\
& 1\epsilon'_7 \otimes \epsilon'_7{}^{1,3,3}123151, \\
& 321\epsilon'_7 \otimes \epsilon'_7{}^{1,3,3}I^{\oplus 2}, \\
& 1\epsilon'_7 \otimes \epsilon'_7{}^{1,4,5}1
\end{aligned}$$

$$\begin{aligned}
& 2321\epsilon'_7 \otimes \epsilon'_7{}^{1,3,3}1, \\
& I^{\oplus 2}\epsilon'_7 \otimes \epsilon'_7{}^{1,3,3}{}^{3,2}, \\
& I^{\oplus 2}\epsilon'_7 \otimes \epsilon'_7{}^{1,3,3}I^{\oplus 3},
\end{aligned}$$

$$\begin{aligned}
& 1\epsilon'_7 \otimes \epsilon'_7{}^{1,3,3}123215 \\
& I^{\oplus 2}\epsilon'_7 \otimes \epsilon'_7{}^{1,3,3}{}^{3,3} \\
& 1\epsilon'_7 \otimes \epsilon'_7{}^{1,4,4}I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& {}^{2,3}1\epsilon_7 \otimes \mathfrak{so}1, \\
& 1\epsilon_7 \otimes \mathfrak{so}1{}^{3,2}32, \\
& I^{\oplus 2}\epsilon_7 \otimes \mathfrak{so}1{}^{1,4,2}
\end{aligned}$$

$$\begin{aligned}
& {}^{3,3}1\epsilon_7 \otimes \mathfrak{so}1, \\
& I^{\oplus 2}\epsilon_7 \otimes \mathfrak{so}1{}^{1,3,2}{}^{3,2}, \\
& 1\epsilon_7 \otimes \mathfrak{so}13{}^{1,4,2}
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 3}\epsilon_7 \otimes \mathfrak{so}1{}^{1,3,2} \\
& I^{\oplus 2}\epsilon_7 \otimes \mathfrak{so}13{}^{1,3,2}
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{so} \otimes \mathfrak{so}{}^{1,1}{}^{2,2}, \\
& 1\mathfrak{so} \otimes \mathfrak{so}132{}^{1,1}{}^2
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{so} \otimes \mathfrak{so}{}^{1,1}{}^{2,3}, \\
& 1\mathfrak{so} \otimes \mathfrak{so}I^{\oplus 2}{}^{1,1}
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{so} \otimes \mathfrak{so}{}^{1,1}{}^{2,4} \\
& 31\mathfrak{so} \otimes \mathfrak{so}1{}^{1,1}
\end{aligned}$$

$$\begin{aligned}
& {}^{2,2}1\epsilon_6 \otimes \mathfrak{so}1, \\
& 31\epsilon_6 \otimes \mathfrak{so}1{}^{31}{}^{2,2}
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2}\epsilon_6 \otimes \mathfrak{so}1{}^{1,2,2} \\
& 1\epsilon_6 \otimes \mathfrak{so}1{}^{1,3,2}
\end{aligned}$$

$$1\epsilon_6 \otimes \mathfrak{so}13{}^{1,2,2}$$

$$I^{\oplus 2}\epsilon'_7 \otimes \mathfrak{so}1,{}^{1,3,2}$$

$$1\epsilon'_7 \otimes \mathfrak{so}13,{}^{1,3,2}$$

$$1\epsilon'_7 \otimes \mathfrak{so}1{}^{1,4,2}$$

2.3.7 SSgLgIS(181)

$$\begin{aligned}
& {}^{2,4}1\epsilon_8 \otimes \epsilon_8{}^{5,5}11223151315, \\
& {}^{3,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}15,5 \\
& {}^{3,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}112231512, \\
& {}^{4,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}15,5 \\
& {}^{4,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}112231512, \\
& 313221\epsilon_8 \otimes \epsilon_8{}^{1,5,5}15,5 \\
& {}^{5,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}11223151, \\
& 313221\epsilon_8 \otimes \epsilon_8{}^{1,5,5}112231512, \\
& I^{\oplus 4}\epsilon_8 \otimes \epsilon_8{}^{1,5,5}11223151315, \\
& I^{\oplus 5}\epsilon_8 \otimes \epsilon_8{}^{1,5,5}11223151,
\end{aligned}$$

$$\begin{aligned}
& {}^{3,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}15,3 \\
& 313221\epsilon_8 \otimes \epsilon_8{}^{1,5,5}15,3 \\
& {}^{3,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}11223151, \\
& 313221\epsilon_8 \otimes \epsilon_8{}^{1,5,5}15,4 \\
& {}^{4,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}11223151, \\
& I^{\oplus 5}\epsilon_8 \otimes \epsilon_8{}^{1,5,5}15,5 \\
& 313221\epsilon_8 \otimes \epsilon_8{}^{1,5,5}1122313, \\
& 313221\epsilon_8 \otimes \epsilon_8{}^{1,5,5}11223151, \\
& I^{\oplus 5}\epsilon_8 \otimes \epsilon_8{}^{1,5,5}1I^{\oplus 5}, \\
& 21513221\epsilon_8 \otimes \epsilon_8{}^{1,5,5}112231512,
\end{aligned}$$

$$\begin{aligned}
& {}^{3,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}15,4 \\
& I^{\oplus 5}\epsilon_8 \otimes \epsilon_8{}^{1,5,5}15,3 \\
& {}^{4,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}15,4 \\
& I^{\oplus 5}\epsilon_8 \otimes \epsilon_8{}^{1,5,5}15,4 \\
& {}^{5,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}15,5 \\
& {}^{5,5}1\epsilon_8 \otimes \epsilon_8{}^{5,5}112231512 \\
& I^{\oplus 5}\epsilon_8 \otimes \epsilon_8{}^{1,5,5}1122313 \\
& 1\epsilon_8 \otimes \epsilon_8{}^{1,5,5}I^{\oplus 2}11223151322 \\
& I^{\oplus 5}\epsilon_8 \otimes \epsilon_8{}^{1,5,5}112231512 \\
& 1513221\epsilon_8 \otimes \epsilon_8{}^{1,5,5}112231512
\end{aligned}$$

$$\begin{aligned}
& 1513221 \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^1 1223151, \\
& \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 24,2}, \\
& I^{\oplus 5} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 24,2}, \\
& 1513221 \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 24,2}, \\
& \otimes \mathbf{e}_8^{4,5} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 2}, \\
& \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 3}, \\
& 313221 \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 3}, \\
& 5131513221 \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 3}, \\
& I^{\oplus 4} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 2} 1223151, \\
& I^{\oplus 5} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 3}, \\
& I^{\oplus 2} \mathbf{e}_8^{2,4} \otimes \mathbf{e}_8^{5,5} I^{\oplus 24,2}, \\
& I^{\oplus 2} 3221 \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} I^{\oplus 2} 122315, \\
& I^{\oplus 3} \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} I^{\oplus 2} 1223151, \\
& I^{\oplus 3} \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
& 1\epsilon_7^1 \otimes \epsilon_8^{4,5} 1223151315, \\
& I^{\oplus 2} 1\epsilon_7^1 \otimes \epsilon_8^{4,5} 1^5 4, \\
& I^{\oplus 2} 1\epsilon_7^1 \otimes \epsilon_8^{4,5} 122313, \\
& I^{\oplus 2} 1\epsilon_7^1 \otimes \epsilon_8^{4,5} 1\epsilon_8^1 I^{\oplus 5}, \\
& I^{\oplus 2} 1\epsilon_7^1 \otimes \epsilon_8^{4,5} 1223151, \\
& 2321\epsilon_7^1 \otimes \epsilon_8^{4,5} I^{\oplus 2} I^{\oplus 2}, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} 1^5 3, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} 122313, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} 1223151, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} I^{\oplus 3} I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned} & 1\epsilon_7^{1,3,3} \otimes \epsilon_7^{1,3,3} 1231513, \\ & I^{\oplus 3} \epsilon_7^{1,3,3} \otimes \epsilon_7^{1,3,2}, \\ & I^{\oplus 2} \epsilon_7^{1,3,3} \otimes \epsilon_7^{1,3,3} 1223, \\ & I^{\oplus 2} \epsilon_7^{1,3,3} \otimes \epsilon_7^{1,3,3} 123215, \end{aligned}$$

[illegible]

$$\begin{aligned}
& 321 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8^{1,4,2}, \\
& I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8^{1,5,5}, \\
& 321 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8 I^{\oplus 4}, \\
& I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8 12231512, \\
& I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes I^{\oplus 2,4,2}, \\
& I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8 I^{\oplus 2} \otimes I^{\oplus 4}, \\
& 1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,5,4}, \\
& 1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8 I^{\oplus 5}, \\
& 1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8 I^{\oplus 2,4,2},
\end{aligned}$$

$$\begin{aligned} & \begin{smallmatrix} 2,3 & 1 & 3,3 & 1 & 3,2 \\ \otimes & \mathfrak{e}_7 & \otimes & \mathfrak{e}_7 & \otimes, \end{smallmatrix} \\ & \begin{smallmatrix} 3,3 & 1 & 3,3 & 1 & 3,3 \\ \otimes & \mathfrak{e}_7 & \otimes & \mathfrak{e}_7 & \otimes, \end{smallmatrix} \\ & 321\mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} 12315, \\ & I^{\oplus 2}\mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} 123151, \end{aligned}$$

$$\begin{array}{l}
{}^{4,5}_1 \otimes \mathfrak{e}_8 \otimes {}^{5,5} I^{\oplus 24,2} \otimes \mathfrak{e}_8 \otimes \\
313221 \otimes \mathfrak{e}_8 \otimes {}^{1,5,5} I^{\oplus 24,2} \otimes \mathfrak{e}_8 \otimes \\
{}^{2,4}_1 \otimes \mathfrak{e}_8 \otimes {}^{5,5} I^{\oplus 2} \otimes \mathfrak{e}_8 \otimes 1223151 \\
{}^{3,5}_1 \otimes \mathfrak{e}_8 \otimes {}^{5,5} I^{\oplus 3} \otimes \mathfrak{e}_8 \otimes I^{\oplus 3} \\
{}^{5,5}_1 \otimes \mathfrak{e}_8 \otimes {}^{5,5} I^{\oplus 2} \otimes \mathfrak{e}_8 \otimes I^{\oplus 4} \\
313221 \otimes \mathfrak{e}_8 \otimes {}^{1,5,5} I^{\oplus 2} \otimes \mathfrak{e}_8 \otimes I^{\oplus 4} \\
I^{\oplus 3} \otimes \mathfrak{e}_8 \otimes {}^{1,5,5} I^{\oplus 2} \otimes \mathfrak{e}_8 \otimes 12231513 \\
21513221 \otimes \mathfrak{e}_8 \otimes {}^{1,5,5} I^{\oplus 2} \otimes \mathfrak{e}_8 \otimes I^{\oplus 4} \\
1513221 \otimes \mathfrak{e}_8 \otimes {}^{1,5,5} I^{\oplus 2} \otimes \mathfrak{e}_8 \otimes 122315 \\
{}^1 1513221 \otimes \mathfrak{e}_8 \otimes {}^{1,5,5} I^{\oplus 3} \otimes \mathfrak{e}_8 \otimes I^{\oplus 3} \\
I^{\oplus 2} \otimes \mathfrak{e}_8 \otimes {}^{2,4}_1 \otimes \mathfrak{e}_8 \otimes {}^{5,5} I^{\oplus 3} \otimes \mathfrak{e}_8 \otimes I^{\oplus 3} \\
I^{\oplus 2} \otimes \mathfrak{e}_8 \otimes {}^{I^{\oplus 2},5,5} I^{\oplus 2} \otimes \mathfrak{e}_8 \otimes 12231513 \\
I^{\oplus 4} \otimes \mathfrak{e}_8 \otimes {}^{I^{\oplus 2},5,5} I^{\oplus 3} \otimes \mathfrak{e}_8 \otimes I^{\oplus 3}
\end{array}$$

$$\begin{array}{l}
I^{\oplus 2} \mathbf{1}_{\mathbf{e}_7}^{4,5} \otimes \mathbf{1}_{\mathbf{e}_8}^{5,3} \\
\mathbf{1}_{\mathbf{e}_7}^{1,4,5} \otimes \mathbf{e}_8^{I^{\oplus 3}} \quad 1223 \\
2321 \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{I^{\oplus 3}} \quad 1 \\
\mathbf{1}_{\mathbf{e}_7}^{1,4,5} \otimes \mathbf{e}_8^{I^{\oplus 2}} \quad 1223151 \\
321 \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{I^{\oplus 3}} \quad I^{\oplus 2} \\
I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{I^{\oplus 3}} \quad I^{\oplus 3} \\
\mathbf{1}_{\mathbf{e}_7}^{1,5,5} \otimes \mathbf{e}_8^{1,5,5} \\
\mathbf{1}_{\mathbf{e}_7}^{1,5,5} \otimes \mathbf{e}_8^{1,12231512} \\
\mathbf{1}_{\mathbf{e}_7}^{1,5,5} \otimes \mathbf{e}_8^{I^{\oplus 2}} \quad I^{\oplus 4}
\end{array}$$

$$\begin{array}{c} \begin{array}{ccccc} 2,3 & 1 & 3,3 & 1 & 3,3 \\ \otimes & \mathbf{e}_7 & \otimes & \mathbf{e}_7 & \otimes \end{array} \\ I^{\oplus 3} \mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} \\ 1 \mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} I^{\oplus 2} 1232 \\ I^{\oplus 3} \mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} I^{\oplus 3} 1 \end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^{2,3}{}_1 {}^{3,3}{}_1 I^{\oplus 2}, \\
I^{\oplus 3} {}^1{}_7 \otimes {}^{3,3}{}_1 I^{\oplus 2}, \\
I^{\oplus 2} {}^1{}_7 \otimes {}^{4,4}{}_1 I^{\oplus 2},
\end{array}
&
\begin{array}{l}
{}^{3,3}{}_1 {}^{3,3}{}_1 I^{\oplus 2}, \\
I^{\oplus 2} {}^1{}_7 \otimes {}^{3,3}{}_1 I^{\oplus 2}, \\
{}^1{}_7 \otimes {}^{5,4}{}_1 I^{\oplus 2},
\end{array}
&
\begin{array}{l}
I^{\oplus 2} {}^1{}_7 \otimes {}^{3,3}{}_1 I^{\oplus 2} 123 \\
{}^1{}_7 \otimes {}^{4,4}{}_1 123 \\
{}^1{}_7 \otimes {}^{5,5}{}_1 1
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,5}{}_8 \otimes {}^{5,3}{}_8, \\
{}^1{}_6 \otimes {}^{3,5}{}_8 122313, \\
{}^1{}_6 \otimes {}^{3,5}{}_8 1223151, \\
{}^1{}_6 \otimes {}^{3,5}{}_8 I^{\oplus 2}, \\
31 {}^1{}_6 \otimes {}^{3,5}{}_8 I^{\oplus 3}
\end{array}
&
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,5}{}_8 \otimes {}^{5,4}{}_8, \\
{}^1{}_6 \otimes {}^{3,5}{}_8 I^{\oplus 5}, \\
{}^1{}_6 \otimes {}^{3,5}{}_8 I^{\oplus 24,2}, \\
{}^1{}_6 \otimes {}^{3,5}{}_8 I^{\oplus 3},
\end{array}
&
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,5}{}_8 \otimes {}^{5,5}{}_8 \\
{}^1{}_6 \otimes {}^{3,5}{}_8 12231512 \\
31 {}^1{}_6 \otimes {}^{3,5}{}_8 \otimes {}^{4,2}{}_8 \\
31 {}^1{}_6 \otimes {}^{3,5}{}_8 I^{\oplus 4}
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^1{}_7' \otimes {}^{3,3}{}_1 1223, \\
{}^1{}_7' \otimes {}^{3,3}{}_1 123151, \\
{}^1{}_7' \otimes {}^{3,3}{}_1 I^{\oplus 2} 123, \\
I^{\oplus 2} {}^1{}_7' \otimes {}^{3,3}{}_1 I^{\oplus 2},
\end{array}
&
\begin{array}{l}
2321 {}^1{}_7' \otimes {}^{3,3}{}_1 1, \\
I^{\oplus 2} {}^1{}_7' \otimes {}^{3,3}{}_1 3,2, \\
321 {}^1{}_7' \otimes {}^{3,3}{}_1 I^{\oplus 2} 1, \\
{}^1{}_7' \otimes {}^{4,4}{}_1 I^{\oplus 2} 1,
\end{array}
&
\begin{array}{l}
{}^1{}_7' \otimes {}^{3,3}{}_1 123215 \\
I^{\oplus 2} {}^1{}_7' \otimes {}^{3,3}{}_1 3,3 \\
I^{\oplus 2} {}^1{}_7' \otimes {}^{3,3}{}_1 I^{\oplus 3} 1 \\
{}^1{}_7' \otimes {}^{4,5}{}_1 1
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^1{}_6 \otimes {}^{2,2}{}_6 123, \\
I^{\oplus 2} {}^1{}_6 \otimes {}^{2,2}{}_6 \otimes {}^{2,2}{}_6, \\
{}^1{}_6 \otimes {}^{2,2}{}_6 I^{\oplus 2} 13, \\
31 {}^1{}_6 \otimes {}^{2,2}{}_6 I^{\oplus 2} 1,
\end{array}
&
\begin{array}{l}
{}^1{}_6 \otimes {}^{2,2}{}_6 13151, \\
31 {}^1{}_6 \otimes {}^{2,2}{}_6 1, \\
31 {}^1{}_6 \otimes {}^{2,2}{}_6 1315, \\
{}^1{}_6 \otimes {}^{3,3}{}_6 1,
\end{array}
&
\begin{array}{l}
{}^{2,2}{}_6 \otimes {}^{2,2}{}_6 \otimes {}^{2,2}{}_6 \\
I^{\oplus 2} {}^1{}_6 \otimes {}^{2,2}{}_6 I^{\oplus 2} 1 \\
31 {}^1{}_6 \otimes {}^{2,2}{}_6 \otimes {}^{2,2}{}_6 \\
{}^1{}_6 \otimes {}^{3,3}{}_6 1
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^1{}_7' \otimes {}^{4,5}{}_8 \otimes {}^{5,3}{}_8, \\
{}^1{}_7' \otimes {}^{4,5}{}_8 122313, \\
{}^1{}_7' \otimes {}^{4,5}{}_8 1223151, \\
{}^1{}_7' \otimes {}^{4,5}{}_8 I^{\oplus 3}
\end{array}
&
\begin{array}{l}
{}^1{}_7' \otimes {}^{4,5}{}_8 \otimes {}^{5,4}{}_8, \\
{}^1{}_7' \otimes {}^{4,5}{}_8 I^{\oplus 5}, \\
{}^1{}_7' \otimes {}^{4,5}{}_8 I^{\oplus 24,2}, \\
{}^1{}_7' \otimes {}^{4,5}{}_8 1
\end{array}
&
\begin{array}{l}
{}^1{}_7' \otimes {}^{4,5}{}_8 \otimes {}^{5,5}{}_8 \\
{}^1{}_7' \otimes {}^{4,5}{}_8 12231512 \\
{}^1{}_7' \otimes {}^{4,5}{}_8 I^{\oplus 2}
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,3}{}_7 \otimes {}^{3,2}{}_7, \\
{}^1{}_6 \otimes {}^{3,3}{}_7 I^{\oplus 2}, \\
31 {}^1{}_6 \otimes {}^{3,4}{}_7 1,
\end{array}
&
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,3}{}_7 \otimes {}^{3,3}{}_7, \\
31 {}^1{}_6 \otimes {}^{3,3}{}_7 I^{\oplus 2} 1, \\
{}^1{}_6 \otimes {}^{3,5}{}_7 1
\end{array}
&
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,3}{}_7 I^{\oplus 3} \\
{}^1{}_6 \otimes {}^{3,4}{}_7 I^{\oplus 2} 1
\end{array}
\end{array}$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}'_7 123,$$

$$I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}'_7 1,$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 1,$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}'_7 1,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} 1$$

2.4 3 node(1907)

2.4.1 gLgLg(11)

$$\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{so},$$

$$\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7$$

$$\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}'_7 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}'_7$$

$$\mathfrak{so} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_6 \otimes \mathfrak{so}$$

$$\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

2.4.2 gLg(8)

$$\mathfrak{so} \otimes \overset{2,4}{\mathfrak{e}_8} \otimes \overset{I^{\oplus 3} 4,2}{\mathfrak{so}}$$

$$\mathfrak{so} \otimes \overset{2,4}{\mathfrak{e}_8} \otimes \overset{I^{\oplus 2} 5,5}{\mathfrak{e}_7'}$$

$$\mathfrak{so} \otimes \overset{2,4}{\mathfrak{e}_8} \otimes \overset{I^{\oplus 2} 5,4}{\mathfrak{e}_6}$$

$$\mathfrak{so} \otimes \overset{2,3}{\mathfrak{e}_7} \otimes \overset{1 \ 3,2}{\mathfrak{so}}$$

$$\mathfrak{so} \otimes \overset{2,2}{\mathfrak{e}_6} \otimes \overset{1 \ 2,2}{\mathfrak{so}}$$

$$\mathfrak{e}_7' \otimes \overset{5,5}{\mathfrak{e}_8} \otimes \overset{1 \ 5,5}{\mathfrak{e}_7'}$$

$$\mathfrak{e}_6 \otimes \overset{4,5}{\mathfrak{e}_8} \otimes \overset{1 \ 5,5}{\mathfrak{e}_7'}$$

$$\mathfrak{e}_6 \otimes \overset{4,5}{\mathfrak{e}_8} \otimes \overset{1 \ 5,4}{\mathfrak{e}_6}$$

2.4.3 Sg(170)

$$513221\mathfrak{e}_7 \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}},$$

$$\overset{2,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{4,2}{\mathfrak{so}},$$

$$\overset{2,4}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}},$$

$$\overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}},$$

$$2151321\mathfrak{e}_7 \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}},$$

$$\overset{1}{151321\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}}$$

$$\overset{3,5}{\mathfrak{e}_8} \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}},$$

$$313221\mathfrak{e}_8 \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}},$$

$$\overset{1}{1513221\mathfrak{e}_8} \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}},$$

$$I^{\oplus 3} \overset{4,5}{\mathfrak{e}_7} \otimes \overset{5,3}{\mathfrak{e}_8} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \overset{4,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{e}_7} \otimes \mathfrak{so},$$

$$\overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{4,2}{\mathfrak{so}},$$

$$\overset{3,4}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}},$$

$$I^{\oplus 4} \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \mathfrak{so},$$

$$\overset{1}{2321\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}},$$

$$\overset{4,5}{\mathfrak{e}_8} \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}},$$

$$I^{\oplus 5} \overset{5,4}{\mathfrak{e}_8} \otimes \overset{3,2}{\mathfrak{e}_7} \otimes \mathfrak{so},$$

$$\overset{2,3}{\mathfrak{e}_7} \otimes \overset{4,5}{\mathfrak{e}_8} \otimes \overset{5,3}{\mathfrak{so}},$$

$$I^{\oplus 2} \overset{5,5}{\mathfrak{e}_7} \otimes \overset{5,3}{\mathfrak{e}_8} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_7 \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}}$$

$$I^{\oplus 3} \overset{3,3}{\mathfrak{e}_7} \otimes \overset{4,2}{\mathfrak{e}_7} \otimes \mathfrak{so}$$

$$\overset{4,4}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}}$$

$$31321\mathfrak{e}_7 \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}}$$

$$1512321\mathfrak{e}_7 \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}}$$

$$\overset{5,5}{\mathfrak{e}_8} \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}}$$

$$21513221\mathfrak{e}_8 \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}}$$

$$\overset{3,3}{\mathfrak{e}_7} \otimes \overset{4,5}{\mathfrak{e}_8} \otimes \overset{5,3}{\mathfrak{so}}$$

$$\begin{array}{ccccc} 5,5 & 5,3 & 2,2 & & \\ \otimes e_8 & \otimes e_6 & \otimes s_0 & & \\ 21513221e_8 & \otimes e_6 & \otimes s_0 & & \\ 2,2 & 3,5 & 5,3 & & \\ \otimes e_6 & \otimes e_8 & \otimes s_0 & & \end{array}$$

$$\begin{array}{c} I^{\oplus 3} \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{so} \\ \begin{array}{ccc} 3,3 & 2,2 & \\ \otimes & \otimes & \end{array} \\ 5131 \mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \\ \begin{array}{ccc} 3,3 & 3,2 & \\ \otimes & \otimes & \end{array} \\ \begin{array}{ccc} 2,2 & 3,3 & 3,3 \\ \otimes & \otimes & \otimes \end{array} \end{array} \mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$\begin{array}{l} 2321\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,4} \otimes \mathfrak{e}_7 \\ 321\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_7 \\ 31\mathfrak{e}_6^{3,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6 \end{array}$$

$$\begin{array}{l} 2321\epsilon'_7 \otimes^{3,3} \epsilon_7 \otimes^{4,4} \epsilon_7 \\ 151321\epsilon_7 \otimes^{3,3} \epsilon_7 \otimes^{4,4} \epsilon'_7 \\ 321\epsilon'_7 \otimes^{3,3} \epsilon_7 \otimes^{4,5} \epsilon_7 \end{array}$$

$$\begin{array}{l} I^{\oplus 3} \mathbf{e}_7 \otimes^{3,3} \mathbf{e}'_7 \otimes^{3,2} \mathbf{50} \\ I^{\oplus 2} \mathbf{e}'_7 \otimes^{3,3} \mathbf{e}_7 \otimes^{4,2} \mathbf{50} \\ I^{\oplus 3} \mathbf{e}'_7 \otimes^{3,3} \mathbf{e}_7 \otimes^{3,3} \mathbf{50} \end{array}$$

$$\begin{array}{c} 151321e_7 \otimes^{3,3} e_7 \otimes^{4,3} e_6 \\ 321e_7 \otimes^{4,4} e_7 \otimes^{3,3} e_6 \end{array}$$

$$315131\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}'_7$$

$$321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}'_7,$$

$$I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}'_7,$$

$$15131\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}'_7,$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}_6$$

$$\otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}'_7$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}_6,$$

$$321\mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$151321\mathfrak{e}_7 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}_6$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$151321\mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$2231\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_7,$$

$$321\mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$231\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_7$$

$$5131513221\mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$51321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$321\mathfrak{e}_6 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$31\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_6,$$

$$231\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_6,$$

$$31\mathfrak{e}_6 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$15131\mathfrak{e}_6 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_6$$

$$321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$15131\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \bigcirc \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$15131\mathfrak{e}_6 \otimes \mathfrak{e}_6 \bigcirc \mathfrak{e}_6$$

$$2313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$51231513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$31513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$321513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$15131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$2313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$51231513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$31513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$321513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$15131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$51321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_6 \bigcirc \mathfrak{e}_6 \otimes \mathfrak{so}$$

$$I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}$$

$$\begin{array}{c} 5131\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ 1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array}$$

$$\begin{array}{c} 2,2 \quad 3,5 \quad 5,5 \\ \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \end{array}$$

$$I^{\oplus 2} \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\begin{array}{c} 3151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7, \\ 151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \end{array}$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7,$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7' \otimes \mathfrak{so}$$

$$5131\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7',$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$321\mathfrak{e}_7' \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$231\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_7',$$

$$31\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_7'$$

$$3151321\mathfrak{e}_7'(\otimes \mathfrak{e}_7')^2,$$

$$321\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7',$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7' \otimes \mathfrak{so}$$

$$51321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7'$$

$$321\mathfrak{e}_7' \otimes \mathfrak{e}_7' \otimes \mathfrak{e}_6$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$2321\mathfrak{e}_7'(\otimes \mathfrak{e}_7')^2$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{so}$$

2.4.4 SgLgLgS(589)

$$I^{\oplus 3} \mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,3}{\otimes}$$

$$I^{\oplus 2} \mathbf{e}'_7 \otimes^{3,3} \mathbf{e}_7 \otimes^{4,4} \mathbf{e}_7 I^{\oplus 2}$$

$$\begin{array}{l}
2,3 \quad 3,3 \quad 4,5 \quad 5,4 \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \\
2,3 \quad 3,3 \quad 4,5 \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 I^{\oplus 5} \\
3,3 \quad 3,3 \quad 4,5 \quad 4,2 \\
3221 \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \\
3,3 \quad 3,3 \quad 4,5 \quad 5,3 \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \\
3,3 \quad 3,3 \quad 4,5 \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 122313 \\
3,3 \quad 3,3 \quad 4,5 \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 1223151 \\
I^{\oplus 3} \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \\
2,3 \quad 3,3 \quad 4,5 \\
23151321 \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 I^{\oplus 2} \\
3,3 \quad 4,5 \\
3151321 \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 I^{\oplus 3} \\
3,3 \quad 4,5 \\
512321 \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 I^{\oplus 4} \\
2,3 \quad 4,5 \quad 5,4 \quad 3,2 \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \mathbf{e}_7 \otimes \\
3,3 \quad 4,5 \quad 5,4 \quad 3,3 \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \mathbf{e}_7 \otimes \\
I^{\oplus 3} \mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \mathbf{e}_7 I^{\oplus 3} \\
3,3 \quad 4,5 \quad 5,5 \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \mathbf{e}_7 I^{\oplus 2}
\end{array}$$

$$\begin{array}{l}
\begin{array}{ccccc} 2,3 & 4,5 & 5,5 & & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 122315123 & \end{array} \\
\begin{array}{ccccc} 2,3 & 4,5 & 5,5 & & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 12231513151 & \end{array} \\
\begin{array}{ccccc} 3,3 & 4,5 & 5,5 & & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 122315123 & \end{array} \\
\begin{array}{ccccc} 3,3 & 4,5 & 5,5 & & \\ \otimes \mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 12231513151 & \end{array} \\
\begin{array}{ccccc} & 4,5 & 5,5 & 5,5 & \\ 51321\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & \otimes & \end{array} \\
\begin{array}{ccccc} & 4,5 & 5,5 & & \\ I^{\oplus 3}\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & I^{\oplus 6} & \end{array} \\
\begin{array}{ccccc} & 4,5 & 5,5 & & 1 \\ I^{\oplus 3}\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 12231513 & \end{array} \\
\begin{array}{ccccc} & 4,5 & 5,5 & & \\ 51321\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 12231512 & \end{array} \\
\begin{array}{ccccc} & 5,5 & 5,5 & & \\ I^{\oplus 2}\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & I^{\oplus 6} & \end{array} \\
\begin{array}{ccccc} & 5,5 & 5,5 & & 1 \\ I^{\oplus 2}\mathbf{e}_7 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_8 & 12231513 & \end{array}
\end{array}$$

$$31513221\mathbf{e}_8 \otimes \mathbf{e}_8 \otimes \mathbf{e}_8 \otimes \mathbf{e}_8 12231513151, \quad 15131513221\mathbf{e}_8 \otimes \mathbf{e}_8 \otimes \mathbf{e}_8 \otimes \mathbf{e}_8 12231513151$$

$$\begin{array}{l}
I^{\oplus 2} \mathfrak{e}'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \\
I^{\oplus 2} \mathfrak{e}'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 122313 \\
I^{\oplus 2} \mathfrak{e}'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 5} \\
1 \mathfrak{e}'_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 12315 \\
I^{\oplus 2} \mathfrak{e}'_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 I^{\oplus 3} \\
1 \mathfrak{e}'_7 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 I^{\oplus 3}
\end{array}$$

$$\begin{array}{l}
150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 122315123 \\
150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 12231513151 \\
3150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \\
223150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 I^{\oplus 3} \\
3150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 12231512 \\
3150 \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} 5013
\end{array}$$

$$\begin{array}{ccccccc}
1c'_7 & \otimes^{3,3} & c'_7 & \otimes^{3,3} & e_7 & 123215 \\
\otimes^{2,3} & c'_7 & \otimes^{3,3} & e_7 & \otimes^{3,3} & c'_7 & \otimes^{3,3} \\
\otimes^{3,3} & c'_7 & \otimes^{3,3} & e_7 & \otimes^{3,3} & c'_7 & \otimes^{3,3} \\
321c'_7 & \otimes^{3,3} & c'_7 & \otimes^{3,3} & e_7 & I^{\oplus 2} \\
I^{\oplus 3} & c'_7 & \otimes^{3,3} & e_7 & \otimes^{3,3} & c'_7 & I^{\oplus 3}
\end{array}$$

$$\begin{array}{cccc}
2,2 & 1,1 & 1,1 & 2,3 \\
\otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \\
I^{\oplus 2} \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} \\
3,2 & 1,1 & 1,1 & 2 \\
\otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \mathfrak{so} 132 \\
4,2 & 1,1 & 1,1 & 2 \\
\otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \mathfrak{so} 132 \\
I^{\oplus 2} \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} & I^{\oplus 2}
\end{array}$$

$$\begin{array}{c}
1\epsilon_6^{\otimes 3,3} \otimes \epsilon_7' \otimes \epsilon_7 I^{\oplus 3} \\
\otimes \epsilon_6^{2,2} \otimes \epsilon_7^{3,3} \otimes \epsilon_7' \otimes \epsilon_7^{3,3} \\
I^{\oplus 2} \epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7' \otimes \epsilon_7^{3,3} \\
5131 \epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7' I^{\oplus 2} \\
1\epsilon_6^{3,3} \otimes \epsilon_7^{4,4} \otimes \epsilon_7' 1
\end{array}$$

$$\begin{array}{l} I^{\oplus 2} \mathfrak{e}'_7 \otimes^{4,5} \mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 122315123 \\ I^{\oplus 2} \mathfrak{e}'_7 \otimes^{4,5} \mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 12231513151 \\ 1\mathfrak{e}'_7 \otimes^{5,5} \mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 122315123 \\ 1\mathfrak{e}'_7 \otimes^{5,5} \mathfrak{e}_8 \otimes^{5,5} \mathfrak{e}_8 12231513151 \end{array}$$

$$\begin{array}{l} 150 \overset{2,3}{\otimes} e_7 \overset{3,3}{\otimes} e_7 \overset{4,4}{\otimes} \\ 150 \overset{2,3}{\otimes} e_7 \overset{3,3}{\otimes} e_7 12313 \\ 150 \overset{2,3}{\otimes} e_7 \overset{3,3}{\otimes} e_7 1232151 \end{array}$$

$$\begin{array}{l} 150^{2,2} \otimes e_6^{2,2} \otimes e_6^{3,4} \\ 150^{2,2} \otimes e_6^{2,2} \otimes e_6 I^{\oplus 3} \\ 150^{2,2} \otimes e_6^{2,2} \otimes e_6^{13151} \end{array}$$

$$\begin{array}{c} 150 \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_7 123151 \\ 3150 \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_7 I^{\oplus 3} \end{array}$$

$$\begin{array}{cc} 1_{50}^{1,1} \otimes 5_0^{2,3} & e_7 I^{\oplus 3} \\ 1_{50}^{1,1} \otimes 5_0^{2,4} & e_7 I^{\oplus 2} \end{array}$$

$$15131\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}'_7 1$$

$$\begin{array}{c} \begin{array}{ccc} \begin{array}{c} 2,2 \\ \otimes \end{array} \mathbf{e}_6 & \begin{array}{c} 2,2 \\ \otimes \end{array} \mathbf{e}_6 & \begin{array}{c} 3,3 \\ \otimes \end{array} \mathbf{e}'_7 I^{\oplus 2}, \\ I^{\oplus 2} \mathbf{e}_6 & \begin{array}{c} 2,2 \\ \otimes \end{array} \mathbf{e}_6 & \begin{array}{c} 3,4 \\ \otimes \end{array} \mathbf{e}'_7 1, \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}'_7 I^{\oplus 2},$$

$$1 \mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes} \mathbf{e}_6 1$$

$$\begin{array}{ccc} 2,2 & 2,2 & 3,4 \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}'_7 1 \end{array}$$

$$\begin{array}{c} 5131e_6 \otimes^{3,5} e_8 \otimes^{5,4} e'_7 1, \\ \otimes^{2,2} e_6 \otimes^{3,5} e_8 \otimes^{5,5} e'_7 1, \\ 1e_6 \otimes^{4,5} e_8 \otimes^{5,5} e'_7 1 \end{array}$$

$$\begin{array}{ccc} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 & \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 & \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 I^{\oplus 2}, \\ I^{\oplus 2} \mathbf{e}_6 & \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 & \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 1, \end{array}$$

$$\begin{array}{ccccc} I^{\oplus 2} \mathbf{e}_6 & \overset{3,5}{\otimes} & \mathbf{e}_8 & \overset{5,4}{\otimes} & \mathbf{e}'_7 I^{\oplus 2} \\ 1 \mathbf{e}_6 & \overset{4,5}{\otimes} & \mathbf{e}_8 & \overset{5,4}{\otimes} & \mathbf{e}'_7 I^{\oplus 2} \end{array}$$

$$\begin{aligned} & 31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8 1223132, \\ & 31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8 12231513215, \end{aligned}$$

$$3150 \otimes^{2,4} \epsilon_8 \otimes^{5,5} \epsilon_8 I^{\oplus 6},$$

$$3150 \otimes^{2,4} \epsilon_8 \otimes^{5,5} \epsilon_8 12231513,$$

$$\begin{array}{l} 3150 \otimes^{2,4} \epsilon_8 \otimes^{5,5} \epsilon_8 122315123 \\ 3150 \otimes^{2,4} \epsilon_8 \otimes^{5,5} \epsilon_8 12231513151 \end{array}$$

$$\begin{array}{l} 1s0 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} e_6 123, \\ 31s0 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} e_6 I^{\oplus 2}, \end{array}$$

$$\begin{array}{l} 150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} e_6 13151, \\ 150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e_6 13, \end{array}$$

$$\begin{array}{l} 3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \epsilon_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \epsilon_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \\ 3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \epsilon_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \epsilon_6 1 \end{array}$$

$$\begin{array}{l} 1_{50}^{1,1} \otimes 5_0^{2,2} \otimes e_6^{2,2}, \\ 1_{50}^{1,1} \otimes 5_0^{2,3} \otimes e_6 1, \end{array}$$

$$\begin{array}{cc} 1,1 & 2,2 \\ 1s_0 \otimes s_0 & \otimes e_6 I^{\oplus 2}, \\ 2,2 & 2,2 \\ 1s_0 \otimes e_6 & \otimes s_0 1 \end{array}$$

$$31\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,2} \oplus \mathfrak{e}_6 1$$

$$I^{\oplus 2} \mathbf{e}'_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}'_7 I^{\oplus 2},$$

$$I^{\oplus 2} \mathbf{e}'_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}'_7 1,$$

$$1e'_7 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e'_7 1$$

$$150 \otimes^{2,4} e_8 \otimes^{5,4} e'_7 123,$$

$$3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e'_7 I^{\oplus 2},$$

$$3150 \otimes^{2,4} e_8 \otimes^{5,5} e'_7 1$$

$$150 \otimes^{2,3} e_7 \otimes^{3,3} e'_7 \otimes^{3,2},$$

$$150 \otimes^{2,3} e_7 \otimes^{3,3} e'_7 \otimes^{3,3},$$

$$150 \otimes^{2,3} e_7 \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$150 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e'_7 I^{\oplus 2},$$

$$3150 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e'_7 1,$$

$$150 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e'_7 1$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix},$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_6 I^{\oplus 2}$$

$$1e'_7 \overset{3,3}{\otimes} e'_7 \overset{3,3}{\otimes} e'_7 123,$$

$$I^{\oplus 2} \mathbf{e}'_7 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes} \mathbf{e}'_7 I^{\oplus 2}$$

$$1e_6 \otimes^{3,3} e'_7 \otimes^{3,3} e'_7 I^{\oplus 2},$$

$$31e_6 \overset{3,3}{\otimes} e'_7 \overset{3,3}{\otimes} e'_7 1$$

2.4.5 SgLIgIg(65)

$$\begin{aligned} & 5131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \\ & 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \\ & 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \\ & 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 \\ & I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 \end{aligned}$$

$$\begin{aligned} & 5131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ & 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ & 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ & 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \\ & I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, \\ & 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, \\ & 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, \\ & I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \\ & 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \end{aligned}$$

$$\begin{aligned} & 321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ & 1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{aligned}$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7,$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$51321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$5131\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}$$

$$31\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_6,$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_6$$

$$2321\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7,$$

$$321\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7$$

$$2321\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6,$$

$$315131\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6$$

$$I^{\oplus 2}\epsilon'_7 \otimes \epsilon_7 \otimes \mathfrak{so}$$

$$5131513221\epsilon_8 \otimes \epsilon_8 \otimes \epsilon_7$$

$$321\epsilon'_7 \otimes \epsilon_7 \otimes \epsilon_6$$

$$31\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$31\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_6$$

$$3151321\epsilon_7 \otimes \epsilon_7 \otimes \epsilon'_7$$

$$2321\epsilon'_7 \otimes \epsilon_7 \otimes \epsilon'_7$$

$$23151321\epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \mathfrak{so}$$

$$1\epsilon'_7 \otimes \epsilon_8 \otimes \mathfrak{so}$$

$$1\epsilon'_7 \otimes \epsilon_8 \otimes \epsilon'_7$$

$$1\epsilon_6 \otimes \epsilon_7 \otimes \mathfrak{so}$$

2.4.6 SgLIgLgS(128)

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7 123151,$$

$$321\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7 I^{\oplus 2},$$

$$I^{\oplus 2}\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7,$$

$$I^{\oplus 2}\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7 I^{\oplus 3}$$

$$I^{\oplus 2}\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8,$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 122313,$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 1223151$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8,$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 I^{\oplus 5},$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 12231512$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 1223132,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 12231513215,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 I^{\oplus 6},$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 12231513,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 122315123$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 12231513151$$

$$1\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 123,$$

$$I^{\oplus 2}\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6,$$

$$1\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 13151,$$

$$I^{\oplus 2}\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 I^{\oplus 2},$$

$$\otimes \epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6$$

$$31\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 1315$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7 \otimes \epsilon_7,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7 I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7 \otimes \epsilon_7,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7 I^{\oplus 3}$$

$$1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7 123,$$

$$1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7 1$$

$$I^{\oplus 2}\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7 I^{\oplus 2},$$

$$I^{\oplus 2}\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7 1$$

$$1\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7,$$

$$31\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_7 I^{\oplus 2}$$

$$1\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7,$$

$$1\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_7 I^{\oplus 3}$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_6 \otimes \epsilon_6,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_6 I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_6 1$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_7 I^{\oplus 2},$$

$$31\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_7 1,$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_7 1$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \mathfrak{so} 1,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7' I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7' 1$$

$$1\epsilon'_7 \otimes \epsilon_8 \otimes \epsilon_7 I^{\oplus 2},$$

$$1\epsilon'_7 \otimes \epsilon_8 \otimes \epsilon_7 1$$

$$1\epsilon'_7 \otimes \epsilon_7 \otimes \epsilon'_7 123,$$

$$I^{\oplus 2} \epsilon'_7 \otimes \epsilon_7 \otimes \epsilon'_7 I^{\oplus 2}$$

$$1\epsilon_6 \otimes \epsilon_7 \otimes \epsilon'_7 I^{\oplus 2},$$

$$31\epsilon_6 \otimes \epsilon_7 \otimes \epsilon'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\epsilon'_7 \otimes \epsilon_8 \otimes \epsilon'_7 1$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon'_7 1$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_6 1$$

$$1\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_6 1$$

2.4.7 SSgLg(80)

$$513221 \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6,$$

$$\otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6,$$

$$\otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6$$

$$\otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6,$$

$$313221 \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6,$$

$$I^{\oplus 5} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6$$

$$21513221 \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6,$$

$$1513221 \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6,$$

$$I^{\oplus 2} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6$$

$$I^{\oplus 4} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6,$$

$$I^{\oplus 3} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6$$

$$\otimes \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so},$$

$$\otimes \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so},$$

$$\otimes \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so}$$

$$313221 \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so},$$

$$I^{\oplus 5} \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so},$$

$$21513221 \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so}$$

$$1513221 \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so},$$

$$I^{\oplus 4} \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so}$$

$$I^{\oplus 3} \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so}$$

$$51321 \epsilon_7 \otimes \epsilon_7 \otimes \mathfrak{so},$$

$$1\epsilon_7 \otimes \epsilon_7 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \epsilon_7 \otimes \epsilon_7 \otimes \mathfrak{so}$$

$$31513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2},$$

$$1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_6^{5,5}$$

$$2321 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,4},$$

$$321 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5}$$

$$2321 \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2,$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5}$$

$$2321 \mathfrak{e}_7^1 \otimes \mathfrak{e}_6^{3,3} \otimes \mathfrak{e}_6^{2,2},$$

$$31 \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_7^{3,5}$$

$$513221 I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{5,5}$$

$$321 \mathfrak{e}_7^1 \otimes \mathfrak{so}^{3,2} \otimes \mathfrak{so}^{1,1}$$

$$31 \mathfrak{e}_6^1 \otimes \mathfrak{so}^{2,2} \otimes \mathfrak{so}^{1,1}$$

$$31513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7'^{3,3}$$

$$231513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_6^{5,5}$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{so}^{5,3}$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_7'^{3,3} \otimes \mathfrak{so}^{3,2}$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_6^{3,3} \otimes \mathfrak{so}^{2,2}$$

[illegible]

[illegible]

[illegible]

$$\begin{array}{c}
I^{\oplus 2} \epsilon_7^{r^{\oplus 2} \, 3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{1 \, 1232} \\
1 \epsilon_7^{1 \, 4,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,2} \\
321 \epsilon_7^{1 \, 3,3} \otimes \epsilon_7^{4,4} \otimes \epsilon_7^1 \\
I^{\oplus 2} \epsilon_7^{1 \, 3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_7^1
\end{array}$$

$$\begin{array}{l}
1 \quad 3,3 \quad 4,5 \\
1\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 1223151315 \\
2,4 \quad 1 \quad 5,4 \quad 3,3 \quad 3,3 \\
\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \\
I^{\oplus 4} 1 \quad 5,4 \quad 3,3 \quad 3,3 \\
\mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \\
I^{\oplus 2} 1 \quad 3,3 \quad 4,5 \quad 5,5 \\
\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \\
I^{\oplus 2} 1 \quad 3,3 \quad 4,5 \\
\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 122313 \\
2321\mathfrak{e}_7 \quad 1 \quad 3,3 \quad 4,5 \\
\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 I^{\oplus 3} \\
I^{\oplus 2} 1 \quad 3,3 \quad 4,5 \\
\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 12231512 \\
I^{\oplus 4} 1 \quad 5,4 \quad 3,3 \\
\mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 I^{\oplus 3} \\
I^{\oplus 3} I^{\oplus 2} 5,4 \quad 3,3 \quad 3,2 \\
\mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \\
I^{\oplus 2} 3221 \quad 5,4 \quad 3,3 \\
\mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 I^{\oplus 2} \\
I^{\oplus 3} I^{\oplus 2} 5,4 \quad 3,3 \\
\mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 I^{\oplus 3} \\
I^{\oplus 2} 1 \quad 4,5 \quad 5,4 \quad 3,3 \\
\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \\
1 \quad 5,5 \quad 5,4 \quad 3,3 \\
1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \\
1 \quad 5,5 \quad 5,5 \\
1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 I^{\oplus 2}
\end{array}$$

$$\begin{array}{l}
1^{3,5} \quad 5,5 \\
1\epsilon_6^1 \otimes \epsilon_8 \otimes \epsilon_8 122315123 \\
\\
1^{3,5} \quad 5,5 \\
1\epsilon_6^1 \otimes \epsilon_8 \otimes \epsilon_8 12231513151 \\
\\
5,5 \quad 1 \quad 5,5 \quad 5,3 \quad 2,2 \\
\otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6 \otimes \\
\\
21513221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6^{2,2} \\
\\
3,5 \quad 1 \quad 5,5 \quad 5,3 \\
\otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6 I^{\oplus 2} \\
\\
313221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6 I^{\oplus 2} \\
\\
21513221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6 I^{\oplus 2} \\
\\
I^{\oplus 2} \otimes \epsilon_8^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6^{2,2} \\
\\
I^{\oplus 2} \otimes \epsilon_8^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6 I^{\oplus 2} \\
\\
I^{\oplus 3} \otimes \epsilon_8^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6 1315
\end{array}$$

$$I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}'_7 1$$

$$\begin{array}{l}
1 \quad 3,3 \quad 3,3 \quad 4,4 \\
1\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_7 \otimes \\
1 \quad 3,3 \quad 3,3 \\
1\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7 12313 \\
1 \quad 3,3 \quad 3,3 \quad 1 \\
1\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7 1232 \\
2,3 \quad 1 \quad 3,3 \quad 3,3 \quad 2,2 \\
\otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_6 \otimes \\
2,3 \quad 1 \quad 3,3 \quad 3,3 \\
\otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_6 I^{\oplus 2} \\
I^{\oplus 2} \quad 1 \quad 3,3 \quad 3,3 \\
\epsilon_7 \otimes \epsilon_7 \otimes \epsilon_6 1315 \\
I^{\oplus 2} \quad 1 \quad 3,3 \quad 4,3 \\
\epsilon_7 \otimes \epsilon_7 \otimes \epsilon_6 1
\end{array}$$

$$\begin{array}{l} \begin{array}{ccc} 5,5 & 1 & 4,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} \end{array} 1 \\ 21513221 \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} 1 \\ \begin{array}{ccc} 2,4 & 1 & 4,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} \end{array} 13 \\ I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} 1 \\ I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} 132 \end{array}$$

$$\begin{array}{l}
I^{\oplus 3} \mathbf{e}_8^{1 \oplus 2, 5, 4} \otimes \mathbf{e}_7^{3, 3} \otimes \mathbf{e}_6 1 \\
1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3, 5} \otimes \mathbf{e}_7^{5, 4} \otimes \mathbf{e}_6^{3, 2} \\
1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4, 5} \otimes \mathbf{e}_6^{5, 3} \mathbf{1315} \\
1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_6^{5, 3} \otimes \mathbf{e}_6^{2, 2} \\
I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4, 5} \otimes \mathbf{e}_6^{5, 4} 1
\end{array}$$

$$\begin{array}{l} 1\epsilon_7^{1,3,3} \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{123} \\ 1\epsilon_7^{1,3,3} \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{13151} \\ 31\epsilon_6^{1,2,2} \otimes \epsilon_6^{3,3} \otimes \epsilon_7^{I^{\oplus 2}} \end{array}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1 \oplus 2} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}1$$

$$\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{1 \oplus 2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}1,$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 \oplus 2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}1,$$

$$31 \mathfrak{e}_6^{31 \oplus 2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}_7^{1 \oplus 3,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1,$$

$$1 \mathfrak{e}_7^{1 \oplus 3,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}13,$$

$$1 \mathfrak{e}_7^{1 \oplus 4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

$$321 \mathfrak{e}_7^{1 \oplus 4,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}1,$$

$$I^{\oplus 2} \mathfrak{e}_7^{1 \oplus 4,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}13,$$

$$1 \mathfrak{e}_7^{1 \oplus 5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}13$$

$$1 \mathfrak{e}_7'^{1 \oplus 4,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7' I^{\oplus 2},$$

$$1 \mathfrak{e}_7'^{1 \oplus 4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7' 1$$

$$1 \mathfrak{e}_6^{1 \oplus 3,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}13,$$

$$31 \mathfrak{e}_6^{1 \oplus 3,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 \oplus 3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}1$$

$$1 \mathfrak{e}_7'^{1 \oplus 4,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}13$$

$$1 \mathfrak{e}_7'^{1 \oplus 3,3} \otimes \mathfrak{e}_7'^{3,3} \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1 \mathfrak{e}_7'^{1 \oplus 3,3} \otimes \mathfrak{e}_7'^{3,3} \otimes \mathfrak{e}_6 1$$

$$1 \mathfrak{e}_7'^{1 \oplus 3,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

$$1 \mathfrak{e}_6^{1 \oplus 3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}1$$

$$1 \mathfrak{e}_6^{1 \oplus 2,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

[illegible]

69

$$I^{\oplus 2} \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2} I^{\oplus 2},$$

$$31 \mathbf{e}_6^{31} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2} \mathbf{e}_6^1,$$

$$31 \mathbf{e}_6^{31} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2} I^{\oplus 2}$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_8^{4,5} \mathbf{e}_8^1,$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_8^{4,5} I^{\oplus 4},$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_8^{4,5} I^{\oplus 3}$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_7^{5,4} I^{\oplus 2},$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_7^{5,5} \mathbf{e}_7^1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \mathbf{e}_7^1,$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \mathbf{e}_7^1,$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 3}$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2}$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2},$$

$$31 \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_7^{3,3} \mathbf{e}_7^1,$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_7^{3,4} \mathbf{e}_7^1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{3,3} \mathbf{e}_6^1$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2},$$

$$I^{\oplus 2} \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2}$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} \otimes \mathbf{e}_7^{5,4} I^{\oplus 2},$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} \otimes \mathbf{e}_7^{5,5} \mathbf{e}_7^1$$

$$1 \mathbf{50} \otimes \mathbf{50}^{1,1} \otimes \mathbf{50}^{1,1} \mathbf{1}$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_7^{5,4} \mathbf{e}_7^1$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \mathbf{e}_7^1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} \otimes \mathbf{e}_7^{5,4} \mathbf{e}_7^1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2}$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_7^{3,3} \mathbf{e}_7^1$$

2.4.10 SSgLIgIg(25)

$$1513221 \mathbf{e}_8^{1\ 5,5} \otimes \mathbf{e}_8^{1\ 5,4} \otimes \mathbf{e}_7',$$

$$I^{\oplus 4} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^1 \otimes \mathbf{e}_7'^{5,5},$$

$$I^{\oplus 2} \begin{smallmatrix} 3221 \\ \mathbf{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathbf{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}'_7,$$

$$I^{\oplus 3} I^{\oplus 2}_{\mathfrak{e}_8} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}'_7$$

$$\begin{array}{ccc} 2,4 & 1 & 5,5 \\ \otimes & \otimes & \otimes \\ \mathbf{e}_8 & \mathbf{e}_8 & \mathbf{e}'_7 \end{array}$$

$$1513221 \mathbf{e}_8^{1 \ 5,5} \otimes \mathbf{e}_8^{1 \ 5,3} \otimes \mathbf{e}_6,$$

$$I^{\oplus 4} \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_6,$$

$$I^{\oplus 2} \begin{smallmatrix} 3221 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_6,$$

$$I^{\oplus 3} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_6$$

$$\begin{array}{ccc} 2,4 & 1 & 5,5 \\ \otimes & \otimes & \otimes \\ \mathfrak{e}_8 & \mathfrak{e}_8 & \mathfrak{e}_6 \end{array}$$

$$\begin{matrix} 2,4 & 1 & 5,5 & I^{\oplus 2} & 4,2 \\ \otimes \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{so}, \end{matrix}$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes I^{\oplus 2} 5,5 \otimes I^{\oplus 2} 4,2 \otimes 50$$

$$I^{\oplus 4} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,5 \\ \mathfrak{e}_8 \end{smallmatrix} I^{\oplus 2} \begin{smallmatrix} 4,2 \\ \mathfrak{e}_8 \end{smallmatrix} \otimes \mathfrak{so},$$

$$513221^{I \oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$31513221 \mathbf{e}_8^{1, 5, 5} \otimes \mathbf{e}_8^{1, 5, 4} \otimes \mathbf{e}_7,$$

$$1513221 \mathbf{e}_8^{1 \ 5,5} \otimes \mathbf{e}_8^{1 \ 5,5} \otimes \mathbf{e}_7,$$

$$I^{\oplus 2} \begin{smallmatrix} 3221 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_7$$

$$I^{\oplus 2} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{1,3,2} \otimes \mathfrak{so}$$

$$321\mathbf{e}_7 \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \otimes \mathbf{e}_6$$

$$2321 \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7'$$

$$1_{\mathfrak{e}_7} \otimes I^{4,5} \oplus 1_{\mathfrak{e}_8} \otimes I^{4,2}$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{1}{\otimes} \overset{5,5}{e_7'}$$

$$1 \overset{1}{\mathbf{e}}_7 \overset{4,5}{\otimes} \overset{1}{\mathbf{e}}_8 \overset{5,4}{\otimes} \mathbf{e}_6$$

$$1e_7' \overset{1}{\otimes} \overset{3,3}{e_7} \overset{1}{\otimes} \overset{3,2}{e_7} \otimes \mathfrak{so}$$

$$1_{\mathfrak{e}_6}^{1,2,2} \otimes 1_{\mathfrak{e}_6}^{1,2,2} \otimes \mathfrak{so}$$

2.4.11 SSgLIgLgS(96)

$$321 \mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7'1,$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7 I^{\oplus 3},$$

$$I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7' I^{\oplus 2}$$

$$\otimes \mathbf{e}_8^{2,4} \otimes \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_6^{1,5,3},$$

$$I^{\oplus 4} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_6^{1,5,3},$$

$$I^{\oplus 3} \mathbf{e}_8^{I^{\oplus 2}} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_6^{1,5,3}$$

$$I^{\oplus 3} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_6^{1,5,3}13,$$

$$I^{\oplus 2} \mathbf{e}_8^{I^{\oplus 2}} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_6^{1,5,3}13$$

$$\otimes \mathbf{e}_8^{2,4} \otimes \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_7^{1,5,4},$$

$$I^{\oplus 4} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_7^{1,5,4},$$

$$I^{\oplus 3} \mathbf{e}_8^{I^{\oplus 2}} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_7^{1,5,4}$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2},$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2} I^{\oplus 2},$$

$$31 \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2}$$

$$I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8^{1,4,2} \otimes \mathfrak{so}1,$$

$$1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,4,2} \otimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{1,3,3},$$

$$1 \mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{1,3,3}$$

$$1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8^{1,5,4} I^{\oplus 2},$$

$$1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_7^{1,5,5}$$

$$1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7'1$$

$$1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8^{1,5,3} \otimes \mathbf{e}_6^{1,5,3}$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7' I^{\oplus 2}$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{1,3,3}$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} \otimes \mathbf{e}_8^{1,4,2} \otimes \mathfrak{so}1$$

2.4.12 SSgLIgLIgS(22)

$$\begin{array}{lll}
1\epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} 122315132, & \begin{smallmatrix} 2,4\ 1\ 5,5\ 1\ 5,5\ 1\ 4,2 \\ \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \end{smallmatrix}, & I^{\oplus 4} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \\
3221\epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} 122315, & I^{\oplus 2} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} 12231513, & I^{\oplus 3} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} 1223151 \\
I^{\oplus 4} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 4}, & \begin{smallmatrix} 2,4\ 1\ 5,5\ 1\ 5,5\ 1\ 5,5\ I^{\oplus 3} \\ \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \end{smallmatrix} I^{\oplus 2}, & I^{\oplus 3} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 2} 1223 \\
1513221\epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 2}, & I^{\oplus 4} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 3} I^{\oplus 2}, & I^{\oplus 2} I^{\oplus 2} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 2} 1223 \\
I^{\oplus 3} \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 5,5\ 1\ 5,5\ 1} I^{\oplus 3} I^{\oplus 2} & &
\end{array}$$

$$\begin{array}{lll}
1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 4,5\ 1\ 5,5\ 1\ 4,2}, & 1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 4,5\ 1\ 5,5\ 1} I^{\oplus 4}, & 1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_8^{1\ 4,5\ 1\ 5,5\ 1} I^{\oplus 3} I^{\oplus 2}
\end{array}$$

$$\begin{array}{ll}
1\epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7^{1\ 3,3\ 1\ 3,3\ 1} 123, & I^{\oplus 2} \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7^{1\ 3,3\ 1\ 3,3\ 1} I^{\oplus 2} 1
\end{array}$$

$$1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7^{1\ 4,5\ 1\ 5,4\ 1} 1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7^{1\ 3,3\ 1\ 3,3\ 1} I^{\oplus 2} 1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7^{1\ 3,3\ 1\ 3,3\ 1} 1$$

$$1\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6^{1\ 2,2\ 1\ 2,2\ 1} 1$$

2.5 4 node(1730)

2.5.1 gLgLG(16)

$$\mathfrak{so}(\otimes \epsilon_6)^{2\ 2,2} \otimes \epsilon_6^{3,3}, \quad \mathfrak{so}(\otimes \epsilon_6)^{2\ 2,2} \bigcirc \epsilon_6^{3,3}$$

$$\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8^{3,5\ 5,5\ 5,5} \otimes \epsilon_6$$

$$\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8^{3,5\ 5,5\ 5,3} \otimes \mathfrak{so}$$

$$\mathfrak{so}(\otimes \epsilon_7)^{2\ 3,3} \otimes \mathfrak{so}^{3,3}$$

$$50 \overset{2,3}{\otimes} e_7 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_7$$

$$50 \begin{matrix} 2,3 \\ \otimes \end{matrix} e_7 \begin{matrix} 3,3 \\ \otimes \end{matrix} e_7 \begin{matrix} 4,4 \\ \otimes \end{matrix} e'_7$$

$$50 \overset{2,3}{\otimes} e_7 \overset{3,3}{\otimes} e_7 \overset{4,3}{\otimes} e_6$$

$$\mathfrak{so}^{2,3} \otimes \mathfrak{e}_7^{3,3} \oplus \mathfrak{e}_7'^{3,3} \otimes \mathfrak{e}_6$$

$$\mathfrak{so}\left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6\right)^2 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_7$$

$$\mathfrak{so}\left(\bigotimes_{\mathbb{C}} \mathfrak{e}_6\right)^2 \bigotimes_{\mathbb{C}} \mathfrak{e}_7'$$

$$\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,3} \oplus \mathfrak{e}_7 \otimes \mathfrak{so}^{3,2}$$

$$\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,2} \oplus \mathfrak{e}_6 \otimes \mathfrak{so}^{2,2}$$

$$\mathfrak{e}_6 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^2 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$e_6 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_7'$$

$$\begin{array}{ccccc} & 3,3 & & 4,5 & & 5,4 \\ e_6 & \otimes & e_7 & \otimes & e_8 & \otimes & e_6 \end{array}$$

2.5.2 gLIgLgIg(11)

$$e_6 \overset{3,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_7',$$

$$e_7' \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 1 \quad 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e_6$$

$$\mathfrak{so} \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} I^{\oplus 2} \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$50 \begin{matrix} 2,4 \\ \otimes \end{matrix} e_8 \begin{matrix} 5,4 \\ \otimes \end{matrix} e_7 \begin{matrix} 3,3 \\ \otimes \end{matrix} e_6$$

$$e_7 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_7'$$

$$\mathbf{e}_7 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_6$$

$$e'_7 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50$$

$$e'_7 \overset{4,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e'_7$$

$$\mathfrak{e}_6 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} \mathfrak{so} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$\mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_6$$

$$e_6 \overset{3,3}{\otimes} e_7 \overset{1}{\otimes} e_7 \overset{3,3}{\otimes} e_7 \overset{3,2}{\otimes} 50$$

2.5.3 gLIgLIgLg(3)

$$50 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_7$$

$$50 \quad \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \quad \begin{smallmatrix} 1 \quad 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \quad \begin{smallmatrix} 1 \quad 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}'_7$$

$$50 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ e_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ e_8 \end{smallmatrix} \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} e_6$$

2.5.4 SgLgLgLg(175)

$$\begin{array}{l} 3,5 \quad 5,5 \quad 5,4 \quad 3,3 \\ \otimes e_8 \otimes e_8 \otimes e_7 \otimes e_6, \\ 313221e_8 \quad 5,5 \quad 5,4 \quad 3,3 \\ \otimes e_8 \otimes e_7 \otimes e_6, \\ 1513221e_8 \quad 5,5 \quad 5,4 \quad 3,3 \\ \otimes e_8 \otimes e_7 \otimes e_6, \\ I^{\oplus 3}_{e_7 \otimes e_8 \otimes e_8 \otimes e_6}, \end{array}$$

$$\begin{array}{cccc}
4,5 & 5,5 & 5,4 & 3,3 \\
\otimes \epsilon_8 & \otimes \epsilon_8 & \otimes \epsilon_7 & \otimes \epsilon_6, \\
I^{\oplus 5} \epsilon_8 & \otimes \epsilon_8 & \otimes \epsilon_7 & \otimes \epsilon_6, \\
2,3 & 4,5 & 5,5 & 5,5 \\
\otimes \epsilon_7 & \otimes \epsilon_8 & \otimes \epsilon_8 & \otimes \epsilon_6, \\
I^{\oplus 2} \epsilon_7 (\otimes \epsilon_8)^2 & \otimes \epsilon_6
\end{array}$$

$$\begin{array}{cccc} 5,5 & 5,5 & 5,4 & 3,3 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6 \\ 21513221\mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6 \\ 3,3 & 4,5 & 5,5 & 5,5 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_6 \end{array}$$

$$1e_7 \begin{smallmatrix} 4,4 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} so,$$

$$\begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$\begin{matrix} 3,4 \\ \otimes \end{matrix} \mathfrak{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{e}_7)^2 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{so}$$

$$I^{\oplus 4} \mathbf{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{so}$$

$$\begin{array}{c} 315131\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^2\overset{3,3}{\otimes}\epsilon_7 \\ \overset{2,2}{\otimes}\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^2\overset{3,5}{\otimes}\epsilon_7 \\ 31\epsilon_6\overset{3,3}{\otimes}\epsilon_7\overset{3,3}{\otimes}\epsilon_6\overset{2,2}{\otimes}\epsilon_6 \end{array}$$

$$\begin{array}{c} 321e_7 \otimes^{3,3} e_7' \otimes^{3,3} e_7 \otimes^{3,3} e_6 \\ 1e_6 (\otimes^{3,3} e_7)^2 \otimes^{4,4} e_7' \end{array}$$

$$\begin{array}{c} 2321e_7 \otimes^{3,3} e'_7 \otimes^{3,3} e_7 \otimes^{3,3} e'_7 \\ I^{\oplus 2} e'_7 (\otimes^{3,3} e_7)^2 \otimes^{4,4} e'_7 \end{array}$$

$$151321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^2\overset{4,4}{\otimes}\mathbf{e}_7$$

$$I^{\oplus 3}\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^2\overset{4,5}{\otimes}\mathbf{e}_7$$

$$I^{\oplus 2} e'_7 \left(\bigotimes e_7 \right)^2 \bigotimes e_7$$

$$\begin{array}{l} 321513221\epsilon_8(\overset{5,5}{\otimes}\epsilon_8)^2\overset{5,5}{\otimes}\epsilon_6 \\ 15131513221\epsilon_8(\overset{5,5}{\otimes}\epsilon_8)^2\overset{5,5}{\otimes}\epsilon_6 \end{array}$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes \epsilon_8)^2 \otimes \mathfrak{so} \\ 15131513221\epsilon_8(\otimes \epsilon_8)^2 \otimes \mathfrak{so} \end{array}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\overset{(3,3)}{\otimes} \mathbf{e}_7 \right)^2 \overset{(4,3)}{\otimes} \mathbf{e}_6$$

$$31\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{e}_6$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$231513221\mathfrak{e}_8 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2,$$

$$223151321\mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_8,$$

$$2321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7$$

$$2321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7,$$

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7$$

$$1513221\mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$I^{\oplus 2}\mathfrak{e}_7 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \bigcirc \mathfrak{e}_6$$

$$I^{\oplus 2}\mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \bigcirc \mathfrak{e}_6$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}'_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}'_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so}$$

$$I^{\oplus 3}\mathfrak{e}'_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so}$$

$$5131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6$$

$$321\mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}'_7,$$

$$15131\mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}'_7,$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}'_7$$

$$I^{\oplus 2}\mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}'_7$$

$$I^{\oplus 2} \mathfrak{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$321\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}'_7$$

$$51321\mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{4,2}{\otimes} \mathbf{50}$$

$$5131\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so}$$

$$1\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,3}{\otimes} \mathfrak{e}_7 \overset{3,2}{\otimes} \mathfrak{so}$$

$$150 \overset{2,4}{\otimes} e_8 \overset{5,3}{\otimes} e_6 \overset{2,2}{\otimes} 50$$

$$1\mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_6$$

$$31\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2\overset{2,4}{\otimes}\mathfrak{e}_7$$

$$1\mathfrak{so}(\overset{1,1}{\otimes} \mathfrak{so})^2 \overset{2,3}{\otimes} \mathfrak{e}_6$$

$$1e'_7 \overset{4,5}{\otimes} e_8 \overset{5,4}{\otimes} e_7 \overset{3,3}{\otimes} e_6$$

$$I^{\oplus 2} \mathfrak{e}_6 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \oplus \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$1e'_7 \left(\overset{5,5}{\otimes} e_8 \right)^2 \overset{5,5}{\otimes} e_6$$

$$I^{\oplus 2} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 I^{\oplus 2}$$

$$\begin{array}{ccccccccccc} 2,2 & 3,5 & 5,5 & 5,5 & & & & & & & \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & & & & & & & \\ & & & & 122315123 & & & & & & \end{array}$$

$$\begin{array}{cccccccc} 2,2 & 3,5 & 5,5 & 5,5 \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & 12231513151 \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 122315123$$

$$I^{\oplus 2} \mathbf{e}_6^{3,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} 12231513151$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 122315123$$

$$1e_6 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 12231513151$$

$$1e'_7 \otimes^{3,3} e_7 \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{5,5}$$

$$1e_7' \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 12231512$$

$$I^{\oplus 2} \mathbf{e}'_7 \otimes^{4,5} \mathbf{e}_8 \otimes^{5,5} \mathbf{e}_8 \otimes^{5,4} \mathbf{e}_7 \otimes^{3,3}$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^2 \overset{5,4}{\otimes} e_7 \overset{3,3}{\otimes}$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} e_7 I^{\oplus 2}$$

$$1\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes}$$

$$1e_6 \overset{2,2}{\otimes} e_6 \overset{3,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 12231512$$

$$31\mathbf{e}_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 I^{\oplus 4}$$

$$I^{\oplus 2} \mathfrak{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$\begin{array}{c} 2,2 \\ \otimes \mathfrak{so} \end{array} \left(\begin{array}{c} 1,1 \\ \otimes \mathfrak{so} \end{array} \right)^2 \begin{array}{c} 1,1 \\ \otimes \mathfrak{so} \end{array} \begin{array}{c} 2,4 \\ \otimes \end{array}$$

$$\begin{array}{c} 3,2 \\ \otimes \mathfrak{so} \end{array} \left(\begin{array}{c} 1,1 \\ \otimes \mathfrak{so} \end{array} \right)^2 \begin{array}{c} 1,1 \\ \otimes \mathfrak{so} \end{array} \begin{array}{c} 2,3 \\ \otimes \end{array}$$

$$I^{\oplus 2} \mathfrak{so} \left(\begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \right)^2 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix}$$

$$I^{\oplus 2} \mathfrak{so} \left(\begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \right)^2 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \otimes \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix}$$

$$\begin{matrix} 2,2 & (2,2) & 3,5 & 4,2 \\ \otimes \mathbf{e}_6 & (\otimes \mathbf{e}_6)^2 & \otimes \mathbf{e}_8 & \otimes \end{matrix}$$

[illegible]

$$3150 \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 2,4 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} 12231513215, \quad 3150 \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 2,4 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} 12231513, \quad 3150 \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 2,4 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} 12231513151$$

$$\begin{array}{lll} 150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7}^{5,4} 12315, & 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7}^{5,4} \otimes_{\mathfrak{e}_7}^{3,2}, & 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{5,4} \otimes_{\mathfrak{e}_7}^{3,3} \\ 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7}^{5,4} I^{\oplus 3}, & 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7} I^{\oplus 2} & \end{array}$$

$$\begin{array}{lll} 1\epsilon_6^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{(4,5)} \otimes \epsilon_8^{(4,2)}, & 1\epsilon_6^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{(4,5)} I^{\oplus 4}, & 31\epsilon_6^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{(4,5)} I^{\oplus 3} \\ 1\epsilon_6^{(3,5)} \otimes \epsilon_8^{(5,4)} \otimes \epsilon_7^{(3,3)} \otimes \epsilon_7 I^{\oplus 2}, & 31\epsilon_6^{(3,5)} \otimes \epsilon_8^{(5,4)} \otimes \epsilon_7^{(3,3)} \otimes \epsilon_7 1 & \end{array}$$

$$\begin{array}{lll}
150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6 1315, & 3150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6^{2,2}, & 3150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6 I^{\oplus 2} \\
3150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_6 1 & &
\end{array}$$

$$\begin{array}{lll} 1\overset{1,1}{\mathfrak{s}}\overset{2,4}{\mathfrak{o}}\overset{5,4}{\mathfrak{e}}_8\overset{3,2}{\mathfrak{e}}_7\otimes, & 1\overset{1,1}{\mathfrak{s}}\overset{2,4}{\mathfrak{o}}\overset{5,4}{\mathfrak{e}}_8\overset{3,3}{\mathfrak{e}}_7\otimes, & 1\overset{1,1}{\mathfrak{s}}\overset{2,4}{\mathfrak{o}}\overset{5,4}{\mathfrak{e}}_8\overset{3,3}{\mathfrak{e}}_7I^{\oplus 3} \\ 1\overset{1,1}{\mathfrak{s}}\overset{2,4}{\mathfrak{o}}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\mathfrak{e}}_7I^{\oplus 2} & & \end{array}$$

$$I^{\oplus 2} e_7' \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{5,4} e_7' I^{\oplus 2}, \quad I^{\oplus 2} e_7' \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{5,5} e_7' 1, \quad 1 e_7' (\otimes^{5,5} e_8)^2 \otimes^{5,5} e_7' 1,$$

$$150 \overset{2,4}{\otimes} \overset{5,4}{e_8} \overset{3,3}{\otimes} \overset{3,2}{e_7} \otimes, \quad 150 \overset{2,4}{\otimes} \overset{5,4}{e_8} \overset{3,3}{\otimes} \overset{3,3}{e_7} \otimes, \quad 150 \overset{2,4}{\otimes} \overset{5,4}{e_8} \overset{3,3}{\otimes} \overset{3,3}{e_7} I^{\oplus 3}$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} e_7' \otimes, \quad 150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7' \otimes, \quad 150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7' I^{\oplus 3}$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} 8 \overset{5,3}{\otimes} 6 \overset{2,2}{\otimes}, \quad 150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} 8 \overset{5,3}{\otimes} 6 I^{\oplus 2}, \quad 150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} 8 \overset{5,4}{\otimes} 6 1$$

$$150 \left(\begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \right)^2 \otimes \epsilon_7 I^{\oplus 2}, \quad 3150 \left(\begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \right)^2 \otimes \epsilon_7 1, \quad 150 \left(\begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \right)^2 \otimes \epsilon_7 1$$

$$1\epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7 I^{\oplus 2}, \quad 31\epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7 1, \quad 1\epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_7 1$$

$$\begin{array}{cc} \begin{array}{c} 2,2 \\ \otimes \mathbf{e}_6 \end{array} (\begin{array}{c} 2,2 \\ \otimes \mathbf{e}_6 \end{array})^2 \begin{array}{c} 3,3 \\ \otimes \mathbf{e}'_7 \end{array} 1, & I^{\oplus 2} \mathbf{e}_6 (\begin{array}{c} 2,2 \\ \otimes \mathbf{e}_6 \end{array})^2 \begin{array}{c} 3,3 \\ \otimes \mathbf{e}'_7 \end{array} 1 \end{array}$$

$$3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ e_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,4 \\ e_8 \end{smallmatrix} \otimes e_7' I^{\oplus 2}, \quad 3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ e_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,5 \\ e_8 \end{smallmatrix} \otimes e_7' 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,5} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2},$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_6^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{so}^{3,2} I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{so}^{2,2} I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{so}^{1,1})^2 \otimes \mathfrak{e}_7^{2,3} I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{so}^{1,1})^2 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,4} I^{\oplus 2}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2}$$

2.5.6 SgLgLIgLg(57)

$$\begin{array}{lll}
5131513221e_8 \otimes e_8 \otimes e_8 \otimes e_7, & \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7, & \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7 \\
\begin{array}{c} 5,5 \quad 5,5 \quad 1 \quad 5,4 \\ \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7, \end{array} & \begin{array}{c} 3,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,5 \\ \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7, \end{array} & \begin{array}{c} 4,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,5 \\ \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7 \end{array} \\
\begin{array}{c} 5,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,5 \\ \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7, \end{array} & 313221e_8 \otimes e_8 \otimes e_8 \otimes e_7, & I^{\oplus 5}e_8 \otimes e_8 \otimes e_8 \otimes e_7 \\
\begin{array}{c} 5,5 \quad 5,5 \quad 1 \quad 5,5 \\ \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_7, \end{array} & \begin{array}{c} 1 \quad 5,5 \quad 5,5 \quad 1 \quad 5,5 \\ 1513221e_8 \otimes e_8 \otimes e_8 \otimes e_7 \end{array} &
\end{array}$$

$$\begin{array}{lll} \otimes_{\epsilon_8}^{3,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{e'_7}^{5,4}, & \otimes_{\epsilon_8}^{4,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{e'_7}^{5,4}, & \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{e'_7}^{5,4} \\ 313221\epsilon_8 \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{e'_7}^{5,4}, & I^{\oplus 5}\epsilon_8 \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{e'_7}^{5,4}, & 21513221\epsilon_8 \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{e'_7}^{5,4} \\ 1513221\epsilon_8 \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^{5,5} \otimes_{\epsilon_8}^1 \otimes_{e'_7}^{5,4} & & \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 3,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ 1 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} & \begin{array}{l} 4,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ 5,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} & \begin{array}{l} 5,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \\ \\ 5,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} \end{array}$$

$$I^{\oplus 2} \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}, \quad 31 \epsilon_6^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,4}, \quad 1 \epsilon_6^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,5}$$

$$1 \epsilon_7^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}$$

$$\begin{array}{lll} 3221\epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7', & 2321\epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7, & 512321\epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7' \\ 151321\epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7' & & \end{array}$$

$$\begin{array}{ccc} \begin{array}{cccc} 2,3 & 3,3 & 3,3 & 1,3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6, \end{array} & \begin{array}{cccc} 3,3 & 3,3 & 3,3 & 1,3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6, \end{array} & I^{\oplus 3} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \end{array}$$

$$I^{\oplus 2} e_7^{4,5} \otimes e_8^{5,5} \otimes e_8^1 \otimes e_7^{5,4}, \quad 1e_7^{4,5} \otimes e_8^{5,5} \otimes e_8^1 \otimes e_7^{5,5}, \quad 1e_7^{5,5} \otimes e_8^{5,5} \otimes e_8^1 \otimes e_7^{5,4}$$

$$321e_7^{4,5} \otimes e_8^{5,5} \otimes e_8^1 \otimes e_7^{5,4}, \quad I^{\oplus 2}e_7^{4,5} \otimes e_8^{5,5} \otimes e_8^1 \otimes e_7^{5,5}, \quad 1e_7^{5,5} \otimes e_8^{5,5} \otimes e_8^1 \otimes e_7^{5,5}$$

$$I^{\oplus 2} \mathfrak{e}'_7 \otimes^{3,3} \mathfrak{e}_7 \otimes^{3,3} \mathfrak{e}_7 \otimes^1 \mathfrak{e}_7 \otimes^{3,3} \mathfrak{e}_6, \quad 31 \mathfrak{e}_6 \otimes^{3,3} \mathfrak{e}_7 \otimes^{3,3} \mathfrak{e}_7 \otimes^1 \mathfrak{e}_7 \otimes^{3,3} \mathfrak{e}'_7$$

$$321\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2}, \quad 15131\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2}$$

$$1\epsilon_6 \otimes \epsilon_8^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,4},$$

$$1\epsilon_7' \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}$$

$$I^{\oplus 2} \epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}^{4,2}$$

$$321\epsilon_7' \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^1 \otimes \epsilon_7^{3,3}$$

$$3151321\epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^1 \otimes \epsilon_7^{3,3}$$

$$1\mathfrak{so}^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{I^{\oplus 2}} \otimes \mathfrak{so}^{4,2}$$

$$1\mathfrak{so}^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,5}$$

$$1\mathfrak{so}^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,4}$$

$$1\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^1 \otimes \mathfrak{so}^{1,1}$$

$$1\epsilon_7' \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,4}$$

$$1\epsilon_7' \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}^{4,2}$$

$$1\epsilon_6 \otimes \epsilon_8^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}$$

$$1\epsilon_6 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^1 \otimes \epsilon_6^{3,3}$$

$$1\epsilon_6 \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}^{4,2}$$

2.5.7 SgLIgLG(28)

$$\begin{aligned}
& \begin{array}{l} \begin{array}{c} 3,5 \quad 5,5 \quad 1 \quad 4,2 \quad 1,1 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} 4,5 \quad 5,5 \quad 1 \quad 4,2 \quad 1,1 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} 5,5 \quad 5,5 \quad 1 \quad 4,2 \quad 1,1 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} \end{array} \\ \\ \begin{array}{c} 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} \end{array} \\ \\ \begin{array}{c} 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \end{array} \quad \begin{array}{c} 2,4 \quad 5,5 \quad 1 \quad 5,5 \quad 5,5 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \end{array} \quad \begin{array}{c} I^{\oplus 4} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 \end{array} \\ \\ \begin{array}{c} 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \end{array} \quad \begin{array}{c} 2,4 \quad 5,5 \quad 1 \quad 5,5 \quad 5,4 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \end{array} \quad \begin{array}{c} I^{\oplus 4} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} \\ \\ \\ \\ \begin{array}{c} I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} 1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 31513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7, \end{array} \quad \begin{array}{c} 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \end{array} \\ \\ \\ \\ \begin{array}{c} I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \end{array} \\ \\ \\ \\ \begin{array}{c} 2321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7 \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so} \end{array} \end{aligned}$$

2.5.8 SgLIgLgLgS(93)

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'1, \quad 1\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 3}, \quad I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1$$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1, \quad I^{\oplus 4}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1, \quad I^{\oplus 3}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 13$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' I^{\oplus 2}, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes, \quad 1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 I^{\oplus 2}, \quad 31\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 1$$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1, \quad I^{\oplus 4}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 1, \quad 1\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 13$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' I^{\oplus 2}, \quad 1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 1$$

2.5.9 SgLIgLIgLg(3)

$$I^{\oplus 4} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so}$$

$$1e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{4,2}{\otimes} 50$$

2.5.10 SgLIgLIgLgS(1)

$$150 \overset{2,4}{\otimes} \overset{1}{e_8} \overset{5,5}{\otimes} \overset{1}{e_8} \overset{4,2}{\otimes} 501$$

2.5.11 SSgLgLgLg(76)

$$\begin{matrix} 5,5 & 1 & (5,5) & 5,5 \\ \otimes & \mathbf{e}_8 & \otimes \mathbf{e}_8 & \otimes \mathbf{e}_6 \end{matrix}$$

$$21513221\mathbf{e}_8^1(\otimes^{5,5}\mathbf{e}_8)^2\otimes^{5,5}\mathbf{e}_6$$

$$I^{\oplus 4} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_6 \end{smallmatrix}$$

$${}^{5,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^2 {}^{5,3} \otimes \mathfrak{so}$$

$$21513221\mathbf{e}_8(\overset{1}{\otimes}\overset{5,5}{\mathbf{e}_8})^2\overset{5,3}{\otimes}\mathbf{so}$$

$$I^{\oplus 4} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8)^2 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$I^{\oplus 3} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_6$$

$$1 \mathbf{e}_7 \begin{pmatrix} 5,5 \\ \otimes \end{pmatrix} \mathbf{e}_8 \begin{pmatrix} 5,5 \\ \otimes \end{pmatrix} \mathbf{e}_6$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^2 \bigotimes \mathbf{so}$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,4}{\otimes} \overset{3,3}{e_7} \otimes e_6$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,4}{\otimes} \overset{3,3}{e_7} \otimes e_6$$

$$1e_7' \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} e_7 \begin{pmatrix} 4,5 \\ \otimes \end{pmatrix} e_7$$

$$1e_7' \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} e_7 \begin{pmatrix} 4,5 \\ \otimes \end{pmatrix} e_7$$

$$1513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2}, \quad I^{\oplus 2} 3221 \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6, \quad 1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,5}$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2} 5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7^1 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}_7^{1'} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_6^{4,3}$$

$$321 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7^{3,3})^2,$$

$$1 \mathfrak{e}_7^{1'} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{4,4}$$

$$321 \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{4,4},$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{4,5}$$

$$31 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,4},$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,5}$$

$$31 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,3},$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,4}$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7^{1'} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{so}^{3,3}$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_6^{3,3},$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \bigcirc \mathfrak{e}_6^{3,3}$$

$$1513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1'},$$

$$I^{\oplus 2} 3221 \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1'}$$

$$1513221 \mathfrak{e}_8^1 \otimes \mathfrak{so}^{4,2} (\otimes \mathfrak{so}^{1,1})^2,$$

$$I^{\oplus 2} 3221 \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,3,3} (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6^{4,3}$$

$$513221 I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$321 \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6$$

$$31513221 \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_8^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,5}$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_8^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}^{5,3}$$

$$1 \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7'$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{3,3}$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{so}^{3,2} \otimes \mathfrak{so}^{1,1}$$

$$1 \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}$$

$$1 \mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}$$

$$1 \mathfrak{e}_6^{1,3,3} (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so}^{3,3}$$

$$1 \mathfrak{e}_6^{1,2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

2.5.12 SSgLgLgLgS(471)

$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} 12231513215,$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} 122315131,$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} 12231513151$
$I^{\oplus 5} \epsilon_8^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 3},$	$21513221 \epsilon_8^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 3},$	$1513221 \epsilon_8^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 3}$
$I^{\oplus 2} \epsilon_8^{2,4} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 \otimes,$	$I^{\oplus 4} \epsilon_8^{I^{\oplus 2}\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 \otimes,$	$I^{\oplus 3} \epsilon_8^{I^{\oplus 3}\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 \otimes$
$I^{\oplus 2} \epsilon_8^{2,4} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 \otimes,$	$I^{\oplus 2} \epsilon_8^{2,4} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 3},$	$I^{\oplus 4} \epsilon_8^{I^{\oplus 2}\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 \otimes$
$I^{\oplus 3} \epsilon_8^{I^{\oplus 3}\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 \otimes,$	$I^{\oplus 4} \epsilon_8^{I^{\oplus 2}\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 3},$	$I^{\oplus 3} \epsilon_8^{I^{\oplus 3}\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 3}$
$1 \epsilon_7^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5} 1223132,$	$1 \epsilon_7^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5} I^{\oplus 6},$	$1 \epsilon_7^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5} 122315123$
$1 \epsilon_7^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5} 12231513215,$	$1 \epsilon_7^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5} 12231513,$	$1 \epsilon_7^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5} 12231513151$
$\otimes \epsilon_8^{3,5\ 1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 2},$	$\otimes \epsilon_8^{4,5\ 1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 2},$	$\otimes \epsilon_8^{5,5\ 1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 2}$
$313221 \epsilon_8^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 2},$	$I^{\oplus 5} \epsilon_8^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 2},$	$21513221 \epsilon_8^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 2}$
$1513221 \epsilon_8^{1\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 2},$	$I^{\oplus 2} \epsilon_8^{2,4} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 2},$	$I^{\oplus 4} \epsilon_8^{I^{\oplus 2}\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 2}$
$I^{\oplus 3} \epsilon_8^{I^{\oplus 3}\ 5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 2}$		

[illegible]

$$31\epsilon_6^{31}(\otimes \epsilon_6)^{2,2} \otimes \epsilon_6 I^{\oplus 3}$$

$$31\epsilon_6^{31}(\otimes \epsilon_6)^{2,2} \otimes \epsilon_6 13151^1$$

$$\begin{aligned}
& 1\overset{1}{\mathbf{e}}_7' \otimes \overset{4,5}{\mathbf{e}}_8 \otimes \overset{5,5}{\mathbf{e}}_8 \otimes \overset{5,5}{\mathbf{e}}_8 122315123 \\
& 1\overset{1}{\mathbf{e}}_7' \otimes \overset{4,5}{\mathbf{e}}_8 \otimes \overset{5,5}{\mathbf{e}}_8 \otimes \overset{5,5}{\mathbf{e}}_8 12231513151 \\
& \overset{5,5}{\mathbf{e}}_8 \otimes \overset{5,5}{\mathbf{e}}_8 (\otimes \overset{5,5}{\mathbf{e}}_8)^2 \otimes \overset{5,4}{\mathbf{e}}_7' I^{\oplus 2} \\
& 21513221\overset{1}{\mathbf{e}}_8 (\otimes \overset{5,5}{\mathbf{e}}_8)^2 \otimes \overset{5,4}{\mathbf{e}}_7' I^{\oplus 2} \\
& I^{\oplus 4} \overset{1}{\mathbf{e}}_8 (\otimes \overset{5,5}{\mathbf{e}}_8)^2 \otimes \overset{5,4}{\mathbf{e}}_7' I^{\oplus 2} \\
& \overset{4,5}{\mathbf{e}}_8 \otimes \overset{5,5}{\mathbf{e}}_8 (\otimes \overset{5,5}{\mathbf{e}}_8)^2 \otimes \overset{5,5}{\mathbf{e}}_7' 1 \\
& I^{\oplus 5} \overset{1}{\mathbf{e}}_8 (\otimes \overset{5,5}{\mathbf{e}}_8)^2 \otimes \overset{5,5}{\mathbf{e}}_7' 1 \\
& I^{\oplus 2} \overset{2,4}{\mathbf{e}}_8 (\otimes \overset{5,5}{\mathbf{e}}_8)^2 \otimes \overset{5,5}{\mathbf{e}}_7' 1
\end{aligned}$$

$$\begin{array}{l}
1^{3,3} \quad 4,5 \quad 5,5 \quad 5,5 \\
1\overline{e}_7 \otimes \overline{e}_7 \otimes \overline{e}_8 \otimes \overline{e}_8 \otimes \\
1^{3,3} \quad 4,5 \quad 5,5 \\
1\overline{e}_7 \otimes \overline{e}_7 \otimes \overline{e}_8 \otimes \overline{e}_8 \otimes 12231512 \\
2,4 \quad 1,5,5 \quad 5,4 \quad 3,3 \\
\otimes \overline{e}_8 \otimes \overline{e}_8 \otimes \overline{e}_7 \otimes \overline{e}_7 I^{\oplus 2} \\
I^{\oplus 2} 1^{5,5} \quad 5,4 \quad 3,3 \\
\overline{e}_8 \otimes \overline{e}_8 \otimes \overline{e}_7 \otimes \overline{e}_7 1232 \\
I^{\oplus 3} I^{\oplus 2} 5,5 \quad 5,4 \quad 3,3 \\
\overline{e}_8 \otimes \overline{e}_8 \otimes \overline{e}_7 \otimes \overline{e}_7 I^{\oplus 2} \\
I^{\oplus 2} 1^{4,5} \quad 5,5 \quad 5,4 \\
\overline{e}_7 \otimes \overline{e}_8 \otimes \overline{e}_8 \otimes \overline{e}_7 I^{\oplus 3} \\
1^{5,5} \quad 5,4 \\
1\overline{e}_7 (\otimes \overline{e}_8)^2 \otimes \overline{e}_7 I^{\oplus 3}
\end{array}$$

$$\begin{aligned}
& I^{\oplus 3} \textcolor{blue}{e}_8^{\textcolor{blue}{1} \textcolor{blue}{5} \textcolor{blue}{4}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{3} \textcolor{blue}{3}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{3} \textcolor{blue}{3}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{3} \textcolor{blue}{2}} \\
& I^{\oplus 2} \textcolor{blue}{e}_8^{\textcolor{blue}{1} \textcolor{blue}{5} \textcolor{blue}{4}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{3} \textcolor{blue}{3}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{3} \textcolor{blue}{3}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{1} \textcolor{blue}{2} \textcolor{blue}{2} \textcolor{blue}{3}} \\
& 2321 \textcolor{blue}{e}_7^{\textcolor{blue}{1} \textcolor{blue}{3} \textcolor{blue}{3}} (\otimes \textcolor{blue}{e}_7^{\textcolor{blue}{4} \textcolor{blue}{5}})^2 \otimes \textcolor{blue}{e}_8 I^{\oplus 2} \\
& I^{\oplus 2} \textcolor{blue}{e}_8^{\textcolor{blue}{1} \textcolor{blue}{5} \textcolor{blue}{4}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{3} \textcolor{blue}{3}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{3} \textcolor{blue}{3}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{1} \textcolor{blue}{2} \textcolor{blue}{3} \textcolor{blue}{1} \textcolor{blue}{5} \textcolor{blue}{1}} \\
& I^{\oplus 2} \textcolor{blue}{e}_8^{\textcolor{blue}{1} \textcolor{blue}{5} \textcolor{blue}{4}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{3} \textcolor{blue}{3}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{3} \textcolor{blue}{3}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{3} \textcolor{blue}{3}} \\
& 1 \textcolor{blue}{e}_7^{\textcolor{blue}{1} \textcolor{blue}{4} \textcolor{blue}{5}} \otimes \textcolor{blue}{e}_8^{\textcolor{blue}{5} \textcolor{blue}{4}} \otimes \textcolor{blue}{e}_7^{\textcolor{blue}{3} \textcolor{blue}{3}} \otimes \textcolor{blue}{e}_7 I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1\mathbf{e}_6^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_7 \otimes^{4,4} \\
& 1\mathbf{e}_6^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_7 12313 \\
& 1\mathbf{e}_6^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_7 1232151 \\
& \otimes^{3,3} \mathbf{e}_7^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_6^{2,2} \\
& \otimes^{3,3} \mathbf{e}_7^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_6 I^{\oplus 2} \\
& I^{\oplus 2} \mathbf{e}_7^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_6 I^{\oplus 2}
\end{aligned}$$

$$\begin{array}{l}
1_{\mathfrak{e}_8}^{1,5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} 1232 \\
I^{\oplus 3} 1_{\mathfrak{e}_8}^{1,5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \\
I^{\oplus 2} 1_{\mathfrak{e}_8}^{1,5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} 13151 \\
I^{\oplus 2} 1_{\mathfrak{e}_8}^{1,5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2}
\end{array}$$

$$\begin{array}{l} \otimes^{5,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{4,2} \mathfrak{so}13 \\ I^{\oplus 5} \otimes^{5,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{4,2} \mathfrak{so}13 \\ I^{\oplus 2} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{4,2} \mathfrak{so}13 \end{array}$$

$$\begin{array}{c} I^{\oplus 3} \mathbf{e}_8^{1 \oplus 2, 5, 5} \otimes \mathbf{e}_8^{5, 4} \otimes \mathbf{e}_7^{3, 3} \otimes \mathbf{e}'_7 1 \\ 1 \mathbf{e}'_7^{1, 4, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{5, 4} \otimes \mathbf{e}_7 I^{\oplus 3} \\ I^{\oplus 2} \mathbf{e}_7^{1, 4, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}'_7 1 \end{array}$$

$$\begin{array}{cccc}
{}^{5,5}_1 \otimes \mathfrak{e}_8 & {}^{5,5} \otimes \mathfrak{e}_8 & {}^{4,2} \otimes \mathfrak{so} & {}^{1,1} \otimes \mathfrak{so} \\
21513221 \otimes \mathfrak{e}_8 & {}^1 \otimes \mathfrak{e}_8 & {}^{5,5} \otimes \mathfrak{so} & {}^{4,2} \otimes \mathfrak{so} \\
I^{\oplus 4} \otimes \mathfrak{e}_8 & {}^{I^{\oplus 2}} \otimes \mathfrak{e}_8 & {}^{5,5} \otimes \mathfrak{so} & {}^{4,2} \otimes \mathfrak{so}
\end{array}$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} \overset{5,5}{\otimes} \overset{5,4}{\mathbf{e}_8} \otimes \mathbf{e}_7 I^{\oplus 3}$$

$$31\mathbf{e}_6^1(\otimes^{2,2}\mathbf{e}_6)^2\otimes^{3,3}\mathbf{e}_71,$$

$$1\mathbf{e}_6^{(2,2)}(\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_7 1$$

$$\otimes^{2,3} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^2 \otimes^{3,2} \mathfrak{so}1,$$

$$\otimes^{3,3} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^2 \otimes^{3,2} \mathfrak{so}1,$$

$$I^{\oplus 3} \mathfrak{e}_7^{(3,3)} (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so} 1$$

$$I^{\oplus 2} \mathfrak{e}_7^{\oplus 2} \left(\bigotimes_{\mathfrak{e}_7}^{3,3} \right)^2 \bigotimes_{\mathfrak{e}_7}^{3,2} \mathfrak{so}1$$

$$I^{\oplus 3} \mathbf{e}_8^1 \otimes \mathbf{e}_7^{5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{3,3} 1,$$

$$I^{\oplus 2} I_8^{\oplus 2} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \otimes \mathfrak{e}_6 1,$$

$$I^{\oplus 2} \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{13}$$

$$1e_7 \overset{1}{\otimes} \overset{3,3}{e_7} \overset{4,5}{\otimes} \overset{5,3}{e_8} \otimes e_6 1$$

$$I^{\oplus 2} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^1,$$

$$1e_7 \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3,3}{\otimes} \overset{2,2}{e_6} \overset{2,2}{\otimes} e_6 13,$$

$$1 \overset{1}{\mathbf{e}}_6 (\overset{3,3}{\otimes} \overset{2}{\mathbf{e}}_7)^2 \overset{3,3}{\otimes} \overset{2,2}{\mathbf{e}}_6 \otimes$$

$$1\mathfrak{e}_6^{1(3,3)}(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6 I^{\oplus 2}$$

$${}^{2,2} \otimes \mathfrak{e}_6 ({}^{2,2} \otimes \mathfrak{e}_6)^2 {}^{2,2} \otimes \mathfrak{so}1,$$

$$I^{\oplus 2} \mathfrak{e}_6 \binom{2,2}{\otimes} \mathfrak{e}_6)^2 \binom{2,2}{\otimes} \mathfrak{so}1,$$

$$31\mathfrak{e}_6^{31}(\otimes \mathfrak{e}_6)^{2,2} \otimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \oplus \mathfrak{so}13,$$

$$1e_7^{(1)}(\otimes e_8^{(5,5)})^2 \otimes so_{13}^{(4,2)}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1,$$

$$1e_7 \overset{1}{\otimes} \overset{5,5}{e_8} \overset{4,2}{\otimes} \overset{1,1}{so} \otimes so1$$

$$1e_7' \otimes_{4,5} e_8 \otimes_{5,5} e_8 \otimes_{5,4} e_7' I^{\oplus 2},$$

$$1e_7' \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_7' 1$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}'_7 \left(\bigotimes^{3,3} \mathfrak{e}_7 \right)^2 \bigotimes^{3,2} \mathfrak{so} 1$$

$$1e_7' \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{4,2} so_{13}$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,3}{\otimes} \overset{2,2}{e_6} \otimes e_6 1$$

$$1e'_7 \begin{smallmatrix} 1 \\ 4,5 \end{smallmatrix} \otimes e_8 \begin{smallmatrix} 4,2 \\ \end{smallmatrix} \otimes so \begin{smallmatrix} 1,1 \\ \end{smallmatrix} \otimes so1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7' \otimes \epsilon_7'1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_6 \otimes \epsilon_61$$

$$1\epsilon_7 \otimes \mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so}13$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\epsilon_6(\otimes \epsilon_7)^2 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6(\otimes \epsilon_6)^2 \otimes \epsilon_7'1$$

2.5.13 SSgLgIgS(149)

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8122313,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8122313,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 I^{\oplus 5},$$

$$21513221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$1513221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

[illegible]

$$\begin{array}{lll}
1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 122313, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 5}, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 12231512 \\
1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 1223151, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 24,2}, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 2} \\
1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} I^{\oplus 3} & & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 4}
\end{array}$$

$$\begin{array}{lll}
1_{\mathbf{e}_7}^{1,3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,4,2}, & 1_{\mathbf{e}_7}^{1,3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{I^{\oplus 4}}, & 1_{\mathbf{e}_7}^{1,3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{I^{\oplus 3}} \\
I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,4} \otimes \mathbf{e}_7^{I^{\oplus 2} 1}, & I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_7^{1,1}, & 1_{\mathbf{e}_7}^{1,5,5} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_7^{5,5} \otimes \mathbf{e}_7^{1,1}
\end{array}$$

$$\begin{array}{lll}
1\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} 1 & 2321\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} 1 & 321\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} I^{\oplus 2} \\
I^{\oplus 2}\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} I^{\oplus 3} & I^{\oplus 2}\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} I^{\oplus 2} & 1\epsilon_7^{(3,3)} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,4} 1
\end{array}$$

$$\begin{array}{lll} \begin{array}{l} 2,2 \\ \otimes \end{array} \mathfrak{e}_6 \begin{pmatrix} 2,2 \\ \otimes \end{pmatrix} \mathfrak{e}_6 \begin{array}{l} 2,2 \\ \otimes \end{array} \mathfrak{e}_6 \begin{array}{l} 1 \\ \otimes \end{array} \mathfrak{e}_6 \begin{array}{l} 2,2 \\ \otimes \end{array} \mathfrak{e}_6, \\ 31 \mathfrak{e}_6 \begin{pmatrix} 31 \\ \otimes \end{pmatrix} \mathfrak{e}_6 \begin{array}{l} 2,2 \\ \otimes \end{array} \mathfrak{e}_6 \begin{array}{l} 1 \\ \otimes \end{array} \mathfrak{e}_6 \begin{array}{l} 2,2 \\ \otimes \end{array} \mathfrak{e}_6, \end{array} \quad \begin{array}{l} I^{\oplus 2} \mathfrak{e}_6 \begin{pmatrix} 2,2 \\ \otimes \end{pmatrix} \mathfrak{e}_6 \begin{array}{l} 2,2 \\ \otimes \end{array} \mathfrak{e}_6 \begin{array}{l} 1 \\ \otimes \end{array} \mathfrak{e}_6, \\ 31 \mathfrak{e}_6 \begin{pmatrix} 31 \\ \otimes \end{pmatrix} \mathfrak{e}_6 \begin{array}{l} 2,2 \\ \otimes \end{array} \mathfrak{e}_6 \begin{array}{l} 1 \\ \otimes \end{array} \mathfrak{e}_6 \begin{array}{l} I^{\oplus 2} \\ \otimes \end{array} \mathfrak{e}_6 1 \end{array} \quad \begin{array}{l} I^{\oplus 2} \mathfrak{e}_6 \begin{pmatrix} 2,2 \\ \otimes \end{pmatrix} \mathfrak{e}_6 \begin{array}{l} 2,2 \\ \otimes \end{array} \mathfrak{e}_6 \begin{array}{l} I^{\oplus 2} \\ \otimes \end{array} \mathfrak{e}_6 1 \end{array}$$

$$\begin{array}{lll}
1 \overset{1}{\mathbf{e}}_7 \overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7 \overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7 \overset{I^{\oplus 2}}{\otimes} 1, & I^{\oplus 2} \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7)^2 \overset{3,3}{\otimes} \overset{1}{\mathbf{e}}_7 \overset{3,2}{\otimes}, & I^{\oplus 2} \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7)^2 \overset{3,3}{\otimes} \overset{1}{\mathbf{e}}_7 \overset{3,3}{\otimes} \\
I^{\oplus 2} \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7)^2 \overset{3,3}{\otimes} \overset{I^{\oplus 3}}{\mathbf{e}}_7 1, & I^{\oplus 2} \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}_7)^2 \overset{3,3}{\otimes} \overset{I^{\oplus 2}}{\mathbf{e}}_7 I^{\oplus 2} &
\end{array}$$

$$\begin{array}{lll} 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{1,3,2}{\otimes}, & 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{1,3,3}{\otimes}, & 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{1^{\oplus 3}}{1} \\ 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{1^{\oplus 2}}{I^{\oplus 2}} & & \end{array}$$

$$1\mathbf{e}_6^{(2,2)}(\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_8^{I^{\oplus 3}} 1, \quad 1\mathbf{e}_6^{(2,2)}(\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_8^{I^{\oplus 2}} I^{\oplus 2}, \quad 31\mathbf{e}_6^{(2,2)}(\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_8^{I^{\oplus 2}} 1$$

$$1e_7' \otimes e_8 \otimes e_8 \otimes e_7^{\oplus 2} \quad 1e_7' \otimes e_8 \otimes e_8 \otimes e_7^{\oplus 1}$$

$$1 \overset{1}{\mathbf{e}}_7' \otimes \overset{3,3}{\mathbf{e}}_7 \otimes \overset{3,3}{\mathbf{e}}_7' \otimes \overset{3,3}{\mathbf{e}}_7 \otimes \overset{1}{\mathbf{e}}_7 1, \quad I^{\oplus 2} \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \mathbf{e}_7)^2 \otimes \overset{3,3}{\mathbf{e}}_7 I^{\oplus 2}$$

$$1e_7'(\otimes^{3,3} e_7)^2 \otimes^{4,5} e_8 I^{\oplus 3}, \quad 1e_7'(\otimes^{3,3} e_7)^2 \otimes^{4,5} e_8 I^{\oplus 2}$$

$$1e_6^{1\ 3,5} \otimes e_8^{5,5} \otimes e_8^{5,4} I^{\oplus 2} \quad 1e_6^{1\ 3,5} \otimes e_8^{5,5} \otimes e_8^{5,5} \otimes e_7^1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}'_71$$

$$1\mathfrak{e}_6\overset{1}{\otimes}\overset{3,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}'_71$$

$$1\mathfrak{e}_6\overset{1}{\otimes}\overset{3,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,3}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_61$$

$$1\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}'_7I^{\oplus 2}1$$

$$1\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_61$$

$$1\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^2\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_71$$

2.5.14 SSgLgLIgLg(21)

$$1513221\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_7,$$

$$I^{\oplus 2}3221\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,4}{\mathfrak{e}}_7,$$

$$\overset{2,4}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\mathfrak{e}}_7$$

$$I^{\oplus 4}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_7,$$

$$I^{\oplus 3}I^{\oplus 2}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\mathfrak{e}}_7$$

$$\overset{2,4}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,4}{\mathfrak{e}}'_7,$$

$$I^{\oplus 4}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,4}{\mathfrak{e}}'_7,$$

$$I^{\oplus 3}I^{\oplus 2}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,4}{\mathfrak{e}}'_7$$

$$\overset{2,4}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,3}{\mathfrak{e}}_6,$$

$$I^{\oplus 4}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,3}{\mathfrak{e}}_6,$$

$$I^{\oplus 3}I^{\oplus 2}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,3}{\mathfrak{e}}_6$$

$$I^{\oplus 2}\overset{1}{\mathfrak{e}}_7\overset{3,3}{\otimes}\overset{3,3}{\mathfrak{e}}_7\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_7\overset{3,3}{\mathfrak{e}}_6$$

$$321\overset{1}{\mathfrak{e}}_7\overset{3,3}{\otimes}\overset{3,3}{\mathfrak{e}}_7\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_7\overset{3,3}{\mathfrak{e}}'_7$$

$$31\overset{1}{\mathfrak{e}}_6\overset{2,2}{\otimes}\overset{2,2}{\mathfrak{e}}_6\overset{2,2}{\otimes}\overset{1}{\mathfrak{e}}_6\overset{2,2}{\mathfrak{e}}_6$$

$$I^{\oplus 3} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{1}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 1$$

$$321 \overset{1}{e_7} \overset{3,3}{\otimes} e_7 \overset{3,3}{\otimes} \overset{1}{e_7} \overset{3,3}{\otimes} e_7 1,$$

$$1e_7 \otimes^{1,5,5} e_8 \otimes^{5,5,1} e_8 \otimes^{4,2} \mathfrak{so}1$$

$$I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 1$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,5}{\otimes} \overset{1}{e_8} \overset{5,4}{\otimes} e_7 1$$

$$1e_7' \otimes e_8 \otimes e_8 \otimes \text{sol}$$

$$1e_7' \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3,3}{\otimes} \overset{1}{e_7} \overset{3,3}{\otimes} e_7' 1$$

$$1e_6 \otimes^{1,3,5} e_8 \otimes^{5,5} e_8 \otimes^{1,4,2} \text{sol}$$

$$1 \overset{1}{\mathbf{e}}_6 \overset{2,2}{\otimes} \overset{2,2}{\mathbf{e}}_6 \overset{1}{\otimes} \overset{2,2}{\mathbf{e}}_6 1$$

2.5.16 SSgLgLIgLIgS(15)

[illegible]

$$1e_7 \otimes^{1,4,5} e_8 \otimes^{5,5} e_8 \otimes^{1,5,5} e_8 \otimes^{I^{\oplus 2}} I^{\oplus 2}$$

$$I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7^1$$

$$1e_7' \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3,3}{\otimes} \overset{1}{e_7} \overset{3,3}{\otimes} \overset{1}{e_7} 1$$

$$I^{\oplus 3} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_6^{5,3} \mathbf{1}, \quad I^{\oplus 2} \mathbf{e}_8^{1 \oplus 2, 5,5} \otimes \mathbf{e}_8^{1, 5,5} \otimes \mathbf{e}_6^{5,3} \mathbf{1}, \quad I^{\oplus 2} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_6^{5,3} \mathbf{1} \mathbf{3}$$

$$1e_7^1 \otimes e_7^{3,3} \otimes e_7^1 \otimes e_7^{3,3} \otimes e_7^{3,3} \otimes e_7^{3,2}, \quad 1e_7^1 \otimes e_7^{3,3} \otimes e_7^1 \otimes e_7^{3,3} \otimes e_7^{3,3} \otimes e_7^{3,3}, \quad 1e_7^1 \otimes e_7^{3,3} \otimes e_7^1 \otimes e_7^{3,3} \otimes e_7^{3,3} \otimes e_7 I^{\oplus 3}$$

$$I^{\oplus 3} \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{5,4} \otimes \mathbf{e}'_7 1, \quad I^{\oplus 2} I^{\oplus 2} \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{5,4} \otimes \mathbf{e}'_7 1$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{1}{\otimes} \overset{5,5}{e_8} \overset{4,2}{\otimes} so1$$

$$1\mathbf{e}_7 \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{3,3}{\otimes} \mathbf{e}_7' I^{\oplus 2}$$

$$1 \overset{1}{\mathfrak{e}_7} \overset{3,3}{\otimes} \overset{1}{\mathfrak{e}_7} \overset{3,3}{\otimes} \overset{3,3}{\mathfrak{e}_7} \overset{3,3}{\otimes} \mathfrak{e}_6 1$$

2.5.19 SSgLIgLgLIgS(15)

$$\begin{array}{lll}
1\epsilon_8^{1\ 5,5\ 1\ 5,5\ 5,5\ 1} 12231513, & I^{\oplus 3}\epsilon_8^{1\ 5,5\ 1\ 5,5\ 5,5\ 1\ 4,2} \otimes, & 3221\epsilon_8^{1\ 5,5\ 1\ 5,5\ 5,5} I^{\oplus 3}\epsilon_8 1 \\
I^{\oplus 2}\epsilon_8^{1\ 5,5\ 1\ 5,5\ 5,5\ 1} 1223151, & I^{\oplus 3}\epsilon_8^{1\ 5,5\ 1\ 5,5\ 5,5\ 1} \epsilon_8 I^{\oplus 4}, & I^{\oplus 2}I^{\oplus 2}\epsilon_8^{5,5\ 1\ 5,5\ 5,5\ 1\ 4,2} \otimes \\
I^{\oplus 2}\epsilon_8^{1\ 5,5\ 1\ 5,5\ 5,5} I^{\oplus 2}\epsilon_8 1223, & 3221\epsilon_8^{1\ 5,5\ 1\ 5,5\ 5,5} I^{\oplus 2}\epsilon_8 I^{\oplus 2}, & I^{\oplus 2}I^{\oplus 2}\epsilon_8^{5,5\ 1\ 5,5\ 5,5\ 1} \epsilon_8 I^{\oplus 4} \\
I^{\oplus 3}\epsilon_8^{1\ 5,5\ 1\ 5,5\ 5,5} I^{\oplus 3}\epsilon_8 I^{\oplus 2}, & I^{\oplus 2}I^{\oplus 2}\epsilon_8^{5,5\ 1\ 5,5\ 5,5} I^{\oplus 3}\epsilon_8 I^{\oplus 2} &
\end{array}$$

$$I^{\oplus 3} \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{5,4,1} \otimes \mathbf{e}_7^1, \quad I^{\oplus 2} I^{\oplus 2} \mathbf{e}_8^{5,5,1} \otimes \mathbf{e}_8^{5,5,1} \otimes \mathbf{e}_8^{5,4,1} \otimes \mathbf{e}_7^1$$

$$1 \mathbf{e}_7 \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{3,3}{\otimes} I^{\oplus 2} \mathbf{e}_7 1$$

$$1e_7^{1\ 3,3} \otimes e_7^{1\ 3,3} \otimes e_7^{3,3} \otimes e_7'^1 1$$

2.5.20 SSgLIgLIgLg(2)

$$I^{\oplus 3} \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}, \quad I^{\oplus 2} I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}$$

2.6 5 node(1332)

2.6.1 gLgLGgLG(17)

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,3} (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,3}$$

$$\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{3,3}$$

$$\mathfrak{so}^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7')^2$$

$$\mathfrak{so}^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2}$$

$$\mathfrak{so}^{2,3} \otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7^{4,4}$$

$$\mathfrak{so}^{2,2} (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7^{3,4}$$

$$\mathfrak{so}^{2,2} (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7^{3,3}$$

$$\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{3,3}$$

$$\mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2 \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,5}$$

$$\mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2 \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_6^{5,4}$$

$$\mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7'$$

$$\mathfrak{e}_6^{5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7'$$

$$\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_7'$$

$$\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_6$$

2.6.2 $\mathfrak{gLgLIgLgLg}(4)$

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_7$$

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_7'$$

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so}$$

2.6.3 $\mathfrak{gLgLgLgLg}(10)$

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7'$$

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so} \left(\overset{1,1}{\otimes} \mathfrak{so} \right)^2$$

$$\mathfrak{e}_7 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so}$$

$$\mathfrak{e}_7' \overset{4,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so}$$

$$\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^2 \overset{5,5}{\otimes} \mathfrak{e}_7'$$

$$\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^2 \overset{5,4}{\otimes} \mathfrak{e}_6$$

$$e'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 2 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} 50$$

$$\mathfrak{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} \mathfrak{so} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^1 \left(\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6 \right)^2 \otimes \mathfrak{so}^{2,2}$$

2.6.4 gLIgLGLIgLG(1)

$$50 \otimes_{\mathbb{F}_8}^{2,4} \mathbb{F}_8 \otimes_{\mathbb{F}_8}^{5,5} \mathbb{F}_8 \otimes_{\mathbb{F}_8}^{5,5} \mathbb{F}_8 \otimes_{\mathbb{F}_8}^{1,5,4} \mathbb{F}_7$$

2.6.5 SgLgLgLgLg(133)

$$\begin{aligned} & 3,5 \otimes \mathfrak{e}_8 \otimes \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^2 \otimes \begin{pmatrix} 5,3 \\ \otimes \mathfrak{e}_6 \end{pmatrix} \otimes \begin{pmatrix} 2,2 \\ \otimes \mathfrak{e}_6 \end{pmatrix}, \\ & 313221 \mathfrak{e}_8 \otimes \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^2 \otimes \begin{pmatrix} 5,3 \\ \otimes \mathfrak{e}_6 \end{pmatrix} \otimes \begin{pmatrix} 2,2 \\ \otimes \mathfrak{e}_6 \end{pmatrix}, \\ & \begin{pmatrix} 1 \\ 1513221 \mathfrak{e}_8 \end{pmatrix} \otimes \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^2 \otimes \begin{pmatrix} 5,3 \\ \otimes \mathfrak{e}_6 \end{pmatrix} \otimes \begin{pmatrix} 2,2 \\ \otimes \mathfrak{e}_6 \end{pmatrix}, \\ & 1 \mathfrak{e}_6 \otimes \begin{pmatrix} 4,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix} \otimes \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^2 \otimes \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_6 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} & \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^2 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6, \\ & I^{\oplus 5} \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^2 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6, \\ & \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^2 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_6, \end{aligned}$$

$$\begin{array}{l} \begin{array}{c} 5,5 \\ \otimes \end{array} \mathfrak{e}_8 \begin{array}{c} 5,5 \\ (\otimes \mathfrak{e}_8)^2 \end{array} \otimes \begin{array}{c} 5,3 \\ \mathfrak{e}_6 \end{array} \otimes \begin{array}{c} 2,2 \\ \mathfrak{e}_6 \end{array} \\ 21513221 \mathfrak{e}_8 \begin{array}{c} 5,5 \\ (\otimes \mathfrak{e}_8)^2 \end{array} \otimes \begin{array}{c} 5,3 \\ \mathfrak{e}_6 \end{array} \otimes \begin{array}{c} 2,2 \\ \mathfrak{e}_6 \end{array} \\ I^{\oplus 2} \mathfrak{e}_6 \begin{array}{c} 3,5 \\ \otimes \end{array} \mathfrak{e}_8 \begin{array}{c} 5,5 \\ (\otimes \mathfrak{e}_8)^2 \end{array} \otimes \begin{array}{c} 5,5 \\ \mathfrak{e}_6 \end{array} \end{array}$$

$$\begin{aligned} & \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{50}, \\ & \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{50}, \\ & 2151321 \mathbf{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{50}, \\ & \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} 151321 \mathbf{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{50} \end{aligned}$$

$$\begin{aligned} & 3,4 \quad (3,3) \quad 3,3 \\ & \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}, \\ & I^{\oplus 4} \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}, \\ & 2321 \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}, \end{aligned}$$

$$\begin{array}{l} 4,4 \quad (3,3) \quad 3,3 \\ \otimes e_7 (\otimes e_7)^3 \otimes 50 \\ 31321 e_7 (3,3) \quad 3,3 \\ 31321 e_7 (\otimes e_7)^3 \otimes 50 \\ 1512321 e_7 (3,3) \quad 3,3 \\ 1512321 e_7 (\otimes e_7)^3 \otimes 50 \end{array}$$

$$\begin{aligned} & 3,5 \quad 5,5 \quad 4,2 \quad (1,1) \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2, \\ & 313221 \mathfrak{e}_8 \quad 5,5 \quad 4,2 \quad (1,1) \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2, \\ & 1513221 \mathfrak{e}_8 \quad 5,5 \quad 4,2 \quad (1,1) \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2, \end{aligned}$$

$$\begin{array}{ccccccc}
4,5 & 5,5 & 4,2 & (1,1) \\
\otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & (\otimes \mathfrak{so})^2, \\
I^{\oplus 5} \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & (\otimes \mathfrak{so})^2, \\
1,1 & 2,4 & 5,5 & 5,3 \\
\mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 \otimes \mathfrak{so}
\end{array}$$

$$\begin{array}{c} 5,5 \quad 5,5 \quad 4,2 \quad (1,1) \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \\ 21513221 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \end{array}$$

$$\begin{array}{l} 3,5 \\ \otimes \mathbf{e}_8 \end{array} \begin{array}{l} (5,5) \\ \otimes \mathbf{e}_8 \end{array}^2 \begin{array}{l} 5,4 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{l} 3,3 \\ \otimes \mathbf{e}_7' \end{array}, \\ 313221 \mathbf{e}_8 \begin{array}{l} (5,5) \\ \otimes \mathbf{e}_8 \end{array}^2 \begin{array}{l} 5,4 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{l} 3,3 \\ \otimes \mathbf{e}_7' \end{array}, \\ 1513221 \mathbf{e}_8 \begin{array}{l} (5,5) \\ \otimes \mathbf{e}_8 \end{array}^2 \begin{array}{l} 5,4 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{l} 3,3 \\ \otimes \mathbf{e}_7' \end{array}$$

$$\begin{array}{c} 4,5 \\ \otimes \mathbf{e}_8 \end{array} \begin{array}{c} 5,5 \\ (\otimes \mathbf{e}_8)^2 \end{array} \begin{array}{c} 5,4 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{c} 3,3 \\ \otimes \mathbf{e}'_7, \\ I^{\oplus 5} \mathbf{e}_8 \end{array} \begin{array}{c} 5,5 \\ (\otimes \mathbf{e}_8)^2 \end{array} \begin{array}{c} 5,4 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{c} 3,3 \\ \otimes \mathbf{e}'_7, \end{array}$$

$$\begin{array}{c} \begin{array}{ccccc} 5,5 & & 5,5 & & 5,4 & & 3,3 \\ \otimes & \mathbf{e}_8 & (\otimes \mathbf{e}_8)^2 & \otimes & \mathbf{e}_7 & \otimes & \mathbf{e}'_7 \end{array} \\ 21513221 \mathbf{e}_8 (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_7 \otimes \mathbf{e}'_7 \end{array}$$

$$2313221\mathbf{e}_8(\overset{5,5}{\otimes}\mathbf{e}_8)^3\overset{5,5}{\otimes}\mathbf{e}_6,$$

$$I^{\oplus 6} \mathbf{e}_8 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^3 \overset{5,5}{\otimes} \mathbf{e}_6,$$

$$321513221\mathbf{e}_8\left(\bigotimes^{5,5}\mathbf{e}_8\right)^3\bigotimes^{5,5}\mathbf{e}_6$$

$$51231513221\epsilon_8^{(5,5)}(\otimes \epsilon_8)^3 \otimes \epsilon_6, \quad 31513221\epsilon_8^{(5,5)}(\otimes \epsilon_8)^3 \otimes \epsilon_6, \quad 15131513221\epsilon_8^{(5,5)}(\otimes \epsilon_8)^3 \otimes \epsilon_6$$

$$\begin{array}{lll} 2313221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & I^{+6}\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & 321513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so} \\ 51231513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & {}^131513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & 15131513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so} \end{array}$$

$$\begin{array}{lll} \begin{array}{c} 2,3 \\ \otimes \mathbf{e}_7 \end{array} \otimes \begin{array}{c} 4,5 \\ \otimes \mathbf{e}_8 \end{array} (\otimes \mathbf{e}_8)^2 \otimes \begin{array}{c} 5,5 \\ \otimes \mathbf{e}_6 \end{array}, & \begin{array}{c} 3,3 \\ \otimes \mathbf{e}_7 \end{array} \otimes \begin{array}{c} 4,5 \\ \otimes \mathbf{e}_8 \end{array} (\otimes \mathbf{e}_8)^2 \otimes \begin{array}{c} 5,5 \\ \otimes \mathbf{e}_6 \end{array}, & I^{\oplus 3} \mathbf{e}_7 \otimes \begin{array}{c} 4,5 \\ \otimes \mathbf{e}_8 \end{array} (\otimes \mathbf{e}_8)^2 \otimes \begin{array}{c} 5,5 \\ \otimes \mathbf{e}_6 \end{array} \\ I^{\oplus 2} \mathbf{e}_7 (\otimes \mathbf{e}_8)^3 \otimes \begin{array}{c} 5,5 \\ \otimes \mathbf{e}_6 \end{array} & & \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 2,3 \\ \otimes \mathfrak{e}_7 \end{array} \otimes \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array}, & \begin{array}{l} 3,3 \\ \otimes \mathfrak{e}_7 \end{array} \otimes \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array}, & I^{\oplus 3} \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_7 \end{array} \otimes \begin{array}{l} 5,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array} \\ I^{\oplus 2} \begin{array}{l} 5,5 \\ \otimes \mathfrak{e}_7 \end{array} (\otimes \mathfrak{e}_8)^3 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array} & & \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 2,3 \\ \otimes \mathbf{e}_7 (\otimes \mathbf{e}_7)^2 (\otimes \mathbf{e}_7')^2, \end{array} & \begin{array}{l} 3,3 \\ \otimes \mathbf{e}_7 (\otimes \mathbf{e}_7)^2 (\otimes \mathbf{e}_7')^2, \end{array} & 321 \mathbf{e}_7' (\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7' \otimes \mathbf{e}_7 \\ I^{\oplus 3} \mathbf{e}_7 (\otimes \mathbf{e}_7)^2 (\otimes \mathbf{e}_7')^2 & & \end{array}$$

$$3221\epsilon_7(\otimes \epsilon_7)^2 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3}, \quad 512321\epsilon_7(\otimes \epsilon_7)^2 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3}, \quad 151321\epsilon_7(\otimes \epsilon_7)^2 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3}$$

$$I^{\oplus 2}\epsilon_7'(\otimes \epsilon_7)^3 \otimes \epsilon_7^{4,4}$$

$$321\mathfrak{e}_6\binom{2,2}{\otimes\mathfrak{e}_6}^3\otimes\mathfrak{e}_7, \quad 15131\mathfrak{e}_6\binom{2,2}{\otimes\mathfrak{e}_6}^3\otimes\mathfrak{e}_7, \quad \binom{2,2}{\otimes\mathfrak{e}_6}\binom{2,2}{\otimes\mathfrak{e}_6}^3\otimes\mathfrak{e}_7$$

$$\begin{array}{lll}
31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,5}, & \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3}, & I^{\oplus 2} \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3} \\
1\mathfrak{e}_6^{4,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3} & &
\end{array}$$

$$\begin{array}{lll}
1e_7^{3,3} \otimes e_7'({}^{3,3}(\otimes e_7)^2 \otimes \mathfrak{so})^{3,2}, & {}^{2,3} \otimes e_7'({}^{3,3}(\otimes e_7)^3 \otimes \mathfrak{so})^{3,3}, & {}^{3,3} \otimes e_7'({}^{3,3}(\otimes e_7)^3 \otimes \mathfrak{so})^{3,3} \\
I^{\oplus 3} e_7'({}^{3,3}(\otimes e_7)^3 \otimes \mathfrak{so})^{3,3} & &
\end{array}$$

$$\begin{array}{ccc} \begin{array}{c} 2,3 \\ \otimes \end{array} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7^3 \begin{array}{c} 4,4 \\ \otimes \end{array} \mathbf{e}_7, & \begin{array}{c} 3,3 \\ \otimes \end{array} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7^3 \begin{array}{c} 4,4 \\ \otimes \end{array} \mathbf{e}_7, & I^{\oplus 3} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7^3 \begin{array}{c} 4,4 \\ \otimes \end{array} \mathbf{e}_7 \end{array}$$

$$\begin{array}{ccc} \overset{2,3}{\otimes} e_7 (\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_6 \overset{2,2}{\otimes} e_6, & \overset{3,3}{\otimes} e_7 (\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_6 \overset{2,2}{\otimes} e_6, & I^{\oplus 3} e_7 (\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_6 \overset{2,2}{\otimes} e_6 \end{array}$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7', \quad 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7', \quad 1 \mathfrak{e}_7 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7',$$

$$I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7'$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}_7 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}_7' (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7' (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} (\otimes \mathfrak{so})^2,$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} (\otimes \mathfrak{so})^2$$

$$31 \mathfrak{so} (\otimes \mathfrak{so})^3 \otimes \mathfrak{e}_7,$$

$$1 \mathfrak{so} (\otimes \mathfrak{so})^3 \otimes \mathfrak{e}_7$$

$$31 \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7,$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7' (\otimes \mathfrak{e}_7)^2 (\otimes \mathfrak{e}_7')^2$$

$$I^{\oplus 2} \mathfrak{e}_7' (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$5131513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7$$

$$3221\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}_6$$

$$321\mathfrak{e}_7\overset{4,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}_7'\overset{3,3}{\otimes}\mathfrak{e}_7$$

$$23151321\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^3\overset{4,5}{\otimes}\mathfrak{e}_8$$

$$2315131\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^3\overset{3,5}{\otimes}\mathfrak{e}_8$$

$$1\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{so}\overset{2,3}{\otimes}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{so}$$

$$1\mathfrak{so}\overset{1,1}{\otimes}\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^3\overset{2,3}{\otimes}\mathfrak{e}_7'$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^3\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{e}_7'\overset{4,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7'$$

$$1\mathfrak{e}_7'\overset{4,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,3}{\otimes}\mathfrak{e}_6\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$1\epsilon'_7 \otimes^{4,5} \epsilon_8 \otimes^{4,2} \mathfrak{so}(\otimes^{1,1} \mathfrak{so})^2$$

$$1e'_7 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{4,2}{\otimes} so \overset{1,1}{\otimes} so$$

$$1e'_7(\overset{3,3}{\otimes} e_7)^2 \overset{4,5}{\otimes} e_8 \overset{4,2}{\otimes} 50$$

$$1e_7 \overset{3,3}{\otimes} e_6 (\overset{2,2}{\otimes} e_6)^2 \overset{2,2}{\otimes} 50$$

$$1\mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}'_7$$

$$1\mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,3}{\otimes} \mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6$$

$$1\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so}(\overset{1,1}{\otimes} \mathfrak{so})^2$$

$$1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^3\overset{4,4}{\otimes}\mathbf{e}_7$$

$$1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2(\overset{3,3}{\otimes}\mathbf{e}_7')^2$$

$$1e_6(\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_6 \overset{2,2}{\otimes} e_6$$

$$1\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{4,2}{\otimes} \mathbf{so} \overset{1,1}{\otimes} \mathbf{so}$$

$$1513221\epsilon_8 \otimes^{4,2} \mathfrak{so}(\otimes^{1,1} \mathfrak{so})^3$$

2.6.6 SgLgLgLgLgS(407)

$$\begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix},$$

$$\begin{matrix} 2,4 \\ \otimes \end{matrix} \mathbf{e}_7 \left(\begin{matrix} 3,3 \\ \otimes \end{matrix} \mathbf{e}_7 \right)^3 \begin{matrix} 3,3 \\ \otimes \end{matrix} \mathbf{e}_7 \begin{matrix} 4,5 \\ \otimes \end{matrix},$$

$$\otimes^{2,4} \mathbf{e}_7 (\otimes^{3,3} \mathbf{e}_7)^3 \otimes^{3,3} \mathbf{e}_7 1231512,$$

$$\begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 4,3 \\ \otimes \end{smallmatrix},$$

$$I^{\oplus 4} \mathbf{e}_7 \left(\bigotimes^{3,3} \mathbf{e}_7 \right)^3 \bigotimes^{3,3} \mathbf{e}_7 \bigotimes^{4,2},$$

$$\overset{2,4}{\otimes} \mathbf{e}_7 (\overset{3,3}{\otimes} \mathbf{e}_7)^3 \overset{3,3}{\otimes} \mathbf{e}_7 1232,$$

$$\begin{array}{ccccccc} 2,4 & & (3,3) & & 3,3 & & 4,4 \\ \otimes & e_7 & (\otimes e_7)^3 & \otimes & e_7 & \otimes & \end{array}$$

$$\overset{2,4}{\otimes} \mathbf{e}_7 (\overset{3,3}{\otimes} \mathbf{e}_7)^3 \overset{3,3}{\otimes} \mathbf{e}_7 12313$$

$$\begin{matrix} 2,4 \\ \otimes \end{matrix} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7)^3 \begin{matrix} 3,3 \\ \otimes \end{matrix} \mathbf{e}_7 1232151$$

$$31513221\epsilon_8(\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5} 12231513151, \quad 15131513221\epsilon_8(\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5} 12231513151$$

$$\begin{array}{l}
{}^{2,2}_{\otimes}\mathfrak{e}_6({}^{3,3}_{\otimes}\mathfrak{e}_7)^3 \otimes {}^{3,3}_{\otimes}\mathfrak{e}_7 \otimes {}^{4,4}_{\otimes}\mathfrak{e}_8 \\
{}^{2,2}_{\otimes}\mathfrak{e}_6({}^{3,3}_{\otimes}\mathfrak{e}_7)^3 \otimes {}^{3,3}_{\otimes}\mathfrak{e}_7 12313 \\
{}^{2,2}_{\otimes}\mathfrak{e}_6({}^{3,3}_{\otimes}\mathfrak{e}_7)^3 \otimes {}^{3,3}_{\otimes}\mathfrak{e}_7 1232151 \\
I^{\oplus 2}\mathfrak{e}_6({}^{3,3}_{\otimes}\mathfrak{e}_7)^3 \otimes {}^{3,3}_{\otimes}\mathfrak{e}_7 \otimes {}^{4,3}_{\otimes}\mathfrak{e}_8 \\
I^{\oplus 2}\mathfrak{e}_6({}^{3,3}_{\otimes}\mathfrak{e}_7)^3 \otimes {}^{3,3}_{\otimes}\mathfrak{e}_7 I^{\oplus 4} \\
I^{\oplus 2}\mathfrak{e}_6({}^{3,3}_{\otimes}\mathfrak{e}_7)^3 \otimes {}^{3,3}_{\otimes}\mathfrak{e}_7 1232^1
\end{array}$$

$$\begin{array}{l}
\begin{array}{l} 2,2 \\ \otimes \end{array} \epsilon_6 \begin{array}{l} 3,5 \\ \otimes \end{array} \epsilon_8 (\begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8)^2 \begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8 122315123 \\
\begin{array}{l} 2,2 \\ \otimes \end{array} \epsilon_6 \begin{array}{l} 3,5 \\ \otimes \end{array} \epsilon_8 (\begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8)^2 \begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8 12231513151 \\
I^{\oplus 2} \begin{array}{l} 3,5 \\ \otimes \end{array} \epsilon_6 \begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8 (\begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8)^2 \begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8 122315123 \\
I^{\oplus 2} \begin{array}{l} 3,5 \\ \otimes \end{array} \epsilon_6 \begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8 (\begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8)^2 \begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8 12231513151 \\
1 \begin{array}{l} 4,5 \\ \otimes \end{array} \epsilon_6 \begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8 (\begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8)^2 \begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8 122315123 \\
1 \begin{array}{l} 4,5 \\ \otimes \end{array} \epsilon_6 \begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8 (\begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8)^2 \begin{array}{l} 5,5 \\ \otimes \end{array} \epsilon_8 12231513151
\end{array}$$

$$\begin{array}{l}
1\epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \\
1\epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{12231512} \\
2,3 \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \\
I^{\oplus 3} \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \\
3,3 \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7^{5,5} I^{\oplus 2}
\end{array}$$

$$\begin{array}{l} \begin{array}{ccccc} 2,2 & (1,1) & & 1,1 & 2,4 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^3 & \otimes & \mathfrak{so} & \otimes \end{array} \\ \begin{array}{ccccc} 3,2 & (1,1) & & 1,1 & 2,3 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^3 & \otimes & \mathfrak{so} & \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & & 1,1 & & 2,3 \\ (\otimes \mathfrak{so})^3 & \otimes & \mathfrak{so} & \otimes & \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & & 1,1 & & 2,4 \\ (\otimes \mathfrak{so})^3 & \otimes & \mathfrak{so} & \otimes & \end{array} \end{array}$$

$$\begin{array}{l} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \otimes \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 3} \\ I^{\oplus 2} \mathbf{e}_6 \otimes \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \otimes \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 3} \\ 1 \mathbf{e}_6 \otimes \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \otimes \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \otimes \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \\ 1 \mathbf{e}_6 \otimes \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \otimes \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 2} \end{array}$$

$$\begin{aligned} & I^{\oplus 2} \epsilon_7'{}^{4,5} \otimes \epsilon_8^{(5,5)}{}^2 \otimes \epsilon_8 122315123 \\ & I^{\oplus 2} \epsilon_7'{}^{4,5} \otimes \epsilon_8^{(5,5)}{}^2 \otimes \epsilon_8 12231513151 \\ & 1\epsilon_7'{}^{(5,5)} \otimes \epsilon_8^{(5,5)}{}^3 \otimes \epsilon_8 122315123 \\ & 1\epsilon_7'{}^{(5,5)} \otimes \epsilon_8^{(5,5)}{}^3 \otimes \epsilon_8 12231513151 \end{aligned}$$

$$\begin{array}{l} 1\mathbf{so}^{2,3} \otimes \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{4,4} \\ 1\mathbf{so}^{2,3} \otimes \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{12313} \\ 1\mathbf{so}^{2,3} \otimes \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1232151} \end{array}$$

$$\begin{array}{l} 1\mathfrak{so}(\overset{2,2}{\otimes}\mathfrak{e}_6)^3 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,4}{\otimes} \\ 1\mathfrak{so}(\overset{2,2}{\otimes}\mathfrak{e}_6)^3 \overset{2,2}{\otimes} \mathfrak{e}_6 I^{\oplus 3} \\ 1\mathfrak{so}(\overset{2,2}{\otimes}\mathfrak{e}_6)^3 \overset{2,2}{\otimes} \mathfrak{e}_6 13151 \end{array}$$

$$I^{\oplus 2} \mathbf{e}'_7{}^{4,5} \otimes \mathbf{e}_8{}^{(5,5)} \otimes \mathbf{e}_7 I^{\oplus 3}$$

$$\begin{array}{c} 3,3 \\ \otimes \mathbf{e}_7 (\otimes \mathbf{e}_7)^3 \otimes \mathbf{e}_8 I^{\oplus 3} \\ 512321 \mathbf{e}_7 (\otimes \mathbf{e}_7)^3 \otimes \mathbf{e}_8 I^{\oplus 2} \end{array}$$

$$\begin{array}{c} 2,3 \\ \otimes \end{array} e'_7 \left(\begin{array}{c} 3,3 \\ \otimes \end{array} e_7 \right)^3 \begin{array}{c} 3,3 \\ \otimes \end{array} e'_7 \begin{array}{c} 3,3 \\ \otimes \end{array}$$

$$I^{\oplus 3} \mathbf{e}'_7 (\overset{(3,3)}{\otimes} \mathbf{e}_7)^3 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes}$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 122315123$$

$$31_{\mathfrak{so}} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{4,2} \mathfrak{so}13$$

$$2231so(\overset{1,1}{\otimes}so)^3\overset{2,4}{\otimes}e_81$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2\overset{2,4}{\otimes}\mathfrak{e}_8\overset{4,2}{\otimes}\mathfrak{so}1$$

$$\begin{matrix} 2,2 & 3,3 & 3,3 & 3,3 \\ \otimes e_6 & (\otimes e_7)^3 & \otimes e'_7 & \otimes \end{matrix}$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix}$$

$$315131\mathbf{e}_6\left(\overset{2,2}{\otimes}\mathbf{e}_6\right)^3\overset{3,5}{\otimes}\mathbf{e}_81$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\bigotimes \mathbf{e}_6 \right)^3 \bigotimes \mathbf{e}_8 I^{\oplus 3}$$

$$\begin{matrix} 2,2 & 3,5 & & 5,5 \\ \otimes \mathbf{e}_6 & \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8)^2 & \otimes \mathbf{e}'_7 1 \end{matrix}$$

$$1e_6 \overset{4,5}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} e_7' 1$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{5,3}{\otimes} \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^2 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^2 \otimes^{5,5} e_8 122315123$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^2 \otimes^{5,5} e_8 12231513151$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^2 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7' 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^3 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \mathfrak{so} 13$$

$$1\mathfrak{so} (\otimes \mathfrak{so})^{1,1} \otimes \mathfrak{e}_7^{2,3} 1$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} 1$$

2.6.7 SgLgLgLIg(27)

$$\otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$\otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$\otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4}$$

$$313221\mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$I^{\oplus 5} \mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$21513221\mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4}$$

$$1513221\mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4}$$

$$\otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7'^{1,3,3},$$

$$\otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7'^{1,3,3},$$

$$321\mathfrak{e}_7' (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3}$$

$$I^{\oplus 3} \mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7'^{1,3,3}$$

$$3221\mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3},$$

$$512321\mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3},$$

$$151321\mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3}$$

$$I^{\oplus 2} \mathfrak{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \right)^2 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6$$

$$1e_7(\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} \overset{5,4}{\otimes} e_7$$

$$I^{\oplus 2} \mathbf{e}'_7 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^2 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{1}{\otimes} \overset{3,3}{\otimes} \mathbf{e}'_7$$

$$31e_6(\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_7 \overset{1}{\otimes} \overset{3,3}{\otimes} e_7$$

$$150 \overset{2,4}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_7$$

$$150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e'_7$$

$$150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} e_6$$

$$1e'_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,4}{\otimes} e_7$$

$$1e_7 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} \overset{4,2}{50}$$

$$1e_6 \overset{3,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,4}{\otimes} e_7$$

$$1e_6 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7$$

2.6.8 SgLgLIgLgLg(4)

$$\begin{array}{ccccc} 2,4 & 5,5 & 5,5 & 1 & 5,5 & 4,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} \end{array}, \quad \begin{array}{ccccc} I^{\oplus 4} & 5,5 & 5,5 & 1 & 5,5 & 4,2 \\ \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} \end{array}$$

$$1_{50} \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50$$

$$1e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{4,2}{\otimes} 50$$

2.6.9 SgLgLIgLgS(1)

$$150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} 501$$

2.6.10 SgLIgLgLgLg(12)

$$\begin{array}{lll} \begin{array}{cccccc} 2,4 & 5,5 & 1 & 5,5 & 4,2 & 1,1 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so}, \end{array} & \begin{array}{cccccc} I^{\oplus 4} & 5,5 & 1 & 5,5 & 4,2 & 1,1 \\ \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so}, \end{array} & \begin{array}{cccc} 2,4 & 1 & 5,5 & 5,3 \\ 1\mathfrak{so} & \otimes \mathfrak{e}_8 & (\otimes \mathfrak{e}_8)^2 & \otimes \mathfrak{so} \end{array} \end{array}$$

$$3221\mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7^{5,4}, \quad I^{\oplus 3} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7^{5,5}$$

$$3221\epsilon_8^{5,5} \otimes \epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_6^{5,3}, \quad I^{\oplus 3} \epsilon_8^{5,5} \otimes \epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_6^{5,4}$$

$$513221e_8 \otimes e_8^{5,5} \otimes e_8^{1,5} (\otimes e_8^{5,5})^2 \otimes e_8^{4,2} \otimes e_8^{5,5}$$

$$3221\mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^2 \overset{5,5}{\otimes} \mathbf{e}_7$$

$$150 \otimes^{2,4} \mathfrak{e}_8 \otimes^1 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{5,5} \mathfrak{e}_6$$

$$1e_7 \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{4,2} 50 \otimes^{1,1} 50$$

$$1e_7 \otimes^{3,3} e_7 (\otimes^{3,3} e_7)^2 \otimes^{3,2} so$$

2.6.11 SgLIgLgLgLgS(43)

$$\begin{array}{lll}
3221\epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{4,2}}, & I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{5,3}}, & I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{5,4}} \\
I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{5,5}}, & 3221\epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes I^{\oplus 4}, & I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes 122313 \\
I^{\oplus 2} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes 1223151315, & I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes I^{\oplus 5}, & I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes 12231512 \\
I^{\oplus 3} \epsilon_8^{\overline{5,5}} \otimes \epsilon_8^{\overline{1,5,5}} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{\overline{5,5}} \otimes 12231511 & &
\end{array}$$

$$\begin{array}{lll} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{4,2} \mathfrak{so}1, & 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} 1223132, & I^{\oplus 4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{4,2} \mathfrak{so}1 \\ 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 I^{\oplus 6}, & 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 122315123, & 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 12231513215 \\ 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8^{12231513}, & 1\mathfrak{so} \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^1 (\otimes_{\mathfrak{e}_8}^{5,5})^2 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8^{12231513151} \end{array}$$

$$\begin{aligned}
& 3221 \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 1, & 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 1232, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 123 \\
& I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 I^{\oplus 2}, & I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 1
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{e}_7 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so} 1, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes \\
& 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 I^{\oplus 3}, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 \otimes, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 I^{\oplus 2}, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 1
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7 \otimes, & 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7 \otimes, & 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7 I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 1, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 13
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7' I^{\oplus 2}, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7' 1
\end{aligned}$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7' 1$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so} 13$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6 1$$

2.6.12 SgLIgLgLIg(1)

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

2.6.13 SgLIgLgLIgS(1)

$$1 \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} 1$$

$$31\mathfrak{e}_6(\otimes\mathfrak{e}_6)^3\otimes\mathfrak{e}_7,$$

$$1\mathfrak{e}_6(\otimes\mathfrak{e}_6)^3\otimes\mathfrak{e}_7$$

$$1513221\mathfrak{e}_8(\otimes\mathfrak{e}_8)^2\otimes\mathfrak{e}_7\otimes\mathfrak{e}_7,$$

$$I^{\oplus 2}{}^{3221}\mathfrak{e}_8(\otimes\mathfrak{e}_8)^2\otimes\mathfrak{e}_7\otimes\mathfrak{e}_7$$

$$I^{\oplus 2}{}^1\mathfrak{e}_8\otimes\mathfrak{e}_6(\otimes\mathfrak{e}_6)^2\otimes\mathfrak{so}$$

$$I^{\oplus 2}{}^1\mathfrak{e}_7(\otimes\mathfrak{e}_7)^3\otimes\mathfrak{e}_7$$

$$I^{\oplus 2}{}^1\mathfrak{e}_7'(\otimes\mathfrak{e}_7)^3\otimes\mathfrak{so}$$

$$I^{\oplus 2}{}^1\mathfrak{e}_7(\otimes\mathfrak{e}_7)^2(\otimes\mathfrak{e}_7')^2$$

$$I^{\oplus 2}{}^1\mathfrak{e}_7(\otimes\mathfrak{e}_7)^2\otimes\mathfrak{e}_6\otimes\mathfrak{e}_6$$

$$3221\mathfrak{e}_8\otimes\mathfrak{so}(\otimes\mathfrak{so})^3$$

$$1\mathfrak{e}_7\otimes\mathfrak{e}_8\otimes\mathfrak{e}_8\otimes\mathfrak{e}_7\otimes\mathfrak{e}_7'$$

$$1\mathfrak{e}_7\otimes\mathfrak{e}_8\otimes\mathfrak{e}_8\otimes\mathfrak{e}_6\otimes\mathfrak{e}_6$$

$$1\mathfrak{e}_7'\otimes\mathfrak{e}_8(\otimes\mathfrak{e}_8)^2\otimes\mathfrak{e}_6$$

$$1\mathfrak{e}_7'\otimes\mathfrak{e}_8(\otimes\mathfrak{e}_8)^2\otimes\mathfrak{so}$$

$$1\mathfrak{e}_7\otimes\mathfrak{e}_8\otimes\mathfrak{so}(\otimes\mathfrak{so})^2$$

$$1\mathfrak{e}_7\otimes\mathfrak{e}_7\otimes\mathfrak{e}_8\otimes\mathfrak{so}\otimes\mathfrak{so}$$

$$\begin{aligned} I^{\oplus 4} \mathbf{e}_8^{\otimes 2} \left(\bigotimes \mathbf{e}_8 \right)^3 \bigotimes \mathbf{e}_8^{12231513} \\ I^{\oplus 3} \mathbf{e}_8^{\otimes 3} \left(\bigotimes \mathbf{e}_8 \right)^3 \bigotimes \mathbf{e}_8^{122315123} \\ I^{\oplus 3} \mathbf{e}_8^{\otimes 3} \left(\bigotimes \mathbf{e}_8 \right)^3 \bigotimes \mathbf{e}_8^{12231513151} \end{aligned}$$

[illegible]

$$\begin{array}{l}
2,3 \quad 1 \quad 3,3 \quad 3,3 \quad 4,4 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes \\
2,3 \quad 1 \quad 3,3 \quad 3,3 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 12313 \\
2,3 \quad 1 \quad 3,3 \quad 3,3 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 1232151 \\
I^{\oplus 3} \quad 1 \quad 3,3 \quad 3,3 \quad 4,2 \\
\mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes \\
3,3 \quad 1 \quad 3,3 \quad 3,3 \quad 4,5 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes \\
3,3 \quad 1 \quad 3,3 \quad 3,3 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 1231512
\end{array}$$

$$\begin{array}{l} 31\epsilon_6^{31}(\otimes \epsilon_6^{2,2})^3 \otimes \epsilon_6^{2,2} I^{\oplus 3} \\ 31\epsilon_6^{31}(\otimes \epsilon_6^{2,2})^3 \otimes \epsilon_6^{2,2} 13151^1 \end{array}$$

$$\begin{aligned}
& 1 \overset{1}{\epsilon}'_7 \otimes \overset{4,5}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^2 \overset{5,5}{\otimes} \epsilon_8 122315123 \\
& 1 \overset{1}{\epsilon}'_7 \otimes \overset{4,5}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^2 \overset{5,5}{\otimes} \epsilon_8 12231513151 \\
& \overset{5,5}{\otimes} \epsilon_8 (\overset{5,5}{\otimes} \epsilon_8)^3 \overset{5,4}{\epsilon}'_7 I^{\oplus 2} \\
& 21513221 \overset{1}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^3 \overset{5,4}{\otimes} \epsilon'_7 I^{\oplus 2} \\
& I^{\oplus 4} \overset{1}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^3 \overset{5,4}{\otimes} \epsilon'_7 I^{\oplus 2} \\
& \overset{4,5}{\otimes} \epsilon_8 (\overset{5,5}{\otimes} \epsilon_8)^3 \overset{5,5}{\otimes} \epsilon'_7 1 \\
& I^{\oplus 5} \overset{1}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^3 \overset{5,5}{\otimes} \epsilon'_7 1 \\
& I^{\oplus 2} \overset{2,4}{\otimes} \overset{5,5}{\epsilon}_8 (\overset{5,5}{\otimes} \epsilon_8)^3 \overset{5,5}{\otimes} \epsilon'_7 1
\end{aligned}$$

$$\begin{array}{l}
{}^{2,3}{}_{1}\otimes e_7'({}^{3,3}\otimes e_7')^3\otimes {}^{3,3}{}_{3}\otimes e_7'^{3,3}\otimes \\
{}^{2,3}{}_{1}\otimes e_7'({}^{3,3}\otimes e_7')^3\otimes {}^{3,3}{}_{3}\otimes e_7'I^{\oplus 3} \\
I^{\oplus 3}e_7'({}^{3,3}\otimes e_7')^3\otimes {}^{3,3}{}_{3}\otimes e_7'^{3,3}\otimes \\
I^{\oplus 2}e_7'^1({}^{3,3}\otimes e_7')^3\otimes {}^{3,3}{}_{3}\otimes e_7'^{4,4}\otimes \\
I^{\oplus 2}e_7'^1({}^{3,3}\otimes e_7')^3\otimes {}^{3,3}{}_{3}\otimes e_7'12313 \\
I^{\oplus 2}e_7'^1({}^{3,3}\otimes e_7')^3\otimes {}^{3,3}{}_{3}\otimes e_7'1232151 \\
I^{\oplus 2}e_7'^1({}^{3,3}\otimes e_7')^3\otimes {}^{3,3}{}_{3}\otimes e_7'^{3,2}\otimes
\end{array}$$

$$\begin{aligned}
& 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^3 \otimes^{3,3} \epsilon_7 \otimes^{4,4} \\
& 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^3 \otimes^{3,3} \epsilon_7 12313 \\
& 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^3 \otimes^{3,3} \epsilon_7 1232151 \\
& \otimes^{3,3} \epsilon_7 (\otimes^{3,3} \epsilon_7)^3 \otimes^{3,3} \epsilon_6 \otimes^{2,2} \\
& \otimes^{3,3} \epsilon_7 (\otimes^{3,3} \epsilon_7)^3 \otimes^{3,3} \epsilon_6 I^{\oplus 2} \\
& I^{\oplus 2} \epsilon_7^{1,3,3} (\otimes \epsilon_7)^3 \otimes^{3,3} \epsilon_6 I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned} & I^{\oplus 3} \mathbf{e}_8^{I^{\oplus 2} \binom{5,5}} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \mathbf{1}^{\binom{5,4}{3,3}} \\ & I^{\oplus 2} \mathbf{e}_7 \mathbf{1}^{\binom{4,5}} \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_7 I^{\oplus 3} \\ & \mathbf{1} \mathbf{e}_7 (\otimes \mathbf{e}_8)^3 \otimes \mathbf{e}_7 I^{\oplus 3} \end{aligned}$$

$$\begin{array}{l} \otimes^{5,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{4,2} \mathfrak{so}13 \\ 21513221 \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{4,2} \mathfrak{so}13 \\ I^{\oplus 4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{4,2} \mathfrak{so}13 \end{array}$$

$$\begin{array}{l} \begin{array}{c} 5,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} \end{array} \\ 21513221 \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} \\ I^{\oplus 4} \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} \end{array}$$

$$\begin{aligned} & 2321^1_{\mathbf{e}_7}(\otimes^{3,3}_{\mathbf{e}_7})^3 \otimes^{4,5}_{\mathbf{e}_8} \mathbf{e}_8 1 \\ & I^{\oplus 2} \mathbf{e}_8^1 \otimes^{5,4}_{\mathbf{e}_7}(\otimes^{3,3}_{\mathbf{e}_7})^2 \otimes^{3,3}_{\mathbf{e}_7} \otimes^{3,2}_{\mathbf{e}_7} \\ & I^{\oplus 2} \mathbf{e}_7^1(\otimes^{3,3}_{\mathbf{e}_7})^3 \otimes^{4,5}_{\mathbf{e}_8} \mathbf{e}_8 I^{\oplus 3} \end{aligned}$$

$$\begin{aligned} & 1_{\mathbf{e}_6}^{1, 3, 5} \otimes \mathbf{e}_8^{(5, 5)} \otimes \mathbf{e}_7^{5, 4} I^{\oplus 3} \\ & 1_{\mathbf{e}_7}^{1, (5, 5)} \otimes \mathbf{e}_8^{5, 3} \otimes \mathbf{e}_6^{2, 2} \\ & I^{\oplus 2} \mathbf{e}_7^{1, 4, 5} \otimes \mathbf{e}_8^{(5, 5)} \otimes \mathbf{e}_6^{5, 4} \end{aligned}$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^2 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$I^{\oplus 2} e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^2 \otimes^{5,5} e_7' 1$$

$$\begin{aligned}
& 1\mathfrak{e}_8 \otimes \mathfrak{e}_6(\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6^{2,2} 123, & 1\mathfrak{e}_6(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_8^{3,5} I^{\oplus 3}, & 1\mathfrak{e}_8 \otimes \mathfrak{e}_6(\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6^{2,2} 13151 \\
& I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{e}_6(\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2}, & I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{e}_6(\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2}, & 31\mathfrak{e}_6(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_8^{3,5} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{so}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,2}, & 1\mathfrak{so}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,3}, & 1\mathfrak{so}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,4} \\
& 1\mathfrak{so}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,2} 132, & 1\mathfrak{so}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} I^{\oplus 2} &
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2}, & 1\mathfrak{e}_6 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}'_7^{5,4} I^{\oplus 2}, & 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,3} I^{\oplus 2} \\
& 1\mathfrak{e}_6 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}'_7^{5,5} \otimes \mathfrak{e}'_7 1, & 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,4} 1 &
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} \otimes \mathfrak{e}'_7^{3,2}, & 1\mathfrak{e}_6(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} \otimes \mathfrak{e}'_7^{3,3}, & 1\mathfrak{e}_6(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} I^{\oplus 3} \\
& I^{\oplus 2} \mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} \otimes \mathfrak{e}_6^{2,2}, & I^{\oplus 2} \mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6^{3,3} I^{\oplus 2} &
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2} 1, & \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2} 1, & I^{\oplus 3} \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2} 1 \\
& I^{\oplus 2} \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2} 1 & &
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 3} 1\mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \otimes \mathfrak{so}^{1,1} 1, & 1\mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \otimes \mathfrak{so}^{1,1} 132, & I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \otimes \mathfrak{so}^{1,1} 1 \\
& I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \otimes \mathfrak{so}^{1,1} 13 & &
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}'_7^{3,3} \otimes \mathfrak{e}_7^{3,3} 1, & I^{\oplus 2} \mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} \otimes \mathfrak{e}'_7^{3,2}, & I^{\oplus 2} \mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} \otimes \mathfrak{e}'_7^{3,3} \\
& I^{\oplus 2} \mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} I^{\oplus 3} & &
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathfrak{e}_6(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{so}^{2,2} 1, & I^{\oplus 2} 1\mathfrak{e}_6(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{so}^{2,2} 1, & 31\mathfrak{e}_6(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{so}^{2,2} 1
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_8 \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}'_7^{3,3} 123, & 1\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_8^{4,5} I^{\oplus 3}, & I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}'_7^{3,3} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2}, & 1\mathfrak{e}_6 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,3} I^{\oplus 2}, & 1\mathfrak{e}_6 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,4} 1
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6^{3,3} 1, & 1\mathfrak{e}_8 \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6^{3,3} 13 &
\end{aligned}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1,$$

$$1\mathfrak{e}_7^{1,5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2}13,$$

$$1\mathfrak{e}_7^{1,5,5}(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2},$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7^{5,5}1$$

$$1\mathfrak{e}_6^{1,3,3}(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6^{3,3} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{e}_6^{1,3,3}(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6^{3,3} I^{\oplus 2}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,3,3}(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2}1$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_71$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{e}_6^{1,3,3}(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2}1$$

$$1\mathfrak{e}_6^{1,2,2}(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7^{3,3}1$$

2.6.16 SSgLgLgLgLIgS(137)

[illegible]

$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5\ 1\ 5,3},$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5\ 1\ 5,4},$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5\ 1\ 5,5}$
$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5\ 1} 122313,$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5\ 1} I^{\oplus 5},$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5\ 1} 12231512$
$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5\ 1} 1223151,$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5\ 1} I^{\oplus 2,4,2} \otimes \epsilon_8,$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5\ 1} I^{\oplus 2} \otimes \epsilon_8 I^{\oplus 4}$
$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5\ 1} I^{\oplus 3},$	$1\epsilon_7^{(5,5)} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5\ 1\ 5,3},$	$1\epsilon_7^{(5,5)} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5\ 1\ 5,4}$
$1\epsilon_7^{(5,5)} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5\ 1\ 5,5},$	$1\epsilon_7^{(5,5)} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5\ 1} 122313,$	$1\epsilon_7^{(5,5)} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5\ 1} I^{\oplus 5}$
$1\epsilon_7^{(5,5)} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5\ 1} 12231512,$	$1\epsilon_7^{(5,5)} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5\ 1} 1223151,$	$1\epsilon_7^{(5,5)} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5\ 1} I^{\oplus 2,4,2}$
$1\epsilon_7^{(5,5)} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5\ 1} I^{\oplus 2} \otimes \epsilon_8 I^{\oplus 4},$	$1\epsilon_7^{(5,5)} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5\ 1} I^{\oplus 3}$	

$$\begin{array}{ccc} \begin{array}{c} 2,3 \quad 1 \quad 3,3 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{c} 3,3 \\ (\otimes \mathbf{e}_7)^3 \end{array} \begin{array}{c} 3,3 \quad 1 \quad 3,2 \\ \otimes \mathbf{e}_7 \otimes, \end{array} & \begin{array}{c} 2,3 \quad 1 \quad 3,3 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{c} 3,3 \\ (\otimes \mathbf{e}_7)^3 \end{array} \begin{array}{c} 3,3 \quad 1 \quad 3,3 \\ \otimes \mathbf{e}_7 \otimes, \end{array} & I^{\oplus 3} \begin{array}{c} 1 \quad 3,3 \\ \mathbf{e}_7 \end{array} \begin{array}{c} 3,3 \\ (\otimes \mathbf{e}_7)^3 \end{array} \begin{array}{c} 3,3 \quad 1 \quad 3,2 \\ \otimes \mathbf{e}_7 \otimes \end{array} \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 3,3 \ 1 \\ \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes, \end{array} & \begin{array}{l} I^{\oplus 3} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^3 \otimes \mathfrak{e}_7^1 \otimes, \end{array} & \begin{array}{l} I^{\oplus 3} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^3 I^{\oplus 3} 1 \end{array} \\ \begin{array}{l} 2,3 \ 1 \\ \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 I^{\oplus 2}, \end{array} & \begin{array}{l} \mathfrak{e}_7^3 \otimes \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^3 I^{\oplus 2}, \end{array} & \begin{array}{l} I^{\oplus 3} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^3 I^{\oplus 2} \end{array} \\ I^{\oplus 2} \mathfrak{e}_7^2 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^3 I^{\oplus 2} & & \end{array}$$

$$\begin{array}{lll}
1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8{}^{5,5} \otimes \mathbf{e}_8{}^{1,5,3}, & 1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8{}^{5,5} \otimes \mathbf{e}_8{}^{1,5,4}, & 1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8{}^{5,5} \otimes \mathbf{e}_8{}^{1,5,5} \\
1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8{}^{5,5} \otimes \mathbf{e}_8{}^{122313}, & 1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8{}^{5,5} \otimes \mathbf{e}_8{}^{I^{\oplus 5}}, & 1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8{}^{5,5} \otimes \mathbf{e}_8{}^{12231512} \\
1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8{}^{5,5} \otimes \mathbf{e}_8{}^{1223151}, & 1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8{}^{5,5} \otimes \mathbf{e}_8{}^{I^{\oplus 2,4,2}}, & 1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8{}^{5,5} \otimes \mathbf{e}_8{}^{I^{\oplus 2}} \\
1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8{}^{5,5} \otimes \mathbf{e}_8{}^{I^{\oplus 3}}, & & 1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8({}^{5,5} \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8{}^{5,5} \otimes \mathbf{e}_8{}^{I^{\oplus 4}}
\end{array}$$

$$\begin{array}{lll}
1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,5,3}, & 1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,5,4}, & 1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,5,5} \\
1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{122313}, & 1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{I^{\oplus 5}}, & 1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{12231512} \\
1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1223151}, & 1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{I^{\oplus 2,4,2}}, & 1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{I^{\oplus 2}} \\
1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{I^{\oplus 3}}, & &
\end{array}$$

$$\begin{array}{lll} \begin{array}{l} 2,2 \quad 1 \quad 2,2 \\ \otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_6 \otimes, \\ \\ 31 \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_6 \otimes, \end{array} & \begin{array}{l} I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_6 \otimes, \\ \\ 31 \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_6 \otimes I^{\oplus 2} \end{array} & \begin{array}{l} I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_6 \otimes I^{\oplus 2} \\ \\ \end{array} \end{array}$$

$$I^{\oplus 2} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \otimes \mathbf{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 1,3,2 \\ \otimes \end{smallmatrix}, \quad I^{\oplus 2} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \otimes \mathbf{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 1,3,3 \\ \otimes \end{smallmatrix}, \quad I^{\oplus 2} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \otimes \mathbf{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix}$$

$$\begin{array}{lll} 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^3\overset{3,3}{\otimes}\mathbf{e}_7\overset{1,3,2}{\otimes}, & 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^3\overset{3,3}{\otimes}\mathbf{e}_7\overset{1,3,3}{\otimes}, & 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^3\overset{3,3}{\otimes}\mathbf{e}_7\overset{I^{\oplus 3}}{1} \\ 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^3\overset{3,3}{\otimes}\mathbf{e}_7\overset{I^{\oplus 2}}{I^{\oplus 2}} & & \end{array}$$

$$I^{\oplus 2} e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^2 \otimes e_7^{\oplus 2} 1, \quad I^{\oplus 2} e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^2 \otimes e_7^{\oplus 2} 1, \quad 1 e_7^1 (\otimes e_8^{5,5})^3 \otimes e_7^{\oplus 2} 1$$

$$321 \mathbf{e}_7 (\otimes \mathbf{e}_7)^3 \otimes \mathbf{e}_8 1, \quad I^{\oplus 2} \mathbf{e}_7 (\otimes \mathbf{e}_7)^3 \otimes \mathbf{e}_8 1^{\oplus 2}$$

$$1e_7' \otimes e_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes e_8 \end{smallmatrix} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} I^{\oplus 2} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} e_7 1,$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7^{5,4} I^{\oplus 2},$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7^{5,5} \epsilon_7^1$$

$$1\epsilon_6^1 (\otimes \epsilon_6^{2,2})^3 \otimes \epsilon_8^{3,5} I^{\oplus 2},$$

$$31\epsilon_6^1 (\otimes \epsilon_6^{2,2})^3 \otimes \epsilon_8^{3,5} \epsilon_8^1$$

$$I^{\oplus 2} \epsilon_7'^1 (\otimes \epsilon_7^{3,3})^3 \otimes \epsilon_7'^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so}^1 (\otimes \mathfrak{so}^{1,1})^3 \otimes \mathfrak{so}^{1,1} \epsilon_7^1$$

$$1\epsilon_7'^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7'^{5,4} \epsilon_7^1$$

$$1\epsilon_7'^1 (\otimes \epsilon_7^{3,3})^3 \otimes \epsilon_8^{4,5} I^{\oplus 2}$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7^{5,4} \epsilon_7^1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_6^{5,3} \epsilon_6^1$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^3 \otimes \epsilon_7'^{3,3} I^{\oplus 2}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^3 \otimes \epsilon_6^{3,3} \epsilon_6^1$$

2.6.17 SSgLgLgLIg(9)

$$\otimes \epsilon_8^{2,4} \epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \epsilon_8^1 \otimes \epsilon_7^{5,4},$$

$$I^{\oplus 4} \epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \epsilon_8^1 \otimes \epsilon_7^{5,4},$$

$$I^{\oplus 3} \epsilon_8^{I^{\oplus 2}} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \epsilon_8^1 \otimes \epsilon_7^{5,4}$$

$$I^{\oplus 2} \epsilon_7^1 (\otimes \epsilon_7^{3,3})^2 \otimes \epsilon_7^{3,3} \epsilon_7^1 \otimes \epsilon_7'^{3,3}$$

$$321\epsilon_7^1 (\otimes \epsilon_7^{3,3})^2 \otimes \epsilon_7^{3,3} \epsilon_7^1 \otimes \epsilon_7^{3,3}$$

$$231513221\epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \epsilon_8^1 \otimes \epsilon_8^{5,5}$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,5}{\otimes} \overset{5,5}{e_8} \overset{1}{\otimes} \overset{5,4}{e_7}$$

$$1e_7' \overset{1}{\otimes} \overset{3,3}{e_7} \overset{2}{\otimes} \overset{3,3}{e_7} \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3,3}{e_7}'$$

$$1\mathbf{e}_6^{1(2,2)}(\otimes \mathbf{e}_6)^{2(2,2)} \otimes \mathbf{e}_6^{1(2,2)} \otimes \mathbf{e}_6^{2,2}$$

2.6.18 SSgLgLgLIgLgS(25)

$\begin{array}{l} \begin{array}{c} 3,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \end{array} \\ \\ 313221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \\ \\ \begin{array}{c} 1 \\ 1513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \end{array} \\ \\ I^{\oplus 3} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1 \end{array}$	$\begin{array}{l} \begin{array}{c} 4,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \end{array} \\ \\ I^{\oplus 5} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \\ \\ I^{\oplus 2} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \\ \\ I^{\oplus 3} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1 \end{array}$	$\begin{array}{l} \begin{array}{c} 5,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1 \end{array} \\ \\ 21513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1 \\ \\ I^{\oplus 4} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1 \end{array}$
$\begin{array}{l} 31513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1, \\ \\ 1513221 \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 I^{\oplus 2}, \\ \\ I^{\oplus 2} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 I^{\oplus 2}, \end{array}$	$\begin{array}{l} \begin{array}{c} 2,4 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 I^{\oplus 3}, \end{array} \\ \\ I^{\oplus 4} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 I^{\oplus 3}, \\ \\ I^{\oplus 3} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 I^{\oplus 3} \end{array}$	$\begin{array}{l} I^{\oplus 3} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1223 \\ \\ I^{\oplus 2} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 1223 \\ \\ \end{array}$

$$I^{\oplus 2} \mathbf{e}_7^{1\ 4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1\ 4,2} \oplus 1 \mathbf{e}_7^{1\ 5,5} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1\ 4,2} \oplus$$

$$I^{\oplus 2} \mathbf{e}_7 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^2 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{1}{\otimes} \overset{3,3}{\otimes} \mathbf{e}_7 1$$

$$1_{\mathfrak{e}_7}^{1, 4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \otimes 1_{\mathfrak{e}_8}^{1, 5,5} \otimes \mathfrak{e}_8^{5,5} I^{\oplus 3}$$

$$1e_7' \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,5}{\otimes} \overset{5,5}{e_8} \overset{1}{\otimes} \overset{4,2}{e_8} \otimes so1$$

$$1 \overset{1}{e}'_7 (\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_7 \overset{1}{\otimes} \overset{3,3}{e_7} 1$$

$$1\mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}1$$

2.7.2 Sg6(81)

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7'$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} (\otimes \mathfrak{so})^{1,1} \otimes \mathfrak{e}_7^{2,3}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$1\mathfrak{e}_6 (\otimes \mathfrak{e}_7)^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7' \otimes \mathfrak{e}_7^{3,3}$$

2.7.3 Sg6S(376)

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,2},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1231512},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{123151},$$

$$\otimes \mathfrak{e}_7^{3,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3},$$

$$I^{\oplus 4} \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,2},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1232},$$

$$\otimes \mathfrak{e}_7^{3,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3},$$

$$I^{\oplus 4} \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3},$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,4}$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{12313}$$

$$\otimes \mathfrak{e}_7^{2,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1232151}$$

$$\otimes \mathfrak{e}_7^{3,4} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,4}$$

$$31321\mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3}$$

$$\begin{aligned}
& \otimes_{\mathbb{C}}^{\substack{2,3 \\ 2,3}} \mathbb{C}'_7 (\otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7)^4 \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7 \otimes_{\mathbb{C}}^{\substack{4,4 \\ 4,4}} \\
& \otimes_{\mathbb{C}}^{\substack{2,3 \\ 2,3}} \mathbb{C}'_7 (\otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7)^4 \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7 12313 \\
& \otimes_{\mathbb{C}}^{\substack{2,3 \\ 2,3}} \mathbb{C}'_7 (\otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7)^4 \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7 1232151 \\
& I^{\oplus 3} \mathbb{C}'_7 (\otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7)^4 \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7 \otimes_{\mathbb{C}}^{\substack{4,2 \\ 4,2}} \\
& \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}'_7 (\otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7)^4 \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7 \otimes_{\mathbb{C}}^{\substack{4,5 \\ 4,5}} \\
& \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}'_7 (\otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7)^4 \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7 1231512 \\
& \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}'_7 (\otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7)^4 \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7 123151^1 \\
& I^{\oplus 3} \mathbb{C}'_7 (\otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7)^4 \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7 \otimes_{\mathbb{C}}^{\substack{4,5 \\ 4,5}} \\
& I^{\oplus 3} \mathbb{C}'_7 (\otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7)^4 \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7 1231512 \\
& I^{\oplus 3} \mathbb{C}'_7 (\otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7)^4 \otimes_{\mathbb{C}}^{\substack{3,3 \\ 3,3}} \mathbb{C}_7 123151^1
\end{aligned}$$

$$\begin{array}{l}
{}^{2,3} \otimes \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 122315123 \\
{}^{2,3} \otimes \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 12231513151 \\
{}^{3,3} \otimes \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 122315123 \\
{}^{3,3} \otimes \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 12231513151 \\
I^{\oplus 3} \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 122315123 \\
I^{\oplus 3} \mathfrak{e}_7 \otimes {}^{4,5} \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^3 \otimes {}^{5,5} \mathfrak{e}_8 12231513151 \\
I^{\oplus 2} \mathfrak{e}_7 ({}^{5,5} \otimes \mathfrak{e}_8)^4 \otimes {}^{5,5} \mathfrak{e}_8 122315123 \\
I^{\oplus 2} \mathfrak{e}_7 ({}^{5,5} \otimes \mathfrak{e}_8)^4 \otimes {}^{5,5} \mathfrak{e}_8 12231513151
\end{array}$$

$$\begin{aligned}
& 2313221\mathbf{e}_8(\otimes\mathbf{e}_8)^4 \otimes^{5,5} \mathbf{e}_8 122315123 \\
& 2313221\mathbf{e}_8(\otimes\mathbf{e}_8)^4 \otimes^{5,5} \mathbf{e}_8 12231513151 \\
& I^{\oplus 6}\mathbf{e}_8(\otimes\mathbf{e}_8)^4 \otimes^{5,5} \mathbf{e}_8 12231513215 \\
& 321513221\mathbf{e}_8(\otimes\mathbf{e}_8)^4 \otimes^{5,5} \mathbf{e}_8 122315123 \\
& 321513221\mathbf{e}_8(\otimes\mathbf{e}_8)^4 \otimes^{5,5} \mathbf{e}_8 12231513151 \\
& 15131513221\mathbf{e}_8(\otimes\mathbf{e}_8)^4 \otimes^{5,5} \mathbf{e}_8 12231513215 \\
& 15131513221\mathbf{e}_8(\otimes\mathbf{e}_8)^4 \otimes^{5,5} \mathbf{e}_8 12231513151
\end{aligned}$$

$$\begin{array}{ccccccc} 2,2 & & 3,3 & & 3,3 & & 4,4 \\ \otimes \mathbf{e}_6 & (& \otimes \mathbf{e}_7 &)^4 & \otimes & \mathbf{e}_7 & \otimes \end{array}$$

$$\begin{array}{l} I^{\oplus 3} \mathbf{e}_7 \otimes \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\otimes \mathbf{e}_8)^3 \otimes \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \otimes \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \\ I^{\oplus 3} \mathbf{e}_7 \otimes \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\otimes \mathbf{e}_8)^3 \otimes \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 3} \\ I^{\oplus 3} \mathbf{e}_7 \otimes \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\otimes \mathbf{e}_8)^3 \otimes \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 2} \end{array}$$

$$\begin{array}{l} 1\textbf{so} \overset{2,3}{\otimes} \textbf{e}_7(\overset{3,3}{\otimes} \textbf{e}_7)^3 \overset{3,3}{\otimes} \overset{4,4}{\textbf{e}_7} \\ 1\textbf{so} \overset{2,3}{\otimes} \textbf{e}_7(\overset{3,3}{\otimes} \textbf{e}_7)^3 \overset{3,3}{\otimes} \textbf{e}_7 12313 \\ 1\textbf{so} \overset{2,3}{\otimes} \textbf{e}_7(\overset{3,3}{\otimes} \textbf{e}_7)^3 \overset{3,3}{\otimes} \textbf{e}_7 1232151 \end{array}$$

$$\begin{array}{l} 150 \binom{2,2}{\otimes \epsilon_6}^4 \otimes \epsilon_6 \otimes \epsilon_6 \binom{3,4}{\otimes} \\ 150 \binom{2,2}{\otimes \epsilon_6}^4 \otimes \epsilon_6 \binom{2,2}{\otimes} I^{\oplus 3} \\ 150 \binom{2,2}{\otimes \epsilon_6}^4 \otimes \epsilon_6 \binom{2,2}{\otimes} 13151^1 \end{array}$$

$$I^{\oplus 2} e_7^{\begin{smallmatrix} 4,5 \end{smallmatrix}} \otimes e_8^{\begin{smallmatrix} 5,5 \end{smallmatrix}} (\otimes e_8)^3 \otimes e_7 I^{\oplus 3}$$

$$1 e_7^{\begin{smallmatrix} 5,5 \end{smallmatrix}} (\otimes e_8)^4 \otimes e_7 I^{\oplus 3}$$

$$1s^0 \otimes^{1,1} s^0 \otimes^{2,4} e_8 (\otimes e_8)^2 \otimes^{5,5} e_8 122315123$$

$$\begin{array}{l} I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes \mathbf{e}_7 \right)^4 \bigotimes \mathbf{e}'_7 \bigotimes \\ I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes \mathbf{e}_7 \right)^4 \bigotimes \mathbf{e}'_7 I^{\oplus 3} \end{array}$$

$$\begin{array}{ccccccc} 2,2 & 3,5 & (5,5) & (5,5) \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & (\otimes \mathfrak{e}_8)^3 & \otimes \mathfrak{e}'_7 1 \\ 1\mathfrak{e}_6 & \otimes \mathfrak{e}_8 & (\otimes \mathfrak{e}_8)^3 & \otimes \mathfrak{e}'_7 1 \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^3 \overset{5,3}{\otimes} \mathbf{e}_6 I^{\oplus 2}$$

$$\begin{matrix} 2,2 \\ \otimes \end{matrix} \mathfrak{e}_6 \begin{matrix} 3,5 \\ \otimes \end{matrix} \mathfrak{e}_8 \left(\begin{matrix} 5,5 \\ \otimes \end{matrix} \mathfrak{e}_8 \right)^3 \begin{matrix} 5,4 \\ \otimes \end{matrix} \mathfrak{e}_6 1,$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,4}{\otimes} \mathfrak{e}_6 1,$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^3 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$\overset{2,2}{\otimes} \mathbf{e}_6 (\overset{3,3}{\otimes} \mathbf{e}_7)^4 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,2}{\otimes},$$

$$\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} e_6 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7^4 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7' \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix},$$

$$\begin{matrix} 2,2 \\ \otimes \end{matrix} \mathfrak{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{e}_7)^4 \begin{matrix} 3,3 \\ \otimes \end{matrix} \mathfrak{e}'_7 I^{\oplus 3}$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\bigotimes^{3,3} \mathbf{e}_7 \right)^4 \bigotimes^{3,3} \mathbf{e}_7' \bigotimes^{3,2},$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^4 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix},$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^4 \overset{3,3}{\otimes} \mathbf{e}'_7 I^{\oplus 3}$$

$$3221e_7(\overset{3,3}{\otimes} e_7)^4 \overset{4,5}{\otimes} e_8 1,$$

$$512321\mathbf{e}_7(\otimes^{3,3}\mathbf{e}_7)^4\otimes^{4,5}\mathbf{e}_81,$$

$$151321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^4\overset{4,5}{\otimes}\mathbf{e}_81$$

$$\bigotimes^{2,3} \mathfrak{e}_7 \left(\bigotimes^{3,3} \mathfrak{e}_7 \right)^4 \bigotimes^{4,5} \mathfrak{e}_8 I^{\oplus 2},$$

$$\otimes^{3,3} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^4 \otimes^{4,5} \mathfrak{e}_8 I^{\oplus 2},$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes^{3,3} \mathbf{e}_7 \right)^4 \bigotimes^{4,5} \mathbf{e}_8 I^{\oplus 2}$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,5} \mathfrak{e}_8 1223132,$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,5} \mathfrak{e}_8 I^{\oplus 6},$$

$$3150 \otimes^{2,4} e_8(\otimes^{5,5} e_8)^3 \otimes^{5,5} e_8 122315123$$

$$3150 \otimes^{2,4} e_8(\otimes^{5,5} e_8)^3 \otimes^{5,5} e_8 12231513215,$$

$$3150 \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{(5,5)}{(\otimes \mathfrak{e}_8)^3} \overset{5,5}{\otimes} \mathfrak{e}_8 12231 \overset{1}{513},$$

$$3150 \overset{2,4}{\otimes} e_8(\overset{5,5}{\otimes} e_8)^3 \overset{5,5}{\otimes} e_8 12231513151$$

$$321\mathfrak{e}_6\left(\overset{2,2}{\otimes}\mathfrak{e}_6\right)^4\overset{3,5}{\otimes}\mathfrak{e}_81,$$

$$15131\mathfrak{e}_6\binom{2,2}{\otimes\mathfrak{e}_6}^4\binom{3,5}{\otimes\mathfrak{e}_8}1,$$

$$\begin{matrix} 2,2 \\ \otimes \end{matrix} \mathfrak{e}_6 \begin{pmatrix} 2,2 \\ \otimes \end{pmatrix} \mathfrak{e}_6)^4 \begin{matrix} 3,5 \\ \otimes \end{matrix} \mathfrak{e}_8 I^{\oplus 2}$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \right)^4 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 I^{\oplus 2}$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^3 \otimes^{5,4} e_7 \otimes^{3,2},$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^3 \otimes^{5,4} e_7 \otimes^{3,3},$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^3 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,5} \mathfrak{e}_7 I^{\oplus 2}$$

$$150 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \right)^2 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix},$$

$$1_{\mathfrak{so}} \otimes^{1,1} \mathfrak{so} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{5,4} \mathfrak{e}_7 \otimes^{3,3},$$

$$1_{50} \overset{1,1}{\otimes} 5_0 \overset{2,4}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^2 \overset{5,4}{\otimes} e_7 I^{\oplus 3}$$

$$1_{\mathfrak{so}} \otimes^{1,1} \mathfrak{so} \otimes^{2,4} \mathfrak{e}_8 \left(\otimes^{5,5} \mathfrak{e}_8 \right)^2 \otimes^{5,5} \mathfrak{e}_7 I^{\oplus 2}$$

$$\begin{array}{c} 2,2 \\ \otimes \end{array} \mathfrak{e}_6 \left(\begin{array}{c} 3,3 \\ \otimes \end{array} \mathfrak{e}_7 \right)^4 \begin{array}{c} 3,3 \\ \otimes \end{array} \mathfrak{e}_6 \begin{array}{c} 2,2 \\ \otimes \end{array},$$

$$I^{\oplus 2} \mathfrak{e}_6 \left(\overset{3,3}{\otimes} \mathfrak{e}_7 \right)^4 \overset{3,3}{\otimes} \mathfrak{e}_6 \overset{2,2}{\otimes},$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^4 \overset{3,3}{\otimes} \mathbf{e}_6 I^{\oplus 2}$$

$$I^{\oplus 2} \mathbf{e}'_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^3 \overset{5,4}{\otimes} \mathbf{e}'_7 I^{\oplus 2},$$

$$I^{\oplus 2} \mathbf{e}'_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^3 \overset{5,5}{\otimes} \mathbf{e}'_7 1,$$

$$1e'_7 \left(\overset{5,5}{\otimes} e_8 \right)^4 \overset{5,5}{\otimes} e'_7 1$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,3} \mathfrak{e}_6 \otimes^{2,2},$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,3} \mathfrak{e}_6 I^{\oplus 2},$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,4} \mathfrak{e}_6 1$$

$$150 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^3 \otimes^{3,3} e'_7 \otimes^{3,2},$$

$$1_{\mathfrak{so}} \otimes^{2,3} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^3 \otimes^{3,3} \mathfrak{e}'_7 \otimes^{3,3},$$

$$1_{\mathfrak{so}} \otimes^{2,3} \mathfrak{e}_7 \left(\otimes^{3,3} \mathfrak{e}_7 \right)^3 \otimes^{3,3} \mathfrak{e}'_7 I^{\oplus 3}$$

$$150 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \right)^2 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} e_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix},$$

$$1_{\mathfrak{so}} \otimes^{1,1} \mathfrak{so} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{5,3} \mathfrak{e}_6 I^{\oplus 2},$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^2 \overset{5,4}{\otimes} e_6 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^4 \otimes \mathfrak{e}_8 I^{\oplus 3},$$

$$231\mathfrak{so}(\otimes \mathfrak{so})^4 \otimes \mathfrak{e}_8 1,$$

$$31\mathfrak{so}(\otimes \mathfrak{so})^4 \otimes \mathfrak{e}_8 I^{\oplus 2}$$

$$321\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 1,$$

$$I^{\oplus 2}\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 I^{\oplus 2}$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6 \otimes,$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{e}_6(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 I^{\oplus 2},$$

$$31\mathfrak{e}_6(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{e}_6)^4 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so} 13$$

2.7.4 SSg6(44)

$$\otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$313221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 5}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$21513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$1513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 4}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$I^{\oplus 3}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathbf{e}_7^1 \left(\bigotimes \mathbf{e}_7 \right)^{3,3} \bigotimes \mathbf{e}_8^{4,5} I^{\oplus 2}$$

$$1e_6^{1,3,5} \otimes e_8^{(5,5)} (\otimes e_8)^3 \otimes e_7'^{5,4} I^{\oplus 2},$$

$$1e_6^{(1)}(\otimes e_7^{(3,3)})^4 \otimes e_7'^{(3,3)} \otimes,$$

$$\otimes^{3,3} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^4 \otimes^{3,2} \mathfrak{so} 1,$$

$$1\mathbf{e}_6^{(1)}(\otimes \mathbf{e}_6^{(2,2)})^4 \otimes \mathbf{e}_8^{(3,5)} I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_6^{(1)} (\otimes^{2,2} \mathfrak{e}_6)^4 \otimes^{2,2} \mathfrak{so} 1,$$

$$I^{\oplus 2} \mathbf{e}'_7 ({}^{3,3} \bigotimes \mathbf{e}_7)^4 {}^{3,3} \bigotimes \mathbf{e}'_7 \bigotimes,$$

$$1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^3 \otimes \mathbf{e}_6^{5,3} I^{\oplus 2},$$

$$1e_8 \otimes^{1,4,2} so(\otimes^{1,1} so)^3 \otimes^{1,1} so_{13}$$

$$1e_7^{(1)}(\otimes e_8^{(5,5)})^4 \otimes so_{13}^{(4,2)}$$

$$1e_7 \binom{5,5}{\otimes} e_8)^3 \binom{4,2}{\otimes} so \binom{1,1}{\otimes} so1$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^3 \otimes^{5,5} e_7' 1$$

$$1e_8^{1,5,4} \otimes e_7^{(3,3)} (\otimes e_7)^3 \otimes e_7'^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6^{1(3,3)}(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_6^{3,3} I^{\oplus 2}$$

$$1 \overset{1}{\mathbf{e}}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^3 \overset{3,3}{\otimes} \mathbf{e}_6 1$$

$$1\mathbf{e}_7' \overset{1}{\otimes} \overset{4,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^2 \overset{4,2}{\otimes} \mathbf{so} \overset{1,1}{\otimes} \mathbf{so} 1$$

$$1\mathfrak{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{4,2}{\otimes} \mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so} 1$$

2.7.6 SSg6IS(135)

$${}^{3,5}_1 \otimes \mathbf{e}_8 ({}^{5,5} \otimes \mathbf{e}_8)^4 {}^{5,5}_1 \otimes \mathbf{e}_8 \otimes$$

$${}^{3,5}{}^1_1 \otimes_8 ({}^{5,5} \otimes_8)^4 {}^{5,5}{}^1_1 \otimes_8 12231512$$

$${}^{4,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^4 {}^{5,5}_1 \otimes \mathfrak{e}_8 \otimes$$

$${}^{4,5} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 ({}^{5,5} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8)^4 {}^{5,5} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 12231512$$

$$313221 \mathbf{e}_8^1 (\otimes^{5,5} \mathbf{e}_8)^4 \otimes^{5,5} \mathbf{e}_8 \otimes^{1,5,5}$$

$${}^{5,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5}_1 \otimes \mathfrak{e}_8)^4 {}^{5,5}_1 \otimes \mathfrak{e}_8 1223151$$

$$313221\mathbf{e}_8^1(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8^{5,5} 12231512$$

$$I^{\oplus 5} \mathbf{e}_8^1 (\bigotimes \mathbf{e}_8^{5,5})^4 \bigotimes \mathbf{e}_8^{5,5} \mathbf{e}_8^1 12231512$$

$$1513221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8^{(5,5)} 12231512$$

$${}^{4,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^4 {}^{5,5} I^{\oplus 2} {}^{4,2} \otimes$$

$$I^{\oplus 5} \mathbf{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \right)^4 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} I^{\oplus 24,2} \begin{smallmatrix} \otimes \\ \mathbf{e}_8 \end{smallmatrix} \otimes$$

$$\begin{array}{l}
3,5 \quad 1 \quad 5,5 \quad 5,5 \quad I^{\oplus 2} \\
\otimes \mathfrak{c}_8 (\otimes \mathfrak{c}_8)^4 \otimes \mathfrak{c}_8 I^{\oplus 4} \\
4,5 \quad 1 \quad 5,5 \quad 5,5 \quad I^{\oplus 3} \\
\otimes \mathfrak{c}_8 (\otimes \mathfrak{c}_8)^4 \otimes \mathfrak{c}_8 I^{\oplus 3} \\
313221 \mathfrak{c}_8 (\otimes \mathfrak{c}_8)^4 \otimes \mathfrak{c}_8 I^{\oplus 2} \\
21513221 \mathfrak{c}_8 (\otimes \mathfrak{c}_8)^4 \otimes \mathfrak{c}_8 I^{\oplus 2} \\
21513221 \mathfrak{c}_8 (\otimes \mathfrak{c}_8)^4 \otimes \mathfrak{c}_8 I^{\oplus 3} \\
I^{\oplus 4} \mathfrak{c}_8 (\otimes \mathfrak{c}_8)^4 \otimes \mathfrak{c}_8 \otimes \\
I^{\oplus 4} \mathfrak{c}_8 (\otimes \mathfrak{c}_8)^4 \otimes \mathfrak{c}_8 I^{\oplus 3}
\end{array}$$

$$\begin{aligned}
& I^{\oplus 1} \mathbf{e}_7^{\otimes 1, 4, 5} \otimes \mathbf{e}_8^{\otimes 5, 5} \otimes \mathbf{e}_8^{\otimes 3, 5, 5} \otimes \mathbf{e}_8^{\otimes 1, 5, 5} \\
& I^{\oplus 2} \mathbf{e}_7^{\otimes 1, 4, 5} \otimes \mathbf{e}_8^{\otimes 5, 5} \otimes \mathbf{e}_8^{\otimes 3, 5, 5} \otimes \mathbf{e}_8^{\otimes 1, 2, 2, 3, 1, 5, 1, 2} \\
& I^{\oplus 2} \mathbf{e}_7^{\otimes 1, 4, 5} \otimes \mathbf{e}_8^{\otimes 5, 5} \otimes \mathbf{e}_8^{\otimes 3, 5, 5} \otimes \mathbf{e}_8^{\otimes 1, 2, 2, 3, 1, 5, 1, 2} \otimes I^{\oplus 4} \\
& 1 \mathbf{e}_7^{\otimes 1, 5, 5} \otimes \mathbf{e}_8^{\otimes 4, 5, 5} \otimes \mathbf{e}_8^{\otimes 1, 5, 4} \\
& 1 \mathbf{e}_7^{\otimes 1, 5, 5} \otimes \mathbf{e}_8^{\otimes 4, 5, 5} \otimes \mathbf{e}_8^{\otimes 1, 2, 2, 3, 1, 5, 1, 2} \otimes I^{\oplus 5} \\
& 1 \mathbf{e}_7^{\otimes 1, 5, 5} \otimes \mathbf{e}_8^{\otimes 4, 5, 5} \otimes \mathbf{e}_8^{\otimes 1, 2, 2, 3, 1, 5, 1, 2} \otimes I^{\oplus 2, 4, 2}
\end{aligned}$$

$$\begin{array}{l} I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes_{\mathbf{e}_7}^{(3,3)} \right)^4 \bigotimes_{\mathbf{e}_7}^{(3,3)} \mathbf{e}_7^1 \bigotimes \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes_{\mathbf{e}_7}^{(3,3)} \right)^4 \bigotimes_{\mathbf{e}_7}^{(3,3)} I^{\oplus 3} \mathbf{e}_7^1 \\ I^{\oplus 3} \mathbf{e}_7^1 \left(\bigotimes_{\mathbf{e}_7}^{(3,3)} \right)^4 \bigotimes_{\mathbf{e}_7}^{(3,3)} I^{\oplus 2} \mathbf{e}_7^1 \end{array}$$

$$\begin{array}{l} 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^3 \epsilon_8^{5,5} 1^{5,5} \epsilon_8 \otimes \\ 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^3 \epsilon_8^{5,5} 1^{5,5} \epsilon_8 12231512 \\ 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} (\otimes \epsilon_8)^3 \epsilon_8^{5,5} I^{\oplus 2} \epsilon_8 I^{\oplus 4} \end{array}$$

$$1_{\mathfrak{e}_6}^{1,3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5}$$

$$1_{\mathfrak{e}_6}^{1,3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,2,2,3,1,5,1,2}$$

$$\begin{aligned}
& 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} 1223151, & 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} I^{\oplus 2} 4,2, & 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} I^{\oplus 2} \\
& 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} I^{\oplus 3} I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathfrak{e}_6^{2,2} (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} \otimes, & I^{\oplus 2} \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} \otimes, & I^{\oplus 2} \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2} \\
& 31\mathfrak{e}_6^{3,1} (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} \otimes, & 31\mathfrak{e}_6^{3,1} (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2} 1
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 3} \\
& I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 3} \\
& 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2} 1, & I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,5} 1, & 1\mathfrak{e}_7^1 (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{e}_7^{5,5} 1
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2} 1, & 1\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,5} 1
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2} 1, & 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,5} 1
\end{aligned}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_8^{4,5} 1$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2} 1$$

$$1\mathfrak{so}^1 (\otimes \mathfrak{so}^{1,1})^4 \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} 1$$

$$1\mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_8^{4,5} 1$$

$$1\mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} (\otimes \mathbf{e}_8^{5,5})^3 \otimes \mathbf{e}_7^{5,4} \mathbf{e}_7'^1 1$$

$$1 \mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,3} 1$$

$$1 \mathbf{e}_6^1 (\otimes \mathbf{e}_7^{3,3})^4 \otimes \mathbf{e}_7'^1 I^{\oplus 2}$$

$$1 \mathbf{e}_6^{1(3,3)} (\otimes \mathbf{e}_7)^4 \mathbf{e}_6^{3,3} 1$$

$$1\mathbf{e}_6^{(1)}(\otimes^{(2,2)}\mathbf{e}_6)^4\otimes^{(3,5)}\mathbf{e}_81$$

2.7.7 gLIg5(3)

$$50 \otimes^{2,4} e_8 \otimes^{1,5,5} (e_8 \otimes e_8)^2 \otimes^{5,4} e_7 \otimes^{3,3} e_7$$

$$e_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50$$

$$e_7^{3,3} \otimes e_7^1 \left(\otimes e_7^{3,3} \right)^3 \otimes e_7^{3,2} \text{ so}$$

2.7.8 SgLIg5(5)

$$I^{\oplus 3} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8^{4,2})^2 \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}, \quad 150 \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8^{5,3})^3 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^1 \left(\otimes \mathbf{e}_8^{5,5} \right)^3 \otimes \mathbf{e}_7^{5,5}$$

$$I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 \left(\otimes \mathfrak{e}_8^{5,5} \right)^3 \otimes \mathfrak{e}_6^{5,4}$$

$$150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,5} \mathfrak{e}_6$$

2.7.9 SgLIg3LIgLg(1)

$$I^{\oplus 2} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so}$$

2.7.15 SSg4LIgLg(3)

$$I^{\oplus 2} \mathbf{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7$$

$$31513221 \mathbf{e}_8^1 (\otimes \mathbf{e}_8^{5,5})^3 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5}$$

$$1e_7' \overset{1}{\overset{(3,3)}{\otimes}} e_7)^3 \overset{3,3}{\otimes} e_7 \overset{1}{\otimes} \overset{(3,3)}{\otimes} e_7$$

2.7.16 SSg4LIgLgS(20)

$$\begin{array}{lll}
\begin{array}{l}
3,5 \overset{1}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{so}1, \\
313221 \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{so}1, \\
1513221 \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{so}1, \\
I^{\oplus 3} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{so}1
\end{array}
&
\begin{array}{l}
4,5 \overset{1}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{so}1, \\
I^{\oplus 5} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{so}1, \\
I^{\oplus 2} \overset{2,4}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{so}1,
\end{array}
&
\begin{array}{l}
5,5 \overset{1}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{so}1 \\
21513221 \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{so}1 \\
I^{\oplus 4} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{so}1
\end{array}
\end{array}$$

$$\begin{array}{lll} 1513221 \epsilon_8^1 (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_8^1, & \otimes \epsilon_8^{2,4} \otimes \epsilon_8^1 (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_8^1 I^{\oplus 2}, & I^{\oplus 2} \epsilon_8^{3221} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_8^1 \\ I^{\oplus 4} \epsilon_8^1 (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_8^1 I^{\oplus 2}, & I^{\oplus 3} \epsilon_8^{I^{\oplus 2}} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_8^1 I^{\oplus 2} & \end{array}$$

$$I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^2 \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^1 \otimes \mathfrak{so}1, \quad \mathbf{1e}_7^1 (\otimes \mathbf{e}_8^{5,5})^3 \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^1 \otimes \mathfrak{so}1$$

$$1\mathbf{e}_7^{1\ 4,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^2 \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1\ 5,5} \otimes \mathbf{e}_8 I^{\oplus 2}$$

$$1e_7' \otimes e_8(\overset{1}{\otimes} \overset{4,5}{e_8})^2 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} \overset{4,2}{so1}$$

$$1\mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}1$$

2.7.17 SSg4LIgLIgS(4)

$$2,4 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix}^3 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 1, \quad I^{\oplus 4} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix}^3 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 1, \quad I^{\oplus 3} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix}^3 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 1$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} \overset{5,5}{e_8} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^3 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^3 \otimes \mathfrak{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^3 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^3 \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^2 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

2.8 n≥7 node(10,217)

2.8.1 g(n)(47)

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^9 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^9 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^8 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^8 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^7 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^7 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^6 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^6 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \otimes \mathfrak{e}_6$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{14} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{14} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{13} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{13} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{12} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{12} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{11} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{11} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{10} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{10} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,3} \left(\otimes \mathfrak{e}_7 \right)^9 \otimes \mathfrak{so}^{3,3}$$

$$\mathfrak{so}\left(\begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}\right)^8 \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$\mathfrak{so}\left(\begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}\right)^7 \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$\mathfrak{so}\left(\begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}\right)^6 \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$\mathfrak{so}\left(\begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}\right)^5 \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$\mathfrak{so}\left(\begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}\right)^{15} \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$\mathfrak{so}\left(\begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}\right)^{14} \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$\mathfrak{so}\left(\begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}\right)^{13} \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$\mathfrak{so}\left(\begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}\right)^{12} \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$\mathfrak{so}\left(\begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}\right)^{11} \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$\mathfrak{so}\left(\begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}\right)^{10} \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$\mathfrak{so} \begin{smallmatrix} 2,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}^5 \begin{smallmatrix} 4,5 \\ \otimes \mathfrak{e}_8 \end{smallmatrix}$$

$$\mathfrak{so}\left(\begin{smallmatrix} 2,2 \\ \otimes \mathfrak{e}_6 \end{smallmatrix}\right)^6 \begin{smallmatrix} 3,5 \\ \otimes \mathfrak{e}_8 \end{smallmatrix}$$

$$\mathfrak{so} \begin{smallmatrix} 1,1 \\ \otimes \mathfrak{so} \end{smallmatrix} \begin{smallmatrix} 2,4 \\ \otimes \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{smallmatrix}^2 \begin{smallmatrix} 5,4 \\ \otimes \mathfrak{e}_7 \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}$$

$$\mathfrak{e}_6\left(\begin{smallmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{smallmatrix}\right)^9 \begin{smallmatrix} 5,5 \\ \otimes \mathfrak{e}_6 \end{smallmatrix}$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^8 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^7 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^6 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^5 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{15} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{14} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{13} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{12} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{11} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{10} \overset{5,5}{\otimes} \mathfrak{e}_6$$

2.8.2 Sg(n)(549)

$$13221\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$1223151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$2151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$\overset{1}{15}1321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so}$$

$$13151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$2221\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$\overset{1}{23}21\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$123151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so}$$

$$31321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so}$$

$$1512321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so}$$

$$13221\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so},$$

$$1223151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so},$$

$$13151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so},$$

$$2221\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so},$$

$$123151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so}$$

$$31321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so}$$

$$1512321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^8\overset{3,3}{\otimes}\mathfrak{so}$$

$$\begin{array}{l} 123151321\mathbf{e}_7(\otimes \mathbf{e}_7)^7 \otimes \mathbf{so}^{3,3} \\ 31321\mathbf{e}_7(\otimes \mathbf{e}_7)^7 \otimes \mathbf{so}^{3,3} \\ 1512321\mathbf{e}_7(\otimes \mathbf{e}_7)^7 \otimes \mathbf{so}^{3,3} \end{array}$$

$$\begin{array}{l} 123151321\epsilon_7(\otimes \epsilon_7)^6 \otimes \mathfrak{so}^{3,3} \\ 31321\epsilon_7(\otimes \epsilon_7)^6 \otimes \mathfrak{so}^{3,3} \\ 1512321\epsilon_7(\otimes \epsilon_7)^6 \otimes \mathfrak{so}^{3,3} \end{array}$$

$$\begin{array}{l} 123151321\mathbf{e}_7(\otimes\mathbf{e}_7)^5\otimes\mathbf{so} \\ 31321\mathbf{e}_7(\otimes\mathbf{e}_7)^5\otimes\mathbf{so} \\ 1512321\mathbf{e}_7(\otimes\mathbf{e}_7)^5\otimes\mathbf{so} \end{array}$$

$$\begin{array}{l} 123151321\epsilon_7(\otimes\epsilon_7)^{15}\otimes\mathfrak{so}^{3,3} \\ 31321\epsilon_7(\otimes\epsilon_7)^{15}\otimes\mathfrak{so}^{3,3} \\ 1512321\epsilon_7(\otimes\epsilon_7)^{15}\otimes\mathfrak{so}^{3,3} \end{array}$$

$$\begin{array}{l} 123151321\epsilon_7(\overset{3,3}{\otimes}\epsilon_7)^{14}\overset{3,3}{\otimes}\mathfrak{so} \\ 31321\epsilon_7(\overset{3,3}{\otimes}\epsilon_7)^{14}\overset{3,3}{\otimes}\mathfrak{so} \\ 1512321\epsilon_7(\overset{3,3}{\otimes}\epsilon_7)^{14}\overset{3,3}{\otimes}\mathfrak{so} \end{array}$$

$$123151321e_7(\overset{3,3}{\otimes} e_7)^{13} \overset{3,3}{\otimes} 50$$

$$\begin{array}{lll} 2313221\epsilon_8(\otimes \epsilon_8)^{5,5}{}^{13} \otimes \mathfrak{so}^{5,3}, & 222221\epsilon_8(\otimes \epsilon_8)^{5,5}{}^{13} \otimes \mathfrak{so}^{5,3}, & 321513221\epsilon_8(\otimes \epsilon_8)^{5,5}{}^{13} \otimes \mathfrak{so}^{5,3} \\ 51231513221\epsilon_8(\otimes \epsilon_8)^{5,5}{}^{13} \otimes \mathfrak{so}^{5,3}, & \overset{1}{3}1513221\epsilon_8(\otimes \epsilon_8)^{5,5}{}^{13} \otimes \mathfrak{so}^{5,3}, & 15131513221\epsilon_8(\otimes \epsilon_8)^{5,5}{}^{13} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{array}{lll} 2313221\epsilon_8(\otimes \epsilon_8)^{12} \otimes \epsilon_6, & 222221\epsilon_8(\otimes \epsilon_8)^{12} \otimes \epsilon_6, & 321513221\epsilon_8(\otimes \epsilon_8)^{12} \otimes \epsilon_6 \\ 51231513221\epsilon_8(\otimes \epsilon_8)^{12} \otimes \epsilon_6, & 31513221\epsilon_8(\otimes \epsilon_8)^{12} \otimes \epsilon_6, & 15131513221\epsilon_8(\otimes \epsilon_8)^{12} \otimes \epsilon_6 \end{array}$$

$$\begin{array}{lll} 2313221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}, & 222221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}, & 321513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so} \\ 51231513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}, & 31513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}, & 15131513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{lll} 2313221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6, & 222221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6, & 321513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6 \\ 51231513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6, & 31513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6, & 15131513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6 \end{array}$$

$$\begin{array}{lll} 2313221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{so}, & 222221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{so}, & 321513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{so} \\ 51231513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{so}, & 31513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{so}, & 15131513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{lll} 2313221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6, & 222221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6, & 321513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6 \\ 51231513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6, & 31513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6, & 15131513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6 \end{array}$$

$$\begin{array}{lll} 2313221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so}, & 222221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so}, & 321513221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so} \\ 51231513221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so}, & 31513221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so}, & 15131513221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so} \end{array}$$

$$\begin{array}{lll} 31\mathfrak{s}\mathfrak{o} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^{5,5})^9 \otimes \mathfrak{e}_6^{5,5}, & 131\mathfrak{e}_6 \otimes \mathfrak{e}_8^{3,5}(\otimes \mathfrak{e}_8^{5,5})^9 \otimes \mathfrak{s}\mathfrak{o}^{5,3}, & 21\mathfrak{e}_6 \otimes \mathfrak{e}_8^{3,5}(\otimes \mathfrak{e}_8^{5,5})^9 \otimes \mathfrak{s}\mathfrak{o}^{5,3} \\ 1\mathfrak{e}_6^{4,5} \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8^{5,5})^9 \otimes \mathfrak{s}\mathfrak{o}^{5,3} & & \end{array}$$

$$\begin{array}{lll}
31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{e}_6^{5,5}, & 131\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{so}^{5,3}, & 21\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{so}^{5,3} \\
1\mathfrak{e}_6^{4,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{so}^{5,3} & &
\end{array}$$

$$31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_6^{7,5,5}, \quad 131\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{so}^{7,5,3}, \quad 21\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_6^{7,5,3}$$

$$1\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^7 \overset{5,3}{\otimes} \mathfrak{so}$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^6 \otimes^{5,5} e_6,$$

$$131\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{so}^{5,3},$$

$$21\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^6 \overset{5,3}{\otimes} \mathfrak{so}$$

$$1\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^6 \overset{5,3}{\otimes} \mathfrak{so}$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 \left(\otimes^{5,5} \mathfrak{e}_8 \right)^5 \otimes^{5,5} \mathfrak{e}_6,$$

$$131e_6 \overset{3,5}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^5 \overset{5,3}{\otimes} so,$$

$$21\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^5 \overset{5,3}{\otimes} 50$$

$$1\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^5 \overset{5,3}{\otimes} \mathfrak{so}$$

$$3150 \overset{2,4}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^4 \overset{5,5}{\otimes} \mathfrak{e}_6,$$

$$131\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^4 \overset{5,3}{\otimes} \mathfrak{so},$$

$$21\mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,3}{\otimes} \mathbf{so}$$

$$1\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^4 \overset{5,3}{\otimes} \mathfrak{so}$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{14} \otimes^{5,5} \mathfrak{e}_6,$$

$$131\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{14} \overset{5,3}{\otimes} \mathfrak{so},$$

$$21\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{14} \overset{5,3}{\otimes} \mathfrak{so}$$

$$1\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{14} \overset{5,3}{\otimes} \mathfrak{so}$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{13} \otimes^{5,5} \mathfrak{e}_6,$$

$$131\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{so}^{5,3},$$

$$21\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{13} \overset{5,3}{\otimes} \mathfrak{so}$$

$$1\mathfrak{e}_6^{4,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{so}^{5,3}$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{12} \otimes^{5,5} \mathfrak{e}_6,$$

$$131\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3},$$

$$21\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{12} \overset{5,3}{\otimes} \mathfrak{so}$$

$$1\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^{12} \overset{5,3}{\otimes} \mathfrak{so}$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{11} \otimes^{5,5} \mathfrak{e}_6,$$

$$131\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{so}^{5,3},$$

$$21\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{11} \overset{5,3}{\otimes} \mathfrak{so}$$

$$1\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{11} \overset{5,3}{\otimes} \mathfrak{so}$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{10} \otimes^{5,5} \mathfrak{e}_6,$$

$$131\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{10} \overset{5,3}{\otimes} \mathfrak{so},$$

$$21\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{10} \overset{5,3}{\otimes} \mathfrak{so}$$

$$1\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^{10} \overset{5,3}{\otimes} \mathfrak{so}$$

$$1321\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \overset{5,5}{\otimes} \mathbf{e}_6,$$

$$12321\mathbf{e}_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^9 \otimes^{5,5} \mathbf{e}_6,$$

$$221\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$21\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{10}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$221e_7 \otimes e_8 \begin{pmatrix} 4,5 \\ 5,5 \end{pmatrix} (\otimes e_8)^9 \otimes 50 \begin{pmatrix} 5,3 \end{pmatrix}$$

$$221\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^8 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$221e_7 \otimes e_8 \binom{5,5}{\otimes e_8}^8 \otimes 50$$

$$221\mathbf{e}_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^7 \otimes^{5,5} \mathbf{e}_6$$

$$221e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,3} so$$

$$221\mathbf{e}_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^6 \otimes^{5,5} \mathbf{e}_6$$

$$221e_7 \otimes^{4,5} e_8 \left(\otimes^{5,5} e_8 \right)^6 \otimes^{5,3} so$$

$$221e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^5 \otimes^{5,5} e_6$$

$$221e_7 \otimes^{4,5} e_8 \left(\otimes^{5,5} e_8 \right)^5 \otimes^{5,3} so$$

$$221\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$21\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^5\overset{5,5}{\otimes}\mathbf{e}_6$$

$$1321e_7 \otimes^{4,5} e_8 \left(\otimes^{5,5} e_8 \right)^4 \otimes^{5,3} so,$$

$$12321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^4 \otimes^{5,3} so,$$

$$221e_7 \overset{4,5}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^4 \overset{5,3}{\otimes} so$$

$$21e_7(\otimes^{5,5} e_8)^5 \otimes^{5,3} so$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{13} \otimes^{5,5} e_6,$$

$$12321e_7 \overset{4,5}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^{13} \overset{5,5}{\otimes} e_6,$$

$$221e_7 \overset{4,5}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^{13} \overset{5,5}{\otimes} e_6$$

$$21\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{14}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{13} \otimes^{5,3} \mathfrak{so},$$

$$12321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{13} \otimes^{5,3} \mathfrak{so},$$

$$221e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{13} \otimes^{5,3} so$$

$$21e_7(\otimes^{5,5} e_8)^{14} \otimes^{5,3} so$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{12} \otimes^{5,5} e_6,$$

$$12321e_7 \overset{4,5}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^{12} \overset{5,5}{\otimes} e_6,$$

$$221\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^{12} \overset{5,5}{\otimes} \mathbf{e}_6$$

$$21\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{13}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{12} \otimes^{5,3} \mathfrak{so},$$

$$12321e_7 \otimes^{4,5} e_8 \left(\otimes^{5,5} e_8 \right)^{12} \otimes^{5,3} \mathfrak{so},$$

$$221\mathfrak{e}_7 \otimes^{4,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{12} \otimes^{5,3} \mathfrak{so}$$

$$21e_7(\overset{5,5}{\otimes} e_8)^{13} \overset{5,3}{\otimes} 50$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{11} \otimes^{5,5} e_6,$$

$$12321\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{11} \overset{5,5}{\otimes} \mathbf{e}_6,$$

$$221\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{11} \overset{5,5}{\otimes} \mathbf{e}_6$$

$$21\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{12}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{11} \otimes^{5,3} \mathfrak{so},$$

$$12321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{11} \otimes^{5,3} \mathfrak{so},$$

$$221\mathfrak{e}_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^{11} \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$21e_7(\overset{5,5}{\otimes} e_8)^{12} \overset{5,3}{\otimes} 50$$

$$1321\mathbf{e}_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^{10} \otimes^{5,5} \mathbf{e}_6,$$

$$12321\mathbf{e}_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^{10} \otimes^{5,5} \mathbf{e}_6,$$

$$221\mathbf{e}_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \right)^{10} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_6$$

$$21\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{11}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{10} \otimes^{5,3} \mathfrak{so},$$

$$12321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{10} \otimes^{5,3} \mathfrak{so},$$

$$221\mathfrak{e}_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^{10} \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$21e_7(\overset{5,5}{\otimes} e_8)^{11} \overset{5,3}{\otimes} 50$$

$$151321\mathbf{e}_7\left(\overset{3,3}{\otimes}\mathbf{e}_7\right)^5\overset{4,5}{\otimes}\mathbf{e}_8$$

$$221e'_7(\overset{3,3}{\otimes} e_7)^9 \overset{3,3}{\otimes} 50$$

$$221e'_7(\overset{3,3}{\otimes} e_7)^8 \overset{3,3}{\otimes} \mathfrak{so}$$

$$221e'_7(\overset{3,3}{\otimes} e_7)^7 \overset{3,3}{\otimes} 50$$

$$221\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^6\overset{4,5}{\otimes}\mathbf{e}_8$$

$$221e'_7(\otimes^{3,3} e_7)^6 \otimes^{3,3} \mathfrak{so}$$

$$221e'_7(\otimes^{3,3} e_7)^5 \otimes^{3,3} \mathfrak{so}$$

$$221\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{15}\overset{3,3}{\otimes}\mathbf{so}$$

$$221\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{14}\overset{3,3}{\otimes}\mathbf{so}$$

$$221e'_7(\otimes^{3,3} e_7)^{13} \otimes^{3,3} 50$$

$$221\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{12}\overset{3,3}{\otimes}\mathbf{so}$$

$$221\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{11}\overset{3,3}{\otimes}\mathbf{so}$$

$$221\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{10}\overset{3,3}{\otimes}\mathbf{so}$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \overset{5,5}{\otimes} \mathbf{e}_6$$

[illegible]

$$321\mathfrak{e}_6(\otimes \mathfrak{e}_6)^5 \otimes \mathfrak{e}_8, \quad 15131\mathfrak{e}_6(\otimes \mathfrak{e}_6)^5 \otimes \mathfrak{e}_8$$

$$221e_7 \otimes e_8^{4,5} (\otimes e_8^{5,5})^{14} \otimes s_0^{5,3}, \quad 21e_7 (\otimes e_8^{5,5})^{15} \otimes s_0^{5,3}$$

$$21\mathfrak{e}'_7 \otimes^{4,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^9 \otimes^{5,5} \mathfrak{e}_6, \quad 1\mathfrak{e}'_7 (\otimes^{5,5} \mathfrak{e}_8)^{10} \otimes^{5,5} \mathfrak{e}_6$$

$$21e_7' \otimes e_8^{4,5} (\otimes e_8^{5,5})^9 \otimes s_0, \quad 1e_7' (\otimes e_8^{5,5})^{10} \otimes s_0$$

$$21e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^8 \otimes^{5,5} e_6, \quad 1e_7' (\otimes^{5,5} e_8)^9 \otimes^{5,5} e_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^8 \otimes^{5,3} 50,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^9 \overset{5,3}{\otimes} \mathfrak{so}$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,5} e_6,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^8 \overset{5,5}{\otimes} e_6$$

$$21e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,3} \mathfrak{so},$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^8 \overset{5,3}{\otimes} 50$$

$$21\mathfrak{e}'_7 \otimes^{4,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^6 \otimes^{5,5} \mathfrak{e}_6,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^7 \overset{5,5}{\otimes} e_6$$

$$21e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^6 \otimes^{5,3} \mathfrak{so},$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^7 \overset{5,3}{\otimes} so$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^5 \otimes^{5,5} e_6,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^6 \overset{5,5}{\otimes} e_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^5 \otimes^{5,3} 50,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^6 \overset{5,3}{\otimes} \mathfrak{so}$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^4 \otimes^{5,5} e_6,$$

$$1\mathbf{e}'_7(\overset{5,5}{\otimes}\mathbf{e}_8)^5\overset{5,5}{\otimes}\mathbf{e}_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^4 \otimes^{5,3} \mathfrak{so},$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^5 \overset{5,3}{\otimes} so$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{14} \otimes^{5,3} \mathfrak{so},$$

$$1e'_7 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \right)^{15} \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} 50$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{13} \otimes^{5,5} e_6,$$

$$1\mathbf{e}'_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{14}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{13} \otimes^{5,3} \mathfrak{so},$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^{14} \overset{5,3}{\otimes} 50$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{12} \otimes^{5,5} e_6,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^{13} \overset{5,5}{\otimes} e_6$$

$$21e'_7 \otimes e_8 \left(\otimes e_8 \right)^{12} \otimes s_0,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^{13} \overset{5,3}{\otimes} 50$$

$$21\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{4,5}{\otimes \mathfrak{e}_8}^{11} \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}'_7 \binom{5,5}{\otimes \mathfrak{e}_8}^{12} \otimes \mathfrak{e}_6$$

$$21\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{4,5}{\otimes \mathfrak{e}_8}^{11} \otimes \mathfrak{so},$$

$$1\mathfrak{e}'_7 \binom{5,5}{\otimes \mathfrak{e}_8}^{12} \otimes \mathfrak{so}$$

$$21\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{4,5}{\otimes \mathfrak{e}_8}^{10} \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}'_7 \binom{5,5}{\otimes \mathfrak{e}_8}^{11} \otimes \mathfrak{e}_6$$

$$21\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \binom{4,5}{\otimes \mathfrak{e}_8}^{10} \otimes \mathfrak{so},$$

$$1\mathfrak{e}'_7 \binom{5,5}{\otimes \mathfrak{e}_8}^{11} \otimes \mathfrak{so}$$

$$21\mathfrak{e}_6 \otimes \mathfrak{e}_8 \binom{3,5}{\otimes \mathfrak{e}_8}^{14} \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \binom{4,5}{\otimes \mathfrak{e}_8}^{14} \otimes \mathfrak{e}_6$$

$$131\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^9 \otimes \mathfrak{so},$$

$$21\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^9 \otimes \mathfrak{so}$$

$$131\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^8 \otimes \mathfrak{so},$$

$$21\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^8 \otimes \mathfrak{so}$$

$$131\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^7 \otimes \mathfrak{so},$$

$$21\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^7 \otimes \mathfrak{so}$$

$$131\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^6 \otimes \mathfrak{so},$$

$$21\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^6 \otimes \mathfrak{so}$$

$$131\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^5 \otimes \mathfrak{so},$$

$$21\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^5 \otimes \mathfrak{so}$$

$$131\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^{15} \otimes \mathfrak{so},$$

$$21\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^{15} \otimes \mathfrak{so}$$

$$131\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^{14} \otimes \mathfrak{so},$$

$$21\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^{14} \otimes \mathfrak{so}$$

$$131\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^{13} \otimes \mathfrak{so},$$

$$21\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^{13} \otimes \mathfrak{so}$$

$$131\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^{12} \otimes \mathfrak{so},$$

$$21\mathfrak{e}_6 \binom{3,3}{\otimes \mathfrak{e}_7}^{12} \otimes \mathfrak{so}$$

$$131\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{11}\overset{3,3}{\otimes}\mathfrak{so},$$

$$21\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{11}\overset{3,3}{\otimes}\mathfrak{so}$$

$$131\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{10}\overset{3,3}{\otimes}\mathfrak{so},$$

$$21\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{10}\overset{3,3}{\otimes}\mathfrak{so}$$

$$131\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^6\overset{3,5}{\otimes}\mathfrak{e}_8,$$

$$21\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^6\overset{3,5}{\otimes}\mathfrak{e}_8$$

$$321\mathfrak{e}'_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^5\overset{4,5}{\otimes}\mathfrak{e}_8$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^9\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^7\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^6\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^5\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^4\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{14}\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{13}\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{11}\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8 \begin{smallmatrix} 2,4 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix}^{5,5} 10 \begin{smallmatrix} 5,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$31\mathfrak{so} \begin{smallmatrix} 1,1 \\ (\otimes \mathfrak{so}) \end{smallmatrix}^6 \begin{smallmatrix} 2,4 \\ \otimes \mathfrak{e}_8 \end{smallmatrix}$$

$$31\mathfrak{e}_6 \begin{smallmatrix} 3,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^5 \begin{smallmatrix} 4,5 \\ \otimes \mathfrak{e}_8 \end{smallmatrix}$$

$$231\mathfrak{so} \begin{smallmatrix} 1,1 \\ (\otimes \mathfrak{so}) \end{smallmatrix}^5 \begin{smallmatrix} 2,4 \\ \otimes \mathfrak{e}_8 \end{smallmatrix}$$

$$21\mathfrak{e}_7' \begin{smallmatrix} 3,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^6 \begin{smallmatrix} 4,5 \\ \otimes \mathfrak{e}_8 \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \begin{smallmatrix} 2,4 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix}^3 \begin{smallmatrix} 5,4 \\ \otimes \mathfrak{e}_7 \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{e}_7 \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^9 \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^8 \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^7 \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^6 \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^5 \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^4 \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^{14} \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^{13} \begin{smallmatrix} 3,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{12} \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{11} \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{10} \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^9 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^9 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^8 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^8 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^7 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^7 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^6 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^6 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^5 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^5 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{e}_6^{5,5}$$

2.8.3 Sg(n)S(3795)

[illegible]

$$\begin{array}{c} 3131\epsilon_6(\otimes^{2,2}\epsilon_6)^{14}\otimes^{2,2}\epsilon_613151 \\ 15131\epsilon_6(\otimes^{2,2}\epsilon_6)^{14}\otimes^{2,2}\epsilon_613151 \end{array}$$

$$\begin{aligned}
& 1321\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_613151321 \\
& 221\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_61231 \\
& 1321\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_613151^1 \\
& 12321\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_6131513221 \\
& 12321\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_61313 \\
& 12315131\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_613151321 \\
& 221\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_613151321 \\
& 12315131\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_613151^1 \\
& 221\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_6131513221 \\
& 122315131\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_613151^1 \\
& 3131\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_61315131 \\
& 221\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_6122 \\
& 221\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_613151^1 \\
& 3131\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_613151^1 \\
& 15131\mathbf{e}_6(\otimes\mathbf{e}_6)^{13}{}^{2,2}\otimes\mathbf{e}_613151^1
\end{aligned}$$

$$\begin{aligned}
& 1321\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{12}\overset{2,2}{\otimes}\mathbf{e}_613151321 \\
& 221\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{12}\overset{2,2}{\otimes}\mathbf{e}_61231 \\
& 1321\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{12}\overset{2,2}{\otimes}\mathbf{e}_613151^1 \\
& 12321\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{12}\overset{2,2}{\otimes}\mathbf{e}_6131513221 \\
& 12321\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{12}\overset{2,2}{\otimes}\mathbf{e}_61313 \\
& 12315131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{12}\overset{2,2}{\otimes}\mathbf{e}_613151321 \\
& 221\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{12}\overset{2,2}{\otimes}\mathbf{e}_613151321 \\
& 12315131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{12}\overset{2,2}{\otimes}\mathbf{e}_613151^1 \\
& 221\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{12}\overset{2,2}{\otimes}\mathbf{e}_6131513221 \\
& 122315131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{12}\overset{2,2}{\otimes}\mathbf{e}_613151^1 \\
& 3131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{12}\overset{2,2}{\otimes}\mathbf{e}_61315131
\end{aligned}$$

$$1321e_7'(\overset{3,3}{\otimes} e_7)^7 \overset{3,3}{\otimes} e_7 12231, \quad 1321e_7'(\overset{3,3}{\otimes} e_7)^7 \overset{3,3}{\otimes} e_7 12315131, \quad 1321e_7'(\overset{3,3}{\otimes} e_7)^7 \overset{3,3}{\otimes} e_7 123151321$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$150(\otimes_{\mathfrak{e}_6}^{2,2})^{12} \otimes_{\mathfrak{e}_6}^{2,2} 131513221,$$

$$150(\otimes_{\mathfrak{e}_6}^{2,2})^{12} \otimes_{\mathfrak{e}_6}^{2,2} 1313,$$

$$\begin{aligned} & 150(\otimes \mathfrak{e}_6)^{12} \otimes \mathfrak{e}_6 1315131, \\ & 150(\otimes \mathfrak{e}_6)^{12} \otimes \mathfrak{e}_6 131512, \end{aligned}$$

$$\begin{array}{l} 150 \left(\otimes \epsilon_6 \right)^{12} \otimes^{2,2} \epsilon_6 122 \\ 150 \left(\otimes \epsilon_6 \right)^{12} \otimes^{2,2} \epsilon_6 13151 \end{array}$$

$$\begin{aligned} &150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 1231, \\ &150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 131513221, \\ &150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 1313, \end{aligned}$$

$$\begin{aligned} &150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 12321, \\ &150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 1315131, \\ &150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 131512, \end{aligned}$$

$$\begin{array}{l} 150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 13151321 \\ 150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 122 \\ 150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 13151 \end{array}$$

$$\begin{aligned} &1\mathfrak{so}^{(2,2)}(\otimes \mathfrak{e}_6)^{10} \otimes^{2,2} \mathfrak{e}_6 1231, \\ &1\mathfrak{so}^{(2,2)}(\otimes \mathfrak{e}_6)^{10} \otimes^{2,2} \mathfrak{e}_6 131513221, \\ &1\mathfrak{so}^{(2,2)}(\otimes \mathfrak{e}_6)^{10} \otimes^{2,2} \mathfrak{e}_6 1313, \end{aligned}$$

$$\begin{aligned} & 150 \left(\otimes \epsilon_6 \right)^{10} \otimes \epsilon_6^{2,2} 12321, \\ & 150 \left(\otimes \epsilon_6 \right)^{10} \otimes \epsilon_6^{2,2} 1315131, \\ & 150 \left(\otimes \epsilon_6 \right)^{10} \otimes \epsilon_6^{2,2} 131512, \end{aligned}$$

$$\begin{aligned} &150(\otimes \epsilon_6)^{10} \otimes \epsilon_6 13151321 \\ &150(\otimes \epsilon_6)^{10} \otimes \epsilon_6 122 \\ &150(\otimes \epsilon_6)^{10} \otimes \epsilon_6 13151 \end{aligned}$$

$$\begin{aligned} & 21\mathbf{e}'_7{}^{4,5} \mathbf{e}_8(\otimes \mathbf{e}_8)^9 \otimes \mathbf{e}_7 1231, \\ & 1\mathbf{e}'_7(\otimes \mathbf{e}_8)^{5,5} \otimes \mathbf{e}_7 1231, \\ & 21\mathbf{e}'_7{}^{4,5} \mathbf{e}_8(\otimes \mathbf{e}_8)^9 \otimes \mathbf{e}_7^{5,5} 12, \end{aligned}$$

$$\begin{aligned} & 21e_7' \otimes^{4,5} e_8 \otimes^{5,5} e_8^9 \otimes^{5,4} e_7 12321, \\ & 1e_7' \otimes^{5,5} e_8^{10} \otimes^{5,4} e_7 12321, \\ & 1e_7' \otimes^{5,5} e_8^{10} \otimes^{5,5} e_7 12 \end{aligned}$$

$$\begin{array}{l} 21\epsilon'_7 \otimes^{4,5} \epsilon_8 (\otimes^{5,5} \epsilon_8)^9 \otimes^{5,4} \epsilon_7 122 \\ 1\epsilon'_7 (\otimes^{5,5} \epsilon_8)^{10} \otimes^{5,4} \epsilon_7 122 \end{array}$$

$$\begin{aligned} & 21\epsilon_7^{\prime 4,5} \epsilon_8 (\otimes \epsilon_8)^8 \otimes \epsilon_7^{5,4} 1231, \\ & 1\epsilon_7^{\prime 5,5} (\otimes \epsilon_8)^9 \otimes \epsilon_7^{5,4} 1231, \\ & 21\epsilon_7^{\prime 4,5} \epsilon_8 (\otimes \epsilon_8)^8 \otimes \epsilon_7^{5,5} 12, \end{aligned}$$

$$\begin{aligned} & 21e_7' \otimes^{4,5} e_8 \otimes^{5,5} e_8^8 \otimes^{5,4} e_7 12321, \\ & 1e_7' \otimes^{5,5} e_8^9 \otimes^{5,4} e_7 12321, \\ & 1e_7' \otimes^{5,5} e_8^9 \otimes^{5,5} e_7 12 \end{aligned}$$

$$\begin{array}{l} 21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^8 \otimes^{5,4} e_7 122 \\ 1e'_7 (\otimes^{5,5} e_8)^9 \otimes^{5,4} e_7 122 \end{array}$$

$$\begin{aligned} & 21c_7^{\prime 4,5} \otimes c_8^{\left(\otimes c_8\right)^{5,5}} \otimes c_7^{\left(\otimes c_7\right)^{5,4}} 1231, \\ & 1c_7^{\prime 5,5} \left(\otimes c_8\right)^{8,5,4} \otimes c_7 1231, \\ & 21c_7^{\prime 4,5} \otimes c_8^{\left(\otimes c_8\right)^{5,5}} \otimes c_7^{\left(\otimes c_7\right)^{5,5}} 12, \end{aligned}$$

$$\begin{aligned} & 21e_7' \otimes^{4,5} e_8 \otimes^{5,5} (e_8)^7 \otimes^{5,4} e_7 12321, \\ & 1e_7' \otimes^{5,5} (e_8)^8 \otimes^{5,4} e_7 12321, \\ & 1e_7' \otimes^{5,5} (e_8)^8 \otimes^{5,5} e_7 12 \end{aligned}$$

$$\begin{array}{l} 21\mathbf{e}'_7 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^7 \otimes \mathbf{e}_7^{5,4} 122 \\ 1\mathbf{e}'_7 (\otimes \mathbf{e}_8^{5,5})^8 \otimes \mathbf{e}_7^{5,4} 122 \end{array}$$

$$\begin{aligned} & 21c_7^{\prime 4,5} c_8(\otimes c_8)^6 \otimes c_7^{5,4} 1231, \\ & 1c_7^{\prime 5,5} (\otimes c_8)^7 \otimes c_7^{5,4} 1231, \\ & 21c_7^{\prime 4,5} c_8(\otimes c_8)^6 \otimes c_7^{5,5} 12, \end{aligned}$$

$$\begin{aligned} & 21e_7' \otimes^{4,5} e_8 \otimes^{5,5} (e_8)^6 \otimes^{5,4} e_7 12321, \\ & 1e_7' \otimes^{5,5} (e_8)^7 \otimes^{5,4} e_7 12321, \\ & 1e_7' \otimes^{5,5} (e_8)^7 \otimes^{5,5} e_7 12 \end{aligned}$$

$$\begin{array}{l} 21\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^6 \otimes \epsilon_7^{5,4} 122 \\ 1\epsilon'_7 (\otimes \epsilon_8^{5,5})^7 \otimes \epsilon_7^{5,4} 122 \end{array}$$

$$\begin{aligned} & 21\mathbf{e}'_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^5 \otimes^{5,4} \mathbf{e}_7 1231, \\ & 1\mathbf{e}'_7 (\otimes^{5,5} \mathbf{e}_8)^6 \otimes^{5,4} \mathbf{e}_7 1231, \end{aligned}$$

$$\begin{aligned} & 21\epsilon_7' \otimes^{4,5} \epsilon_8 (\otimes^{5,5} \epsilon_8)^5 \otimes^{5,4} \epsilon_7 12321, \\ & 1\epsilon_7' (\otimes^{5,5} \epsilon_8)^6 \otimes^{5,4} \epsilon_7 12321, \end{aligned}$$

$$\begin{array}{l} 21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^5 \otimes^{5,4} e_7 122 \\ 1e'_7 (\otimes^{5,5} e_8)^6 \otimes^{5,4} e_7 122 \end{array}$$

$$3150 \overset{2,4}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{12} \overset{4,2}{\otimes} 5013$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^{11} \otimes^{4,2} 5013$$

$$31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{5,5} 12231513215, \quad 31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 12231513, \quad 31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{5,5} 12231513151$$

$$3150 \otimes c_8^{2,4} (\otimes c_8^{5,5})^8 \otimes c_8^{5,5} 12231513215, \quad 3150 \otimes c_8^{2,4} (\otimes c_8^{5,5})^8 \otimes c_8^{5,5} \otimes c_8^1 12231513, \quad 3150 \otimes c_8^{2,4} (\otimes c_8^{5,5})^8 \otimes c_8^{5,5} 12231513151$$

$$3150 \otimes c_8^{2,4}(\otimes c_8^{5,5})^7 \otimes c_8^{5,5} 12231513215, \quad 3150 \otimes c_8^{2,4}(\otimes c_8^{5,5})^7 \otimes c_8^{5,5} \overset{1}{\otimes} c_8 12231513, \quad 3150 \otimes c_8^{2,4}(\otimes c_8^{5,5})^7 \otimes c_8^{5,5} 12231513151$$

$$31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{e}_8^{5,5} 12231513215, \quad 31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{e}_8^{5,5} \overset{1}{\otimes} \mathfrak{e}_8 12231513, \quad 31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{e}_8^{5,5} 12231513151$$

$$31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{5,5} 12231513215, \quad 31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 12231513, \quad 31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 12231513151$$

$$31\text{so}^{2,4} \otimes \text{e}_8^{(5,5)} (\otimes \text{e}_8)^4{}^{5,5} \otimes \text{e}_8 12231513215, \quad 31\text{so}^{2,4} \otimes \text{e}_8^{(5,5)} (\otimes \text{e}_8)^4 \otimes \text{e}_8^1 12231513, \quad 31\text{so}^{2,4} \otimes \text{e}_8^{(5,5)} (\otimes \text{e}_8)^4 \otimes \text{e}_8^{5,5} 12231513151$$

$$3150^{2,4} \otimes e_8^{(5,5)} \otimes e_8^{13} \otimes e_8^{5,5} 12231513215, \quad 3150^{2,4} \otimes e_8^{(5,5)} \otimes e_8^{13} \otimes e_8^{5,5} 12231513, \quad 3150^{2,4} \otimes e_8^{(5,5)} \otimes e_8^{13} \otimes e_8^{5,5} 12231513151$$

$$3150 \otimes \epsilon_8^{2,4} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_8^{5,5} 1223132, \quad 3150 \otimes \epsilon_8^{2,4} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_8^{5,5} 122222, \quad 3150 \otimes \epsilon_8^{2,4} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_8^{5,5} 122315123$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{14} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 122$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{13} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 122$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{12} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 122$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{11} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 122$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{10} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 122$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^9 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^8 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^7 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^6 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^5 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^4 \overset{5,4}{\otimes} e_6 1$$

$$1_{50} \overset{1,1}{\otimes} 5_0 \overset{2,4}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^3 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^{13} \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^{12} \overset{5,4}{\otimes} e_6 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}'_7 12,$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}'_7 12,$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{e}'_7 12,$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}'_7 12,$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}'_7 12,$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^5 \otimes \mathfrak{e}_8 12,$$

$$31\mathfrak{so}(\otimes \mathfrak{so})^5 \otimes \mathfrak{e}_8 1$$

$$131\mathfrak{e}_6(\otimes \mathfrak{e}_6)^5 \otimes \mathfrak{e}_8 1,$$

$$21\mathfrak{e}_6(\otimes \mathfrak{e}_6)^5 \otimes \mathfrak{e}_8 1$$

$$222221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{14} \otimes \mathfrak{e}_8 122222$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^5 \otimes \mathfrak{e}_8 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^9 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^8 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^7 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^6 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^5 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{14} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{13} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{12} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{11} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{10} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^9 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^8 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^7 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^6 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^5 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^{15} \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^{14} \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^{13} \otimes \mathfrak{so}^{2,2} 1$$

2.8.4 SSg(n)(356)

$$\begin{aligned} &131513221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{5,5}, \\ &313221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{5,5}, \\ &1513221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{5,5}, \\ &221 \mathbf{e}_8^{221(5,5)} \otimes \mathbf{e}_6^{5,5} \end{aligned}$$

$$\begin{aligned} &1231513221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_6, \\ &22221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_6, \\ &21^{13221} \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_6, \end{aligned}$$

$$\begin{array}{l} 12231513221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{5,5} \\ 21513221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{5,5} \\ 2221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{5,5} \end{array}$$

$$\begin{aligned} &131513221 \mathbf{c}_8(\otimes \mathbf{c}_8)^9 \otimes^{5,3} \mathbf{so}, \\ &313221 \mathbf{c}_8(\otimes \mathbf{c}_8)^9 \otimes^{5,3} \mathbf{so}, \\ &1513221 \mathbf{c}_8(\otimes \mathbf{c}_8)^9 \otimes^{5,3} \mathbf{so}, \\ &221 \mathbf{c}_8(\otimes \mathbf{c}_8)^9 \otimes^{5,3} \mathbf{so} \end{aligned}$$

$$\begin{aligned} &1231513221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^9 \otimes^{5,3} \mathfrak{so}, \\ &22221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^9 \otimes^{5,3} \mathfrak{so}, \\ &21^{13221} \epsilon_8^{(5,5)} (\otimes \epsilon_8)^9 \otimes^{5,3} \mathfrak{so}, \end{aligned}$$

$$\begin{array}{l} 12231513221 \mathfrak{c}_8(\otimes \mathfrak{c}_8)^9 \otimes \mathfrak{s}\mathfrak{o} \\ 21513221 \mathfrak{c}_8(\otimes \mathfrak{c}_8)^9 \otimes \mathfrak{s}\mathfrak{o} \\ 2221 \mathfrak{c}_8(\otimes \mathfrak{c}_8)^9 \otimes \mathfrak{s}\mathfrak{o} \end{array}$$

$$\begin{aligned} &131513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_6, \\ &313221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_6, \\ &1513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_6, \\ &221 \mathbf{c}_8^{221} (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_6 \end{aligned}$$

$$\begin{aligned} &1231513221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{e}_6, \\ &22221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{e}_6, \\ &21^{13221} \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{e}_6, \end{aligned}$$

$$\begin{array}{l} 12231513221 \mathbf{e}_8(\otimes \mathbf{e}_8)^8 \otimes \mathbf{e}_6^{5,5} \\ 21513221 \mathbf{e}_8(\otimes \mathbf{e}_8)^8 \otimes \mathbf{e}_6^{5,5} \\ 2221 \mathbf{e}_8^{21}(\otimes \mathbf{e}_8)^8 \otimes \mathbf{e}_6^{5,5} \end{array}$$

$$\begin{aligned} &131513221 \mathbf{c}_8(\otimes \mathbf{c}_8)^8 \otimes^{5,3} \mathbf{so}, \\ &313221 \mathbf{c}_8(\otimes \mathbf{c}_8)^8 \otimes^{5,3} \mathbf{so}, \\ &1513221 \mathbf{c}_8(\otimes \mathbf{c}_8)^8 \otimes^{5,3} \mathbf{so}, \\ &221 \mathbf{c}_8(\otimes \mathbf{c}_8)^8 \otimes^{5,3} \mathbf{so} \end{aligned}$$

$$\begin{aligned} &1231513221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^8 \otimes \mathbf{so}, \\ &22221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^8 \otimes \mathbf{so}, \\ &21^{13221} \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^8 \otimes \mathbf{so}, \end{aligned}$$

$$\begin{array}{l} 12231513221 \epsilon_8^{(5,5)} \otimes^{5,3} \mathfrak{so} \\ 21513221 \epsilon_8^{(5,5)} \otimes^{5,3} \mathfrak{so} \\ 2221 \epsilon_8^{(5,5)} \otimes^{5,3} \mathfrak{so} \end{array}$$

$$\begin{aligned} &131513221 \mathbf{e}_8^{1, (5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_6^{5,5}, \\ &313221 \mathbf{e}_8^{1, (5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_6^{5,5}, \\ &1513221 \mathbf{e}_8^{1, (5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_6^{5,5}, \\ &221 \mathbf{e}_8^{221, (5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_6^{5,5} \end{aligned}$$

$$\begin{aligned} & 1231513221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_6, \\ & 22221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_6, \\ & 21^{13221} \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_6, \end{aligned}$$

$$\begin{array}{l} 12231513221\mathfrak{e}_8(\otimes\mathfrak{e}_8)^7\otimes\mathfrak{e}_6^{5,5} \\ 21513221\mathfrak{e}_8(\otimes\mathfrak{e}_8)^7\otimes\mathfrak{e}_6^{5,5} \\ 2221\mathfrak{e}_8^{21}(\otimes\mathfrak{e}_8)^7\otimes\mathfrak{e}_6^{5,5} \end{array}$$

$$\begin{aligned} &131513221\mathbf{e}_8(\otimes\mathbf{e}_8)^7\otimes\mathbf{50}, \\ &313221\mathbf{e}_8(\otimes\mathbf{e}_8)^7\otimes\mathbf{50}, \\ &1513221\mathbf{e}_8(\otimes\mathbf{e}_8)^7\otimes\mathbf{50}, \end{aligned}$$

$$\begin{aligned} &1231513221 \mathbf{e}_8(\otimes \mathbf{e}_8)^7 \otimes \mathbf{so}, \\ &22221 \mathbf{e}_8(\otimes \mathbf{e}_8)^7 \otimes \mathbf{so}, \\ &21^{13}221 \mathbf{e}_8(\otimes \mathbf{e}_8)^7 \otimes \mathbf{so}, \end{aligned}$$

$$\begin{array}{l} 12231513221 \mathfrak{e}_8 \left(\bigotimes \mathfrak{e}_8 \right)^7 \bigotimes \mathfrak{so}^{5,3} \\ 21513221 \mathfrak{e}_8 \left(\bigotimes \mathfrak{e}_8 \right)^7 \bigotimes \mathfrak{so}^{5,3} \\ 2221 \mathfrak{e}_8^{21} \left(\bigotimes \mathfrak{e}_8 \right)^7 \bigotimes \mathfrak{so}^{5,3} \end{array}$$

$$2221\mathfrak{e}_8^{21}(\otimes^{5,5}\mathfrak{e}_8)^{14}\otimes^{5,3}\mathfrak{so}$$

$$\begin{array}{l} 12231513221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}_6^{5,5} \\ 21513221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}_6^{5,5} \\ 2221 \mathfrak{e}_8^{21} (\otimes \mathfrak{e}_8)^{5,5} \otimes \mathfrak{e}_6^{5,5} \end{array}$$

$$\begin{array}{l} 12231513221 \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{so}^{5,3} \\ 21513221 \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{so}^{5,3} \\ 2221 \mathfrak{e}_8^{21(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{array}{l} 12231513221 \epsilon_8^1 (\otimes \epsilon_8)^{5,5} 12 \otimes \epsilon_6^{5,5} \\ 21513221 \epsilon_8^1 (\otimes \epsilon_8)^{5,5} 12 \otimes \epsilon_6^{5,5} \\ 2221 \epsilon_8^{21} (\otimes \epsilon_8)^{5,5} 12 \otimes \epsilon_6^{5,5} \end{array}$$

$$\begin{array}{l} 12231513221 \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3} \\ 21513221 \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3} \\ 2221 \mathfrak{e}_8^{21(5,5)} (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{array}{l} 12231513221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{(5,5)} \\ 21513221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{(5,5)} \\ 2221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{(5,5)} \end{array}$$

$$12231513221\mathbf{e}_8(\bigotimes^{5,5}\mathbf{e}_8)^{11}\bigotimes^{5,3}\mathbf{so}$$

$$\begin{array}{l} 21513221 \epsilon_8^{(1)} (\otimes \epsilon_8)^{11} \otimes \mathfrak{so}^{5,3} \\ 2221 \epsilon_8^{(21)} (\otimes \epsilon_8)^{11} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{array}{l} 12231513221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^{10} \otimes \mathbf{e}_6^{5,5} \\ 21513221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^{10} \otimes \mathbf{e}_6^{5,5} \\ 2221^{21} \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^{10} \otimes \mathbf{e}_6^{5,5} \end{array}$$

$$\begin{array}{l} 12231513221\mathfrak{e}_8^{1(5,5)}(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{so}^{5,3} \\ 21513221\mathfrak{e}_8^{1(5,5)}(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{so}^{5,3} \\ 2221\mathfrak{e}_8^{21(5,5)}(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$221e_7^1(\otimes^{3,3} e_7)^9 \otimes^{3,3} so$$

$$221e_7({}^{3,3}\otimes e_7)^8 {}^{3,3}\otimes \mathfrak{so}$$

$$221e_7^1(\otimes^{3,3} e_7)^7 \otimes^{3,3} so$$

$$221e_7^1(\otimes e_7)^{3,3} \otimes so^{3,3}$$

$$221e_7(\otimes^{3,3} e_7)^5 \otimes^{3,3} so$$

$$221e_7(\overset{1}{\otimes} \overset{3,3}{e_7})^{15} \overset{3,3}{\otimes} so$$

$$21\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{so}^{5,3},$$

$$1\mathfrak{e}_7^1 (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{so}^{5,3}$$

$$321\mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^5 \otimes \mathfrak{e}_8^{4,5}$$

$$31\mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^5 \otimes \mathfrak{e}_8^{3,5}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^9 \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^8 \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^7 \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^6 \otimes \mathfrak{e}_8^{4,5}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^6 \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^5 \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^{15} \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^{14} \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^{13} \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^{12} \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^{11} \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}'_7(\otimes^{3,3}\mathfrak{e}_7)^{10}\otimes^{3,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^9\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^9\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^8\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^8\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^7\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^7\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^6\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^6\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^5\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^5\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^4\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^4\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^{14}\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^7 \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^6 \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^6 \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^5 \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^5 \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^4 \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^4 \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^{14} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^{14} \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^{13} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^{13} \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^{12} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^{12} \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes^{11} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \mathfrak{so}^{5,3}$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_6^{5,5}$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \mathfrak{so}^{5,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^9 \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^8 \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^7 \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^6 \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^5 \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{15} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{14} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{13} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{12} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{11} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{10} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_6^{2,2})^6 \otimes \epsilon_8^{3,5}$$

$$\begin{array}{l} 2221^{21} \mathfrak{e}_8(\overset{5,5}{\otimes} \mathfrak{e}_8)^7 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513 \\ 221^{221} \mathfrak{e}_8(\overset{5,5}{\otimes} \mathfrak{e}_8)^7 \overset{5,5}{\otimes} \mathfrak{e}_8 122315123 \\ 221^{221} \mathfrak{e}_8(\overset{5,5}{\otimes} \mathfrak{e}_8)^7 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513151 \end{array}$$

$$\begin{aligned}
& 131513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 131513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 1231513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 1231513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 12231513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 12231513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 21513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 1223132 \\
& 313221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 313221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 22221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513215 \\
& 21513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122222 \\
& 1513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 21513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 1513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 21^{13221} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 21^{13221} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 2221^{21} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122222 \\
& 2221^{21} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513 \\
& 221^{221} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 221^{221} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151
\end{aligned}$$

$$\begin{aligned} &131513221 \mathbf{e}_8^{\binom{5,5}{\otimes}} \mathbf{e}_8^{\binom{5,5}{\otimes}} \mathbf{e}_8 122315123 \\ &131513221 \mathbf{e}_8^{\binom{5,5}{\otimes}} \mathbf{e}_8^{\binom{5,5}{\otimes}} \mathbf{e}_8 12231513151 \\ &1231513221 \mathbf{e}_8^{\binom{5,5}{\otimes}} \mathbf{e}_8^{\binom{5,5}{\otimes}} \mathbf{e}_8 122315123 \\ &1231513221 \mathbf{e}_8^{\binom{5,5}{\otimes}} \mathbf{e}_8^{\binom{5,5}{\otimes}} \mathbf{e}_8 12231513151 \\ &12231513221 \mathbf{e}_8^{\binom{5,5}{\otimes}} \mathbf{e}_8^{\binom{5,5}{\otimes}} \mathbf{e}_8 122315123 \end{aligned}$$

[illegible]

[illegible]

[illegible]

[illegible]

$$21513221\mathbf{e}_8({}^{5,5}\otimes\mathbf{e}_8)^{13}{}^{5,5}\otimes\mathbf{e}_712$$

$$2221\mathbf{e}_8^{21}(\overset{5,5}{\otimes}\mathbf{e}_8)^{13}\overset{5,5}{\otimes}\mathbf{e}_712$$

$$12231513221 \mathbf{e}_8^1 (\bigotimes^{5,5} \mathbf{e}_8)^{12} \bigotimes^{5,4} \mathbf{e}_7 1231$$

$$21513221\mathbf{e}_8(\otimes\mathbf{e}_8)^{12}\otimes\mathbf{e}_71231$$

$$1231513221 \mathbf{e}_8^1 (\otimes^{5,5} \mathbf{e}_8)^{12} \otimes^{5,4} \mathbf{e}_7 12321$$

$$22221\mathbf{e}_8(\overset{5,5}{\otimes}\mathbf{e}_8)^{12}\overset{5,4}{\otimes}\mathbf{e}_712321$$

$$131513221 \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^{12} \bigotimes \mathbf{e}_7 122$$

$$21 \overset{1}{\mathbf{e}}_7 \otimes \overset{4,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{11} \overset{5,5}{\otimes} \mathbf{e}_8 1223132$$

$$21 \overset{1}{\mathbf{e}}_7 \otimes \overset{4,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{11} \overset{5,5}{\otimes} \mathbf{e}_8 122315123$$

$$21 \overset{1}{e}_7 \otimes \overset{4,5}{e}_8 (\overset{5,5}{e}_8)^{11} \otimes \overset{5,5}{e}_8 12231513151$$

$$1513221 \mathbf{e}_8 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^{12} \overset{5,4}{\otimes} \mathbf{e}_7 122$$

$$221 \mathbf{e}_8 \left(\bigotimes^{5,5} \mathbf{e}_8 \right)^{12} \bigotimes^{5,4} \mathbf{e}_7 1231$$

$$2221\mathbf{e}_8(\bigotimes^{5,5}\mathbf{e}_8)^{12}\bigotimes^{5,4}\mathbf{e}_712321$$

$$221 \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^{12} \bigotimes \mathbf{e}_7 122$$

$$1\mathfrak{e}_7^{1(5,5)}(\otimes \mathfrak{e}_8)^{12} \otimes^{5,5} \mathfrak{e}_8 122315123$$

$$1e_7^{(5,5)}(\otimes e_8)^{12} \otimes e_8^{5,5} 12231513151$$

$$12231513221 \mathbf{e}_8^1 (\bigotimes^{5,5} \mathbf{e}_8)^{12} \bigotimes^{5,5} \mathbf{e}_7 12$$

$$21513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,5}{\otimes}\mathfrak{e}_712$$

$$2221\mathbf{e}_8^{21}(\overset{5,5}{\otimes}\mathbf{e}_8)^{12}\overset{5,5}{\otimes}\mathbf{e}_712$$

$$12231513221 \mathbf{e}_8^1 (\bigotimes^{5,5} \mathbf{e}_8)^{11} \bigotimes^{5,4} \mathbf{e}_7 1231$$

$$21513221\mathbf{e}_8(\overset{1}{\otimes}\overset{5,5}{\mathbf{e}_8})^{11}\overset{5,4}{\otimes}\mathbf{e}_71231$$

$$1231513221 \mathbf{e}_8^1 (\otimes \mathbf{e}_8)^{5,5} {}^{11} \otimes \mathbf{e}_7^{5,4} 12321$$

$$22221\mathbf{e}_8(\overset{5,5}{\otimes}\mathbf{e}_8)^{11}\overset{5,4}{\otimes}\mathbf{e}_712321$$

$$131513221 \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^{11} \bigotimes \mathbf{e}_7 122$$

$$21 \overset{1}{\mathbf{e}}_7 \otimes \overset{4,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{10} \overset{5,5}{\otimes} \mathbf{e}_8 1223132$$

$$21 \overset{1}{\mathfrak{e}_7} \overset{4,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{10} \overset{5,5}{\otimes} \mathfrak{e}_8 122315123$$

[illegible]

$313221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_71231,$	$22221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_71231,$	$21513221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_71231$
$1513221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_71231,$	$313221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712321,$	$22221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712321$
$131513221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_7122,$	$21\mathbf{c}_7 \otimes \mathbf{c}_8(\otimes \mathbf{c}_8)^{13} \otimes \mathbf{c}_81223132,$	$313221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_7122$
$21\mathbf{c}_7 \otimes \mathbf{c}_8(\otimes \mathbf{c}_8)^{13} \otimes \mathbf{c}_8122222,$	$21\mathbf{c}_7 \otimes \mathbf{c}_8(\otimes \mathbf{c}_8)^{13} \otimes \mathbf{c}_8122315123,$	$21\mathbf{c}_7 \otimes \mathbf{c}_8(\otimes \mathbf{c}_8)^{13} \otimes \mathbf{c}_812231513$
$22221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_7122,$	$21513221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_7122,$	$1513221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_7122$
$21^{13221}\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_71231,$	$2221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_71231,$	$221^{221}\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_71231$
$21^{13221}\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712321,$	$21^{13221}\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_7122,$	$2221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712321$
$221^{221}\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712321,$	$2221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_7122,$	$221^{221}\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_7122$
$1\mathbf{c}_7(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_81223132,$	$1\mathbf{c}_7(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_8122222,$	$1\mathbf{c}_7(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_8122315123$
$1\mathbf{c}_7(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_812231513,$	$131513221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712,$	$313221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712$
$22221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712,$	$21513221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712,$	$1513221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712$
$21^{13221}\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712,$	$2221\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712,$	$221^{221}\mathbf{c}_8(\otimes \mathbf{c}_8)^{14} \otimes \mathbf{c}_712$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_8^{5,5} 1223132, \quad 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_8^{5,5} 122222, \quad 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_8^{5,5} 122315123$$

$$21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^5 \otimes \mathbf{e}_6^{5,4} 1$$

$$\begin{aligned} & 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_7^{5,4} 122 \\ & 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_6^{5,3} 131 \\ & 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_6^{5,4} 1 \end{aligned}$$

$$\begin{aligned} & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} 122 \\ & 1\epsilon_7^{1,5,5} (\otimes \epsilon_8)^{14} \otimes \epsilon_6^{5,3} 131 \\ & 21\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,4} 1 \end{aligned}$$

$$\begin{aligned} & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} 122 \\ & 1\epsilon_7^{1,5,5} (\otimes \epsilon_8)^{13} \otimes \epsilon_6^{5,3} 131 \\ & 21\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,4} 1 \end{aligned}$$

$$\begin{array}{l} 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} 122 \\ 1\epsilon_7^{1,5,5} (\otimes \epsilon_8)^{12} \otimes \epsilon_6^{5,3} 131 \\ 21\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,4} 1 \end{array}$$

$$\begin{aligned} & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} 122 \\ & 1\epsilon_7^{1,5,5} (\otimes \epsilon_8)^{11} \otimes \epsilon_6^{5,3} 131 \\ & 21\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,4} 1 \end{aligned}$$

293

$$21e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^8 \otimes e_7'^1$$

$$\begin{array}{l} 1e_7' \otimes e_8^{4,5} (\otimes e_8^{5,5})^7 \otimes e_7^{5,4} 122 \\ 21e_7 \otimes e_8^{4,5} (\otimes e_8^{5,5})^7 \otimes e_7^{5,5} 1 \end{array}$$

$$\begin{array}{l} 1 \overset{1}{e}'_7 \otimes \overset{4,5}{e}_8 (\overset{5,5}{\otimes} \overset{5,5}{e}_8)^6 \otimes \overset{5,4}{e}_7 122 \\ 21 \overset{1}{e}_7 \otimes \overset{4,5}{e}_8 (\overset{5,5}{\otimes} \overset{5,5}{e}_8)^6 \otimes \overset{5,5}{e}'_7 1 \end{array}$$

$$\begin{array}{l} 1e_7' \otimes e_8 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e_7 122 \\ 21e_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_7' 1 \end{array}$$

$$\begin{array}{l} 1e_7' \otimes e_8^{4,5} (\otimes e_8^{5,5})^4 \otimes e_7^{5,4} 122 \\ 21e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^4 \otimes e_7^{5,5} 1 \end{array}$$

$$\begin{array}{l} 1_{\epsilon_7'}^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{13} \otimes \epsilon_7^{5,4} 122 \\ 21_{\epsilon_7}^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{13} \otimes \epsilon_7^{5,5} 1 \end{array}$$

$$\begin{array}{l} 1^1_{e'_7} \otimes e_8(5,5) \otimes e_8^{12} \otimes e_7^{5,4} 122 \\ 21^1_{e_7} \otimes e_8(5,5) \otimes e_8^{12} \otimes e_7^{5,5} 1 \end{array}$$

$$1e_7' \otimes e_8^{(4,5)} (\otimes e_8)^{(5,5)} e_7^{(5,4)} 122$$

$$1_{50}^{1,1}(\otimes 50)^{13} \otimes 50^{1,1} 132,$$

$$1_{\mathfrak{so}}({}^{1,1}\otimes \mathfrak{so})^{13} {}^{1,1}\otimes \mathfrak{so}12$$

$$1_{\mathfrak{so}}({}^{1,1}\otimes \mathfrak{so})^{12} \otimes \mathfrak{so}131,$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{12}\overset{1,1}{\otimes}\mathfrak{so}1321,$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{12} \otimes \mathfrak{so}^{1,1}13221$$

$$1_{\mathfrak{so}}({}^{1,1}\otimes \mathfrak{so})^{12} \otimes \mathfrak{so} 132,$$

$$1\mathfrak{so}^{(1,1)}(\otimes \mathfrak{so})^{12} \otimes \mathfrak{so}^{12}$$

$$150 \binom{1,1}{\otimes 50}^{11} \otimes 131,$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{11} \otimes \mathfrak{so}^{1,1} 1321,$$

$$1\mathfrak{so}(\overset{1}{\otimes}\overset{1,1}{\mathfrak{so}})^{11}\overset{1,1}{\otimes}\mathfrak{so}13221$$

$$150 \binom{1,1}{\otimes so}^{11} \binom{1,1}{\otimes so}^2 132,$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{11} \otimes^{1,1} \mathfrak{so}^{12}$$

$$150 \binom{1,1}{\otimes 50}^{10} \binom{1,1}{\otimes 50}^{131},$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{10} \otimes \mathfrak{so}^{1,1} 1321,$$

$$1\mathfrak{so}^1(\overset{1,1}{\otimes}\mathfrak{so})^{10}\overset{1,1}{\otimes}\mathfrak{so}13221$$

$$150 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50^{10} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50^{132},$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{10} \otimes \mathfrak{so}^{1,1}12$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^9 \otimes^{5,3} e_6 131,$$

$$1e_6^{1,3,5} \otimes e_8^{(5,5)} \otimes e_7^{5,4} 12,$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathbf{e}_6 12$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \overset{5,5}{\otimes} \mathbf{e}'_7 1,$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^9 \otimes^{5,4} e_6 1$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^8 \otimes^{5,3} e_6 131,$$

$$1e_6^{1,3,5} \otimes e_8^{(5,5)} (\otimes e_8)^8 \otimes e_7'^{5,4} 12,$$

$$1e_7' \otimes e_8^{4,5} (\otimes e_8^{5,5})^8 \otimes e_6^{5,3} 12$$

$$1e_6^{1,3,5} \otimes e_8^{(5,5)} (\otimes e_8)^8 \otimes e_{71}'^{5,5},$$

$$1\mathbf{e}'_7 \overset{1}{\otimes} \overset{4,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^8 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,3} e_6 131,$$

$$1e_6 \otimes^{3,5} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,4} e_7' 12,$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,3} e_6 12$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^7 \overset{5,5}{\otimes} \mathbf{e}'_7 1,$$

$$1 \overset{1}{\mathbf{e}}_7' \otimes \overset{4,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^7 \otimes \overset{5,4}{\mathbf{e}}_6 1$$

$$1e_7' \otimes e_8 \binom{5,5}{\otimes e_8}^6 \otimes e_6 131,$$

$$1\mathbf{e}_6 \otimes \mathbf{e}_8 \binom{5,5}{\otimes \mathbf{e}_8}^6 \otimes \mathbf{e}_7' 12,$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathbf{e}_6 12$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^6 \overset{5,5}{\otimes} \mathbf{e}'_7 1,$$

$$1 \overset{1}{\mathbf{e}}_7' \otimes \overset{4,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^6 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1\epsilon_7' \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^5 \otimes \epsilon_6 131,$$

$$1\mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} (\otimes \mathbf{e}_8^{5,5})^5 \otimes \mathbf{e}_7'^{5,4} 12,$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \begin{pmatrix} 5,5 \\ \otimes \mathbf{e}_8 \end{pmatrix}^5 \otimes \mathbf{e}_6 12$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^5 \overset{5,5}{\otimes} \mathbf{e}'_7 1,$$

$$1 \mathbf{e}'_7 \overset{1}{\otimes} \overset{4,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^5 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1e_7' \otimes e_8 \binom{5,5}{\otimes e_8}^4 \otimes e_6 131,$$

$$1\mathbf{e}_6 \otimes \mathbf{e}_8 \binom{5,5}{\otimes \mathbf{e}_8}^4 \otimes \mathbf{e}'_7 12,$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathbf{e}_6 12$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,5}{\otimes} \mathbf{e}'_7 1,$$

$$1 \overset{1}{\mathbf{e}}_7' \otimes \overset{4,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$\begin{array}{lll} 1\epsilon_7' \otimes \epsilon_8(\otimes \epsilon_8)^{13} \otimes \epsilon_6^{5,3} 131, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8(\otimes \epsilon_8)^{13} \otimes \epsilon_7^{5,4} 12, & 1\epsilon_7' \otimes \epsilon_8(\otimes \epsilon_8)^{13} \otimes \epsilon_6^{5,3} 12 \\ 1\epsilon_6^{1,3,5} \otimes \epsilon_8(\otimes \epsilon_8)^{13} \otimes \epsilon_7^{5,5} 1, & 1\epsilon_7' \otimes \epsilon_8(\otimes \epsilon_8)^{13} \otimes \epsilon_6^{5,4} 1 & \end{array}$$

$$\begin{array}{lll} 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_6^{5,3} 131, & 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_7^{5,4} 12, & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_6^{5,3} 12 \\ 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_7^{5,5} 1, & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_6^{5,4} 1 & \end{array}$$

$$\begin{array}{lll} 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{11} \otimes \epsilon_6^{5,3} 131, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{11} \otimes \epsilon_7^{5,4} 12, & 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{11} \otimes \epsilon_6^{5,3} 12 \\ 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{11} \otimes \epsilon_7^{5,5} 1, & 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{11} \otimes \epsilon_6^{5,4} 1 & \end{array}$$

$$\begin{array}{lll} 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,3} 131, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} 12, & 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,3} 12 \\ 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,5} 1, & 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,4} 1 & \end{array}$$

$$\begin{array}{lll} 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^9 \otimes \epsilon_7'^{3,3}1231, & 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^9 \otimes \epsilon_7'^{3,3}12321, & 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^9 \otimes \epsilon_7'^{3,3}122 \\ 21\epsilon_7'^{1\ 3,3}(\otimes \epsilon_7)^9 \otimes \epsilon_6^{3,3}131, & 21\epsilon_7'^{1\ 3,3}(\otimes \epsilon_7)^9 \otimes \epsilon_6^{3,3}12 & \end{array}$$

$$\begin{array}{lll} 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^8 \otimes \epsilon_7'^{3,3}1231, & 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^8 \otimes \epsilon_7'^{3,3}12321, & 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^8 \otimes \epsilon_7'^{3,3}122 \\ 21\epsilon_7'^{1\ 3,3}(\otimes \epsilon_7)^8 \otimes \epsilon_6^{3,3}131, & 21\epsilon_7'^{1\ 3,3}(\otimes \epsilon_7)^8 \otimes \epsilon_6^{3,3}12 & \end{array}$$

$$\begin{array}{lll} 1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_7'1231, & 1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_7'12321, & 1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_7'122 \\ 21\mathbf{e}_7'(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_6131, & 21\mathbf{e}_7'(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_612 & \end{array}$$

$$\begin{array}{lll} 1\mathbf{e}_6^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}6} \otimes^{3,3} \mathbf{e}_7'1231, & 1\mathbf{e}_6^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}6} \otimes^{3,3} \mathbf{e}_7'12321, & 1\mathbf{e}_6^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}6} \otimes^{3,3} \mathbf{e}_7'122 \\ 21\mathbf{e}_7'^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}6} \otimes^{3,3} \mathbf{e}_6131, & 21\mathbf{e}_7'^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}6} \otimes^{3,3} \mathbf{e}_612 & \end{array}$$

$$\begin{array}{lll} 1\mathbf{e}_6^{(1,3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_7'^{3,3}1231, & 1\mathbf{e}_6^{(1,3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_7'^{3,3}12321, & 1\mathbf{e}_6^{(1,3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_7'^{3,3}122 \\ 21\mathbf{e}_7^{(1,3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_6^{3,3}131, & 21\mathbf{e}_7^{(1,3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_6^{3,3}12 & \end{array}$$

$$1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^{15}{}^{3,3} \mathbf{e}_7' 1231, \quad 1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^{15} \otimes \mathbf{e}_7' 12321, \quad 1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^{15} \otimes \mathbf{e}_7' 122$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{15}\overset{3,3}{\otimes}\mathbf{e}_6131,$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{15}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$1e_6^{(3,3)}(\otimes e_7)^{14} \otimes e_7' 1231,$$

$$1e_6^{(3,3)}(\otimes e_7)^{14} \otimes e_7' 12321,$$

$$1e_6^{(3,3)}(\otimes e_7)^{14} \otimes e_7' 122$$

$$21\mathbf{e}'_7 \binom{1}{(3,3)} \binom{3,3}{\otimes \mathbf{e}_7}^{14} \otimes \mathbf{e}_6 131,$$

$$21\overset{1}{\mathbf{e}}_7'(\overset{3,3}{\otimes}\mathbf{e}_7)^{14}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$1\mathbf{e}_6\left(\overset{3,3}{\otimes}\mathbf{e}_7\right)^{13}\overset{3,3}{\otimes}\mathbf{e}_7'1231,$$

$$1e_6^{(3,3)}(\otimes e_7)^{13} \otimes e_7' 12321,$$

$$1e_6^{(3,3)} (\otimes e_7)^{13} \otimes e_7' 122$$

$$21\mathbf{e}'_7 \binom{1}{(3,3)} \binom{3,3}{\otimes \mathbf{e}_7}^{13} \otimes \mathbf{e}_6 131,$$

$$21 \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \mathbf{e}_7)^{13} \overset{3,3}{\otimes} \mathbf{e}_6 12$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{12} \otimes \mathbf{e}_7' 1231,$$

$$1\mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7^{12} \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7' 12321,$$

$$1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^{12} \otimes \mathbf{e}_7^{3,3} 122$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{12}\overset{3,3}{\otimes}\mathbf{e}_6131,$$

$$21\mathbf{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{12}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^{11} \otimes \mathbf{e}_7^{3,3} 1231,$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{11} \otimes \mathbf{e}_7'^{3,3}12321,$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{11} \otimes \mathbf{e}_7' 122$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{11}\overset{3,3}{\otimes}\mathbf{e}_6131,$$

$$21\mathbf{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{11}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$1\mathbf{e}_6 \binom{3,3}{\otimes} \mathbf{e}_7^{10} \binom{3,3}{\otimes} \mathbf{e}_7' 1231,$$

$$1\mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7^{10} \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7' 12321,$$

$$1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^{10} \otimes \mathbf{e}_7^{3,3} 122$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{10}\overset{3,3}{\otimes}\mathbf{e}_6131,$$

$$21\mathbf{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{10}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$1321e_7^1(\otimes^{3,3} e_7)^9 \otimes^{3,2} \mathfrak{so}1,$$

$$12321\mathbf{e}_7^1(\otimes \mathbf{e}_7)^9 \otimes \mathfrak{so}1,$$

$$221\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^9 \otimes \mathbf{so}1$$

$$21\mathfrak{e}_7(3,3 \otimes \mathfrak{e}_7)^9 \otimes \mathfrak{so}1$$

$$1321e_7^{(3,3)}(\otimes e_7)^8 \otimes^{3,2} \mathfrak{so}1,$$

$$12321\mathfrak{e}_7^1(\otimes \mathfrak{e}_7)^{3,3}{}^8 \otimes \mathfrak{so}1,{}^{3,2}$$

$$221\mathbf{e}_7^{(1)}(\otimes \mathbf{e}_7^{(3,3)})^8 \otimes \mathbf{so}1^{(3,2)}$$

$$21\mathbf{e}_7\left(\bigotimes\mathbf{e}_7\right)^8\bigotimes\mathfrak{so}1$$

$$1321e_7^{(3,3)}(\otimes e_7)^7 \otimes so_1,$$

$$12321e_7(\otimes e_7)^7 \otimes so1,$$

$$221\mathbf{e}_7({}^{(3,3)}\otimes\mathbf{e}_7)^7\otimes{}^{(3,2)}\mathbf{so}1$$

$$21\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^7\overset{3,2}{\otimes}\mathbf{sol}$$

$$1321e_7^{(3,3)}(\otimes e_7)^6 \otimes^{3,2} \mathfrak{so}1,$$

$$12321 e_7^1 (\otimes e_7)^{3,3} \otimes so1^{3,2},$$

$$221\mathbf{e}_7({}^{3,3}\otimes\mathbf{e}_7)^6\otimes\mathbf{so}1$$

$$21\mathfrak{e}_7^{21}(\otimes \mathfrak{e}_7^{3,3})^6 \otimes \mathfrak{so}^{3,2}1$$

$$\begin{array}{l} 1321^1_{\mathbf{e}_7}(\otimes \mathbf{e}_7)^5 \otimes^{3,2} \mathbf{sol}, \\ 21^{21}_{\mathbf{e}_7}(\otimes \mathbf{e}_7)^5 \otimes^{3,2} \mathbf{sol} \end{array}$$

$$12321\mathbf{e}_7^1(\otimes \mathbf{e}_7)^{3,3}{}^5 \otimes \mathbf{so}1,{}^{3,2}$$

$$221\mathbf{e}_7({}^1\otimes\mathbf{e}_7)^5\otimes\mathbf{sol}$$

$$\begin{array}{l} 1321e_7^{(3,3)}(\otimes e_7)^{15} \otimes so_1, \\ 21e_7^{(3,3)}(\otimes e_7)^{15} \otimes so_1 \end{array}$$

$$12321\mathbf{e}_7({}^{3,3}\otimes\mathbf{e}_7)^{15}{}^{3,2}\otimes\mathfrak{so}1,$$

$$221\mathfrak{e}_7({}^1(3,3)\otimes\mathfrak{e}_7)^{15}({}^{3,2}\otimes\mathfrak{so}1$$

$$\begin{array}{l} 1321 \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^{14} \otimes \mathbf{so}1, \\ 21 \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^{14} \otimes \mathbf{so}1 \end{array}$$

$$12321\mathbf{e}_7^1(\otimes \mathbf{e}_7)^{3,3}{}^{14} \otimes \mathbf{so}1,{}^{3,2}$$

$$221e_7^{(1)}(\otimes e_7^{(3,3)})^{14} \otimes so1^{(3,2)}$$

$$\begin{array}{l} 1321\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^{13} \otimes \mathbf{sol}, \\ 21\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^{13} \otimes \mathbf{sol} \end{array}$$

$$12321\mathbf{e}_7^{(1)}(\otimes \mathbf{e}_7^{(3,3)})^{13} \otimes \mathbf{so}1,$$

$$221\mathfrak{e}_7({}^1\otimes\mathfrak{e}_7)^{13}{}^{3,2}\otimes\mathfrak{so}1$$

$$\begin{array}{l} 1321e_7^{(3,3)}(\otimes e_7)^{12} \otimes \mathfrak{so}1, \\ 21e_7^{(3,3)}(\otimes e_7)^{12} \otimes \mathfrak{so}1 \end{array}$$

$$12321\mathbf{e}_7({}^{3,3}\otimes\mathbf{e}_7)^{12}{}^{3,2}\otimes\mathfrak{so}1,$$

$$221e_7^{(1)}(\otimes e_7^{(3,3)})^{12} \otimes so1^{(3,2)}$$

$$\begin{array}{l} 1321\epsilon_7^{(3,3)}(\otimes \epsilon_7)^{11} \otimes \mathfrak{so}1, \\ 21\epsilon_7^{(3,3)}(\otimes \epsilon_7)^{11} \otimes \mathfrak{so}1 \end{array}$$

$$12321e_7^1(\otimes^{3,3} e_7)^{11} \otimes^{3,2} so1,$$

$$221e_7^{(1)}(\otimes e_7^{(3,3)})^{11} \otimes so1^{(3,2)}$$

$$\begin{array}{l} 1321_7^{(3,3)} (\otimes_7)^{10} \otimes^{3,2} \mathfrak{so}1, \\ 21_7^{(3,3)} (\otimes_7)^{10} \otimes^{3,2} \mathfrak{so}1 \end{array}$$

$$12321e_7^1(\otimes e_7)^{10} \otimes so1,$$

$$221\mathfrak{e}_7({}^1\otimes\mathfrak{e}_7)^{10}{}^3\otimes\mathfrak{so}1$$

$$22221\mathbf{e}_8^1(\otimes^{5,5}\mathbf{e}_8)^{14}\otimes^{5,5}\mathbf{e}_8122222,$$

$$2221\mathbf{e}_8^{21}(\otimes^{5,5}\mathbf{e}_8)^{14}\otimes^{5,5}\mathbf{e}_8122222,$$

$$221 \overset{221}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^{14} \overset{5,5}{\otimes} \mathfrak{e}_8 122222$$

$$21\mathbf{e}'_7(\otimes \mathbf{e}_7)^9 \otimes \mathbf{e}'_7 1231,$$

$$21 \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \mathbf{e}_7)^9 \overset{3,3}{\otimes} \mathbf{e}_7' 12321,$$

$$21\overset{1}{e}'_7(\overset{3,3}{\otimes} e_7)^9 \overset{3,3}{\otimes} e'_7 122$$

$$21\mathbf{e}'_7(\bigotimes^{3,3}\mathbf{e}_7)^8\bigotimes^{3,3}\mathbf{e}'_71231,$$

$$21\mathbf{e}'_7(\otimes^{3,3}\mathbf{e}_7)^8\otimes^{3,3}\mathbf{e}'_712321,$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^8\overset{3,3}{\otimes}\mathbf{e}'_7122$$

$$21\mathbf{e}'_7(\otimes_{\mathbf{e}_7}^{3,3})^7\otimes_{\mathbf{e}_7}^{3,3}\mathbf{e}'_71231,$$

$$21\mathbf{e}'_7(\otimes_{3,3}\mathbf{e}_7)^7\otimes_{3,3}\mathbf{e}'_712321,$$

$$21e_7'(\otimes^{3,3} e_7)^7 \otimes^{3,3} e_7' 122$$

$$1 \overset{1}{\mathbf{e}}_6 \overset{3,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{13} \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1 \overset{1}{\mathbf{e}}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^{12} \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1 \overset{1}{\mathbf{e}}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^{11} \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1 \overset{1}{\mathbf{e}}_6 \overset{3,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{10} \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$31\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^9\overset{2,2}{\otimes}\mathfrak{so}1$$

$$31\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^8\overset{2,2}{\otimes}\mathfrak{so}1$$

$$31\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^7\overset{2,2}{\otimes}\mathfrak{so}1$$

$$31\mathfrak{e}_6(2,2) \otimes \mathfrak{e}_6(2,2) \otimes \mathfrak{so}(1)$$

$$31\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^5\overset{2,2}{\otimes}\mathfrak{so}1$$

$$31\mathbf{e}_6(\overset{2,2}{\otimes} \mathbf{e}_6)^{15} \overset{2,2}{\otimes} \mathfrak{so}1$$

$$31\mathbf{e}_6(2,2) \otimes \mathbf{e}_6^{14} \otimes \mathbf{so}1$$

$$31\mathbf{e}_6(2,2) \otimes \mathbf{e}_6^{13} \otimes \mathbf{so}1$$

$$31\mathfrak{e}_6^{31}(\otimes \mathfrak{e}_6)^{12} \otimes \mathfrak{so}1$$

$$31\mathfrak{e}_6(2,2) \otimes \mathfrak{e}_6^{11} \otimes \mathfrak{so}(1)$$

$$31\mathfrak{e}_6^{31}(\otimes \mathfrak{e}_6)^{10} \otimes \mathfrak{so}1$$

$$1e_7^{1,5,5}(\otimes e_8)^{10} \otimes so^{4,2} \otimes so^{1,1}$$

$$1e_7^{1(5,5)}(\otimes e_8)^{10} \otimes so_{13}^{4,2}$$

$$1e_7^{1(5,5)}(\otimes e_8)^9 \otimes^{4,2} so \otimes^{1,1} so1$$

$$1e_7^{(5,5)}(\otimes e_8)^9 \otimes so_{13}$$

$$1e_7 \binom{5,5}{\otimes} e_8 \binom{4,2}{\otimes} so \binom{1,1}{\otimes} so1$$

$$1e_7^{(5,5)}(\otimes e_8)^8 \otimes so_{13}^{4,2}$$

$$1e_7^{1(5,5)}(\otimes e_8)^{7(4,2)} \otimes so^{1,1} \otimes so1$$

$$1e_7^{1(5,5)}(\otimes e_8)^7 \otimes so_{13}^{4,2}$$

$$1e_7 \binom{1}{5,5} (\otimes e_8)^6 \binom{4,2}{1,1} \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1e_7^{(1)}(\otimes_{e_8}^{(5,5)})^6 \otimes_{so13}^{(4,2)}$$

$$1e_7(\overset{1}{\otimes} \overset{5,5}{e_8})^5 \overset{4,2}{\otimes} so \overset{1,1}{\otimes} so1$$

$$1e_7^{(1)}(\otimes e_8^{(5,5)})^5 \otimes so_{13}^{(4,2)}$$

$$1e_7(\otimes e_8)^4 \otimes^{4,2} so \otimes^{1,1} so1$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon'_7{}^{5,4} 12,$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon'_7{}^{5,5} 1$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon'_7{}^{5,4} 12,$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon'_7{}^{5,5} 1$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon'_7{}^{5,4} 12,$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon'_7{}^{5,5} 1$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon'_7{}^{5,4} 12,$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon'_7{}^{5,5} 1$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon'_7{}^{5,4} 12,$$

$$1\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon'_7{}^{5,5} 1$$

$$1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^{14} \otimes \epsilon_6^{5,3} 12,$$

$$1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^{14} \otimes \epsilon_6^{5,4} 1$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^9 \otimes \epsilon_6^{3,3} 131,$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^9 \otimes \epsilon_6^{3,3} 12$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^8 \otimes \epsilon_6^{3,3} 131,$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^8 \otimes \epsilon_6^{3,3} 12$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^7 \otimes \epsilon_6^{3,3} 131,$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^7 \otimes \epsilon_6^{3,3} 12$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^6 \otimes \epsilon_6^{3,3} 131,$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^6 \otimes \epsilon_6^{3,3} 12$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^5 \otimes \epsilon_6^{3,3} 131,$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^5 \otimes \epsilon_6^{3,3} 12$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^{15} \otimes \epsilon_6^{3,3} 131,$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^{15} \otimes \epsilon_6^{3,3} 12$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^{14} \otimes \epsilon_6^{3,3} 131,$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^{14} \otimes \epsilon_6^{3,3} 12$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^{13} \otimes \epsilon_6^{3,3} 131,$$

$$1\epsilon_6^{1,3,3} (\otimes \epsilon_7)^{13} \otimes \epsilon_6^{3,3} 12$$

$$1\epsilon_6^1(\otimes \epsilon_7)^{12} \otimes \epsilon_6^{3,3} 131,$$

$$1\epsilon_6^1(\otimes \epsilon_7)^{12} \otimes \epsilon_6^{3,3} 12$$

$$1\epsilon_6^1(\otimes \epsilon_7)^{11} \otimes \epsilon_6^{3,3} 131,$$

$$1\epsilon_6^1(\otimes \epsilon_7)^{11} \otimes \epsilon_6^{3,3} 12$$

$$1\epsilon_6^1(\otimes \epsilon_7)^{10} \otimes \epsilon_6^{3,3} 131,$$

$$1\epsilon_6^1(\otimes \epsilon_7)^{10} \otimes \epsilon_6^{3,3} 12$$

$$21\epsilon_7^1(\otimes \epsilon_7)^9 \otimes \mathfrak{so}^{3,2} 1$$

$$21\epsilon_7^1(\otimes \epsilon_7)^8 \otimes \mathfrak{so}^{3,2} 1$$

$$21\epsilon_7^1(\otimes \epsilon_7)^7 \otimes \mathfrak{so}^{3,2} 1$$

$$21\epsilon_7^1(\otimes \epsilon_7)^6 \otimes \mathfrak{so}^{3,2} 1$$

$$21\epsilon_7^1(\otimes \epsilon_7)^5 \otimes \epsilon_8^{4,5} 1$$

$$21\epsilon_7^1(\otimes \epsilon_7)^5 \otimes \mathfrak{so}^{3,2} 1$$

$$21\epsilon_7^1(\otimes \epsilon_7)^{15} \otimes \mathfrak{so}^{3,2} 1$$

$$21\epsilon_7^1(\otimes \epsilon_7)^{14} \otimes \mathfrak{so}^{3,2} 1$$

$$21\epsilon_7^1(\otimes \epsilon_7)^{13} \otimes \mathfrak{so}^{3,2} 1$$

$$21\epsilon_7^1(\otimes \epsilon_7)^{12} \otimes \mathfrak{so}^{3,2} 1$$

$$21\epsilon_7^1(\otimes \epsilon_7)^{11} \otimes \mathfrak{so}^{3,2} 1$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{10} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^4 \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{so}13$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{so}13$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{so}13$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{so}13$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{so}13$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so}13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^8 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^7 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^7 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^6 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^6 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^5 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^5 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{14} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{13} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{13} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^9 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^8 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^7 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^6 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^5 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{15} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{14} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{13} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{12} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{11} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{10} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^5 \otimes \mathfrak{e}_8^{3,5} 1$$

2.8.6 SSg(n)IS(1294)

$$1\mathbf{e}_7 \binom{5,5}{\otimes} \mathbf{e}_8 \binom{5,5}{\otimes} \mathbf{e}_8 I^{\oplus 2} 12231$$

$$\begin{aligned}
& 21\mathbf{c}_7^1 \otimes \mathbf{c}_8^{4,5} (\otimes \mathbf{c}_8^{5,5})^5 \otimes \mathbf{c}_8^{5,5} \mathbf{1} 12231513221 \\
& 21\mathbf{c}_7^1 \otimes \mathbf{c}_8^{4,5} (\otimes \mathbf{c}_8^{5,5})^5 \otimes \mathbf{c}_8^{5,5} \mathbf{1} 12231512 \\
& 21\mathbf{c}_7^1 \otimes \mathbf{c}_8^{4,5} (\otimes \mathbf{c}_8^{5,5})^5 \otimes \mathbf{c}_8^{5,5} I^{\oplus 2} \mathbf{1} 222 \\
& 1\mathbf{c}_7^1 (\otimes \mathbf{c}_8^{5,5})^6 \otimes \mathbf{c}_8^{5,5} \mathbf{1} 1223151321 \\
& 1\mathbf{c}_7^1 (\otimes \mathbf{c}_8^{5,5})^6 \otimes \mathbf{c}_8^{5,5} \mathbf{1} 12222 \\
& 1\mathbf{c}_7^1 (\otimes \mathbf{c}_8^{5,5})^6 \otimes \mathbf{c}_8^{5,5} I^{\oplus 2} \mathbf{1} 2231
\end{aligned}$$

$$\begin{aligned}
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} 12231513221 \\
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} 1\epsilon_8 12231512 \\
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} I^{\oplus 2} \epsilon_8 1222 \\
& 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} 1\epsilon_8 1223151321 \\
& 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} 1\epsilon_8 12222 \\
& 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} I^{\oplus 2} \epsilon_8 12231
\end{aligned}$$

$$\begin{aligned}
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} 12231513221 \\
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} 1\epsilon_8 12231512 \\
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} I^{\oplus 2} \epsilon_8 1222 \\
& 1\epsilon_7^1 (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} 1223151321 \\
& 1\epsilon_7^1 (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} 12222 \\
& 1\epsilon_7^1 (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} I^{\oplus 2} \epsilon_8 12231
\end{aligned}$$

$$\begin{aligned}
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} 12231513221 \\
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} 12231512 \\
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} I^{\oplus 2} 1222
\end{aligned}$$

[illegible]

$$\begin{array}{l} 221 \mathbf{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^9 \otimes \mathbf{e}_7^{I^{\oplus 3}} 1 \\ 221 \mathbf{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^9 \otimes \mathbf{e}_7^{I^{\oplus 2}} 12 \end{array}$$

$$\begin{array}{l} 221 \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^8 \otimes \mathbf{e}_7^{(3,3)} 1231 \\ 221 \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^8 \otimes \mathbf{e}_7^{(3,3)} I^{\oplus 3} 1 \\ 221 \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^8 \otimes \mathbf{e}_7^{(3,3)} I^{\oplus 2} 12 \end{array}$$

$$\begin{array}{l} 221 \mathbf{e}_7(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_7^{1231} \\ 221 \mathbf{e}_7(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_7^{I^{\oplus 3}} \\ 221 \mathbf{e}_7(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_7^{I^{\oplus 2}} \end{array}$$

$$\begin{array}{l} 221 \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^6 \otimes \mathbf{e}_7^{(3,3)} 1231 \\ 221 \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^6 \otimes \mathbf{e}_7^{(3,3)} I^{\oplus 3} 1 \\ 221 \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^6 \otimes \mathbf{e}_7^{(3,3)} I^{\oplus 2} 12 \end{array}$$

$$\begin{array}{l} 221 \epsilon_7^{(3,3)} \otimes \epsilon_7^{(3,3)} \otimes \epsilon_7^{(1)} 1231 \\ 221 \epsilon_7^{(3,3)} \otimes \epsilon_7^{(3,3)} \otimes \epsilon_7^{(1 \oplus 3)} 1 \\ 221 \epsilon_7^{(3,3)} \otimes \epsilon_7^{(3,3)} \otimes \epsilon_7^{(1 \oplus 2)} 12 \end{array}$$

$$\begin{array}{l} 221\epsilon_7^{(3,3)}(\otimes \epsilon_7)^{15} \otimes \epsilon_7^{(3,3)} 1231 \\ 221\epsilon_7^{(3,3)}(\otimes \epsilon_7)^{15} \otimes \epsilon_7^{(3,3)} I^{\oplus 3} 1 \\ 221\epsilon_7^{(3,3)}(\otimes \epsilon_7)^{15} \otimes \epsilon_7^{(3,3)} I^{\oplus 2} 12 \end{array}$$

$$221 \mathbf{e}_7^1 ({}^{3,3} \otimes \mathbf{e}_7)^{14} {}^{3,3} \otimes \mathbf{e}_7^1 1231$$

$$\begin{aligned} &12321\epsilon_7^{(1,3,3)}(\otimes \epsilon_7)^{14} \otimes \epsilon_7^{(3,3,1)} 12321, \\ &1321\epsilon_7^{(1,3,3)}(\otimes \epsilon_7)^{14} \otimes \epsilon_7^{(3,3)} I^{\oplus 2} 12, \\ &21\epsilon_7^{(21,3,3)}(\otimes \epsilon_7)^{14} \otimes \epsilon_7^{(3,3)} I^{\oplus 2} 12 \end{aligned}$$

$$\begin{aligned} & 221\mathbf{e}_7^{1(3,3)}(\otimes \mathbf{e}_7)^{14} \otimes \mathbf{e}_7^{3,3} 12321, \\ & 12321\mathbf{e}_7^{1(3,3)}(\otimes \mathbf{e}_7)^{14} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2} 12, \end{aligned}$$

$$\begin{array}{l} 221\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^{14} \otimes \mathbf{e}_7^{I \oplus 3} 1 \\ 221\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^{14} \otimes \mathbf{e}_7^{I \oplus 2} 12 \end{array}$$

$$\begin{aligned} &1321 \overset{1}{\mathbf{e}_7} (\overset{3,3}{\otimes} \mathbf{e}_7)^{13} \overset{3,3}{\otimes} \overset{1}{\mathbf{e}_7} 1231, \\ &12321 \overset{1}{\mathbf{e}_7} (\overset{3,3}{\otimes} \mathbf{e}_7)^{13} \overset{3,3}{\otimes} \overset{1}{\mathbf{e}_7} 12321, \\ &1321 \overset{1}{\mathbf{e}_7} (\overset{3,3}{\otimes} \mathbf{e}_7)^{13} \overset{3,3}{\otimes} \mathbf{e}_7 12, \\ &21 \overset{21}{\mathbf{e}_7} (\overset{3,3}{\otimes} \mathbf{e}_7)^{13} \overset{3,3}{\otimes} \mathbf{e}_7 12 \end{aligned}$$

$$\begin{aligned} &1321\epsilon_7^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}}\epsilon_7^{13\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}}12321, \\ &221\epsilon_7^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}}\epsilon_7^{13\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}}\epsilon_7^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}}12321, \\ &12321\epsilon_7^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}}\epsilon_7^{13\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}}\epsilon_7^{1\oplus 2\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}}12, \end{aligned}$$

$$\begin{array}{l} 221 \mathbf{e}_7^1 (\otimes \mathbf{e}_7)^{13} \otimes \mathbf{e}_7^{1231} \\ 221 \mathbf{e}_7^1 (\otimes \mathbf{e}_7)^{13} \otimes \mathbf{e}_7^{1 \oplus 3} \\ 221 \mathbf{e}_7^1 (\otimes \mathbf{e}_7)^{13} \otimes \mathbf{e}_7^{1 \oplus 2} \end{array}$$

$$\begin{aligned} &1321\epsilon_7^1(\otimes \epsilon_7)^{12} \otimes \epsilon_7^1 1231, \\ &12321\epsilon_7^1(\otimes \epsilon_7)^{12} \otimes \epsilon_7^1 12321, \\ &1321\epsilon_7^1(\otimes \epsilon_7)^{12} \otimes \epsilon_7^{I^{\oplus 2}} 12, \\ &21\epsilon_7^{21}(\otimes \epsilon_7)^{12} \otimes \epsilon_7^{I^{\oplus 2}} 12 \end{aligned}$$

$$\begin{aligned} &1321\epsilon_7^1(\otimes \epsilon_7)^{12} \otimes \epsilon_7^{3,3} 12321, \\ &221\epsilon_7^1(\otimes \epsilon_7)^{12} \otimes \epsilon_7^{3,3} 12321, \\ &12321\epsilon_7^1(\otimes \epsilon_7)^{12} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12, \end{aligned}$$

$$\begin{array}{l} 221\mathbf{e}_7^{(3,3)}\left(\otimes\mathbf{e}_7\right)^{12}\otimes\mathbf{e}_7^{3,3}1231 \\ 221\mathbf{e}_7^{(3,3)}\left(\otimes\mathbf{e}_7\right)^{12}\otimes\mathbf{e}_7^{3,3}I^{\oplus 3}1 \\ 221\mathbf{e}_7^{(3,3)}\left(\otimes\mathbf{e}_7\right)^{12}\otimes\mathbf{e}_7^{3,3}I^{\oplus 2}12 \end{array}$$

$$\begin{aligned} & 1321 \epsilon_7^1 (\otimes \epsilon_7)^{11} \otimes \epsilon_7^{3,3} 1231, \\ & 12321 \epsilon_7^1 (\otimes \epsilon_7)^{11} \otimes \epsilon_7^{3,3} 1712321, \\ & 1321 \epsilon_7^1 (\otimes \epsilon_7)^{11} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12, \\ & 21 \epsilon_7^{21} (\otimes \epsilon_7)^{3,3} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$\begin{aligned} &1321\epsilon_7^1(\otimes \epsilon_7)^{11} \otimes \epsilon_7^{3,3} 12321, \\ &221\epsilon_7^1(\otimes \epsilon_7)^{11} \otimes \epsilon_7^{3,3} 12321, \\ &12321\epsilon_7^1(\otimes \epsilon_7)^{11} \otimes \epsilon_7^{3,3} 12, \end{aligned}$$

$$\begin{array}{l} 221\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^{11} \otimes \mathbf{e}_7^{3,3} 1231 \\ 221\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^{11} \otimes \mathbf{e}_7^{3,3} 1^{\oplus 3} \\ 221\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^{11} \otimes \mathbf{e}_7^{3,3} 12^{\oplus 2} \end{array}$$

$$\begin{aligned} & 1321\epsilon_7^{(3,3)}(\otimes \epsilon_7)^{10} \otimes \epsilon_7^{(3,3)} 1231, \\ & 12321\epsilon_7^{(3,3)}(\otimes \epsilon_7)^{10} \otimes \epsilon_7^{(3,3)} 12321, \\ & 1321\epsilon_7^{(3,3)}(\otimes \epsilon_7)^{10} \otimes \epsilon_7^{(3,3)} I^{\oplus 2} 12, \\ & 21\epsilon_7^{(3,3)}(\otimes \epsilon_7)^{10} \otimes \epsilon_7^{(3,3)} I^{\oplus 2} 12 \end{aligned}$$

$$\begin{aligned} &1321\epsilon_7^1(\otimes \epsilon_7)^{10} \otimes \epsilon_7^{3,3} 12321, \\ &221\epsilon_7^1(\otimes \epsilon_7)^{10} \otimes \epsilon_7^{3,3} 12321, \\ &12321\epsilon_7^1(\otimes \epsilon_7)^{10} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12, \end{aligned}$$

$$\begin{array}{l} 221 \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^{10} \otimes \mathbf{e}_7^{3,3} 1231 \\ 221 \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^{10} \otimes \mathbf{e}_7^{3,3} 1^{I \oplus 3} \\ 221 \mathbf{e}_7^{(3,3)} (\otimes \mathbf{e}_7)^{10} \otimes \mathbf{e}_7^{3,3} 12^{I \oplus 2} \end{array}$$

$$\begin{aligned} & 1\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^{13} \otimes \mathbf{e}_8^{5,5} 122315131, \\ & 1\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^{13} \otimes \mathbf{e}_8^{5,5} 12222, \\ & 1\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^{13} \otimes \mathbf{e}_8^{5,5} I^{\oplus 2} 12231, \end{aligned}$$

$$\begin{aligned} & 1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{13} \otimes \mathbf{e}_8^{5,5} 1223151321, \\ & 1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{13} \otimes \mathbf{e}_8^{5,5} 12231512, \\ & 1\mathbf{e}'_7{}^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{13} \otimes \mathbf{e}_8^{5,5} I^{\oplus 2} 1222, \end{aligned}$$

$$\begin{array}{l} 1_{\mathcal{C}_7'}^{1,4,5} \otimes \mathfrak{c}_8^{(5,5)} \otimes \mathfrak{c}_8^{5,5} 122313 \\ 1_{\mathcal{C}_7'}^{1,4,5} \otimes \mathfrak{c}_8^{(5,5)} \otimes \mathfrak{c}_8^{5,5} 1223151 \\ 1_{\mathcal{C}_7'}^{1,4,5} \otimes \mathfrak{c}_8^{(5,5)} \otimes \mathfrak{c}_8^{5,5} I^{\oplus 3} 122 \end{array}$$

$$22221\mathfrak{e}_8^1(\overset{5,5}{\otimes}\mathfrak{e}_8)^{14}\overset{5,5}{\otimes}\mathfrak{e}_8^112222,$$

$$22221\mathbf{e}_8({}^{5,5}\bigotimes\mathbf{e}_8)^{14}{}^{5,5}I^{\oplus 2}\bigotimes\mathbf{e}_81222,$$

$$22221 \mathbf{e}_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \end{smallmatrix}^{14} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} I^{\oplus 3} \\ \otimes \end{smallmatrix} 122$$

$$2221\mathfrak{e}_8(\overset{5,5}{\otimes} \mathfrak{e}_8)^{14} \overset{5,5}{\otimes} \mathfrak{e}_8 I^{\oplus 2} 1222,$$

$$2221\mathbf{e}_8(\overset{5,5}{\otimes}\mathbf{e}_8)^{14}\overset{5,5}{\otimes}I^{\oplus 3}\mathbf{e}_8122,$$

$$221 \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^{14} \bigotimes \mathbf{e}_8^{5,5} I^{\oplus 3} 122$$

$$\begin{aligned} & 131\epsilon_6^{1(2,2)}(\otimes \epsilon_6)^9 \otimes \epsilon_6^{2,2} 131, \\ & 31\epsilon_6^{31(2,2)}(\otimes \epsilon_6)^9 \otimes \epsilon_6^{2,2} 131, \end{aligned}$$

$$\begin{array}{l} 21^1_6 ({}^{2,2} \otimes \mathbf{e}_6)^9 {}^{2,2} \otimes \mathbf{e}_6 131, \\ 31^{31}_6 ({}^{2,2} \otimes \mathbf{e}_6)^9 {}^{2,2} I^{\oplus 2} \mathbf{e}_6 1 \end{array}$$

$$21 \mathbf{e}_6 ({}^{2,2} \otimes \mathbf{e}_6)^9 {}^{2,2} I^{\oplus 2} \otimes \mathbf{e}_6 1$$

$$\begin{aligned} & 131\epsilon_6^{1(2,2)}(\otimes \epsilon_6)^8 \otimes \epsilon_6^{2,2} 131, \\ & 31\epsilon_6^{31(2,2)}(\otimes \epsilon_6)^8 \otimes \epsilon_6^{2,2} 131, \end{aligned}$$

$$\begin{array}{l} 21^1 \mathbf{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \right)^8 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} 1^1 131, \\ 31^{31} \mathbf{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \right)^8 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} I^{\oplus 2} 1 \end{array}$$

$$21^1 \mathbf{e}_6 ({}^{2,2} \otimes \mathbf{e}_6)^8 {}^{2,2} I^{\oplus 2} \otimes \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^{1(2,2)}(\otimes \epsilon_6)^7 \otimes \epsilon_6^{2,2} 131, \\ &31\epsilon_6^{31(2,2)}(\otimes \epsilon_6)^7 \otimes \epsilon_6^{2,2} 131, \end{aligned}$$

$$\begin{array}{l} 21^1 \epsilon_6^{(2,2)} \otimes \epsilon_6^{(2,2)} \otimes 131, \\ 31^{31} \epsilon_6^{(2,2)} \otimes \epsilon_6^{(2,2)} \otimes I^{\oplus 2} 1 \end{array}$$

$$21^1_6 ({}^{2,2} \otimes \mathbf{e}_6)^7 {}^{2,2} I^{\oplus 2} \otimes \mathbf{e}_6 1$$

$$\begin{aligned} & 131\epsilon_6^1(\otimes \epsilon_6)^6 \otimes \epsilon_6^{2,2} 131, \\ & 31\epsilon_6^{31}(\otimes \epsilon_6)^6 \otimes \epsilon_6^{2,2} 131, \end{aligned}$$

$$\begin{array}{l} 21^1 \mathbf{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \right)^6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} 1^1 131, \\ 31^{31} \mathbf{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \right)^6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} I^{\oplus 2} 1 \end{array}$$

$$21 \mathbf{e}_6^1 (\otimes \mathbf{e}_6^{2,2})^6 \otimes \mathbf{e}_6^{2,2} I^{\oplus 2} 1$$

$$\begin{aligned} &131\epsilon_6^{1(2,2)}(\otimes \epsilon_6)^5 \otimes \epsilon_6^{2,2} 131, \\ &31\epsilon_6^{31(2,2)}(\otimes \epsilon_6)^5 \otimes \epsilon_6^{2,2} 131, \end{aligned}$$

$$\begin{array}{l} 21^1_6 ({}^{2,2} \otimes \epsilon_6)^5 {}^{2,2} \otimes \epsilon_6 131, \\ 31^{31}_6 ({}^{2,2} \otimes \epsilon_6)^5 {}^{2,2} I^{\oplus 2} \epsilon_6 1 \end{array}$$

$$21 \mathbf{e}_6 \left(\bigotimes \mathbf{e}_6 \right)^5 \bigotimes \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^1(\otimes \epsilon_6)^{15} \otimes \epsilon_6^1 131, \\ &31\epsilon_6^{31}(\otimes \epsilon_6)^{15} \otimes \epsilon_6^1 131, \end{aligned}$$

$$\begin{array}{l} 21 \mathbf{e}_6^{1(2,2)} (\otimes \mathbf{e}_6)^{15} \otimes \mathbf{e}_6^{2,2} 131, \\ 31 \mathbf{e}_6^{31(2,2)} (\otimes \mathbf{e}_6)^{15} \otimes \mathbf{e}_6^{2,2} I^{\oplus 2} 1 \end{array}$$

$$21 \mathbf{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \right)^{15} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} I^{\oplus 2} \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^1(\otimes \epsilon_6^{2,2})^{14} \otimes \epsilon_6^1 131, \\ &31\epsilon_6^{31}(\otimes \epsilon_6^{2,2})^{14} \otimes \epsilon_6^1 131, \end{aligned}$$

$$\begin{array}{l} 21 \epsilon_6^{1(2,2)} (\otimes \epsilon_6)^{14} \otimes \epsilon_6^{2,2} 131, \\ 31 \epsilon_6^{31(2,2)} (\otimes \epsilon_6)^{14} \otimes \epsilon_6^{2,2} I^{\oplus 2} 1 \end{array}$$

$$21 \mathbf{e}_6 \left(\bigotimes \mathbf{e}_6 \right)^{14} \bigotimes \mathbf{e}_6 \mathbf{1}^{2,2} I^{\oplus 2}$$

$$\begin{aligned} &131\epsilon_6^1(\otimes^{2,2}\epsilon_6)^{13}\otimes^{2,2}\epsilon_6^1131, \\ &31\epsilon_6^{31}(\otimes^{2,2}\epsilon_6)^{13}\otimes^{2,2}\epsilon_6^1131, \end{aligned}$$

$$\begin{array}{l} 21^1 \epsilon_6 \left(\bigotimes \epsilon_6 \right)^{13} \bigotimes^{2,2} \epsilon_6 131, \\ 31^{31} \epsilon_6 \left(\bigotimes \epsilon_6 \right)^{13} \bigotimes^{2,2} I^{\oplus 2} \epsilon_6 1 \end{array}$$

$$21\mathbf{e}_6(\overset{2,2}{\otimes} \mathbf{e}_6)^{13} \overset{2,2}{\otimes} I^{\oplus 2} \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^1(\otimes \epsilon_6^{2,2})^{12} \otimes \epsilon_6^1 131, \\ &31\epsilon_6^{31}(\otimes \epsilon_6^{2,2})^{12} \otimes \epsilon_6^1 131, \end{aligned}$$

$$\begin{array}{l} 21 \mathbf{e}_6 \left(\bigotimes \mathbf{e}_6 \right)^{12} \bigotimes \mathbf{e}_6 131, \\ 31 \mathbf{e}_6 \left(\bigotimes \mathbf{e}_6 \right)^{12} \bigotimes \mathbf{e}_6 1 \end{array}$$

$$21\mathbf{e}_6(\overset{2,2}{\otimes} \mathbf{e}_6)^{12} \overset{2,2}{\otimes} I^{\oplus 2} \mathbf{e}_6 1$$

$$\begin{aligned} 131\epsilon_6^{1(2,2)11} \otimes \epsilon_6^{2,2} 131, \\ 31\epsilon_6^{31(2,2)11} \otimes \epsilon_6^{2,2} 131, \end{aligned}$$

$$\begin{aligned} 21\epsilon_6^{1(2,2)11} \otimes \epsilon_6^{2,2} 131, \\ 31\epsilon_6^{31(2,2)11} \otimes \epsilon_6^{2,2} I^{\oplus 2} 1 \end{aligned}$$

$$21\epsilon_6^{1(2,2)11} \otimes \epsilon_6^{2,2} I^{\oplus 2} 1$$

$$\begin{aligned} 131\epsilon_6^{1(2,2)10} \otimes \epsilon_6^{2,2} 131, \\ 31\epsilon_6^{31(2,2)10} \otimes \epsilon_6^{2,2} 131, \end{aligned}$$

$$\begin{aligned} 21\epsilon_6^{1(2,2)10} \otimes \epsilon_6^{2,2} 131, \\ 31\epsilon_6^{31(2,2)10} \otimes \epsilon_6^{2,2} I^{\oplus 2} 1 \end{aligned}$$

$$21\epsilon_6^{1(2,2)10} \otimes \epsilon_6^{2,2} I^{\oplus 2} 1$$

$$\begin{aligned} 21\epsilon_7^{1(3,3)9} \otimes \epsilon_7^{3,3} 1231, \\ 21\epsilon_7^{1(3,3)9} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)9} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)9} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} 21\epsilon_7^{1(3,3)8} \otimes \epsilon_7^{3,3} 1231, \\ 21\epsilon_7^{1(3,3)8} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)8} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)8} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} 21\epsilon_7^{1(3,3)7} \otimes \epsilon_7^{3,3} 1231, \\ 21\epsilon_7^{1(3,3)7} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)7} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)7} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} 21\epsilon_7^{1(3,3)6} \otimes \epsilon_7^{3,3} 1231, \\ 21\epsilon_7^{1(3,3)6} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)6} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)6} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} 21\epsilon_7^{1(3,3)5} \otimes \epsilon_7^{3,3} 1231, \\ 21\epsilon_7^{1(3,3)5} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)5} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)5} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} 21\epsilon_7^{1(3,3)15} \otimes \epsilon_7^{3,3} 1231, \\ 21\epsilon_7^{1(3,3)15} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)15} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)15} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} 21\epsilon_7^{1(3,3)14} \otimes \epsilon_7^{3,3} 1231, \\ 21\epsilon_7^{1(3,3)14} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)14} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)14} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$21 \overset{1}{e}_7' (\overset{3,3}{\otimes} \mathbf{e}_7)^{13} \overset{3,3}{\otimes} I^{\oplus 3} \mathbf{e}_7 1$$

$$21 \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{12} {}^{3,3} I^{\oplus 3} \otimes \mathbf{e}_7 1$$

$$21_{\mathbf{e}_7'}^1 ({}^{3,3} \otimes \mathbf{e}_7)^{11} {}^{3,3} I^{\oplus 3} \otimes \mathbf{e}_7 1$$

$$21 \overset{1}{e}_7' (\overset{3,3}{\otimes} \mathbf{e}_7)^{10} \overset{3,3}{\otimes} I^{\oplus 3} \mathbf{e}_7 1$$

$$1 \mathbf{e}_6 \begin{pmatrix} 1 & 3,3 \\ & \otimes \mathbf{e}_7 \end{pmatrix}^9 \begin{pmatrix} 3,3 & I^{\oplus 3} \\ & \otimes \mathbf{e}_7 \end{pmatrix} 1$$

$$1 \mathbf{e}_6 \begin{pmatrix} 1 & 3,3 \\ & \otimes \mathbf{e}_7 \end{pmatrix}^8 \begin{pmatrix} 3,3 & I^{\oplus 3} \\ & \otimes \mathbf{e}_7 \end{pmatrix} 1$$

$$1 \mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} I^{\oplus 3} \mathbf{e}_7 1$$

$$1 \mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} I^{\oplus 3} \mathbf{e}_7 1$$

$$1 \mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} I^{\oplus 3} \mathbf{e}_7 1$$

$$1 \mathbf{e}_6 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^{15} \overset{3,3}{\otimes} I^{\oplus 3} \mathbf{e}_7 1$$

$$\begin{array}{lll}
21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8)^{5,5} 1^3 \otimes \mathbf{e}_7^{5,4} I^{\oplus 2} 1, & 21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8)^{5,5} 1^3 \otimes \mathbf{e}_7^{5,5} 1, & 1 \mathbf{e}_7^1 (\otimes \mathbf{e}_8)^{5,5} 1^4 \otimes \mathbf{e}_7^{5,5} 1 \\
21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8)^{5,5} 1^2 \otimes \mathbf{e}_7^{5,4} I^{\oplus 2} 1, & 21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8)^{5,5} 1^2 \otimes \mathbf{e}_7^{5,5} 1, & 1 \mathbf{e}_7^1 (\otimes \mathbf{e}_8)^{5,5} 1^3 \otimes \mathbf{e}_7^{5,5} 1 \\
21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8)^{5,5} 1^1 \otimes \mathbf{e}_7^{5,4} I^{\oplus 2} 1, & 21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8)^{5,5} 1^1 \otimes \mathbf{e}_7^{5,5} 1, & 1 \mathbf{e}_7^1 (\otimes \mathbf{e}_8)^{5,5} 1^2 \otimes \mathbf{e}_7^{5,5} 1 \\
21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8)^{5,5} 1^0 \otimes \mathbf{e}_7^{5,4} I^{\oplus 2} 1, & 21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8)^{5,5} 1^0 \otimes \mathbf{e}_7^{5,5} 1, & 1 \mathbf{e}_7^1 (\otimes \mathbf{e}_8)^{5,5} 1^1 \otimes \mathbf{e}_7^{5,5} 1
\end{array}$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \left(\otimes \mathbf{e}_8 \right)^9 \otimes \mathbf{e}_7^{\oplus 2} 1, \quad 1\mathbf{e}'_7 \otimes \mathbf{e}_8 \left(\otimes \mathbf{e}_8 \right)^9 \otimes \mathbf{e}_7^{\oplus 1} 1$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^8 \otimes \mathbf{e}_7^{\oplus 2} 1, \quad 1\mathbf{e}'_7 \otimes \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^8 \otimes \mathbf{e}_7^{\oplus 1} 1$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} I^{\oplus 2} \mathbf{e}_7 1, \quad 1\mathbf{e}'_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_7 1$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \left(\otimes \mathbf{e}_8 \right)^{5,5} \otimes \mathbf{e}_7 \mathbf{1}, \quad 1\mathbf{e}'_7 \otimes \mathbf{e}_8 \left(\otimes \mathbf{e}_8 \right)^{5,5} \otimes \mathbf{e}_7 \mathbf{1}$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8^5 \otimes \mathbf{e}_7 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} I^{\oplus 2} 1, \quad 1\mathbf{e}'_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8^5 \otimes \mathbf{e}_7 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} 1$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^4 \otimes \mathbf{e}_7^{5,4} I^{\oplus 2} 1, \quad 1\mathbf{e}'_7 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^4 \otimes \mathbf{e}_7^{5,5} 1$$

$$1\epsilon_7' \otimes \epsilon_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \epsilon_8 \begin{smallmatrix} 13 & 5,4 \\ \otimes & \end{smallmatrix} I^{\oplus 2} 1, \quad 1\epsilon_7' \otimes \epsilon_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \epsilon_8 \begin{smallmatrix} 13 & 5,5 \\ \otimes & \end{smallmatrix} \epsilon_7 1$$

$$1\epsilon_7' \otimes \epsilon_8^{(4,5)} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{(12)} \otimes \epsilon_7^{(5,4)} I^{\oplus 2} 1, \quad 1\epsilon_7' \otimes \epsilon_8^{(4,5)} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{(12)} \otimes \epsilon_7^{(5,5)} 1$$

$$1\epsilon_7' \otimes \epsilon_8^{(4,5)} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{(11)} \otimes \epsilon_7^{(5,4)} I^{\oplus 2} 1, \quad 1\epsilon_7' \otimes \epsilon_8^{(4,5)} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{(11)} \otimes \epsilon_7^{(5,5)} 1$$

$$1_{\mathbf{e}'_7} \otimes \mathbf{e}_8^{(4,5)} (\otimes \mathbf{e}_8^{(5,5)})^{10} \otimes \mathbf{e}_7^{(5,4)} I^{\oplus 2} 1, \quad 1_{\mathbf{e}'_7} \otimes \mathbf{e}_8^{(4,5)} (\otimes \mathbf{e}_8^{(5,5)})^{10} \otimes \mathbf{e}_7^{(5,5)} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^8 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1 \mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^7 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^6 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^5 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1 \overset{1}{\mathbf{e}}_6 \otimes \overset{3,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{13} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}}_7 1$$

$$1 \overset{1}{\mathbf{e}}_6 \otimes \overset{3,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{12} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}}_7 1$$

$$1 \overset{1}{\mathbf{e}}_6 \otimes \overset{3,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{11} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}}_7 1$$

$$1 \overset{1}{\mathbf{e}}_6 \otimes \overset{3,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{10} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}}_7 1$$

$$21 \mathbf{e}'_7 \begin{pmatrix} 1 & 3,3 \\ \otimes & \mathbf{e}_7 \end{pmatrix}^8 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} I^{\oplus 2} \mathbf{e}'_7 1$$

$$21 \mathbf{e}'_7 ({}^{3,3} \otimes \mathbf{e}_7)^6 {}^{3,3} I^{\oplus 2} \otimes \mathbf{e}'_7 1$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^5 \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{15} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{14} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{13} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{12} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{11} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{10} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^9 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^8 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^7 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^6 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^5 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^{15} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^{14} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{13}\overset{1,1}{\otimes}\mathfrak{so}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{12}\overset{1,1}{\otimes}\mathfrak{so}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{11}\overset{1,1}{\otimes}\mathfrak{so}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{10}\overset{1,1}{\otimes}\mathfrak{so}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^9\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^7\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^6\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^5\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^4\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^{13}\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^{11}\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^{10}\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^8 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^8 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^7 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^7 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^6 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^6 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{14} \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{13} \otimes \epsilon_7'^{5,4} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{13} \otimes \mathfrak{e}_6^{5,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{e}_7^{5,4} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{e}_6^{5,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{e}_7^{5,4} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{e}_6^{5,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{e}_7^{5,4} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{e}_6^{5,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^9 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^9 \otimes \mathfrak{e}_6^{3,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^8 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^8 \otimes \mathfrak{e}_6^{3,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^7 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^7 \otimes \mathfrak{e}_6^{3,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^6 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^6\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^5\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^5\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{15}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{15}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{14}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{14}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{13}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{13}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{12}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{12}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{11}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{11}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{10}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{10}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

2.8.7 $\mathfrak{gLIg}(n-1)(3)$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^5 \otimes \mathfrak{e}'_7$$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^5 \otimes \mathfrak{e}_6$$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^5 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

2.8.8 $\mathfrak{gLIgLg}(n-4)\mathfrak{LIgLg}(1)$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 2,4 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^4 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8$$

2.8.9 $\mathfrak{SgLIg}(n-1)(26)$

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^3 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 2,4 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^4 \otimes \mathfrak{so}$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^4 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 2,4 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^5 \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 2,4 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^9 \otimes \mathfrak{e}_6$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 2,4 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^9 \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 2,4 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^8 \otimes \mathfrak{e}_6$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 2,4 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^8 \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 2,4 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^7 \otimes \mathfrak{e}_6$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 2,4 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^7 \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\begin{smallmatrix} 2,4 & 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^6 \otimes \mathfrak{e}_6$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^6 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^5 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^4 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{14} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{14} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{13} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{13} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{12} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{12} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{11} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{11} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{10} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{10} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8^{1,5,5})^4 \otimes \mathfrak{e}_7'^{5,5}$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8^{1,5,5})^4 \otimes \mathfrak{e}_6^{5,4}$$

2.8.10 SgLIg(n-3)LIgLg(1)

$$1\mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1(4,2)} \otimes \mathfrak{so}$$

2.8.11 SgLIg(n-3)LIgLgS(11)

$$150 \otimes_{\mathfrak{e}_8}^{2,4} (\otimes_{\mathfrak{e}_8}^{5,5})^9 \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{1,4,2} 501$$

$$150 \otimes_{\mathfrak{e}_8}^{2,4} (\otimes_{\mathfrak{e}_8}^{5,5})^8 \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{1,4,2} 501$$

$$150 \otimes_{\mathfrak{e}_8}^{2,4} (\otimes_{\mathfrak{e}_8}^{5,5})^7 \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{1,4,2} 501$$

$$150 \otimes_{\mathfrak{e}_8}^{2,4} \mathfrak{e}_8 \left(\otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 \right)^6 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 \otimes_{\mathfrak{e}_8}^{4,2} 501$$

$$150 \otimes_{\mathbb{C}} \mathfrak{e}_8 \left(\otimes_{\mathbb{C}} \mathfrak{e}_8 \right)^5 \otimes_{\mathbb{C}} \mathfrak{e}_8 \otimes_{\mathbb{C}} 501$$

$$150 \otimes_{\mathbb{C}} e_8 \left(\otimes_{\mathbb{C}} e_8 \right)^4 \otimes_{\mathbb{C}} e_8 \otimes_{\mathbb{C}} 501$$

$$150 \otimes^{2,4} e_8 \otimes^{1,5,5} (e_8 \otimes e_8)^3 \otimes^{5,5} e_8 \otimes^{1,4,2} 501$$

$$150 \otimes^{2,4} e_8 \otimes^{5,5} (e_8)^{13} \otimes^{5,5} e_8 \otimes^{4,2} 501$$

$$150 \otimes^{2,4} e_8 \otimes^{5,5} (e_8)^{12} \otimes^{5,5} e_8 \otimes^{4,2} 501$$

$$1s0 \otimes^{2,4} e_8 \otimes^{5,5} (e_8)^{11} \otimes^{5,5} e_8 \otimes^{4,2} s01$$

$$150 \otimes^{2,4} e_8 \otimes^{1,5,5} (e_8 \otimes e_8)^{10} \otimes^{5,5} e_8 \otimes^{4,2} 501$$

2.8.12 SSgLIg(n-1)(1)

$$1_{\mathfrak{e}_8}^{1,5,5} \otimes \mathfrak{e}_8^{1,(5,5)} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

2.8.13 SSgLIg(n-1)S(1)

$$1\mathfrak{e}_8 \overset{1}{\otimes} \overset{5,5}{\mathfrak{e}_8} \overset{1}{\otimes} \overset{5,5}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^4 \overset{4,2}{\otimes} \mathfrak{so}1$$

2.8.14 SSg(n-2)LIgLg(6)

$$\begin{array}{c} 2,4 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \quad \otimes \mathfrak{e}_8 \quad \otimes \mathfrak{e}_8, \end{array}$$

$$I^{\oplus 4} \mathbf{e}_8 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^5 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{1}{\otimes} \overset{5,5}{\otimes} \mathbf{e}_8,$$

$$I^{\oplus 3} I^{\oplus 2} \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^5 \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix} \begin{pmatrix} 1 \\ \mathfrak{e}_8 \end{pmatrix} \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}$$

$$1513221 \mathbf{e}_8^1 (\otimes^{5,5} \mathbf{e}_8)^4 \otimes^{5,5} \mathbf{e}_8^1 \otimes^{5,5} \mathbf{e}_8,$$

$$I^{\oplus 2} 3221 \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix}^4 \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 1 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix}$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^4 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} \overset{5,5}{e_8}$$

2.8.15 SSg(n-2)LIgLgS(148)

$$131513221 \epsilon_8^{(5,5)} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{(4,2)} \otimes \mathfrak{so}1, \quad 1231513221 \epsilon_8^{(5,5)} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{(4,2)} \otimes \mathfrak{so}1, \quad 12231513221 \epsilon_8^{(5,5)} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{(4,2)} \otimes \mathfrak{so}1$$

$$313221\epsilon_8(\otimes \epsilon_8)^9 \otimes \epsilon_8 \otimes \mathfrak{so}1, \quad 22221\epsilon_8(\otimes \epsilon_8)^9 \otimes \epsilon_8 \otimes \mathfrak{so}1, \quad 21513221\epsilon_8(\otimes \epsilon_8)^9 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1513221 \mathbf{c}_8 (\otimes \mathbf{c}_8)^9 \otimes \mathbf{c}_8 \otimes \mathbf{so}1, \quad 21^{13221} \mathbf{c}_8 (\otimes \mathbf{c}_8)^9 \otimes \mathbf{c}_8 \otimes \mathbf{so}1, \quad 2221^{21} \mathbf{c}_8 (\otimes \mathbf{c}_8)^9 \otimes \mathbf{c}_8 \otimes \mathbf{so}1$$

$$221^{221} \mathfrak{e}_8 \left(\bigotimes \mathfrak{e}_8 \right)^9 \bigotimes \mathfrak{e}_8 \bigotimes \mathfrak{so} 1$$

$$131513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^8 \epsilon_8^{5,5,1} \otimes \epsilon_8^{4,2} \otimes \mathfrak{so}1, \quad 1231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^8 \otimes \epsilon_8^{5,5,1} \otimes \epsilon_8^{4,2} \otimes \mathfrak{so}1, \quad 12231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^8 \epsilon_8^{5,5,1} \otimes \epsilon_8^{4,2} \otimes \mathfrak{so}1$$

$$313221\epsilon_8^{1\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix}}\epsilon_8^8\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix}\epsilon_8^1\begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix}\mathfrak{so}1, \quad 22221\epsilon_8^{1\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix}}\epsilon_8^8\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix}\epsilon_8^1\begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix}\mathfrak{so}1, \quad 21513221\epsilon_8^{1\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix}}\epsilon_8^8\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix}\epsilon_8^1\begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix}\mathfrak{so}1$$

$$1513221 \otimes_8^{5,5} \otimes_8^{5,5} \otimes_8^{1,4,2} \otimes \mathfrak{so}1, \quad 21^{13221} \otimes_8^{5,5} \otimes_8^{5,5} \otimes_8^{1,4,2} \otimes \mathfrak{so}1, \quad 2221 \otimes_8^{21,5,5} \otimes_8^{5,5} \otimes_8^{1,4,2} \otimes \mathfrak{so}1$$

$$221 \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^8 {}^{5,5} \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1$$

$$131513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^7 \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1, \quad 1231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^7 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1, \quad 12231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^7 \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$313221\epsilon_8^{1\ 5,5}(\otimes\epsilon_8)^7\otimes\epsilon_8^{5,5}\otimes\epsilon_8^{1\ 4,2}\otimes\mathfrak{so}1, \quad 22221\epsilon_8^{1\ 5,5}(\otimes\epsilon_8)^7\otimes\epsilon_8^{5,5}\otimes\epsilon_8^{1\ 4,2}\otimes\mathfrak{so}1, \quad 21513221\epsilon_8^{1\ 5,5}(\otimes\epsilon_8)^7\otimes\epsilon_8^{5,5}\otimes\epsilon_8^{1\ 4,2}\otimes\mathfrak{so}1$$

$$1513221 \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}1, \quad 21^{13221} \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}1, \quad 2221^{21} \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}1$$

$$221 \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^7 {}^{5,5} \otimes \mathfrak{e}_8 \otimes \mathfrak{so} 1$$

$$131513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^6 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1, \quad 1231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^6 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1, \quad 12231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^6 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$313221\epsilon_8^{1(5,5)}(\otimes \epsilon_8)^6 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(4,2)} \otimes \mathfrak{so}1, \quad 22221\epsilon_8^{1(5,5)}(\otimes \epsilon_8)^6 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(4,2)} \otimes \mathfrak{so}1, \quad 21513221\epsilon_8^{1(5,5)}(\otimes \epsilon_8)^6 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(4,2)} \otimes \mathfrak{so}1$$

$${}^1_{15}13221\epsilon_8(\otimes \epsilon_8)^6 \overset{5,5}{\otimes} \overset{5,5}{\epsilon_8} \overset{1}{\otimes} \overset{4,2}{\epsilon_8} \otimes so1, \quad {}^{21}_{13}221\epsilon_8(\otimes \epsilon_8)^6 \overset{5,5}{\otimes} \overset{5,5}{\epsilon_8} \overset{1}{\otimes} \overset{4,2}{\epsilon_8} \otimes so1, \quad {}^{22}_{21}1\epsilon_8(\otimes \epsilon_8)^6 \overset{5,5}{\otimes} \overset{5,5}{\epsilon_8} \overset{1}{\otimes} \overset{4,2}{\epsilon_8} \otimes so1$$

$$\begin{array}{lll} 1513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \mathfrak{so}1, & 21^{13221} \epsilon_8^{5,5} (\otimes \epsilon_8)^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \mathfrak{so}1, & 2221 \epsilon_8^{21,5,5} (\otimes \epsilon_8)^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \mathfrak{so}1 \\ 221^{221} \epsilon_8^{5,5} (\otimes \epsilon_8)^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \mathfrak{so}1 & & \end{array}$$

$$13221\epsilon_8^{1(5,5)}(\otimes \epsilon_8)^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(5,5)} \otimes \epsilon_8 1, \quad 2221\epsilon_8^{1(5,5)}(\otimes \epsilon_8)^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(5,5)} \otimes \epsilon_8 1, \quad 221\epsilon_8^{21(5,5)}(\otimes \epsilon_8)^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(5,5)} \otimes \epsilon_8 1$$

$$21e_7 \otimes e_8^{4,5} (\otimes e_8^{5,5})^9 \otimes e_8^{5,5} \otimes e_8^{1,4,2} \otimes \mathfrak{so}l, \quad 1e_7 (\otimes e_8^{5,5})^{10} \otimes e_8^{5,5} \otimes e_8^{1,4,2} \otimes \mathfrak{so}l$$

$$21e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^8 \otimes e_8^{5,5} \otimes e_8^1 \otimes so1^{4,2}, \quad 1e_7^1 (\otimes e_8^{5,5})^9 \otimes e_8^{5,5} \otimes e_8^1 \otimes so1^{4,2}$$

$$21e_7^{1,4,5} \otimes e_8^{(5,5)} \otimes e_8^{5,5,1,4,2} \otimes \mathfrak{so}l, \quad 1e_7^{(5,5)} \otimes e_8^{5,5,1,4,2} \otimes \mathfrak{so}l$$

$$21\mathbf{e}_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 & 1 & 4,2 \\ \otimes \otimes \otimes \end{smallmatrix} \mathbf{sol}, \quad 1\mathbf{e}_7 \begin{smallmatrix} 1 & 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 & 1 & 4,2 \\ \otimes \otimes \otimes \end{smallmatrix} \mathbf{sol}$$

$$21e_7^{1,4,5} \otimes e_8^{(5,5)} \otimes e_8^{5,5} \otimes e_8^{1,4,2} \otimes \mathfrak{so}l, \quad 1e_7^{(5,5)} \otimes e_8^{5,5} \otimes e_8^{1,4,2} \otimes \mathfrak{so}l$$

$$21\mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8 \otimes \mathbf{so}1, \quad 1\mathbf{e}_7 \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_8 \otimes \mathbf{so}1$$

$$21e_7 \otimes e_8 \otimes e_8^3 \otimes e_8 \otimes \text{sol}, \quad 1e_7 \otimes e_8^4 \otimes e_8 \otimes \text{sol}$$

$$21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{13} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}l^{4,2}, \quad 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^{14} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}l^{4,2}$$

$$21\mathbf{e}_7 \overset{1}{\otimes} \overset{4,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^{12} \overset{5,5}{\otimes} \mathbf{e}_8 \overset{1}{\otimes} \overset{4,2}{\mathbf{so}1}, \quad 1\mathbf{e}_7 (\overset{5,5}{\otimes} \mathbf{e}_8)^{13} \overset{5,5}{\otimes} \mathbf{e}_8 \overset{1}{\otimes} \overset{4,2}{\mathbf{so}1}$$

$$21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}1^{4,2}, \quad 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}1^{4,2}$$

$$21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}l^{4,2}, \quad 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}l^{4,2}$$

$$1e_7' \otimes^{1,4,5} e_8 (\otimes^{5,5} e_8)^9 \otimes^{5,5} e_8 \otimes^{1,4,2} so1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^8 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^7 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^6 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^5 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^4 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^3 \otimes \epsilon_8 \otimes \epsilon_8 1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^3 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^{13} \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^{12} \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^{11} \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon'_7 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^{10} \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^9 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^8 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \left(\otimes \epsilon_8 \right)^7 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^6 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^3 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{13} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

2.8.16 Sg(n-2)LlgLg(15)

$$\otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5},$$

$$\otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5},$$

$$\otimes \epsilon_8^{5,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$313221\epsilon_8 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5},$$

$$I^{\oplus 5} \epsilon_8 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5},$$

$$21513221\epsilon_8 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$1513221\epsilon_8 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$I^{\oplus 2} \epsilon_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5},$$

$$1\epsilon_7 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$5131513221\epsilon_8 (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$321\epsilon_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^3 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$31\epsilon_6 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^3 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$\begin{array}{ll}
1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^9 \otimes \mathbf{e}'_7 12, & 1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^9 \otimes \mathbf{e}'_7 1 \\
1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^8 \otimes \mathbf{e}'_7 12, & 1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^8 \otimes \mathbf{e}'_7 1 \\
1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}'_7 12, & 1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}'_7 1 \\
1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^6 \otimes \mathbf{e}'_7 12, & 1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^6 \otimes \mathbf{e}'_7 1 \\
1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^5 \otimes \mathbf{e}'_7 12, & 1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^5 \otimes \mathbf{e}'_7 1 \\
1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}'_7 12, & 1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}'_7 1 \\
1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^{14} \otimes \mathbf{e}'_7 12, & 1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^{14} \otimes \mathbf{e}'_7 1 \\
1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^{13} \otimes \mathbf{e}'_7 12, & 1\mathbf{50} \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^{13} \otimes \mathbf{e}'_7 1
\end{array}$$

$$150 \otimes^{2,4} e_8 \otimes^1 (e_8 \otimes^{5,5} e_8)^{12} \otimes^{5,5} e_7' 1$$

$$150 \otimes e_8^{2,4} (\otimes e_8^{5,5})^{11} \otimes e_7^{5,5} 1$$

$$150 \otimes e_8^{2,4} (\otimes e_8^{5,5})^{10} \otimes e_7^{5,5} 1$$

$$1\mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{1}{\otimes} \overset{5,5}{\otimes} \mathbf{e}_8 \overset{4}{\otimes} \overset{5,5}{\otimes} \mathbf{e}_7 1$$

$$150 \otimes^{\substack{2,4 \\ 1}} e_8 \left(\otimes^{\substack{5,5 \\ 1}} e_8 \right)^5 \otimes^{\substack{4,2 \\ 1}} so13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^4 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^4 \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^3 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{14} \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{13} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{13} \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{12} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{12} \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{11} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{11} \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{10} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{10} \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8^1)^{5,5}{}^4 \otimes \mathfrak{e}_7'^{5,4}1$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8^1)^{5,5}{}^4 \otimes \mathfrak{e}_6^{5,3}1$$